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The relative chronology of some major Latin sound changes

Parker, Holt Neumon, Ph.D.

Yale University, 1986

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The Relative Chronology of Some Major Latin Sound Changes

A Dissertation
Presented to the Faculty of the Graduate School
of
Yale University
in Candidacy for the Degree of
Doctor of Philosophy

by
Holt Neumon Parker
December 1986

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ABSTRACT

THE RELATIVE CHRONOLOGY OF SOME MAJOR LATIN SOUND CHANGES

Holt Neumon Parker

Yale University

1986

This dissertation attempts a historical ordering of various Latin phonological rules. The starting point is the phonemic split of l into bright and dark allophones. Ch. 1 assembles the evidence for their phonetic quality. Ch. 2 concerns the assimilations of ld, ln, ls and lv to ll. Ch. 3 examines the sources for surface ld, ln, ls and lv, in particular the case of helvus. Ch. 4 and 5 treat consonant clusters: rules of loss of stops before sC in Latin, and the loss of nasals with compensatory lengthening before spirants with parallels in Osco-Umbrian and Celtic (Ch. 4); the loss of liquids (Ch. 5). Part II deals with changes in vowels. Changes in short vowels in initial syllables (Ch. 6). Changes of diphthongs in initial syllables (Ch. 7). Ch. 8 presents the weakening of vowels in medial syllables as an ordered series of rules. Ch. 9 details the process of R-Lowering (i > e, o > u before r in an open syllable) as a single rule. An excursus on dereduplication of the perfects of compound verbs, and a second on mollis and the adjectives in -ui- complete the work.

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PATRI ET MATRI

ATQUE

DIS MANIBUS

MATRIS ILLIUS ET PATRIS HUIUS

JOSEPHINE PARKER

FRITZ FLY

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Preface

> Traditionally, to approach questions of relative chronology is regarded as some kind of a supplementary refinement of the main task, which is conceived as the reconstruction of a proto-language and of a list of innovations that have converted it into its descendant languages. But in fact it is completely impossible to do historical linguistics without taking account of relative chronology, as the correct form of the innovations critically depends on the order in which they applied. (Kiparsky 1965.57).

1) The sound changes that led from Proto-Indo-European to Classical Latin are now fairly well known and codified, due to the researches of generations of previous scholars. However, a large problem still remains to be considered and that is in what order did these laws apply, in other words an investigation of the process of change in the phonology of Latin as a historical system. We are currently at a state in Latin studies which Arlotto (1972.69) describes in discussing the place of relative chronology in historical linguistics:

> We do not always have texts adequately spaced in time to establish the order in which specific changes occurred or all the intermediate steps between two stages of a language ... An even more obvious case is when our starting point is a reconstructed language and we are faced with tracing its history to one of its daughter languages. In many instances, there will be no way to tell us in what order the given changes have occurred. The observed shifts are then listed in some convenient or logical order, grouped together as consonants, vowels, glides, etc.

For Latin, and indeed for most of Indo-European, the listing has been made but the ordering has been at best unsystematically attempted.

2) There exist many handbooks which present the sound changes of Latin in the manner Arlotto describes. In particular the modern scholar is greatly aided by the work of Manu Leumann (1977), which provides the most complete treatment of Latin phonology and morphology to date, as well as

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many pertinent chronological observations. Yet the handbook format, with its phoneme by phoneme listing, prevents many significant generalizations from being drawn. The necessity of organizing the data one rule at a time obscures the fact that, as soon as we consider any individual form to which two or more sound laws have applied, the question must be raised of the order of their application.

3) The Method of Relative Chronology.

> We know that every language is a product of history, an end point in a series of changes ... All changes occur in absolute historical order, whether we can observe them or not. Thus any two changes in a language have occurred one after the other, in partial overlap (i.e., one change begins before another has ended), or simultaneously (complete overlap) ... When the output of one change is the input of another, we can establish relative chronology between them, even when we cannot tell their exact dates. The establishment of relative chronology is one of the prime goals of historical linguistics (Anttila 1972.109).

In building a relative chronology, one therefore has three things to work with: 1) a starting point, that is the reconstructed parent language Proto-Indo-European, 2) a goal, that is Classical Latin, the standard post-Plautine literary language of Rome, and 3) the rules themselves. The method for establishing a relative chronology then appears simple: we find words where two or more sound changes have interacted on a PIE form, apply them in their possible orders to the protoform, and check the results against the attested forms to see which order or orders give the correct results. Kiparsky (1965.58) restates the method thus:

> The phonological changes that a language has undergone between stages A and B should, when applied in the order of their relative chronology to the phonetic forms of stage A, yield stage B.

4) However, the method is complicated by the fact that neither the starting point nor the rules themselves, and occasionally even the endpoint,

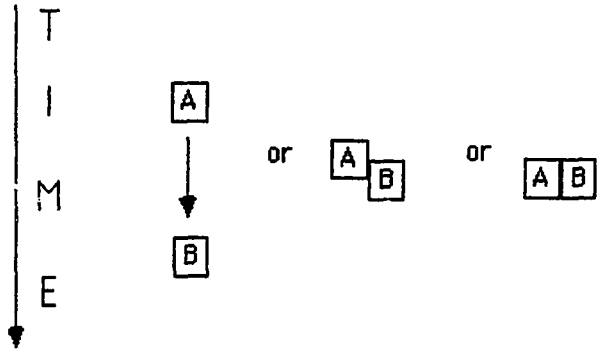
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is unproblematic. Clearly assumptions as to the nature of the PIE form of a Latin word will determine the view held of the changes that led to it, and the reverse is also true. Thus the argumentation of a relative chronology may at times seem distressingly discursive. It may be necessary to establish first the correct protoform of the Latin word under discussion, when this is not agreed on, and this will involve consideration of the relevant forms in the other daughter languages together with any problems or controversies in their phonology. In returning to the Latin word, it may be necessary to establish the sound laws it has undergone by examining their occurrence in other Latin words, together with their protoforms and phonological difficulties. It is only after these points have been discussed that the actual problem of the relative ordering of the rules can be undertaken. In general, in the discussion that follows I have not multiplied examples when the sound change is well-known or agreed on. Listing all the words to which a particularly frequent rule applied, even if possible, would not be directly useful for establishing a relative chronology. However, when the change is disputed or is attested by only a few examples, I have tried to be exhaustive in listing the evidence. Naturally, since a relative chronology depends on the overlapping of rules in individual forms, certain single words will be of greater importance and bear the brunt of closer scrutiny than words which illustrate only a single rule.

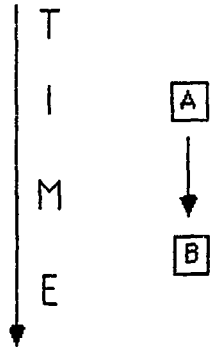
5) There is an additional important observation about Anttila's description of a relative chronology. The fact that the output of one change is the input of another does not allow us to say which of the three situations in real time which he outlines was the one that occurred: one change after the other (either A preceding B or B preceding A), partial overlap or complete overlap. That is, when Rule A feeds (provides the input

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for) Rule B, we can only say that Rule A occurred before or at the same time as Rule B. Graphically a feeding relation can be either:



However, when Rule A bleeds (changes forms which otherwise would have undergone the change of) Rule B, Rule A must be, in an absolute sense, before Rule B, since a change of Rule must be completely over by the time Rule B applies. An example might be the change of oi to ei between l and a labial. This change removes material for the change of oi to ū, and must be completely over with by the time the change of oi to ū occurred since otherwise the form *kloiuos would have become **clūvus and not the correct clīvus. That is, a bleeding relationship can only be of the form:



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though of course in the absence of datable attestations we cannot say how big the space between the two changes might have been. Thus "A feeds B" and "A bleeds B" both mean that A was in some sense "before" B, but while "A feeds B" means that A was over with before, or began before, or began at the same time as B, i.e. chronologically A is greater than or equal to (≥) B, the statement "A bleeds B" means only that A was completely over before B began, i.e. that A > B. Despite the obvious importance of the difference in the claims being made, and the frequent application of the notions of feeding and bleeding in generative phonology, the distinction is often overlooked in discussions of relative chronology. I have tried to be explicit in the text as to what type of relation exists between the rules under discussion, though I have not found it practical in the charts that illustrate the orderings of various rules to have a separate sign for rules related by feeding (before or simultaneous) and those related by bleeding (absolutely before).

6) In this study the sound changes and their orderings are written for the most part using the terminology and notational conventions of generative phonology. A specific model is Sommerstein's Sound Pattern of Ancient Greek (1973). Rules when established are given a convenient name, usually a mnemonic abbreviation (e.g. the change of e to i and o to u before the velar nasal ŋ is called **Rais-iNG**). I have printed these names in **bold** type to facilitate finding them in the text and to separate major rules which are used as structural points in building the relative chronology from minor rules not so used.

7) The use of a generative description necessitates a caveat to the reader. A relative chronology is of course a diachronic ordering, which may be recapitulated in the synchronic description of a language's phonology but

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is distinct from it. The diachronic use of a generative model differs from the synchronic use in both its starting point and in its conception of the rules. The beginning of both is a construct, but in the case of a relative chronology it is a construct arrived at by the comparative method (Birnbaum 1977.6-7 confuses relative chronology with internal reconstruction), which is "characterized by its precedence in time, [but] in transformational generative linguistics by its precedence in the generative process" (Bynon 1977.71). In a relative chronology the rules are conceived of as "historical changes, which must have taken place in real time and space" (ibid. 53), rather than as psychological processes leading from a deep to a surface structure each time a speaker makes an utterance.

8) Theoretical Considerations

Though generative phonology was devised for synchronic analysis, its diachronic application "enables the concept of ordered rules to be used in historical linguistics ..., opening up wholly new possibilities of explaining historical phenomena" (Kiparsky 1970.308). This greater explanatory power is the main justification for using a generative descriptive framework. This is not to say that valuable observations were not made by the previous generation of scholars, or that a relative chronology must be in a generative framework to avoid vagueness, but the notational conventions enable one to be precise and concise as to the relationship between rules, and the specificity of writing rules is in itself an aid both to the presentation of the material and to its investigation. That is, a sound change may be traditionally described in the handbooks as applying to "consonants". The necessity of writing the rule in generative terms forces any investigator to enquire whether this is best described a [+consonantal] or [-syllabic], that is, to ask whether the rule applies to semivowels in addition to stops, and

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to search for examples. It is not that such questions cannot be raised outside a generative framework, it is merely that it is more likely to be raised within one.

9) A further justification for the use of a generative framework is found in Jakobson's principle that "a change is, at its beginning, a synchronic fact" (quoted in Saporta 1965.289). However, this principle opens up another possibility, which has generated considerable literature. As Kiparsky (loc. cit.) notes, "Implicit in saying that sound changes are added rules is the possibility that some sound changes might be added not just at the end of the existing sequence of rules, but before some of the rules in that sequence." It was this possibility, along with those of reordering and rule-loss which occur in some synchronic descriptions, that led Kiparsky in his dissertation (1965.59) to say, "this might make linguistic reconstruction a rather hopeless task." However, as he later wrote (1970.308), "At present all that can be said is that the majority of sound changes are rules added to the end of the existing sequence of phonological rules, in so far as the ordering is fixed at all." Anttila (1972.121) is more definite still: "The only mechanism that seems historically real is the addition of rules [at the end of sequences of existing rules]," a conclusion with which King (1973) agrees. Any exceptions are better explained not as cases of rules applying out of order, that is to historically (and hence derivationally) previous forms which are no longer spoken, but as cases of the analogical reintroduction of features into the underlying form. This topic, however revelant to the concerns of this study, is too large to be dealt with adequately here. I have raised it in order to explain why a generative analysis seems to be the most fruitful methodology for relative chronology, to distinguish it from synchronic analyses, and to answer some possible

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objections to its use. For fuller discussion, besides the works referred to above, see Chafe 1959, 1968; Dressler-Grosu 1972; Halle 1962; King 1969a, 1969b, 1973; Kiparsy 1965, 1968, 1970; Marchand 1956; Saporta 1965; Sommerstein 1977. 218 f., 238-45 (the best summary). For an interesting example of the application of relative chronology in a generative framework to Greek see Miller 1976.

10) Therefore, while a single change is a synchronic fact, the relationship between two changes is a diachronic fact, again the changes entered the language one before the other, partially overlapping, or at the same time, and so we are brought back to a consideration of relative chronology. As Bynon (1977.57) notes:

> Initial construct, rule form and rule order are thus closely interdependent. The phonology of the proto-language, together with the sets of ordered rules by which the linguistic systems of the descendant languages are derived from it, form just as much an intgrated whole, un système où tout se tient, as does the synchronic description of a language.

11) Procedure and Organization.

The changes form a complicated network of orderings. To break into this web and begin unraveling and tracing the strands, one needs a starting point. A good rule from which to begin needs to be one with a wide variety of inputs (rules ordered before it) and a wide variety of effects (rules which can be ordered after it). Accordingly, the entry point chosen for this dissertation is the split of the phoneme /l/ into two allophones, the so-called bright [l] and the dark [ɫ]. Since this rule affects not only consonant clusters but vowel qualities as well, it proves particularly useful as a start for building up the system of changes. The first chapter, therefore, discusses the allophones of l. The second deals with assimilations of l plus consonant to -ll-; the third with new clusters of l plus consonant which do

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not assimilate. Chapters Four and Five concern various groups of consonants and their simplifications, some of which provide new sources of unassimilated -lC-. The second half of the dissertation considers changes in vowels, and follows the traditional presentation of material, beginning with short vowels in initial syllables (Chapter Six), diphthongs in initial syllables (Chapter Seven), and then changes in medial syllables, that is changes dependent on the prehistoric initial accent (Chapter Eight). The special case of changes of vowels in medial syllables before r is kept for the final chapter.

12) As each new rule is established and integrated into the pattern, it in turn can be used to order other changes. Hence a certain amount of redundancy in the discussion is inevitable, as single strands are traced to their endpoints, then retraced to particular branching points which are in turn traced. Establishing a relative chronology is rather like doing a jigsaw puzzle where some of the pieces are of necessity missing (rules that do not overlap or are not crucially ordered). It is partially due to this necessity that the dissertation has been given the qualified title of "The Relative Chronology of Some Major Latin Sound Changes".

13) Previous Work and Results.

The first result of such an investigation is the establishment of a relative chronology itself, and hence an enhanced understanding of what happened in the development of the phonology of Latin. Such an investigation has not been undertaken before. Most of the previous literature consists of scattered remarks on individual groups of two and rarely three rules. Brugmann's Grundriß (1897) contains several, while most of the handbooks, including some of the most recent (Sommer-Pfister 1977, Maniet 1975) offer little in the way of chronological observation. The

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chief exceptions are Buck (1933) and Leumann (1977), who offer many valuable notations, usually by means of cross-references. The only attempt at a general relative chronology is that of Albrecht Goetze's dissertation at Heidelberg (1920/1, publ. 1923). Unfortunately this deals almost exculsively with syncope, and Leumann's brief comment (1977. Allg. Teil 56) is just: "wenig günstig, weil letztere [syncope] unter verschiedenen Bedingungen zu sehr verschiedenen Zeiten erfolgte." In addition, Goetze made use of many outmoded or questionable preforms that weaken the value of some observations. Helmut Rix's article "Die lateinische Syncope als historisches und phonologisches Problem" (1966 = 1973) admirably points out the dangers of attributing to syncope the same regularity as other sound changes, due to the influence of surface phonemic constraints, of the prosodic and rhythmical systems, of morphological transparency and analogy, as well as of different syllistic and dialectal levels. Consequently, I have tried to avoid using syncope as a rung in the ladder of a relative chronology, though I have commented on it when it is relevant for the process under discussion. When syncopes have been discussed, I have tried to be as specific as possibile in identifying their environment. The only exceptions that have been made are for those syncopes which seem to be exceptionless, in particular the syncope of *-gu$\breve{V}$- in initial syllables, of -r$\breve{V}$- and of $\breve{V}$ / VR _ y (see §§7.42-59, 2.23, 3.7), and the examples have been closely examined.

14) A second result can be a more accurate understanding and statement of the rules themselves, their form, environment, exceptions, as well as their sequence. This in turn can permit a third result, which is the traditional linguistic goal of "capturing generalizations" (Kiparsky 1972, Sommerstein 1977.10-15). The usual handbook organization makes this

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difficult. Thus distinct changes are treated together if they share a common outcome, and even a work as attuned to chronological considerations as Buck (1933.147) simply states, "dl, ld, nl, ln, rl, ls become ll" and so misses being able to make the observations that all of the assimilations are not of the same type, and that there are no surface -dl-, -nl-, or -rl- clusters, while there are surface -ld-, -ln- and -ls- clusters, and to draw phonological and chronological conclusions from these data (Chapters 2 and 3). On the other hand, parallel changes, that is single rules, are treated separately if they involve differing sounds, and Leumann (1977.45 and 48-9) does not point out that the raising of e > i before velar ŋ is parallel to the raising of o > u before ŋ (see §6.16-27); nor (1977.50-1) that the lowering of i > e before r from intervocalic s is parallel to the lowering of u > o in the same environment (see Chapter 9).

There remains only the pleasant task of formally giving my thanks to various people.

This dissertation began as a paper under the late Warren Cowgill. The debt I owe him, if evidenced only by the frequency of the citation of his work, cannot be repaid.

My chief thanks are to my advisor, Prof. Alan Nussbaum. He has read several drafts of this work in whole and in part, outlined both problems and approaches to their solutions, commented extensively, provided supporting data, counterexamples, and encouragement. Since his influence has been so great, and it has not been possibile to credit him as fully as I ought and would wish in every instance, it seems to me all the more proper to make the formal statement that he does not necessarily agree with all the

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opinions here expressed, and that I alone am responsible for any errors of fact or interpretation.

My thanks are also due to my readers, Prof. George Goold and Prof. Brent Vine, who were willing to read this work while still in a very rough form and provided help from the orthographic to the linguistic and stylistic.

I would also like to express my gratitude to: Victor Bers, for his administrative aid; Wilkins Poe, for assistance with various computer related problems; to Don Ringe, for valuable discussion on several points; to Mary Kuntz for help with the computer and moral encouragement; and to Barbara Burrell for her support in the final stages of writing and preparation.

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List of Abbreviations

Abbreviations are, in the main, the standard forms. Classical authors are cited according to the abbreviations in The Oxford Classical Dictionary, The Oxford Latin Dictionary, and Liddell-Scott-Jones: A Greek-English Lexicon. Latin inscriptions are usually cited by volume and number in the Corpus Inscriptionum Latinarum, or by the numbering in Diehl's edition (abbreviated D; see Bibliography). Grammarians are cited by the volume and page number in Keil's edition (abbreviated K; see bibliography).

Languages:

Aeol.	Aeolic
Alb.	Albanian
Arm.	Armenian
Av.	Avestan
BS	Balto-Slavic
Dor.	Doric
Fal.	Faliscan
Gmc.	Germanic
Goth.	Gothic
Grk.	Greek
Hitt.	Hittite
Hom.	Homeric
IE	Indo-European
II	Indo-Iranian
Lat.	Latin
Latv.	Latvian
Lith.	Lithuanian
O.	Oscan

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OCS	Old Church Slavic
OE	Old English
OHG	Old High German
Olce.	Old Icelandic
Olr.	Old Irish
ON	Old Norse
O.-U.	Osco-Umbrian
Pael.	Paelignian
Pam.	Pamphilian
PIE	Proto-Indo-European
Plt.	Proto-Italic
Skt.	Sanskrit
Slav.	Slavic
Toch.	Tocharian
U.	Umbrian
Ved.	Vedic
Ven.	Venetic
VL	Vulgar Latin
YAv.	Young Avestan

In addition the following standard abbreviations and signs are used:

C	any consonant (voiced or voiceless)
G	any voiced consonant
K	any velar consonant
N	any nasal
P	any labial consonant
R	any liquid
T	any dental consonant

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V	any vowel (length unspecified)
h_1	e-coloring laryngeal
h_2	a-coloring laryngeal
h_3	o-coloring laryngeal
abl.	ablative
acc.	accusative
adj.	adjective(s)
aor.	aorist
caus.	causative
d.	dative
f.	feminine
fut.	future
g.	genitive
imf.	imperfect
impv.	imperative
inf.	infinitive
instr.	instrumental
loc.	locative
m.	masculine
mid.	medio-passive
n.	nominative
neut.	neuter
pass.	passive
pf.	perfect
pl.	plural
plpf.	pluperfect
pr.	present

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sg.	singular
>	becomes (by regular sound change)
<	comes from (by regular sound change)
-->	becomes (by analogy)
*	denotes a reconstructed or unattested form
**	denotes a form which does not occur, but would be the regular outcome if the preceding rules had applied in the stated order.

Cross-references are to chapter and paragraph.

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Chapter 1: The Allophones of [l].

1) We may take as our starting point in exploring a Latin relative chronology a famous bifurcation in the phonetic path: the two well-known allophones of l. The only evidence we have for the quality of the so-called bright and dark allophones in the prehistoric period, and the period before double consonants were written, comes from the changes they caused in preceding front vowels, which come in part from anaptyxis (Cowgill 1978.35). These changes are easy to summarize: the bright phoneme acts no differently from any other consonant in medial weakening: the [l], a single bright l (l before i) makes for an open syllable, while double -ll- produces a closed syllable. Thus we have open Μασσαλία > Massilia parallel to *ce-cadai > cecidī and closed *fe-fallai > fefellī parallel to τάλαντον > talentum. Nor does a following bright allophone occasion any changes in the vowel of an initial syllable. It is the dark allophone that acts unusually. Descriptively, an original front vowel in initial as well as medial or final syllables, was changed to the corresponding back vowel before an [ɬ], that is an l either word-final or followed by any consonant other than [l] or any vowel other than [i]. Some standard examples are: Doric ἐλαί϶α borrowed as *elaiu̯a > *oɬaiu̯a > *olēu̯a > olīva; Σικελός --> *Sicelos > *Sicoɬos, later Siculus (Leumann 1977:47; Cowgill 1978:35; Buck 1933:82). We clearly have a backing of [e] to [o] before a velar [ɬ] and can reasonably assign the feature [+back] to the dark allophone.

2) That [l] before [e] was also back is clear and this is stated in most of the modern literature. However, it is useful to go over the arguments briefly here. The velar quality before long ē is shown by *ad-al-ēscēns (to the root seen in alō) > *adeɬēscēns (**A-Weakening**, §§8.4-6, 11-20) >

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adolēscens > adulēscēns, and there is no reason to suppose that a short ĕ would be less able to produce a velar ɬ than a long ē (Cowgill 1978.35 n. 18)[1]. For ɬ before ē, the case of Ἡρακλῆς --> Hercles (CIL 1^2.563 [2], 564, 394) > Herceles (CIL 1^2.551 [3]; with anaptyxis) > Hercołes (frequent: CIL 1^2.30, 61 from Praeneste, 1531.8) > Hercules is decisive in showing the velar quality of the [ɬ] (Leumann 1977:141) and since the word would have had an e vowel following the l throughout the paradigm, there seems to be no analogical source for a [ɬ], e.g. dat. *Hercelei > Hercolei > HerculI.

3) Cited as evidence for velar ɬ before ĕ (Cowgill loc. cit.), though less probative, is the verb tolerō. This may be a denominative from a s-stem, which is regularly built to the e-grade of the root. Thus we would have *telh$_2$-es- (Cowgill loc. cit.; Leumann 1977:551), and the Latin tolerō would show a velar ɬ in the backing of the first e to o. This seems likely. However, the neuter s-stem necessary for this analysis is not attested in Latin (though it is in Grk. ἀ-, εὐ-τάλης). Since tolerā-re might also represent the outcome of a denominative *telh$_2$-ro- > *telaro- (cf. τάλαρος

[1] Cowgill cites Meillet 1905-6.238, who notes that the Lith. dialects that have ɬ / _ ē, also have it before ĕ.

[2] One of the series of Etruscan-work objects from Praeneste; that the syncopation here is probably due to Etruscan intermediacy, see Leumann 1977.107, and note below.

[3] Ernout 1957. 31 says, "sans doute forme étrusque". However, Latin anaptyxis is clear (see §8.30). Etruscan shows Herecele (2x) and Herecles (1x) in the oldest texts (5th. c.) The regular form is Ercle (68x; 5th -2nd cc.), see above; Hercele is attested once (Palestrina, 4th c.); Heracle (2x; 4th c.) showing the direct Greek form. See de Simone 1968.33, 72, 76.

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< zero-grade *t]h_2-ro-[4]), this lessens the evidence of tolerō for ɬ /_ ĕ. In addition it could be compared to the recup-erā-re formation which, while its origins are obscure, need not have an etymological e (Leumann 1977.551 and litt.)[5]. Likewise the forms volē-bam, volē-ns, volē-s, while they seem to show the effects of a velar ɬ before ē are too easily analogical to volō, reinforced by volumus, volunt, and the pivotal 3.sg. volt (Watkins' Law), to be unexceptionable evidence. Herculēs then is the best evidence for /_ ĕ.

4) The velar quality of dark ɬ is not in doubt and this has been the standard interpretation since Osthoff 1893.50 f. However, the evidence for a positively palatal pronunciation of bright l is much less certain. Some scholars have stated rather dogmatically that it was palatal (Maniet 1975:27, Juret 1938:14, Niedermann 1953:12, Sommer-Pfister 1977:131, Kieckers 1930:97-8, Kent 1945:59). The typology of Russian is clearly in mind (Sommer-Pfister loc. cit.), whether explicit or not. The best statement is that bright l is "non-velar" (Cowgill 1978:35); Leumann (1977.140-1) calls it "neutrale oder palatale". Typologically [ɬ] is not necessarily opposed to [l']. English has a velar [ɬ] principally in word-final position, e.g. bell [beɬ], versus a plain alveolar lateral [l] in lid [lid] (Ladefoged 1975:149). Nor is -ll- palatal of necessity. It is in Spanish: pollo [pol'o], caballo [kaβal'o], but not in French ville, Italian villa. Note, too, that

[4] Note that there is no direct evidence for such a full-grade, which would be unusual for a -ro- stem (**τέλα-po-). However, Cowgill (1980) explains the anomalous developments of R̥H > V̆RV̆ (vs. the usual RV̄) in such cases, as reflecting an original accented full-grade, with subsequent vowel harmony, e.g. *dhénx-to-s > *θένατος > θάνατος, but *dhn̥x-tó-s > θνητός. See Rix 1976.73. A discussion of this point is beyond the purposes of this study.

[5] In addition tolerāre has been explained simply as an Oppositionsbildung to onerāre (see Walde-Hofmann s.v.), but there is nothing to recommend this view, which involves an unlikely missegmentation..

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Sp. shows an opposition of [ɬ] to [l'], not [ɬ] to both [l'] and [l'l'], as claimed for Latin.

5) Only one piece of evidence from historical change might be taken as indicating a positively palatal pronunciation of the bright l. This is Cowgill's derivation of vīs (1978..25-44). He has correctly explained the form as originating from the lautgesetzlich 2. sg. *ɣel-si > ɣell (with assimilation of ls > lz > ll; see §2.7). The Romans hypercharacterized the now morphologically opaque ɣell by adding the 2. sg. formant -s again. The regular vell, now isolated from the paradigm then becomes the particle vel, a relic form, while in the analogical *ɣells the -ll- undergoes simplification next to another consonant (§3.23) becoming *ɣels. This *ɣels is unique in the language, since it has a plain l before a consonant. The solution, rather than a phonemic split, since l and ɬ are no longer in complementary distribution because of a single form, was to eliminate the feature [+lateral] from the l resulting in the alveolar continuant ļ, and so produce *ɣeļs which then regularly becomes *ɣēs and then vīs. See Cowgill (loc. cit.) for a full discussion. The change of l to the palatal ļ in this one item might be considered as evidence for a genuinely palatal l. However, note that all that is required is that bright l be allophonically distinct from dark ɬ, and Cowgill who uses the term palatal in his discussion of the change (32-33) is himself careful to point out that bright l is best considered "non-velar" (see above). The change of l to ļ does not necessitate a palatal l, though a palatalized l' is what underlies the change in Romance of such things as filius > fil'os > fijos > Sp. hijo (§11), and we can parallel the change of the voiced alveolar continuant l to the voiced alveolar continuant ļ by the elimination of the feature [+lateral] with the change of the voiced alveolar continuant n to ņ by the elimination of the feature

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[+nasal] in the Lesbian treatment of second compensatory lengthening groups, where *pantja > *pansa > παῖσα with the change of n to j.

6) As we have noted there is no reason for assuming a pronunciation of bright l that is palatal in an absolute sense, based on phonological change. However, before going on, we should review three classes of evidence which come from a much later period, that of the late grammarians, developments in Romance languages, and Latin loanwords into languages which distinguish different phonemes of l. This evidence must be used cautiously as several centuries have intervened between the time of the prehistoric sound changes under discussion and our earliest statements or attestation of loan-words. The internal structure of the phoneme [l], the number and distribution of allophones, may have undergone considerable change from the early stage to the later period. In addition the evidence of loanwords is unlikely to tell us about the the Roman pronunciation of Latin, but rather about the local pronunciation of Latin see § 13).

Evidence from the Grammarians

7) The principal sources for the pronunciation of Latin in the Classical age are the statements of the grammarians. These are, however somewhat difficult to interpret, and less orderly than they appear in Allen's Vox Latina (1965) whose observations on them are excellent. Their statements on the pronunciation of l should be reviewed here. The oldest testimony is that of Pliny (A.D. 23/4-75) quoted by Priscian (early 6th c. A.D.):

> L triplicem, ut Plinio videtur, sonum habet: exilem quando geminatur secundo loco posita, ut ille, Metellus; plenum quando finit nomina vel syllabas et quando aliquam habet ante se in eadem

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> syllaba consonantem, ut sol, silva, flavus, clarus; medium in aliis, ut lectum, lectus. (K.2.29.8-12).

It is very difficult to evaluate what Pliny is saying. To take him at his word, we have:

1) l with sonus exilis only as the second member of a geminate l,

2) l with sonus plenus :

/_ # (that is at the end of a word: sol)

_ C (that is at the end of a syllable: silva)

(C) $ C _ (that is after a consonant in the same syllable, presumably in ac-clāmāre, as well as flāvus, and irrespective of what vowel follows,

3) l with sonus medius "elsewhere", which can be i) word initial, the only case for which Pliny gives examples, ii) syllable initial, either after a consonant e.g. ab-luō, or between vowels, e.g. presumably then volō as well as velim. Cowgill (1978.35-6) comments:

> One would like to know a great many things. E.g. did Pliny really pronounce the l of ille considerably more palatal than the l of lectus? Should his statement that l sounds plenus 'quando finit nomina vel syllabas' be taken to mean that it has that sound always in word final, or should Pliny be taken very literally as implying that l at the end of adverbs and conjunctions like vel and semel had the sonus medius in his pronunciation? Or did he just forget to specify what sound final l had in such words?[6] Why does he say that l in flavus or clarus has the sonus plenus when the evidence from Romance languages is rather for a neutral or palatal sound... and does his omission of examples of single intervocalic l mean that l in words like volō, velim has the sonus medius, as his

[6] Pliny's choice of example, sol, seems to imply the former, that l had sonus plenus word-finally, but he simply fails to say anything about vel or semel, which historically at least must have had bright l to account for the e-vowel.

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'elsewhere' would seem to imply, or did he just neglect to include that environment?

For -VlV-, Sturtevant (1945:149) interprets: "We must, apparently suppose that l before vowels had ceased to have velar character by Pliny's time." Cowgill (loc. cit. n. 19) speculates that "perhaps the real rule was that l before e had this sound [medius], which could be explained as a change towards the expected palatal pronunciation of l before all front vowels", and (n. 20) "Was it really the following ē, rather than the preceding consonant that conditioned the velar l in Pliny's pronunciation of flāvus and clārus?"

8) Nor does Consentius (fl. 475) shed much light:

> Romana lingua emendationem habet in hoc quoque distinctione. Nam alicubi pinguius, alicubi debet exilius proferri; pinguius cum vel b sequetur, ut in albo, vel c, ut in pulchro, vel f, ut in adelfis, vel g, ut in alga, vel m, ut in pulmone, vel p, ut in scalpro; exilius autem proferenda est, ubicumque ab ea verbum incipit, ut in lepore, lana, lupo, vel ubi in eodem verbo et prior syllaba in hac finitur et sequens ab ea incipit, ut ille et Allia. (K. 5.394.29-34).

Not only is Consentius poor at capturing generalizations and providing for all environments (what about -ld-, -ls-, -lt-,etc. ?), but some of his statements contradict Pliny's. Here we have an opposition of 1) l exilis word initially, or geminate (the phrasing makes it impossible to tell whether the first, the second, or both l's were considered exilis), and 2) l pinguis / _ C (labials, and some velars at least). Consentius' l pinguis is presumably the same as Pliny's l plenus, but there is no mention of word-final, or post-consonantal position. Nor is there anything which correlates to Pliny's sonus medius, which Pliny assigns to word-initial l and which here is treated togeher with ll. If Pliny really meant that the l of lectus

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was exilis before e, then Consentius denies this, and implies that all initial l's are exiles. Again, nothing is said about intervocalic single l.

9) The other grammarians (assembled by Lindsay 1894:92; a good selection in Kent 1945 and Allen 1965) offer a few more puzzling corrections of errors in pronunciation current in their time. Diomedes (fl. 375?) calls it a mistake to pronounce l "nimium plene" in lūcem or almam. This treats inital l, at least before u, the same as l plenum before consonants (K. 1.454.3). Servius quoted by Donatus (both 4th c. A.D.) calls it a mistake to make a single l (e.g. "Lucius") too tenuis, or a double ll too pinguis. Here the distinction seems to be single l (word initial? before u?) as against geminate ll (K.4.445.12-13). Pompeius (5th c.) says, "debemus dicere largus ut pingue sonet; et si dicas lex, non lex, vitiosa sunt per labdacismum." One is reminded of a dialect joke. Presumably the original lectures demonstrated the different pronunciations, and one MS reads "llex non lex" attempting to make the difference manifest. Perhaps he means that the l in lex is also to be pronounced pinguis, i.e. the fault lies in pronouncing l before e as palatal. He continues:

> item in gemino ll, si volueris pinguius sonare, si dicamus Metellus Catullus, in his etiam agnoscimus gentium vitia; labdacismis scatent Afri, raro est ut aliquis dicat l. (K. 5.286-7).

Here the opposition is 1) l pinguis /* _ (even before e), vs. 2) double ll. He uses 'subtiliter' in opposition to 'pinguis' in a passage quoted below. The fault of the Africans seems to be in pronouncing ll as l. Is this then a length mistake such as English speakers (and many Italians) may make in Tuscan Italian (Lindsay's example 1894.92), rather than a mistake in quality? Consentius says the Greeks are liable to the same error, pronouncing ille "subtilius":

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> nam ecce Graeci subtiliter hunc sonum efferunt. ubi enim dicunt 'ille mihi dixit', sic sonant duae ll primae syllabae, quasi per unum l sermo ipse consistat. contra alii sic pronuntiant 'ille meum comitatus est iter' et 'illum ego per flammas eripui', ut aliquid illic soni etiam consonantis ammiscere videantur, quod pinguissimae prolationis est. (K. 5.394.25-9).

The mistake of the Greeks is clear: not pronouncing double consonants, but the mistake of the "others" is difficult to fathom. Lindsay (loc. cit.) believes it may mean "they gave it the -ly- sound it now has in Spain." This statement is not quite accurate. There is is a genuine biphonemic [ly] which some dialects of Spanish distinguish from the single phoneme [l'] which is the ulimate outcome of -ll- (see below): thus polio 'the disease polio' /polyo/, but pollo 'chicken' /pol'o/ (Ladefoged1975.150). What Consentius seems to hear is some kind of intrusive consonantal sound. As he opposes subtilius to pinguius it seems more likely that he is referring to pronouncing ll as velar ɫ, as if syllabified ill-e, cf. the comments of Isidore on the 'Afri', below.

10) Lindsay (loc. cit.) sums up: "All that seems certain is that initial l had some distinction of sound from the l of Metellus, Catullus, &c." Even this is too optimistic. If we write up the data in a table, we can see the lack of concord.

	Pliny	Consent.	Diomedes	Servius	Pompeius
pinguis/ plenus	/ _ C * _ $C _	/ _ C	/ _ C * _ (u)	-l- (/* _ ?) (/ _ u ?)	-l- ? (/* _ {a, e} ?)
exilis/ tenuis	-ll-	-ll- /* _		-ll-	-ll-

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There seems to be a consensus that -ll- is different from -lC-. There is also a hint that single (initial?) l is different from double -ll-. Perhaps the distinctions are of different types. Sletsjøe 1959:147 n. 2 wonders whether we have "une prononciation prolongée resp. raccourcie, donc qu'il y avait ici différence qualitative et quantitative en même temps." Isidore (c. A.D. 602-36) supports a single vs. double distinction:

> Labdacismus est, si pro una L duo pronuntientur, ut Afri faciunt, sicut 'colloquium' pro 'conloquium'; vel quotiens unam L exilius, duo largius proferimus. Quod contra est; nam unam largius, duo exilius proferre debemus.

The evidence from the grammarians is not clear enough then to support any precise conclusions. In particular it does not suggest any specifically palatal pronunciation for bright l, nor that the original distribution of allophones as shown by the sound changes was maintained. In any case it is clear that with the grammarians we are dealing with changes in the system of pronunciation not of the period of the sound changes which we are interested in but rather with those that led to the Romance languages, and it is to these that we shall now turn.

IIi. Developments in Romance Languages

11) Romance developments do not support a specifically palatal pronunciation of [l]. The majority of palatal l's which show up later are the results of secondary palatalization in various languages, arising primarily from the passage of -li- to -lj-; for example fīlius > It. figlio /fiʎ'ʎo/, etc. For this change evidence already exists in Imperial Latin: e.g. FIIOS = fīlius ,

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where palatalized l' has become j (CIL VI.667; see Leumann 1977:142). It has been proposed that Spanish ll [l'] preserves the palatal coloring of Latin (Ettmayer 1905.653). However, this view entails great difficulties. First, the Spanish ll is a single palatal [l']. Secondly, as mentioned above, no other Romance language shows such a pronunciation: Sp. villa [vil'a], but Fr. ville [vil], It. [villa] (Sturtevant 1945:149). We would have to suppose that every Romance language except Spanish had lost the palatal quality of a l'l'. This is not impossible, but much less likely than an innovation in a single branch. Thirdly, that this is an innovation is shown by the fact that Iberian Romance shares the early palatalization of -lj- to -l'-, which remains in Port., e.g. folia > folha /fol'a/, and develops to [ž] > [χ] in Sp. hoja /oχa/, but does not share the later change of [l'] > [l'], which is found only in Spanish (Bourciez 1946:411 dates it to the 14th c.). Fourthly, this palatalization is part of a larger rule affecting both -nn-, and -ll-. Thus, we have annus > Sp. año, but Port. anno (Lausberg 1956:70). Finally, the developments in S. Italian (Calabrian, Sicilian, Sardinian, etc.) argue against a palatal pronunciation, as [ll] > [ḍḍ], e.g. stēlla > Sic. stidda, while VLat. -lj- > Sic. [g'g'], Sard. [dz]: folia > Sic. foggya, Sard. fodza. Such retroflex outcomes[7] argue for a plain rather than palatal pronunciation of [ll], since it has a different outcome from the palatal [l'l'] < -lj-.

IV. Latin Loanwords

12) Unfortunately, Latin words borrowed into languages which distinguish palatal l' fail to provide any information about the pronunciation of Latin bright l. Albanian, which opposes l, ł, and l', due to its own

[7] See Bourciez 1945.491,494; Lausberg 1956.58,68-9.

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phonological developments can offer no help, despite the large number of Latin loanwords. Thus, while l' does occur in places the Latin had -ll- (so bul'ɛ 'bud' < bulla, kāl'ɛ < caballus; Brugmann 1897:365), since l' is the regular outcome of inherited -ll- < PIE -ln- (loc. cit.), Albanian provides no evidence here. Likewise, as intervocalic l becomes ɫ, (muɫí < molīna· ibid. 438), and sequences of ClV become ClV and so cause palatalization, e.g. in familia > fɛmil'ɛ 'children', Albanian supplies no evidence for the other posible locus of Latin bright l.

13) The Irish evidence is inconclusive, due to both the different chronological layers in the borrowing, and the prevalent influence of both Britannic and Britannic Latin, that is the local pronunciation of Latin (see Vendryes 1902:38 et passim, Thurneysen 1946:565 f., Jackson 1953.122-48, Lewis-Pedersen 1961:56-63). The earliest borrowings were made when Old Irish still retained its final syllables, and show the effects of palatalization, e.g. brevis > breib. However, later borrowings, particularly those influenced by Britannic, treat the Latin loan as simply endingless, e.g. orāti(ō) --> oróit, where the ō also shows the change in Britannic and in the pronunciation of Brittanic Latin of ā to ō (we would expect in Old Irish by straightforward sound changes orātiō > *oratiu > *orthu, indirectly attested by Olr. acc. pl. orthana, Mlr. ortha). Old Irish shows even more drastic loppings off in e.g. ulti(ma) --> uilt or comparatī(vus) -> comparit.

12) No evidence will be provided when Old Irish retains a vowel which causes palatalization within Irish itself. So, molīna --> mulenn shows the e reduction vowel which regularly appears between a palatalized and a plain consonant and tells us nothing about the quality of the Latin l. Likewise pallium > caille, an early borrowing with substitution of c for p, shows e retained after i, thus giving an intra-Irish palatalization. However, -ll- in

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final syllables shows contradictory evidence. Thus, while we have fégil < fidēlis, achtail < actuālis, etc. developing regularly and showing the Old Irish palatalization due to the following i (and so these forms say nothing about the quality of the Latin l), we also have pāpiliō --> pupall, publicus --> puplach, humilis --> humal, etc. which show the reduction vowel regular between plain consonants. A possible explanation might be that l before i was no longer palatal at least in Britannic Latin, so that in later borrowings when the loan is treated as endingless the l is borrowed as plain. However, Irish shows too much fluctuation in its borrowings to permit a clear deduction, cf. adiectīvus --> adiecht beside genitīvus --> genitiu, etc.

14) The evidence suggests a plain pronunciation for -ll- in the Latin that gave its vocabulary to Irish. The forms flagellum --> sroigell (with dissimilation of the first l), cancellus --> caingel, crātella --> greidel, show the e expected between a palatal and a plain consonant; while metallum --> mital(l), crystallum --> cristal(l), sollemnis --> sollaman show the regular a between two plain consonants.

15) In summation, the evidence from the grammarians, Romance, and loanwords are, for the most part, too late, or too uncertain to provide firm clues to the pronunciation of l. However, the evidence in no way supports an absolutely palatal pronunciation for bright l. This, combined with the phonological evidence of the early sound changes, indicates that the oppositon in the time under discussion was between a plain [l] and a conditioned velar allophone [ɫ].

16) This velar allophone appeared in all positions except before l (i.e. in geminates) and before the high vowel i (Leumann 1977.140-1), that is "before consonants other than itself, word final, and when preceded by a vowel and followed by a vowel other than i" (Cowgill 1978.35). Cowgill

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adds the specification "when preceded by a vowel" since word initially and after a consonant we can have no direct knowledge of the quality of the l which is shown only in changes of a preceding vowel[8]. To the claim that l was dark in final position there are five counter-examples, showing a short ĕ before a final l, which have all been thoroughly discussed by Cowgill (loc. cit.), and I will only briefly repeat his conclusions here. The case of vel comes of course from *ŭel-si > vell (Enn. Ann. 340; perhaps Ter. Heaut. 640) > vel, and so not a final non-geminate l at the time when ĕ > ŏ / _ l occurred; see §5. Likewise the nom. for mel and fel were clearly *mell and *fell, with the nom. backformed from the oblique cases mellis and fellis, a process frequent in stems ending in a resonant (Leumann 1977.220; see §§2.12-13, 21; 9. n. 30). The history of mellis is given below (§2.12-13). The case of fellis need not detain us here. It is very likely to be analogical to mellis[9], particularly in that it shows an ĕ-grade in contrast to the ŏ- or zero-grade of its cognates[10] (Cowgill op. cit. 39; §2.21). The adv. semel 'one time, once' is of disputed origin. Cowgill (ibid.) summarizes the situation:

> The adverb semel goes back to something like *sm̥-mĕli, as proposed by Wackernagel 1890:316 = 1953:1.679, or *semalis, as stated by Leumann 1977:494 (cf. also Osthoff 1893:64 for a similar view). Of the two Wackernagel's view seems more likely to be correct. As Wackernagel says, semel 'einmal' cannot have the same origin as simul 'zur gleichen Zeit'. But, in view of the shape of expressions for 'once' in other languages, such as einmal, une

[8] Though Pliny seems to indicate that l was velar (sonus plenus) after a consonant in the same syllable, regardless of the vowel that followed (§6).
[9] Cowgill (op.cit. 39) notes Sommer's observation (1914.108) that fel occurs in Plautus only in passages where it is contrasted with mel.
[10] ŏ-grade in Gmc. OHG, OS, galla, OE gealla, ON gall < PGmc. *gallōn- < *ĝhol-n-; Grk. and Av. ŏ-grade thematic: Grk. χόλος, χολή, Av. zārə-; zero-grade in BS: Lith. tulžìs < *žultis, OCS zlŭčĭ. See §3.8 f.

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fois, ainamna sinþa, viena karta, oenfecht (and probably also ἅ-παξ, sa-krt), semel ought to be a compound of sm- functioning as a cardinal number with a noun meaning 'fois, Mahl', not the neuter of a (presumably possessive) adjective *sm-mēl-i-s, as proposed by Wackernagel. Rather than posit for Latin an i-stem noun *mēli beside the o-stem *mēlom implied by Germanic (Go. mel, Olc. mál, OE mæl, OHG māl, etc.), it seems to me thinkable that semel goes back to *sm-mēlom 'one time', with the same noun as in Germanic, although the absence of relics of the type *semēlum, *semēlu comparable to nīlum and noenu(m) beside nīl, nōn makes me extremely doubtful that this solution is correct either.

Cowgill's objection to this last proposal is sound and for an example of Latin i-stem in nouns for thematic elsewhere cf. *kol-ni-s > collis, but Lith. kálnas (and see Excursus II §3). As Cowgill notes, none of the preforms posited has a final single l, and so semel does not constitute an exception to -l* as having been velar. If semel is a compound of *mēl- the ē was long until iambic shortening applied, a change certainly after the backing of e > o / _ ł. The form subtel is as Lindsay (1894.213) notes apocopated from a *subtēle, a neut. sg. i-stem adj. 'that which is under the heel' (Cowgill 1978.40), and so again had a long ē until shortened before the final resonant.

16) Now that the two allophones of l have been discussed, we can ask whether there are any new sources for l that feed this rule. One new source of an l in Latin is the so-called Sabine[11] l, that is an l arisen from an Italic d under unknown conditions, e.g. dacruma (Liv. Andr. apud Festus 68.10) > lacrima; dingua (Mar. Victor. K 6.26.3) > lingua; oleō beside odor. Note here that if the citations of the grammarians are correct, the change, whatever

[11] This misleading term goes back to Conway IF 2 (1894) 157-67. The usual explanation of dialectal borrowing may be correct, but there is no evidence of such a change in what little there is of Sabine or any other dialect. In addition, while it is understandable that Latin might borrow the words for 'cow' and 'wolf' from some rural or outlying dialect, it is less understandable that the words for 'tear', tongue', and 'smell' should be borrowed. See Leumann 1977.155-6 and litt.

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its nature occurs rather late and in historical Latin. The limited number of forms do not show any phonotactics that might tell something about the change, but one example might furnish some relative chronological information. The word solium 'throne, seat' was connected by the ancients with solus 'alone and solum 'soil'[12], but these are obvious folk-etymologies, and the connection with the root *sed- 'sit' is obvious. Thus solium is from an earlier *sodiom with "Sabine" l (so W.-H., E.-M.). If the preform were *sed-iom, this would place the change of e > o / _ i after the arrival of "Sabine" l and into the historical period. However, nothing points to an e-grade in the preform. Neut. substantives in -ium come from a variety of sources and an e-grade of the root seems at least a possibility, cf. subsidium, etc. (see Leumann 1977.293-6). However, if the preform were *sed-iom, since the l of the resulting solium would be exilis before the i, solium would have to be a secondary derivative (substantive use of adjective) of a thematic noun which is likely to have o-grade: *sod-o- > *soi-o- > *sol-o- --> solium, which accords well with the frequent use of -io- to form derivatives from nouns, e.g. coniunx/coniugium, etc. (Buck 1933.317; Leumann loc. cit.). However, an inherited o-grade is very likely in the face of an exact cognate in Olr. suide < *sodiom, and for o-grade, cf. spolium, corium 'hide' (root *ker- 'cut'), and perhaps folium (despite Grk. φύλλον).[13]

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[12] "Solia appellantur sedilia in quibus non plures singulis possint sedere, ideoque soliar sternere dicuntur qui sellisternium habent, et soli<a>ria vocantur Babylonica, quibus eadem sternuntur. Quae, ut ait Verrius, omnia ducta sunt a solo." Festus 386.1.

[13] See Leumann loc. cit. and Walde-Hoffmann s.vv. These are the only nominal forms of the root *sed- that show an o-grade. The possibility of independant development is less likely, despite the productive nature of -io- in Latin, since in Olr. "the neuter suffix -e [< *-io-] was apparently obsolete by our period, surviving only in old formations" (Thurnneysen 1945.169).

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None of the other forms with Sabine l aid in a relative chronology, e.g. mālus can owe its long ā to an immediate preform of either mazdos or mazlos (cf. Eng. mast, Olce. mastr).

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1) Beginning with these two allophones of l, we can now write our first rule to account for them. We shall set up a rule called **L-Darkening:** l becomes velar before a segment, including word final, if that segment is not l or i, i.e. l > ɫ /_ C (not l), or V (not i). In generative terms:

$$[+\text{lat.}] > \begin{bmatrix} +\text{high} \\ +\text{back} \end{bmatrix} / _ \begin{Bmatrix} [-\text{lat.}] \\ \begin{bmatrix} -\text{high} \\ -\text{front} \end{bmatrix} \end{Bmatrix}$$

The use of the feature [+lat.] implies nothing further about the phonetic qualities of [l] than a lateral articulation, which is well attested by the ancient grammarians; see Allen (1965.32). It picks out [l] from all other phonemes of the language and distinguishes it from the other liquid [r].

2) Already with these two allophones, in classic complementary distribution, we can begin to say something about the relative chronology of Latin. Some rules are crucially ordered before **L-Darkening**. The most obvious set are those that change a cluster of -lC- to -ll-, that is rules which bleed[1] **L-Darkening** by removing potential new material for the rule to work on. Thus we have inf. *vel-se > velle, not *vel-se > **volse > **volle. So Leumann 1977:213 more precisely: "Nach den Beispielen mit e vor ll sind die Wandel ln > ll, auch ls > ll und ld > ll, älter als der Wandel el > ol vor Kons."

3) These rules of Progressive Assimilation which bleed **L-Darkening** will be treated in the following paragraphs. Another class of rules which

[1] For the terms "bleeding", and "feeding", see Sommerstein 1977:117 f., 243 f. for a full discussion and literature. See also Introduction, §5 ff.

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can be ordered before **L-Darkening** are those which feed it, i.e. rules which create new clusters of l plus C or l plus a vowel other than i, and so create new material for **L-Darkening** to work on. These will be considered in Chapter 3.

Progressive Assimilation

4) This transfer forwards of a feature, in this case [+lat.], is quite unusual in Latin, which favors regressive assimilation. Leumann 1977:185 makes the same point, but uses the opposite terminology:

> Normal erfolgte sie "progressive", d. h. als Adaptation des vorangehenden an den folgenden Konsonanten... man spricht von antizipatorischer oder regressiver Angleichung... Seltner, besonders bei l, ist die Angleichung in umgekehrter Richtung (die Magnien leider als progressive Assimilation bezeichnet): ld > ll, ln > ll und lz rz > ll rr.

I shall follow the terminology more widely used in English (Gleason 1961:83, Anttila 1972:73, Bloomfield 1933:373, etc.) and in German (Wendt 1961.228, Sommer-Pfister 1977.197, etc.).

5) There are four clusters which result in new -ll-. These are: 1) -ld-, 2) -ln-, 3) -lz-, and 4) -lu̯-. Though the outcome is the same, and they are usually handled together in the handbooks, we can show that the first two changes occurred at an earlier time than -lz- > -ll- (see below §§3.26-7). In addition, d, n, z, and u̯ do not form a natural class, and we have therefore no reason to presuppose that a single rule covers all their changes to -ll-. Standard examples for -ld- > -ll- are:

*saldō > sallō 'salt' (cf. Goth. saltan, and the p.p.p. salsus, see §§3.22, 5.6)

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*(per)-celdō > percellō (cf. p.p.p. culsus < *cols(s)os < *celssos < *celd-to- (§4.8); clādēs with a different root shape).

This change was first recognized by Meyer-Lübke (KZ 28.171) in 1887, and represents the communis opinio (despite Collinge 1973.264-70).

5) For -ln- many examples occur in the nasal presents:

(PIE *tl-ne-h$_2$- > *tolnă -->) Plt. *tolnō > tollō[2]

*kol-ni-s > collis (cf. Lith. kálnas,[3] Goth. hullaz)

6) It seems reasonable to write these changes as a single rule. Both n and d are voiced dentals, forming a natural class, and the rule is one of assimilation to l, another voiced dental but with lateral articulation. The second rule can be named

LDN Assimilation

$$\begin{bmatrix} +\text{ant.} \\ +\text{cor.} \\ +\text{voice} \\ -\text{cont.} \end{bmatrix} > [+\text{lat.}] \; / \; [\text{lat.}] _$$

i.e. d or n becomes l after an l.

7) Such a rule -ld-, -ln- > -ll- makes us want to include the two other voiced dentals r and z in the rule[4]. While r does pattern with d and n in

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[2] See Watkins 1965.184-6 and Leumann 1977.534 for discussion of the thematicization. See further, §8.7.

[3] With a (and e) secondarily lengthened in accented syllables before tautosyllabic resonants. See Endzelīn 1971.46.

[4] SPE (Chomsky-Halle 1968) defines /r/ as [-anterior], for English in any case. However, for Classical Latin /r/ was clearly described by the grammarians as an apico-dental trill; see Sturtevant 1940.150-1. What r was at the time of these rules is unknowable, though the strange distribution of the phonemes that conditioned the change of (*dh >) *θ > *f : /_ r, /r _, /_ l, /u _, suggests that not only did l have a velar allophone, but that r also may have been velar or uvular, like the modern Fr. r.

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becoming l before l,[5] there are no clusters of -lr- in PIE (Cowgill 1982). The cluster -lz- does become -ll-. However there is a parallel process, by which -rz- also becomes -rr-. For -lz- > -ll- a different rule therefore, one of z being assimilated to a preceding resonant, is operating, i.e. z > R / R _ . This rule can be called:

RZ Assimilation:

$$\begin{bmatrix}+\text{ant}\\+\text{cor.}\\+\text{voice}\\+\text{cont.}\end{bmatrix} > \begin{bmatrix}+\text{son.}\\ \alpha\ \text{lat.}\end{bmatrix} / \begin{bmatrix}+\text{son.}\\ \alpha\text{lat.}\\ +\ \text{cont.}\end{bmatrix} __$$

Standard examples of z assimilating to r or l are:

*vel-se (cf. esse) > *velze > velle

*fer-se > *ferze > ferre

*kolzom > collum (cf. Goth. hals)

*torzeo > torreo (cf. τέρσομαι, Skt. tárṣ-).

We have assumed a stage where the s has already assimilated in voice to the resonant (s > z). For this, see §21. Note, however, that although -ld- and -ln- assimilate to -ll-, -rd- and -rn- do not: g. sg. cordis < *k̂rd-es; cornu < *k̂r-n-u (Leumann 1977.57; Nussbaum 1976.2). Thus it is clear that there are two different rules at work here, even though the outcome, -ll-, is the same in some of the cases.

8) There is a further cluster of l+C that results in -ll-. This is -lu-. This change is disputed by some (Solmsen KZ 38.437; Kieckers 1931.135 calls it "strittig"; Buck 1933.136 "doubtful). The cluster was not common in PIE, so the examples in Latin are few, but nonetheless secure: Lat. soll-ers 'with all the arts' < *sollo- (plus ars, artis), cf. Osc. sullus pl. 'all', so too in

[5] E.g. *stēr-lə > stēlla, (*agre-lo-s > *agr̥los >) *agerlos > agellus; *corōne-lā > *corōnla > corōlla; *sed-lə > sella

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Festus: sollum osce 'totum' (Vetter 1953.374). This sollo- can be compared to Grk. ὅλος. Hom. οὖλος < ὅλϝος. Skt. sárva-, Av. haurva- 'all, entire, whole' < PIE *sol(H)-u̯o-[6]. Lat. salvus, Osc. salavs < *salau̯o- (see §§3.6-7; Excursus II, §6, G), and Olr. slán < *s|H-no-, point to a set root, while the forms with o-vocalism show no trace of the laryngeal. This seems additional evidence for the suggestion first made by de Saussure (1909.511), and later by Meillet (1922.68), that the laryngeal fails to vocalize when separated from an o by a resonant or nasal[7]. For surface -lv- clusters, i.e. those which have arisen after the assimilation of -lu̯- to -ll-, see §§3.6 ff.

9) This seems to be a separate rule from those above. No natural class results from n, d, and u̯, and -lu̯- > -ll- is not paralleled by any changes in -ru̯-: curvus, corvus (Leumann 1977.131). This is also the only one of the rules resulting in an -ll- which can be established for Osco-Umbrian and may perhaps therefore be Common Italic, though the possibility of independent innovation cannot be ruled out[8]. For the other changes there are no sure examples in O.-U. See von Planta 1899.302 for -ln-, 498 for -lz-; no discussion of -ld-; the Latin examples have no cognates in our current corpus. Given that -rz- does not give -rr- in Umbrian but -rs-[9] (frequently written -s showing a "weak" r), the rule of **RZ Assimilation** (rz > rr, lz >

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[6] Hardly by "gémination expressive," Meniet 1975.165.

[7] Not simply "after o-vocalism" (Beekes 1969.238). These are the famous cases of ὀμείχω/μοιχός; ὀλιγός/λοιγός, etc. The example of sal(a)vus, sollo- does not, to the best of my knowledge, appear in the literature, and I owe the observation to Alan Nussbaum.

[8] No examples in the limited Venetic corpus, though if fo.u.vo.s is cognate with fulvus it may show a dissimilation of some type, before or after syncope. Lejeune 1974.136.

[9] The development to -rf- is only from secondary s ; < -rtt- or rts; von Planta 1899.488.

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ll) is specifically Latin (Brugmann 1892.760-1). We can write our third assimilation rule:

LW Assimilation

$$\begin{bmatrix}-\text{con}\\-\text{vocalic}\\+\text{back}\end{bmatrix} > [+\text{lat.}] / [\text{lat.}]_$$

i.e. u̯ > l / l _ . Note that u̯ already agrees with l in being [+voice], [+cont.], [+son].

10) There are two apparent, and much discussed, cases of -ll- arising from new -ld- clusters, which need to be reviewed here, before other sources of -ll- are considered. These are Pollux and mel, mellis. The new -ld- has its source in both cases in a syncope and therefore great caution needs to be exercised in any claims to Regelmäßigkeit or placement in relative chronology. The Latin Pollūcēs (Plaut. Bacch 894), earlier POL<L>OUCES (CIL XIV.4094 = Ernout 1973.30/*44)[10] is a borrowing of the Grk. Πολυδεύκης. It is not a native Latin word, nor does it reflect a Latin treatment of a directly inherited PIE form: πολυ- is specifically Greek. Therefore it is irrelevant, from the Latin point of view, to speculate on a PIE etymology[11] for it (Collinge 1972:264-70 seems methodologically confused here), as it cannot reflect anything but the immediate Greek form. We clearly have then a Grk. Poludeukēs > *Poludoucēs > Poldoucēs (with syncope of the second syllable of a four syllable word[12]) > Polloucēs (with -ld- > -ll-). The inscriptional PODLOUQUEIQUE (CIL 1². fas. 4.2833; see Gordon 1983.76-7 and litt.) has caused Radke (Gl. 42 [1964] 218) to propose

[10] This is on one of the Etruscan-work mirrors found at Praeneste, and so not to be taken as documentation but illustration of a Latin midstage.
[11] Besides the general danger of etymologizing proper names.
[12] Which will be abbreviated sync. 2/4.

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and Collinge (op. cit., who, however, makes no mention of Radke) to hint at a metathesis of -ld- to -dl- which would then assimilate to -ll- (cf. *sed-la > sella, etc.). This change is quite unnecessary as well as unlikely, and is proposed by authors who do not accept that -ld- becomes -ll-. No other -ld- cluster shows any trace of this hypothetical Zwischenstufe, and the change of -ld- to -ll- is sufficient to explain the -ll-. The form PODLOUQUEIQUE is clearly due to scribal error, containing besides the mistake of the metathesis of the -LD- an anticipation of the QU of QUE[13] (so Leumann in the same issue of Glotta, see also 1977.71; Meillet-Vendryes 1966.16; Sommer-Pfister 1977.178).

11) Another aspect of this word needs to be examined. If the syncopation of the second syllable (sync. 2/4) is due to a Latin syncope, we can say that sync. 2/4, of i between l and d at least, fed the **LDN Assimilation** rule[14]. However, the possibility exists that the immediate source of borrowing was not the Greek Πολυδεύκης but rather the Etruscan Poltuke (Walde-Hofmann 1954:330). In Etruscan, apparently, there was no phonemic distinction between voiced and voiceless stops, and Greek voiced stops are usually represented in Etruscan by voiceless stops (de Simone 1968. passim). These voiceless stops are then continued in Latin borrowings from Etruscan, and indeed the presence of such voiceless for voiced stops is prima facie evidence for an Etruscan intermediary. So Γανυμήδης --> Etr. Catmite -->

[13] Again searching for a labio-velar in the proto-form of the Latin betrays a mistake in method. Unless the Latin continues the PIE independently (or else borrowed it before the Greek changes in kᵘ !), there is no source for a labio-velar, certainly no substitution of Latin -kᵘ- for Greek -π-.

[14] See Introduction, for the difficulties in employing syncope rules in a relative chronology.

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Lat. Catamītus.[15] The cluster -lt- is preserved in Latin (multus, etc.), so that had Pollūces had an Etruscan intermediary, the expected form would be **Poltūcēs. Thus a borrowing directly from the Greek seems more reasonable, and Pollouces is likely, therefore to show the results of a Latin, rather than Etruscan syncope. For futher cases of syncope between l and d, see §3.5.

12) The second controversial case under discussion is mel, mellis 'honey'. On the basis of Greek μέλι, μέλιτος (cf. βλίττω 'honey, flavor with honey' < zero-grade *mlit-jō); Hitt. milit, Luv. mallit (< *melit); Goth. miliþ, we can reconstruct a PIE nom. *melit. This would give a PIt. *melid, with voicing of the final stop (cf. OL sied < *(h_1)s-iieh_1-t, OL deded, etc.; Buck 1933.157). The original Italic paradigm therefore would be:

nom. *melid

gen. *melit-es,

and to the nom. a new gen. *melides is formed by analogy. This can then undergo syncope of the second syllable in three syllable words, where the vowel is short and in an open syllable[16]. Thus: *melides > *meldes (sync. 3) > mellis (**LDN Assim.**). To the gen mellis then a new analogical endingless nom. *mell is formed which becomes mel (Cowgill 1978.38, Leumann 1977.220). The collateral form mulsus 'honeyed' also prima facie suggests a final dental (Cowgill loc. cit.). No verbal form of this root exists, but this is, in itself, not a hindrance to taking mulsus from the stem *melit, as there are a number of purely denominative adjectives built with the -to-

[15] The Cata- may be due to folk etymology of κατα-, a perfectly natural interpretation given the connotations. See Leumann 1977.152.
[16] Which will be abbreviated sync. 3.

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suffix[17], and the semantics "supplied with" (Leumann 1977.333) are excellent, mulsus then meaning "supplied with honey." Since the comparative evidence guarantees a *melit, which can also underlie the Latin mel, mellis, the expected denominative would be a *melit-to-s. Morphologically and semantically this is unexceptionable, but is phonologically difficult. A *melittos would presumably give a Latin **melissus[18], though a syncope to *meissos > moɬs(s)os > mulsus is not unthinkable[19]. Nor is mulsus likely to have come from the new stem *meld- (meld-to-s > melssos > mulsus). This would involve assuming first, that the form was not created till after syncope, and while there are many inherited denominative -to- adjectives, the productive form in Latin is -ātus[20]. Secondly, it assumes that secondary -dt- also gives -ss-. This is not attractive particularily in the face of at-tendō, etc. Our choice, if *melit-to-s and *meld-to-s are ruled out, is either to assume a stem *melt- (for which there is no evidence) in addition to *melit, thus melt-to-s > *melssos > moɬs(s)os > mulsus, or else to seek an analogical explanation. The latter is methodologically sounder, and a model lies easily to hand in salsus (so E.-

[17] Contra Ernout-Meillet (1959.395): "même [mulsus] donne à supposer l'existence d'un verbe *mellō qui serait parallèle à sallō".

[18] Whatever the conditionings on the various syncopes in Latin may have been, syncope appears more likely between resonants and voiced stops. See Rix 1966 = 1973.90 ff., esp. 99. In addition, syncope before two consonants, i.e. in closed non-final syllables is extremely uncommon, a fact not noted by Leumann. See Rix loc. cit. 91

[19] For syncope of V̆ /R _ t, cf. virtūs < *viro-tūt-s, with the same formation as servitūs. However, following von Planta (1892.215), Vendryes (1902.181) and Petersen (MSL 22.1-12) suggest that syncope only occurred when at least two morae followed an immediately post-tonic syllable.

[20] Originally to -ā stems: togātus and then to other types, e.g. neut. pl. arma --> armātus, from which type are created many new first conjugation verbs: --> armāre. For C-stems, cf. scelerātus (but original sceles-tus), eburātus, etc. See Leumann 1977.333.

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M., s.v. "la forme mulsus peut être faite d'après salsus). The proposional analogy:

sal 'salt' : salsos :: mel 'honey' : X

would create a *melsos which then becomes mulsus. Note that *melsos > mulsus shows that the early changes of *ls > *lz > ll were over by the time that the new analogical *melsos was created (i.e. *melsos does not become **mellus), but *melsos was created in time to undergo the change of e > o / _ i.

13) The main argument against this account of mel, mellis is that the syncope of -lid- in other cases occurs only late and after the assimilation of -ld- to -ll- is over. Thus: validē beside valdē, calida beside calda. However, it is possible to explain these forms as retained or rather renewed, since they are morphologically transparent as -idus adjectives, while *melides > *meldes > mellis, not being so analyzable, was not subject to analogic restoration on the model of tepidus, etc.[21], and so fed **LDN Assimilation**. All the other examples known to me of syncope of i / l _ d occur in -idus forms and all have attested unsyncopated byforms pointing to

[21] Again the conditioning factors of syncope are perhaps unformulatable, but it is doubtful that a form such as tepidus was ever subject to syncope. The short -ĭ- does not lie after a resonant where syncope seems to be favored, and the phonotactics of the cluster which would have resulted had syncope taken place, here -pd-, almost certainly played a role. See Rix, locc. citt.

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a recent date.[22] On this basis we shall tentatively assign sync. 3 of -lV̆d- (at least in the morphologically opaque cases *melides with i and Poludoukēs with u) before **LDN Assimilation**.

14) We can now turn to new cases of l plus z which assimilate to -ll-. Again these will be due to various syncopes and so must be approached cautiously. We'll begin with cases of syncope of short vowels in the second syllable of four syllable words with samprasāraṇa[23], i.e. forms beginning *CVClVC- > *CVC]C-. An interesting example and one which will be under a good deal of discussion is the superlative *faklisamos > *fak]zamos (sync. 2/4-]) > facillimus[24]. For the stem-shape *fakli-, cf. neut. as adv. *facli > fak] (by apocope) > facił > facul (see §6.4; see also Brugmann 1891.219). There are several points to be made here. The first is that the development of secondary l_2 in Latin was to il; original PIE *l had become ol. This change is guaranteed by the pair pōculum, pōcillum, among others. The first shows *pō-tlom (with the PIE instrumental suffix, Buck 1933.329) > *pōklom (§2.19, on the change of -tl- to -kl-) > pōcolom (see §8.30, on the

[22] So validē, Plaut. Am. 1062, etc.; valdē, Plaut. Merc. 103; caldus in Cato (RR 6.1) and Varro (RR 3.2.1); Lucr. and Cic. know only calidus. For sync. 3 of -lit-, Leumann (1977.96) has only the single example alter, comparing the adv. aliter. However, there is no reason that this should not have come directly from *al-teros with the root of alius < *al-ios, plus the comparative/contrastive suffix (so also E.-M. s.v. alius), while aliter (cf. ἀλλότρ-ιος), which does not syncopate, represents a stem *al-i(o)-. If aliter is simply a fossilized nom. used as adv. and identical in preform to alter (so Leumann 1977.500 following Skutsch 1914 n. on pp. 25-7), it is difficult to see why it should not also have syncopated (cf. the sync. of dexter), since there are no -iter adverbs to provide a model for analogical restoration or retention. Rather aliter itself provides the model for the other (and later) -iter adverbs. See Leumann loc. cit.

[23] Which will be abbreviated sync. 2/4-].

[24] Leumann 1977.143, 497; not from *facil-simos, so Kieckers 1931.143. See Cowgill 1970.113-153, esp. 124 ff. Cf. §7.37

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anaptyxis) > pōculum. Its diminutive *pō-tle-lom became *pōklelom > *pōc‖lom (by sync. 2/4-‖) > pōcillum. In a medial closed syllable both -i- and -e- are preserved, so that an *l_2 > -el- would have given **pōcellum. Note that this allows us to order the Plt. change of -ii- to -kl- before the purely Latin syncope with samprasāraṇa, and equally before the similar syncope in Umbrian.

15) An apparent counterexample, seeming to show l > -el-, is marcellus, takenby W.-H. as from < *marclo-los. However, this form is attested only in a passage in Isidore (Orig. 19.7.2):

> malleus vocatur, quia dum quid calet et molle est, caedit et producit: marcus, malleus maior: et dictus marcus, quod maior sit ad caedendum, et fortior: marcellus mediocris: marculus malleus pusillus.

If we compare marculus beginning from a *mar-tlo-s > *marklos > *markolos > marculus 'hammer'[25], we may expect a dim. *martle-lo-s > *marklelos >*mark‖los > **marcillus. The -ellus dim. is analogic to the such pairs as porculus/ porcellus, with marculus reinterpreted as a dim. and a new noun marcus back-formed to it[26]. However, the forms with a -t- : martellum (variant in some MSS of Isidore), martulus (Pliny, loc. cit.) cannot then represent inherited forms, since -tl- > -kl- is a Plt. change, and preceded any Latin syncope. Rather, as Heraeus (cited in Walde-Hofmann

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[25] The instrumental -tlo- suffix clearly underlies this word, as Niedermann IF 15.105 ff., was the first to observe. Its ultimate root connection remains more obscure. It is difficult to sunder marculus from a *mal- seen in malleus, but Niedermann's suggestion of an original *mal-tlo-s is unlikely, as Latin shows in such formations a regular dissimilation of l ... l to l ... r (e.g. *fulk-tlo-m > *ful(k)klom > fulcrum; for which see §5.28) and the outcome would be a **malcrus. Neither Pliny (7.56.57), Fronto (4.4.3) nor Martial (12.57.6.) show any diminutive sense in the word..

[26] Only Isidore records either of these words and his ranking by size is likely to be artificial. Marcellus is not therefore likely to represent an inherited double dim. like porcellus, though phonologically this is possible.

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1956.37) noted, they are hyper-urbanisms, based on the Vulgar Latin recapitulation of the change of -tl- to -cl- in new clusters arrising from syncope, thus VL veclum beside vetulum. The continuance of these forms with -t- in Romance (Fr. marteau, etc.) and borrowings in Celtic (Bret. morthol, etc.) point also to a VL origin for martellus (Walde-Hoffmann, loc. cit.).

16) Thus l_2 > -il- is confirmed. However, it is clear that r_2 became -er-, not -ir-: *agre-lo-s > *agr$_2$los > *agerlos > agellus. The asymmetry is curious and has caused difficulties for some scholars (Brugmann 1891.218 f.; Goetze 1923.83,119). On the basis of phonetic particulars, we might expect l, which is higher than r, to produce a higher vowel. Sommerstein's system (1977.112)[27] assigns to the the English phoneme l a height of 3 (out of 4) and r a height of 2. However, it is clear that n_2 (also with a height of 2; loc. cit.) became -in-: *pugne-lo-s > pugn$_2$los > pugilnos > pugillus. There are three counterexamples to the development of n_2 to in : columella to columna, *Sabnelo- > Sabellus, and scabellum to scamnum, and on this basis Brugmann (loc. cit.) reconstucts *n_2 > en (as well as l_2 > el)[28]. It is

[27] While Ladefoged 1975.268, 270, 272, who using a system based on percentage of variation in the first formant assigns a height value of 95 to both r and l, and one of 90 to both n and m.

[28] Those who followed his lead are then forced to explain all these cases of -illus/a/um as analogical, e.g. Kent 1945.98-99, who lists facultās as his only example of l > el (see §6.4). In addition if n, r, l all developed an -e- vowel, it is difficult to see where a source for -illus diminutives would ever arise. Kent (loc. cit.) suggests "forms with -ill- ... probably had an i by analogy of other words or forms in which there was an original i." I.e. Kent is suggesting a sequence *pāstni-los > *pāstn$_2$los > *pāstenlos --> *pāstin-los (by analogy to *pāstni-s) > pāstillus. However, as is clear from the examples above, the vast majority of these diminutives are thematics, and it is difficult to believe that the handfull of i-stems, e.g. pāstni-s > pāsnis > pāznis > pānis, dim. *pāstni-los > *pāstn$_2$los > *pāstin-los > pāstillus, could serve for models to replace the regular development of the thematics.

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tempting to argue that n_2 here shows a conditioned reflex after a labial but the form scabillum besides scabellum and *agᵘnelos > *agᵘn₂los > *agᵘinlos > avillus argue against it[29]. These -ell(us) diminutives are regular in thematic nouns and adjectives in -en-, e.g. *asen-o-s > asinus, but dim. *asen-e-lo-s > asellus[30], cf. duenos > bonus, *duene-los > bellus. It is here that we can refer columella, as a dim. not to directly to columna (which would give **columilla) but to the thematic middle participle that underlies columna < *kel-o-m(e)n-ā (with syncope), but dim. *kel-o-men-(e)-lā (with syncope of the thematic vowel)[31]. The pair scabillum beside scabellum strongly suggests analogy, and as Leumann (1977.143) notes, "freilich mag für -illo- statt -ello- oder umgekehrt auch analogische Vertauschung mit in Spiele sein." Scabellum cannot be an inherited double dim. of the porcellus type, since *scab-ne-le-lom > *scabnellos > **scamnellum. Note then, that this form allows us to order sync. 2/4 (in these cases at least) before the change of [+labial] > m / _ [+nasal].

17) Sabellī presents problems for this view. There is no independent evidence for a *Sab(e)ne-lo-, of which Sabellī might be the descendant. Samnium and Sabellī point to the immediate preforms *sabn-io- and

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[29] It would be possible to argue that an etymologically correct scabellum might be replaced by scabillum on the analogy of pugnus : pugillus, signus: sigillus, though for an entirely analogical creation we would expect pugnus : pugillus :: scamnum : X, X = **scamillus. However, this still leaves avillus in the way and on so small a data set it seems unsound to propose a rule conditioned by b only and no other labial.

[30] N.B. that sync. 3, of e / n _ l in any case, preceded the medial weakening of e > i in open syllables (see §8.28).

[31] Leumann 1977:306 seems to suggest something of this sort: "Aus -en-(e)lo-: asellus (asinus); lāmella, columella, auch Cogn. Col. (lāmina, columna);"

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*sabnelo- from a stem *sabn- (< *saβn- <*safn-, as if from a PIE *sabh-[32]), while Sabīnī points to a *sab-, which Niedermann (1953.177) takes as dissimilated from *sabn-īno-. Osc. safini is ambiguous. It can reflect a *saf-īno- (so Brugmann 1897.675, Ernout-Meillet 1959.585) or else a saf-nio- (so Buck 1904.2n., 53) with Oscan anaptyxis. If the latter, for which the Grk. borrowing Σαυνῖται speaks strongly[33], the change of f (> β) > m / _ [+nasal] must have occurred after Oscan anaptyxis,and so independently in Latin and Oscan-Umbrian[34].

18) Sabellī thus is the only counter-example to *n_2 > -in-, and we may take the development of l_2 and n_2 to il and in as assured. Since there are no good cases of m_2 (Leumann 1977.142-3; Goetze 1923.116), this leaves r_2 the odd man out. Oscan and Umbrian provide no examples of l_2 (von Planta

[32] Etymological speculations about proper names are problematica. E.-M. confidently assert a *s(u)e-bho- 'von eigenen Art', comparing Suēbī (as extended-grade), Goth. sibja < *sebh-ja 'relationship'. While "The Relatives" is not an unthinkable tribal name, forms with and without a -w- are the least of the problems here, and the explanation of Walde-Pokorny 2.456, which W.-H. accept, that the -a- of Sabīnī, etc. is due to the Italic borrowing of an Illyrian name with PIE *o > a is fantastic.

[33] I.e. reflecting a *saβn-, which Grk. borrows as saun-, the closest phonetic equivalent; see Leumann 1977. Allg. Teil 25.

[34] E.g. Umb. pelmner, Lat. pulmentum < *pelp-ment-o-.

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1899.226)[35]. There is no feature which unites /l/ and /n/ against /r/[36]. The "rhotacism" (that is the retroflex vowel) of English -er might serve as a parallel for a lower vowel quality before r (Ladefoged 1975.71-2). More germane is the effect of r in medial weakening, where there is a lowering of i > e / _ r (§9.1 ff.). It might be attractive to suppose that r_2 > -ir- parallel to l and n, and then was lowered to er. However as the lowering occurs only in open syllables and r_2 > er is necessarily in closed syllables (e.g. *krinō> *kr_2nō > cernō), this is ruled out.

19) A further point to be established in the derivation of pōcillum and similar forms, is that the Latin -lo- suffix, usually with diminutive force (Buck 1933:328), is joined to thematic roots with the -e- vowel, not -o- as is found in some works (Goetze 1923:83-4; Maniet 1975:105; however Buck loc. cit., von Planta 1899.222 already have it correctly; Brugmann 1891.218 reconstructs with -o- but notes that -e- is likely). This is not only suggested by comparison with other languages: *porkelos > *porcołos > porculus, cf. Lith. paršẽlis, Mod. Germ. Ferkel (Brugmann loc. cit.) but also guaranteed by the double diminutive porcellus < *porke-le-los, with syncope of the third syllable (sync. 3/4). A *porko-lo-los would have given a *porcollos > ** porcullus in a closed syllable (Leumann 1977.143). Thus our

[35] The only possible example known to me is the Osc. town name Abellanús. This has been connected with 'apple' and while Virgil's malifera is suggestive it may be no more than coincidence. Buck suggests *Apro-la (cf. aper) and von Planta *Agᵘno-la, i.e. a city distinguished for wild boar or lambs rather than apples. Aderl. (< *Atro-la) is the only example in Osc. for -rl- (there are none in Umb.) so it is difficult to say what the regular outcome is. The preforms for 'apple' indicated by the daughter languages are various, and there is nothing to recommend W.-H.'s *Ablo-na

[36] SPE assigns Eng. /r/ the feature [-ant.], but this is not the case for the apical-dental trill of Classical Latin. However, if /r/ at the time of the syncopes were uvular (see n. 4), a lower vowel might be expected, though this point is controversial. See Sommerstein 1977.102 and litt. there cited.

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reconstruction of *pōtle-lom > *pōklelom > *pōkllom > pōcillum is secure. Note that the change of PIE -tl- > -kl- has already occurred by the time of syncope. As this change is almost certainly Proto-Italic: *piātlom > *piāklom > piāculum cf . U. pihaklu (§2.14; Leumann 1977:153), this ordering is not surprising. More interesting is the fact that this syncope of -lV-> -l_2- bleeds the anaptyxis of clV_1 > cV_1lV_1, so *pōklelom > pocillum; not *pōklelom > **pokelelom > **pocellum (?) (see below §8.30).

20) Parallel to *faklisamos > *fac$l_2$zamos > *facilzamos > *facillimus, involving Rz > RR after syncope of the second vowel in four syllable words (sync. 2/4) are *pigr-isamo-s > *pig$r_2$zamos > pigerrimus; *pulcr-isamos > *pulc$r_2$zamos> pulcherrimus; *acr-isamo- > *ac$r_2$zamos > acerrimus (Leumann 1977:497; Cowgill 1970:125).

21) It is also clear that s must have voiced to z before the operation of this law of Rz > RR. We cannot state yet with certainty when z voiced in the derivation. At any stage: *faklizamos, fac$l_2$zamos, or facilzamos, a z is possible. That the voicing rules for s eventually cease to operate is clear from fulsī, versus, etc. (for these changes in the relative chronology see §§3.22 ff.). However, what can now be pointed out is that the later change of *z > r had not yet taken place in Latin. Thus, while in the sync. 2/4 of faklisamos a development: *faklizamos > **fakliramos > **fak$l_2$ramos > **facilramos > facillimus is conceivable, with z > r before sync. 2/4 (though no original cases of -lr- are known), the case of maximus is decisive. Starting from *magisamos (Cowgill loc. cit.; cf. Leumann loc. cit. who assembles the data but calls maximum, pessimus <*ped-isamos "unregelmäßigen") we have maksimus by sync. 2/4, where the voiceless s devoices the g. A change of s > z is impossible before this purely Latin syncope takes place, as a **magizamos would go presumably to

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**magzamos. The combination voiced stop plus z is phonotactically impossible in Latin: rēgō, but rēg-sī > rēxī, etc. It would be possible to account for this fact by a redundancy rule (for which see Schane 1973:35 f.; Sommerstein 1977:166 f. with litt.), so that any underlying -gz- generated by the rules would appear on the surface as -ks-. However, a chronological solution is equally probable, i.e. that s had not yet voiced to z in the creation of these forms. That is the regressive devoicing of -gs- to -ks- preceded any rules which voiced s to z. So: *magisamos > *magsamos > *maksamos > maximus. Note that this is a matter of purely Latin and not PIE phonotactics (as in the case of rēxī). If so, we may in all likelihood place the voicing of s / V _ V after sync. 2/4, to the extent that this change is regular. For facillimus, etc. then the derivation will be *faklisamos > *fak|samos (sync 4/2 with samprasāraṇa) > fak|zamos > *fakilzamos (with s > z after the sync. though not necessarily after the change of | to -il-) > facillimus. This diachronic view accords well with the synchronic data: no voiced stops plus s exist, nor is s voiced after n (for the phonological status of which and chronology, see §4.17 ff.), while surface -ls- and -rs- exist only as the result of later changes (§§3.22 ff., 5.4).

22) While it is possible that s had become z before sync. 2/4 (with maximus by a redundancy rule) it can not have gone on and become r. A **magiramos (**mageramos by **R-Lowering**[37]?) would become **magrimus, which while phonotactically possible (aegre, etc.) does not occur. So too *ped-isamos > *petsamos > pessimus, and not to **pedzamos > ??, nor to the phonotactically possible **pediramos > **pedramos > **petrimus??

[37] See Chapter 9.

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23) These seem to be all the cases for new -lz- in quadrisyllables. No cases of noninherited Rz in trisyllables are known to me. Syncope does not occur in trisyllbles, at least in morphologically transparent formations, where analogical restoration is of course a possibility. In -s- stems, a likely source of new Rs, we find for example: holus[38], holeris (attested in its pre-darkened form helus by Paulus-Festus. 89.3) < *ĝhelh$_3$os, *ĝhelh$_3$-es-es, and not **hellis. Syncope of any kind in s-stems is unknown (for rūs/rūris, iūs/iūris, see §7.43). Disyllables are subject to syncope: tris-noi > *tr$_2$znoi > *terrnoi > ternī[39]; *kri-to-s > *kr$_2$tos > certus, and there is even a case where a monosyllable has undergone samprasāraṇa in ter. This syncope of rV̆ in all final syllables, and in initial and medial syllables before an open syllable (i.e. V̆ > Ø / r _ C{V,#}; see below §9.9) changes *tris (Skt. tríḥ) to *tr$_2$s > *terz > terr (cf. terr-unicus; also guaranteed by the scansion of Plautus Bacch.1127) > ter (Brugmann 1897-204; Sommer-Pfister 1977:70, 204; Leumann 1977-142). Note that the voicing of s > z / r _ occurs in final position, even though no voiced segment follows (cf. §2.21). Cowgill (1982) takes ter from *tris by a metathesis[40]. If *tris becomes *tirs, the problem remains of how to get a *ters. Clearly the rule of vowel lowering before r is of no help here, since it operates only in open syllables (cf. *viros > virs > *virr > vir, see §9.9). Maniet (1975.88 with 126) seems to be trying to make use of some such rule. However, in view of the lack of further examples of such a metathesis (*kritos, etc. must undergo syncope to *kr$_2$tos > certus, as a **cirtus by metathesis would simply remain) and keeping in mind the regularity of rV̆s > rs (*agros

[38] For holus, and the anomalous -e- of helvus, see §§3.8-18, below.
[39] Therefore trīnī is analogical to bīnī and other compounds of trī-.
[40] This idea was first proposed by Hermann 1923.265.

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>*agrs > ager) apparently including monosyllables, such a metathesis seems unnecessary. This rV̆ syncope also then feeds r > -er- , which feeds Rz > RR.

24) Further examples of s > z and Rz > RR* in original final position are:

*sals > *salz > *sall > sal;

*pars (cf. Umbr. pars-est) > *parr (guaranteed by meter at Plautus Poen. 376[41]) > par ;

*fars (cf. Umb. fars-io) > *farr > far.

Analogical restoration is seen in fers (either direct to fer, or else parallel to the development of vīs, *ferr is recharacterized as *ferrs > fers, Cowgill 1978.31), uls (Leumann 1977: 220-1), though this might reflect a regular change from *ult-s (cf. ultis). Thus s > z /R _ is not necessarily older than the loss of final short vowels, since we could have: *velsi > *vels > *velz > *vell > vel, as well as *velsi > *velzi > *velz > vel (§1.5).

25) This syncope of rV̆ seems to be regular. A similar syncope exists in Oscan and Umbrian[42]. It is not, however, Proto-Italic as shown by the SAKROS of the Forum Inscription (CIL 1[2].1, c. 500)[43]. Since rV̆ syncope feeds **RZ-Assimilation**, this is good confirmation that **RZ-Assimilation** is a purely Latin-Faliscan change.

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[41] As Leumann points out, this is a nicely fossilized form, since the neut. par est also scans long at Plaut. Pers. 834: nisi si dignust, non opust./et me hau par est.

[42] Standard examples *agros > *agrs > O. ager, *tris-tōs > O. trstus (§5.1; Buck 1904.91). A similar syncope of -ri- to r, which develops to -er- as in Latin may perhaps be seen in Venetic vo.l.te.r.-kon. apparently 'voluntary' if this is correctly analysed as an adj. in -iko- to a substantive *vel-tro-; Lejeune 1974.123-4.

[43] An excellent example of the tentative nature of the assumption that a shared innovation must necessarily belong to the period of the Proto-language. See Buck 1904.61.

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Chapter 3: Feeding Rules

1) We have examined the rules which lead to new double l's--that is, rules which bleed **L-Darkening** by changing an -lC- cluster to -ll- and so keep the l's bright. We need now to look at feeding rules, i.e. rules which create more material for **L-Darkening** to work on. In particular, our attention will be on new surface groups of -ln-, -ld-, -ls-, and -lu̯- which do not assimilate. The rules which lead to these must therefore lie between the earlier assimilations (**LDN-Assimil.**, **LW-Assimil.** and **RZ-Assimil.**) and **L-Darkening** itself. We shall consider first the cases of syncope and vocalization. These processes are, as has been noted, more difficult to establish as lautgesetzlich rules, and therefore less useful as cardinal points in building a relative chronology. When these have been dealt with, we shall examine larger processes involving consonant clusters which resulted in new -ln-, -ld-, -ls-, and -lu̯- .

New -ln- by Syncope

2) A new case of surface -ln- by syncope is ulna. The preform is most likely to have been *olenā. Grk. ὠλένη points to an *ōlĕnā with long ō. However, Grk. also shows an ὠλήν, gen. ὠλένος, or an ὠλῆνος (Hesych.) which seems to point to an originally hysterodynamic paradigm (see Rix 1977.121 f.):

n. ōl-én

a. ōl-én-m̥

g. ōl-n̥-és[1]

[1] The Hesych. ὠλλόν also seems to reflect a mix of an *ōl-n̥- and a thematic.

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Lith. shows both ō- and ŏ-: úole-ktis < *ō-, but alkúnė < *ŏ-. The vowel of the suffix is equally variable. Grk. shows -ĕ -, while Goth aleina < *ŏlĭna, but OHG elina from a Gmc. *ŏlĭna (< PIE *olinā or *olenā). Olr. uilen /ul'en/ < *ŏlĭna or *ŏlīna, while Welsh elin < *ŏlĭna (since *ĭ > y). A long ī would not syncopate in Latin so we seem to have a *olĭna or *oliĕna > *olna > *oɬna > ulna. (Brugmann 1897:153, 424; Leumann 1977:213).

3) A further case of new -ln- by an early syncope is vulnus, vulneris, built to the root *uelh$_2$- (cf. Hitt. ualh- 'strike') like moi-n-os > mūnus[2]. So *uelh$_2$-n-os > *uelanos > *ueɬanos > volnus > vulnus[3].

4) Other examples of -ln- by syncope, are the late balneum, where the unsyncopated balineum is attested in Plautus (Most. 756 etc.), < βαλανεῖον. This sync 2/4 is evidently after vowel weakening of a > i. Yet the syncope of the metrically similar Πολυδεύκης is very old (§2.10-11). Bal(i)neum seems to have been borrowed later. Due to the apparently earlier[4] syncope 3/4 are the tree names pōpulnus < *pōpol-inos, aesculnus < *aiskol-inos and fīculnus, ficulneus < *fĭk-e-l-inos (from the dim. *fĭk-e-los > fīculus); see below (§2.10-11). The later sculna 'mediator, umpire' (Varro cited by

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[2] With the inherited -n-os (-n-es-es) employed in Latin "im juristisch-sozialen Bericht" (Leumann 1977.378), e.g. fūnus, facinus; see Meillet MSL 15.260.

[3] Note that the darkening of l to ɬ can be due to the -a-, and so occur before the sync. 3 / l _ n. Later (§3.17), it will be argued, on the basis of helvus that the rule of l > ɬ was in fact over by the time of sync. of V̆ / V̆R _ u, at least. Leumann 1977.96, 378 gives the preform as *volenus with no explanation, as does Ernout-Meillet 1959.750 who note, however, the set root.

[4] Already pōpulnus in Plaut. Cas. 384; in addition, this syncope between l and n appears exceptionless.

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Gellius 20.11.2) appears to be Etruscan: scvulna[5]. No further cases of -ln- by syncope are known to me[6].

New -ld- by Syncope.

5) Here we find the common valdē < *validē beside validus, calda beside calidus (Leumann 1977:97). The most likely explanation for the differing treatment of *melit > *melid, gen. *melit-es --> *melides (by analogy to nom.) > *meldes > mellis, with a prehistoric sync. 3, is that the morphologically unanalyzable *melides underwent a regular syncope before -ld- > -ll- (see above §2.12-13), while validus etc., were subject to analogical retention or restoration beside the many adj. in -idus (tepidus[7] etc.) where no syncope occurred. These seem to be the only cases of new -ld- by syncope. This restoration is paralleled by the chronological difficulties raised by ornus vs. pōnō (Leumann 1977.99). While *po-sinō becomes *poznō (by sync. 3) > pōnō (with compensatory lengthening), *os-inos with exactly the same structure rather undergoes **S-Voicing** then **Rhotacism** before syncope: *os-inos > *ozinos > *orinus > *ornus[8]. A

[5] The folk etymology repeated since Gellius connecting the word with sequester is unlikely both semantically and morphologically: cf. scurra also from Etr.

[6] For alnus, see below §3.26. Gradenwitz (1904) lists a culna, which I have been unable to find elsewhere.

[7] Again the conditions on syncope are unknown, but it is clear that syncope was favored after or between resonants. It also seems likely that syncope does not, exept in a few cases, lead to consonant clusters which do not otherwise exist (for a single example, cf. pōnō below). Thus in the case of tepidus, etc., these forms are unlikely ever to have been subject to syncope. There will be, therefore, many analogical models for the retention of the -ĭ- in the analyzable cases (see §2.13).

[8] Cf. Falis-cus, but *Falis-inos > *Falizinos > *Falirinos > *Falerinos > Falernus. Here we can make the observation, repeated below (§9.2-3), that this particular syncope does not occur until after both **Rhotacism** and the lowering of vowels before r, which occurs only in open syllables, cf. stĭrpo, exstĭrpo; vĭrga, primivĭrgius, etc.

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possible answer is that OCS jasenĭ might permit a Lat. pre-form *ōsĕnos[9]. Could we assume that ĭ (posĭnō) was syncopated earlier than ĕ (*ōsĕnos)? Arguing against this is the failure of -s- to vocalize till after syncope in *subs-emō > *supsmō[10] > *susmō[11] > *suzmō > sūmō. The answer, rather, seems to lie again in morphological analyzability, with *osinos restored after the other tree names and adjectives in -inos: fāg-inus, frax-inus. This retention/restoration would be an early process for the -i- was particularly subject to syncope between voiced consonants and so seldom made its way to the attested form[12], cf. pōpulnus < *pōpul-inos, etc. (above §3.4).

New -lu̯-

6) It appears that all new clusters of -lu̯- remain unassimilated. This is in keeping with our findings that -lu̯-> -ll- is very early and possibly Proto-Italic. New -lu̯- (outside of consonant clusters, see below Chapters 4, 5.) can arise by three processes: 1) the prehistoric syncope of V̆ / V̆R _ u̯V, 2) the syncope after the OL period of V̆ / V̄R _ u̯V, or 3) irregular metathesis. There are thus many sources for surface -lu̯- clusters, and not all of the words with -lu̯- have cogent etymologies. This has led some scholars to deny -lu̯- > -ll- altogether as a rule for Latin (see §2.8-9 above). Examples of -lu̯- by the early syncope are: *sl̥H-u̯o-s > *salau̯os > salvus (for sollus, see §2.8, Exc. II. §6.6); cf. *h₂erh₃-u̯o-m > *arau̯om > arvum; *se-luu̯ō (cf.

[9] For ŏs- cf. Olce askr, OE æsc, MEng. ash < Gmc. *askiz < *ŏs-kis; Celtic shows an n-stem with no vowel: W. onn-en < *ŏs-n-; Olr. (fem.) huinnius < *ŏs-n-jō(n); Balto-Slavic points to *ōs- : OCS jasenĭ , Lith. úosis < *ōs-is.

[10] See above §2.21,for the late voicing of -s-; cf. *magisamos > *maksamos > maximus.

[11] For loss of C / _ sC, see Chap. 4.

[12] W.-H. claim fāginus (already in Col. 12.45.5, and fāgineus in Cato RR 21.4) as borrowed from Greek. However, the earliest attestations of f for Grk. φ, reflecting the change in Grk. pronunciation, are imperial: Orfeus (D 140, 59 A.D.), Fyllis, Dafne, Trofime (Pompeii); cf. Leumann 1977.162.

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λύω) > *seɬu̯ō[13] > solvō ; *u̯eluu̯ō > *u̯eɬu̯ō > volvō. (cf. ϝελύω). 2) By the later syncope after long vowels *pēleu̯is (Skt. pālavī; see Exc. II §2) > Plt. *pēlou̯is > *pēluu̯is (attested in pēlŭĭm , Laberius 94) > pēlvis; so too, mīlŭŭs (Plaut. Aul. 316, etc.), lārŭă (Plaut. Merc. 981, etc.) > lārva (disyllabic first in Hor. Sat. 1.5.64). Leumann 1977:132 treats these as changes of a vocalic ŭ to u̯, and says "Nach Langvokal ist -lu̯- -ru̯- im Altlatein noch erhalten". Rather, more formally: in the cases after a short vowel, an early (prehistoric, but after the assimilation of -lu̯- to -ll-) syncope occurs of any short vowel; while in the later cases (well into the historic period), after a long vowel, there occurs a syncope of ŭ /R_u̯V, which has resulted from Medial Weakening (see Ch. 8). 3) Metathesis: A change of an original ŭR > Ru̯ is suggested by Grk. αὐλός but Latin alvus; of νεῦρον but nervus; παῦρος but parvus. However, this metathesis is not regelmäßig; so Grk. ταῦρος corresponds to Lat. taurus, while Gaulish shows the metathesized form tarvus (Leumann 1977:101). The etymological connection between αὐλός 'flute, pipe' and alvus 'belly' is also not perfect. (However, cf. alveus 'hollow, cavity'). These seem to be our cases of new -lu̯- not by consonant cluster reduction.

7) There are in addition some nine words, which show a -lv- in Latin, few of which possess a good etymology. I propose treating them very cursorily here, noting simply that if the assimilation of -lu̯- to -ll- holds good, these forms must be by either some type of syncope or else come from a simplified consonant cluster.

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[13] Again the l can become dark (/_ u̯) before the syncope. As will be argued below (§§8 ff.), helvus shows that **L-Darkening** was over before this syncope took place.

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a) calvus: the standard etymology compares Skt. -kulvah, Av. kauruua-[14]. E.-M. (s.v.) says, " L'a peut reposer sur un a «populaire» indoeuropéen." Rather, the vocalism calls to mind that of salvus, and Pokorny (1959.554) notes Skt. áti-kūlva- 'completely bald," and takes the preform as *k$_e$lauo-, *keleuo-. W.-H. (s.v.) also note the set nature of the root. This allows a preform *k]H-uo-[15] > *kalauos > calvus. To this can be compared the Osc. proper name Kalaviis, said by von Planta (1892.258) and Buck (1904.51) to show Osc. anaptyxis[16]. Osc. also has the proper name Kalúvieis and as Buck (1904.189) notes this reflects a -euo- suffix beside a -uo-, the preform here then being *k(e)lH-eu-iio-[17] (see Exc. II, §6).

b) On fulvus E.-M. says,"Le groupe le plus ordinairement rapproché est celui de lit. geĩtas, v.sl. žlĭtŭ (serbe žût) «jaune»; cf. sous fel. Des formes à gh- prépalatal sont signalées sous holus." For this group of words see below. However, it is now to be noted that *ĝh- does not regularly become f-,

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[14] Mayrhofer, s.v. notes that kauruua-, found only in a number of compounds in Yt. 8.21, may not mean 'bald' at all, and that the sense "verstümmelt, kurz" fits equally well.

[15] For the shape cf. *s]H-uo- > *salauos > salvus, see above §2.8 and below Exc. II §6.G.

[16] Cf. Osc. Salaviis, salavs, σαλαϝς and Lat. salvus. It is less likely to show an unsyncopated inherited form, though Oscan syncope is not absolutely regular. For -a- unsyncopated in a medial open syllable, cf. O. Anter-stataí but eestínt < *eks-stahint; this is the only example of unsyncopated a in an open syllable, vs. three examples of syncope (see Buck 1904.58), and the resulting -tt- may have had some influence. Here Buck (ibid.) says "salavs, σαλαϝς 'salvus' owes its form to the oblique cases, for *saluos would give *salus in the O.-U. period." Again the inherited form is also possible. It cannot be entirely ruled out that this is a Latin family name borrowed into Osc. However, since the Osc. form Calavus occurs in Latin inscriptions (Buck ibid.), it is more likely to represent a genuinely Oscan name.

[17] -euio- in Buck's notation.

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precisely as holus shows[18] and for initial f- in Latin the possibilities are *gʷh- (formus), *ĝhu/u̯- (fundō, ferus), *dh- (fūmus) or *bh- (frater). Pokorny (1959.160) points to the correct solution by comparing Lat. flāvus and the Gmc. family of OHG blāo 'blue' but also 'yellow', OE blāw, etc., which he takes from *bh̥i̯-u̯o-s, to fulvus, which he takes from *bh̥l̥-u̯o-s. In laryngealist terms, rather than two versions of the same root, flāvus comes from the zero-grade *bh̥lH-u̯ó-s , while fulvus represents the full grade < *bhélH-(e)u̯o-s (> *felau̯os > foɬu̯os > fulvus). This root shape, however, is not directly comparable to the e-grade *bhleH- shown by the Gmc. cognates[19] or the o-grade of flōrus, Olr. blār , W. blawr < *bhloH-ro-s. The *bhelH-, then represents the familiar TeRT/H- new full-grade built to the zero-grade[20].

c) malva 'mallow': Greek shows a variety of forms μαλάχη, μαλόχη, μολοχά. E.-M. (s.v.) say, "sans doute mot pris à une langue méditerranéenne." Hebrew shows a mallūāḥ (W.-H., s.v.), and the Latin probably reflects a *mal(V̆)u̯ā.

d) pulvīnus 'cushion', also pulvīnar 'a couch equipped with cushions for the gods', etc. E.-M. : "Étymologie indéterminée", W.-H. : "Herkunft unsicher."

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[18] There are interchanges of initial f- and h- which the ancient grammarians who preserve them call "old" ("faedus folus fostis fostia antiqui dicebant" Paul. Fest.) or "Sabelian" (Varro Ling. 6.15; see Leumann 1977.168-9), and color terminology is not an impossible place to find dialectal variants as rūfus beside ruber shows. However, no Latin form with h- is attested beside fulvus. Before dialectal influence is arbitarily asserted, the possiblities of a regular development should be explored.

[19] A zero-grade is impossible for Gmc as RH and R fall together in Gmc. (*p̥lh₁-nó-s > Goth. fulls, etc.)

[20] See Anttila (1969) for a survey of the earlier statements on Schwebeablaut.

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e) pulvis: Latv. pelus (pl.) shows a plain u-stem, while Latv. pelawas (pl.). shows -eu̯o-, and Skt. palāva- points perhaps to an -ou̯o-. There is clearly a connection to pollen (an original n-stem with the n of the zero-grade spread to the root? So W.-H.; cf. n. 33). For Latin a preform of *pel-eu̯i-s, with perhaps substitution of i-stem for thematic, seeem likely. See Excursus II.

f) silva: no known cognates or etymology. The scansion sĭlŭae (Horace Odes 1.23.4) is, of course, a poetic licence and does not point to a *siluu̯a.

g) ulva 'sedge, a marsh plant'. E.-M. (s.v.) give no etymology. W.-H. (813, 23) connect it with alga 'sea weed' which they derive from *ₑlgā, comparing Skt ṛjīṣaḥ 'sticky, slippery, slimy' and various Norwegian dialect words: ulka 'purulent, mold, an unclean woman', Danish ulk 'frog', etc. This etymology is itself very dubious and even if the semantic connection between ulva and alga were convincing (or those between alga and its presumed cognates) this would still leave the initial u- unexplained. As W.-H. note (813) a preform "*olu̯ā [ist] wenig wschl.", and indeed if -lu̯- > -ll-, impossible. Pokorny (1959.305) posits a root *el /*ol 'modrig sein, faulen' and a preform *oleu̯ā. Again, while oleu̯ā would give a Latin ulva, the semantics are not by themselves compelling, and the preform has no support. Rather given the wide variety of forms that plant names frequently exhibit, and the possibility of dialectal borrowing, folk etymology or substratum influence, a clear etymology for ulva is not to be expected readily. It can, in any case, not be used as a counter-example.

h) valvae: E.-M. :"Doit appartenir au groupe de uoluo ; partir de wºluwā ?"; W.-H.: "zu volvō ". The root connnection is almost certainly correct, though the -a- vocalism remains a puzzle.

i) vulva also is most plausibly connected with volvō, with the basic meaning 'wrapper'.

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8) Besides these nine words, there is also helvus which necessitates separate and longer treatment. What has passed unremarked in most previous discussions of this word is the set nature of the root[21]. Greek χλωρός points clearly to a *ĝhlh₃-ró-s: zero-grade of a root *ĝhelh₃- with the usual accent and vocalism of derived -ró- adjectives[22]. Skt. híraṇya 'gold' with -ir- also shows a zero-grade set root, < *ĝhlh₃-enio- (cf. Av. zaranim, OCS zelenŭ), beside the full-grade hari- < *ĝhelh₃-i- (Av. hairi-). The Gmc. data can easily agree with this, as R̄ (< RH) and R fall together in Proto-Gmc. Thus *ĝhlh₃-tó- > Goth. gulþ (cf. *plh₂-nó- > *fulnaz > Goth. fulls), *ĝhelh₃-uo-s > PGmc. *gelwaz > OE geolu > yellow (cf. for treatment of medial laryngeal, the test case *dhugh₂tḗr > *dukter > *duhter > Goth. dauhter). Most important are the Balto-Slavic data: Lith. žãlias, žãles (long ã under the accent), žalvas 'green' and žalsvas 'green-ish' (no accents recorded) from the o-grade; verb in the e-grade inf. žélti, 3. pres. žélia (long under the accent), žéle (long ē preterite, as is usual for -io- pres. and so not probative for a set root). What has not been pointed out are the zero-grade formations, in particular the thematic žìlas, which with its accent and the preconsonantal development of -l- must be from *ĝhlh₃-ó-s > *žil(H)os > žìlas (see Watkins 1965.117 for a clear and succinct statement of the development of laryngeals in BS).

9) A second feature of this root also needs to be noted: it functions in a set of Caland derivatives. Clear evidence is the -ró- suffix of χλωρός, and the compositional -i- of Skt. hari-ná- 'gazelle', hári-ta- 'yellow/green', hári-vat 'drawn by tawny (horses)' (of Indra), etc.; Av. zairi- 'golden',

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[21] Pokorny 1959.429 lists "g'hel- (und ghel- ?)...; g'lela, g'hlē, g'hlō: g'hla-" but invokes the disyllabic form specifically only for Ir. glán < *ghla-.
[22] For RHC in Greek, see Rix 1976.72-4.

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zairita- 'yellow', etc. In addition, for the Caland set note the s-stem neut. *ĝhelh$_3$-os > Lat. helos (helus Paul. ap. Fest. 100) > holos 'green stuff(s), greens, vegetables' (see §3.18, 6.2). The thematic vowel itself can function as part of a Caland set; thus we have *ĝhólh$_3$-o- > χόλος 'bile' and χόλη 'anger', Av. zāra-, and the above mentioned *ĝhlh$_3$-ó-s > Lith. žìlas.[23]

10) As an etymology for helvus Brugmann (1897.324, 334) and Walde-Hofmann (1938.639) proposed a proto-form *ĝhelu̯os and compared Lat. fel, holus, flāvus and fulvus[24]; Lith. želvas, etc. This etymology is impossible since the root is set, and a *ĝhelh$_3$u̯os would give PIt. *helau̯os > *heɬau̯os > *holvos > **hulvus, and even an anit root *ĝhelu̯os would fall foul of the -lu̯- > -ll- assimilation and yield **hellus. Osthoff (1893.59-60), to avoid this latter problem, proposed a *ĝhel-eu̯o-s on the pattern of κενεϝός, etc. (see Exc. II §6, E). Again the set root here would give a *ĝhelh$_3$-eu̯o-s > *heleu̯os [25]> *hoɬvus > **hulvus. An anit root would equally give *heɬeu̯os >

[23] Since a u-stem adj. can also form part of a Caland set, but -ū- nouns are usually isolates in Greek, even those with clear inherited forms (ὗς, etc.; see Risch 1974.65, 75-6), it is unlikely that χέλυς 'tortoise' can be connected with the root *ĝhelh$_3$- (i.e. as if 'the green thing). This etymology was first proposed by Bechtel (cited in W.-H.), and has been reoffered many times subsequently (Brugmann 1904.220; recently Meid IF 63 [1958] 9 n. 9). Given, however, the close agreement of OCS. zely < *g(h)elū, zelĭve, it is possible that the Greek reflects an inherited word from the plain velar root seen in the Baltic and Slavic reflexes, though Schmeja (IF 68 [1963] 40-1) takes it as more likely to be a Fremdwort folk-etymologized to the words for 'green/yellow'.

[24] That all these represent a single ĝh- initial root involves the assumption of dialect influence is the least of the problems. It is neither desirable nor necessary to assume a single root here. See above §7.5.

[25] Or perhaps *heleu̯os. It is not clear whether a laryngeal in the environment /C _ e would color the following e across a morpheme boundary (within a morpheme, cf. the 1. sg. pf. ending -h$_2$e). See remarks of Beekes (1969.181) on Greek. Rix's examples (1976.72), *dh$_3$-ént- > δόντος, etc. are complicated by problems of the proper ablaut grade. For Latin, there appear to be no certain examples.

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*hol(o)yus > **hulyus (or **holyus, if no syncope), since e also darkens l (noted by Cowgill 1978.42).

11) Connection with Lith. geĩsvas is proposed by Leumann 1977.141, E.-M. 1959.291, setting up a preform *ghelsyos. To begin with, the root must have a plain velar (Lith. g- not ž-), and based on the Baltic-Slavic evidence may be aspirated or not. Clear evidence of a plain velar is found only in Baltic (beside the palatal of žilas, etc. and Skt. hari-[26]) and Slavic, where the zero-grade form shows ž < g < g(h) with secondary palatalization (*g(h)ltos > *gĭltos > *žĭltos > žl$_2$to > OCS žlŭtĭ 'yellow'); however the o-grade forms point only to a palatal: *ĝholh$_3$-to- > *zolto- > OCS zlato 'gold'.

12) There is, in addition, one form in the Baltic evidence which speaks against a root-final laryngeal for the set of words which show the initial plain velar. Several of the derivatives in -no- of this root in Balto-Slavic indicate a kind of bird: Russ. žluna < žĭlna < Proto-Sl. *glna, Latv. dzilna 'Specht', Lith. gilna 'Wacholderdrossel' (Pokorny 1959.430). The accent of gilna (not **gìlna) argues against a final laryngeal; contrast pìlna < *plh$_2$-nó-; žìrnis[27] 'pea' < *ĝrh$_2$-nó- (Lat. grānum; OCS zrĭno, Goth. kaurn). It seems then that we are dealing with two different roots: a *ĝhelh$_3$-, which is the base of a set of Caland derivatives, and a *g(h)el-, not definitely attested outside of BS, both of which furnish words for colors ranging from yellow to green to grey, and whose ultimate relationship, if any, is speculative. Of the satem languages, Baltic and Slavic show forms of both roots, Skt. only of *ĝhelh$_3$-. The Gmc. forms (due to the treatment of the laryngeal) could derive from either a *ĝhelh$_3$-, or a *ghel-. Greek χλωρός

[26] E.-M. (loc. cit.) lump all these forms together. Pokorny (loc. cit.) holds them to be variants of a single root, but does distinguish the Lith. g- forms.
[27] Given consistently in Brugmann (1897.418, 474, 568) as žírnis; the vowel is properly marked short (Senn 1966.88, etc.).

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and Ir. glán derive from the laryngeal final *g'helh₃-, as do Latin holus (part of a Caland set) and helvus, as is argued below.

13) Cowgill's objection (loc.cit.) that a *ghelsuos should have given a *helluos > **hellus, based on *molluis > mollis cannot now be maintained with certainty, since the etymology of mollis < *molluis < *molduis appears to be incorrect (see Excursus II) and with it disappears the only putative -llu- cluster in the language.

14) However, his second objection is that, as comparison within Lith. to geĩtas, geĩtonas, etc. (and to Latv. dzeltāns, etc.) shows, geĩsvas derives from a base gelt-[28]. This is fatal to its use in a Latin etymology. A *gheltsuos in Latin would become **hēvus, developing parallel to cēna (for which, and the ordering of these rules, see §§5.1-2).

*χeltsuos	*kertsna	
*helssuos	*kerssna	(-ts- > -ss-: §4.8)
*hessuos	*kessna	**(CsC-Loss**: §4.1 ff., 5.1-2)
*hesuos	*kesna	**(Geminate Simp.**: §3.23)
*hezuos[29]	*kezna	**(S-Voicing**: §§4.3 ff.)
**hēvus	cēna	**(Comp. Lengthening**: §4.4)

15) A connection with Lith. geĩsvas can therefore be ruled out on several grounds: morphologically, since -suo- is not productive outside Baltic; and phonologically, since the preform necessary for BS yields the wrong results

[28] So too Pokorny 1959.429-31, Trautmann 1923.63-4. W.-H. had objected to a preform *ghelsuos on the grounds that "-suo- außerhalb des Baltischen nicht begegnet." Brugmann (1904.205) holds that traces (from s-stems plus a -uo- suffix) are perhaps to be observed elsewhere, but as he notes the suffix is likely due to false segmentation in primary color terms whose stem ended in a -s-. It is only in Baltic that -sva- becomes productive.
[29] For s > z / _ u, cf. *seks-uiroi > *sesuiroi (**CsC-Loss**) > *sezuiroi (**S-Voicing**) > sēvirī (§4.4).

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for Latin. Rather the pair holus (< *ĝhelh₃-os) and helvus must be connected with the root of the Caland set seen in Skt. hari-, Grk χλωρός.

16) Warren Cowgill (ibid.) was the first to suggest invoking the Skt. form to explain Lat. helvus:

> If we are to venture an explanation for helvus (I am reminded of the dictum that not every word deserves an etymology), I would suggest Italic *xeli-u̯o-s, i.e. the stem ĝheli- seen in Skt. hári- 'pale, yellowish', plus a u̯o-suffix added on the model of the other Latin color adjectives such as fulvus, flāvus, rāvus. The *-i- would serve both to keep the l from becoming velar in time to cause the *e to become *o, and to keep the l and u̯ from assimilating; by the time it syncopated, the resulting velar ɬ was no longer able to affect the timbre of the preceding vowel, and the rule assimilating *lu̯ to ll was no longer operating.

17) We can now be precise about the formation of helvus. The root *ĝhelh₃-, in a full-grade Caland compositional form *ĝheli- (with the laryngeal disappearing before the vowel), which therefore originally lacked a -u̯o-suffix, acquired one analogically from the primary -u̯o- color adjectives. Phonologically and chronologically, this works well. A Plt. *xeliu̯os undergoes the sync. of V̆ / V̆R _ u̯ (see above §6), before which we have ordered **L-Darkening**, before which is ordered lu̯ > ll (see §2.8). What we would like to have is another sequence of *Celiu̯V- > *Celu̯V-, to test this ordering of changes, but of course what makes helvus unusual is exactly the e before lC. No other such preform of the shape *Celiu̯V- is known to me, nor ones involving the other consonants which assimilate to l, i.e. *CelinV-, *CelidV- or *CelisV-[30]. We can argue, however, on grounds of probability. The change of -lu̯- > -ll- is, as we have seen very early, and precedes the sync. of V̆ / V̆R _ u̯. What sort of indirect evidence might there

[30] The syncopated vowel must be an original -i-, as any other could result in a velar l at an earlier stage of the derivation.

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be, besides helvus itself, that the change of l > ɬ and so the change of e > o / _ ɬ was over with by the time that this syncope occurred? The -o- of volnus > vulnus, is not good evidence that l > ɬ and e > o did not happen till after syncope, since the l of (*ǵelHnos >) PIt. *ǵelanos could have become velar due to the following -a- at any time before the syncope to *ǵelnos > volnos > vulnus (§3.3 n. 3); likewise for seluyō > solvō. We do, however know that the change of e > o / _ɬ was fed by Medial Weakening of ă to ĕ; so: *ad-alēskēnts > *adelēskēns (**A-Weakening**) > *adolēskēns > adulēscēns (Ch. 8). We also know that the change of e > o / _ ɬ was over with before the raising of the new long monophthongs; thus we have: *elaiua (ἐλαιϝα) > *eleiua (**A-Weakening**) *oɬēua (**Monophthongization**, attested to by Goth. alēw) > olīva, where an -ī- would have prevented the l from becoming velar (§6.2). In addition, of course, the change of e > o is before (bleeds) the raising of e > i in open syllables (Siculus, not **Sicilus; §8.28). None of these examples is probative for the ordering here proposed, especially since various examples of sync.-3 themselves show various orderings[31], but they do add up to e > o / _ ɬ being an early change, certainly nothing argues against it being over by the time this -i- syncopated between l and u. We are left then with the situation familiar in Relative Chronology, in which, because of the rules involved, only a few words in the whole of the language can serve as evidence, or one, or none (see Cowgill 1978.33).

18) Morphologically, this addition of a -uo- suffix to a Caland base can be paralleled in Grk. πελιός 'bruised, livid', πολιός, Myc. po-ri-wo 'grey'. Evidence for the -i- compositional stem is found in Skt. páli-tá-(same formation as hári-tá-), páli-knī (< *pel-i-tn-ih$_2$), cf. Grk. πελιτνός; further,

[31] See §3.5 for pōnō vs. ornus.

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for the Caland set, the s-stem seen in πέλεια < *pél-es-ih$_2$. See Chantraine 1933.123; Risch 1974.169.

19) We can set up a *χeliu̯os then as the preform for Latin helvus. Will it also explain Osc. Heleviі[s], gen. Heleviieís, Pael.[32] Helevis? These forms shown clear anaptyxis, and reflect an immediate preform *helu̯-iios (Buck 1904. 50, 121-2). We have no way of knowing when or under what conditions an -i- would have syncopated in Oscan (Buck 1904. 57). Heleviі[s seems to show the same development as salavs, Kalaviis, with anaptyxis of the vowel from the preceding syllable following an Oscan syncope[33]. There are examples of the less regular (later?) anaptyxis of the vowel of the following syllable after syncope has taken place, e.g. *diē-kelo- > diēklo- > Osc. zicolom. I have been able to find nc counter-examples to prevent taking *χeliu̯os as the Proto-Italic form.

20) From this same root *ĝhelh$_3$- come also words for 'bile'. Cited above are the o-grade thematic nouns χόλος, Av. zāra-. Germanic shows an o-grade root with the ablauting -ōn/-en-/-n- suffix: OHG, OS galla, OE gealla, ON gall[34] < PGmc. *gallōn < goln-ōn[35] < *ĝholh$_3$-ōn. BS, however, seems to show a zero-grade root with no laryngeal and a -tí- or -kí- suffix: Lith. tulžìs, metathesized from *žultìs (cf. Latv. žults) < *ĝ(h)l-tí-s and not

[32] Not Osc. as Sommer-Pfister (1977.56) state.
[33] These are the only two examples of this sequence and the possibility was raised (n. 16) that they reflect unsyncopated forms. Herekleís reflects a pan-Italic syncope with the likelihood of Etruscan intermediacy (§1.2)
[34] Replacing OE gealla, whence Mod.E gall.
[35] The ablauting suffix generated a paradigm nom. *gol-ōn, loc. *gol-en-i, gen. *gol-n-es. This zero-grade oblique stem was generalized throughout the paradigm, so the new nom. *goln-ōn > *gallōn. For this well known process, cf. *sun-ōn, sun-n-es --> nom. *sunnōn > Goth. sunnō. See Brugmann 1906.303. See also §21 below.

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*žùltis (bzw. *žìltis[36]) < *ĝ(h)l̥h₃-tí-s; note Serb. žût (not žùt). Cf. OCS zlŭčĭ, Serb. žûc < *ĝ(h)l̥-ki-s. OCS, however, also shows forms with a plain velar (perhaps the *g(h)el- root posited for OCS žlutu 'yellow', where only the zero-grade forms show clearly a plain velar): so žlŭčĭ, žlŭtĭ apparently from *g(h)l̥-ti-/*g(h)l̥-ki-. This alternation is extremely puzzling, but is clearly an intra-Balto-Slavic phenomenon.

21) Does Latin fel belong here? The first problem is the initial f-. If fel is to be derived from the root *ĝhelh₃-, it must be considered a dialectal borrowing, in the face of holus, helvus. Paul. Fest. 84 does note a folus beside holus (see n. 18). While borrowing is not impossible for a word 'bile' (witness English gall), neither is it as likely as for words specifically connected with the countryside: lupus, bos, etc., and vegetables themselves. Thus it seems odd that city Latin should choose the dialectal form for the word for 'bile', but not for 'greens'. Secondly the -n- stem usually proposed for fel to explain the -ll- of the oblique fellis, comparing the Gmc. data, raises three difficulties: i) Gmc. extends thematic -o- stems to -on- stems so frequently that Gmc. n-stems are not particularly good evidence for an inherited n-stem (e.g. *sṵéĥuros --> Goth. swaíhra, gen. swaíhrans). ii) For this ablauting suffix Latin has generalized either the extended grade of the

[36] The conditions for the appearance of PIE *l̥ and *r̥ as il, ir and ul, ur are unknown. For a full discussion see Kuryłowicz (1956.227-43).

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nominative (type lēnō, lēnōnis) or the full grade (type homō, hominis)[37]. The lack of a nom. *felō can be explained by analogy to mel (so W.-H.), but the Latin paradigms lack the zero-grade of this suffix (*fel-n-) that is supposed to underlie this word (W.-H., E.-M., Leumann 1977.213, etc). The one form that seems to show a zero-grade in n-stems: carō, carnis is accordingly far more likely to be due rather to syncope (of -rV̆-) of a *car-on-es[38]. iii) If fel is from the set root *ĝhelh₃-, we would expect the zero-grade oblique to be *felanis > **fulnis in any case (cf. *ulH-no-s >*ulanos > *ulnos (with sync.-3) > vulnus; see §3). If we invoke the aniṭ root *g(h)el- to get a *fel-n- > fell-, the problem of the lack of zero-grade in the suffix still exists, besides the fact that this root is not clearly attested outside BS, where it is probably secondary, and that BS shows no n-stems for this root or indeed from *ĝhelh₃-. If we are to venture on the problematical business of an etymology for fel, one might suggest the root *ĝhelh₃-, assuming dialectal borrowing to explain the f-. For this root the only certain formation in Latin is, as we have seen, *ĝhelh₃-i-. Perhaps

[37] Because of medial weakening, the original ablaut grade of the oblique in Latin is difficult to determine (so Buck 1933.189, Szemerényi 1970.154). Leumann 1977.360 states dogmatically that it was e-grade, but notes "*-on-es oder -en-es" at p. 363. However, pointing to o-grade is the dim. homunculus < *homon-kel-o-s, and perhaps the U. nom. pl. humuns, which should reflect the normal grade. Leumann (loc. cit.) to explain this proposes for U. an extension of the extended grade to all the strong cases, and thence to the dat. pl. homonus (i.e. passage to the sermōnis type) comparing Paul. Fest. (acc.) hemōnem, Enn. Ann. (nom. pl.) hemōnes, but these are poetic creations and Umbrian -u- is ambiguous between -o- and -ō-.

[38] So too for U. caru, O. g. sg. carneis, U. abl. pl. karnus. Buck 1904.130 holds these forms to continue an inherited zero-grade; so too Ernout 1953.11. Note however, that this claim involves not just saying that the zero-grade in the gen. is inherited, contrary to all the other cases in Italic, but also that it was extended throughout the paradigm of this one word in Plt., of else each branch generalized an otherwise unknown zero-grade in this one item. See Leumann 1977.363.

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then we have a neut. *ĝhelh$_3$-i > *χel(i) > Lat. fel, or an animate *ĝhelh$_3$-i-s > *χelis > *fels > fel(l). In either case, the oblique would be by analogy to mel (as suggested by W.-H., Leumann 1977.213, et. al.).[39]

New l+C arising from Consonant Clusters.

22) The most fruitful source of new clusters of -ld-, -ln-, -lv- and -ls- is from simplification of consonant groups of l+C+ d, n, u, or s, i.e. groups in which the l did not come into contact with d, n, v, or s, until after the rules which assimilated these consonants to l had ceased to apply. There are three rules involved, depending on what consonant has disappeared in between the l and the following unassimilated consonant. The processes (considered in full below) briefly are:

i) R + s + voiced consonant (G): becomes RzG which falls under the rule of Rz > RR (**RZ-Assimilation**, §2.7) and becomes RRG. This in turn gets simplified (**Geminate Simplification**, below) to RG: thus e.g. *alsnos > *alznos > allnos > alnus , parallel to *horsdeiom > *horzdeiom > *horrdeom > hordeum. See §23 below.

ii) A consonant, other than s[40], disappears after r or l and before another consonant: *mulksī > mulsī, (**RkC-Loss**, as opposed to original Rs in *uel-se > velle). See §5.19 ff. below. For the special case of mollis, see Excursus II.

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[39] This possibility was suggested to me by Alan Nussbaum. The only other suggestion, that of envoking the *bhelH- root of flāvus, fulvus (so E. -M.) runs into the difficulty of the e-vocalism of fel. See §7b above.

[40] An s before a voiced consonant goes under i) above, while s before a voiceless consonant falls under the rule of **CsC-Loss**, see §4.2 ff.

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iii) The cluster -ts- became -ss- which simplified (**Geminate Simplification**) to s next to another consonant. Thus -lts- > -lss- > -ls- in *sald-to-s > salssos > salsus. See §5.7 f. below.

23) Let us examine the first process. We have here an old rule: Rz > RR (**RZ-Assimilation**) and a new rule of **Geminate Simplification**. We shall state the rule as broadly as possible first and latter see if its power needs to be restricted: A consonantal geminate is simplified next to another consonant. Cowgill (1978.29) refers to "the general Latin pattern of shortening a long consonant adjacent to another consonant." Thus:

Geminate Simplification

$$\begin{bmatrix} -\text{syl} \\ \alpha \text{ features} \end{bmatrix} \begin{bmatrix} -\text{syl} \\ \alpha \text{ features} \end{bmatrix} > \begin{bmatrix} -\text{syl} \\ \alpha \text{ features} \end{bmatrix} // \begin{Bmatrix} [-\text{syl}] \\ \# \end{Bmatrix}$$ [41]

i.e. $C_1C_1 > C_1$ // C. We specify [-syl] to be able to include /w/ and /y/. Examples are: for final position:

(*tris > *tr_2s >) *terz > terr > ter* (see §2.23).

*hod-ce > *hocce > hocc (Plaut. Mil. 1025; OCC EST: CIL VIII 17938) > hŏc. Note here that the simplification (in final position certainly) is well into the historical period. Note also the masc. *ho-ce > hic with proclitic weakening.[42] Therefore, the raising of o > u in final syllables (and in closed medial syllables) is over before the apocope[43] which in turn feeds **Geminate Simplification** in final position), a fact confirmed by the inscriptional record (so already -us in the Scipio epithaph CIL 1^2.7, c. 200). For **Geminate Simplification** before a consonant some examples are:

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[41] For this form of the rule see Sommerstein 1977.137.
[42] Leumann 1977.468; first established by Skutsch 1914.77 f.
[43] I.e. we do not find neut. *hod-ce > *hocce > hocc > hoc, and then in final position **huc.

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*dis-scindō > discindō,

*ad-scendō > *asscendō > ascendō

*es-sci-t > escit (XII Tables[44] ; -skˆo- present to sum).

After a consonant:

*eks-sistere > existere (many examples from -Css- > -Cs-; see §4.6 f.).

*pelp-mentom > *pelmmentom > *polmentom > pulmentum (many examples from consonants after assimilation; see §5.19 ff.).

*con-ce-cidī > *conccidī (sync. 2/4) > concidī [45].

Exceptions are due in the main to compositional analogy, usually with ad-, occasionally ob- and ex-. So ac-crescō, acclamāre, etc., by analogy to ac-cumulō, etc.; occludō, opprimō , etc., and *eks-fringō > *esfringō > effringō (§4.1; Leumann 1977:192, 210). Here too may fall dissuadeō and assuescō, etc.

24) It is possible that initial geminates underwent the same change, e.g. *dieu-es > *iioues > Iovis, etc. However, here we have no direct metrical evidence for a geminate such as we find for final geminates in Latin, or for initial geminates in Homer: κατὰ (⏑–) μοῖρον : Χ. 331, showing *μμοῖρον < *hmoiron < *smór-iom; Chantraine (1958.176).

25) Thus, we can have new surface RC clusters by either simplifying a *RRC > RC or a *RCC > RC. This simplification must still be occurring after the geminates arise, and in the case of surface -ln-, -ld-, -lv- or -ls- be after the assimilations to l are over. Let us look at some examples:

*pērsnā (cf. Skt pā́rṣṇi-, Goth. fairzna exactly) > *perzna (**S-Voicing**) > *perrna (**RZ-Assimilation**) > perna (**Geminate Simplification**).

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[44] Apud Gellius 20.1.25 = Ernout 1957.114, 119; escunt 121; perhaps in Accius (line 119), loc. cit. 217.

[45] See Excursus I at the end of this chapter.

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*tris-noi > *trsnoi (rV̆ sync) > *terznoi **(S-Voicing**[46]**)** > *terrnoi **(RZ-Assimiliation)** > ternī **(Gem. Simp.)**

*ĝhrsd-(ejom), cf. Germ. Gerste[47], > Plt. *χorsdejom > *horzdeom > *horrdeom **(RZ-Assim.)** > hordeum.

There is even a case with syncope: *per-regō > *perrgō (sync. 3) > pergō **(Gem. Simp)**.

26) The examples above have featured simplifications of -rrC- > -rC- or -Crr- > -Cr-. We have one very interesting example invoving -l- in Lat. alnus. The comparative evidence is complex. Lith. aĨksnis, eĨksnis, Latv. àiksnis, are from *als-no- with "eingefugten k" (Brugmann 1897:766; see also Trautmann 1923.6)[48]. OCS jelĨxa < *elisā, but Pol. olcha, Russ. ol'khá < *alisā; OE alor, OHG elira < *alisā. All the branches show an -s-, so an

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[46] In the derivation of ternī itself, the order of rV̆ sync. and s > z could be reversed; (*tris-noi) > *triznoi > *trznoi > *terznoi. However, the case of *tris > *trs > *terz > ter(r), where the s in final position cannot voice until after sync. brings it in contact with the sonant, argues for the above ordering, at least in final position. The rV̆ sync. (again finally) is late (§2.23, 9.9), but so is the voicing of s (§2.19). We are for the moment treating s > z as a single rule, and not using it as a crucial part of the relative chronology.

[47] We seem to have to do here with a root showing a number of different extensions or formations. The clósest cogante to the Latin is the Germ. Gerste, from a full-grade *ĝhérsdā with a plain -d-. E.-M. maintain that the L. is ambiquous between a -d- or -dh-. However, one would expect a **ĝhrsdh-ejom, to become Plt. *χorsθeom > L. **hosteum. A phonological parallel is *kuzdho- (cf. Goth. huzds 'treasure', Iran. kusda- 'palace') > custōs (Leumann 1977.168). Alb. drith (neut.), drithe are ambiguous but reflect a zero-grade *ĝhrd(h)-. Grk. κριθή, Hom. *κρῑ́θ > κρῖ (neut.) point to *ĝhriH-dh- and Arm. gari to perhaps a metathesised form *ĝhrHi-. This last has been claimed as a Georgian loanword (geri), but the suggestion goes back to Marr (cited in W.-H.), and I am sceptical though I cannot pronounce on the likelihood.

[48] The -k- here is epenthetic in Baltic (see §5.18) and not to be inserted mechanically in the PIE form as is done by Huld, KZ 95 (1981) 304.

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**alinos is ruled out. What appears, at first, to be a good reconstruction for alnus is the *alis- of Gmc., Pol. and Russ., plus the -inos of the tree names (for which see §4). However, an *alisinos would give **alernus:

*alisinos > *alizenos > *alirinos > *alerinos > **alernus, parallel to

*Falisinos > *Falizenos > *Falirinos > *Falerinos > Falernus , and

*vetusinos > *vetizinus > *vetirinus > *veterinus > veternus (see §3.5 n. 8, §9.23). So, the choices for alnus seem to be *alsnos, or *alsinos. The former is to be preferred[49] as having exact cognates in the Baltic material, while there is no comparative evidence for an *alsinos. Therefore the development is *alsnos > *alznos (**S-Voicing**) > *allnos (**RZ-Assim.**) > alnus.

27) Here we can note a most important chronological observation: The stage *allnos passes to alnus by Geminate Simplification, parallel to *perzna > *perrna > perna (§25); it does not undergo -ln- assimilation even though *allnos (> **allus) meets the Structural Description[50]. We can conclude, therefore, that **LND-Assimilation** was over with when **RZ-Assimilation** occurred (i.e. **RZ-Assim.** counterfeeds **LDN Assimilation**). So we can add a new ordering, and summarize our conclusions so far in the following schema (arrows simply indicate precedence in time; changes, such as syncope, which are not being used as building points in the relative chronology are placed in parentheses):

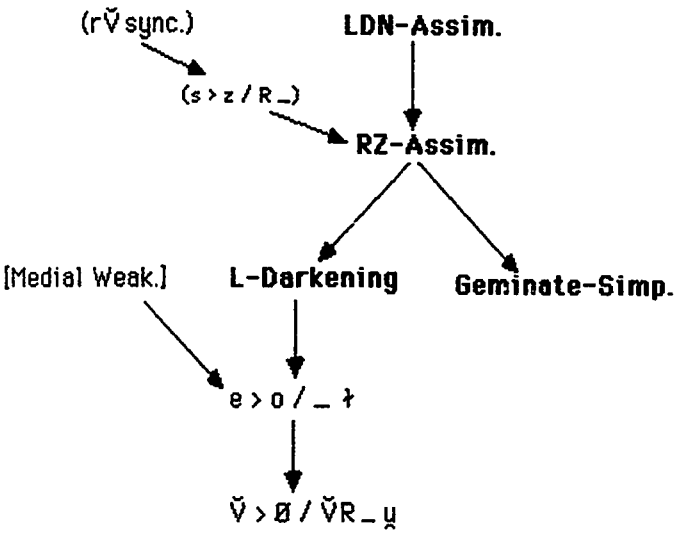
CHART I

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[49] So Leumann 1977.211; Walde-Hoffmann 1938.38, s.v.; Brugmann 1897.766, who, however, also takes cernuus < *cersn- (see Nussbaum 1976.12-14).

[50] N.B. that even if the ultimate preform of alnus were *alsinos, the immediate preform would still be *allnos, since either sequence *alsinos > *alzinos > *allinos > *allnos (with sync. after the assimilation) or *alsinos > *alsnos > *allnos (with sync. before the asssimilation) will yield *allnos.

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Excursus I: Dereduplication in Compound Verbs

1) It is the communis opinio (represented by Buck 1933.292; Leumann 1977.587) that it was precisely the process of syncope (followed, though this is not stated, by **Geminate Reduction**, described in §3.23) that led to the general loss of reduplication in the perfect, beginning in compound verbs and spreading to the simplexes. Thus there is tetulī still attested in Plautus and Ennius, but the compound *at-te-tulī > attulī from which the new perfect tulī is abstracted. Yet there are examples of dereduplication in other Italic languages, suggesting a possible Proto-Italic origin for this morphological feature. Surprisingly, this topic has not been completely investigated. A clear pattern emerges when all the Latin data are arranged according to those compounds of reduplicating perfects i) whichshow a syncopated reduplication syllable, ii) those which show no reduplication at all, iii) those which always show a full reduplication syllable.

2) Clear examples of syncope will show the -CC- retained only after a vowel; after a consonant **Geminate Reduction** will apply. Of the preverbs, only re- and ante- end in a short vowel; only prō-, prae- and dē- end in a long vowel. The examples showing the effects of syncope are:

a) cadō, cecidī: *re-ce-cadai > reccidī (see Cic. Rep. 2.8.14)

b) pariō, peperī: *re-pe-perai > repperī

c) pellō, pepulī: *re-pe-polai > reppulī

d) tundō, tutudī: *re-te-tudai > rettudī

e) tollō, tetulī: *re-te-tolai > rettulī

After consonants, however, the -CC- resulting from the syncope was simplified, e.g.: *con-te-tolai > *conttolī > contulī. So with the other compounds of these four verbs which have preverbs ending in consonants.

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Buck (ibid.) states that this simplification occurs after a long syllable, but offers only examples after a consonant. Forms such as dē-cidī, ante-tulī are too easily analogical to compounds with con- ab- ex-, etc., to be probative, and rather show, of course, the spread of the dereduplication, even to the simplex. There is no general shortening of geminate stops after long vowel. There is a simplification of -ss- to -s- after long vowel or diphthong, but this is quite late (after Cicero's time: Quint. 1.7.20; Leumann 1977.181; Buck 1933.155).

3) For the compounds which show no reduplication syllable, it is necessary to distinguish those with preverbs which end in a vowel and thus might show the -CC- resulting from syncope from those with preverbs ending in consonants. I.e. an *ad-me-mordī is expected to become *ad-mmordī > admordī, while *re-me-mordī (*re-mo-mordī with the analogic spread of the vowel of the root to the reduplication) theoretically might show up as **remmordī, see below). Examples of reduplicating perfects whose compounds never show even a syncopated trace of reduplication are:

a) mordeō, momordī (memordī Enn.): ad-/ dē-, prae-, re-.

b) pendeō, pependī: ad- (not before Apic.), in-/ dē-, prō-.

c) spondeō, spopondī (but Plaut. shows de-spo-pondisse, with reduplication and dissimilation of the -s- as in ste-(s)tī): dē-.

d) tondeō, totondī (dē-to-tondī Varro ap. Priscan[1]); ad-, circum- (part. only), in-/ dē-, re- (part. only).

e) caedō, cecīdī: ab-, circum-, con-, ex-, in-, ob-, sub-/ dē-, re-.

f) parcō, pepercī: com-/ re- (rare; reading at Plaut. Truc. 375-6 uncertain).

[1] This, if it is accurate, may reflect a form before the syncope, or more likely is an analogical isolate.

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g) fellō, fefellī; re-

h) pendō, pependī: dē-, re-

i) tendō, tetendī: re-

From these forms it is clear what distinguishes the compounds which show a syncopated reduplication syllable from those which descriptively show nothing: the former are those with a light root syllable (*re-te-tolai > rettulī); the latter, those with a heavy root syllable (*re-fe-fellai > *reffellai > refellī). What has gone unremarked (both in discussions of dereduplication and of the law itself) is the effect of the Mamilla-Gesetz. The law refers to simplification of geminates that are immediately followed by a long[2] syllable that bears the historical Latin accent. It is well illustrated by the eponymous pair: mamma but mamílla; further, far, fárris but farína; *om-mittō > omíttō; *quatslos > *quaslos > *quazlos > quālus, but dim. *quatsle-lo-s > *quatsllos > *qatsillos > *quassíllos > quasillus[3]. Thus we have *re-ce-cadai > reccidī, versus *re-cē-caidai > *reccaídai > recīdī.

4) There are five possible counter-examples to this formulation: these are verbs which are listed by the standard grammars (Gildersleeve-Lodge 1895.105-6, e.g.) as never showing reduplication in compounds, but which have a short root syllable. However, these are either compounded only with consonant-final preverbs (and so are subject to **Geminate Simplification**), or else the perfect itself in compounds has passed over to a different (non-reduplicating) class than that of the simplex. Thus, we find:

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[2] All the examples show a syllable long by nature or position, though the conditioning factor seems to be the location preceding the accent.

[3] Note that the Mamilla Gesetz occurs after **Rhotacism** is complete. Counterexamples are due to analogy: summíttō, or cúrrēns, curréntis.

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a) canō, cecinī; but compounds in: ad-, con-, in-, inter-, ob-, sub-, and prae-, have a prefect in -cinuī (praececinī, but this is post-classical, Tert. adv. Jud.10, to the analogical prae-canō).

b) pangō, pepigī; but com-, in-pēgī; dē-pangō, re-pangō (no perf.).

c) pungō, pupugī; but com-, inter-, per-punxī; ex-pungāvī (Plaut. Amph. 210), re-pungāre (Caes. BG 3.4)

d) tangō, tetigī; comp. only in ad-, con-, ob-; per- (rare, no perf.).

e) tineō, old tetinī (Pacuv.): abs-tinī (Plaut. Amph. 926), con-tinī (Asin. 582)[4]; all other forms of teneō (even in Plaut.) are found with the new analogical perf. tenuī[5]

These forms, therefore are not counter-examples to assuming that the Mamilla-Gesetz accounts for the dereduplication of compound perfects which do not lose the reduplication through syncope and **Geminate Simplification.**

5) The following perfects always show reduplication in compounds:

a) dō, dedī: ab-, ad-, circum-, con-, abs-, con-, dis- ex-, in-, ob-, per-, pessum-, satis-, sub-, trans-, venum-/ dē-, prae, prō-.

The vast majority of these compounds build their perfects analogically, and are etymologically from *dhē- 'place': abdō, abdere, abdidī 'put away', etc. See Leumann loc. cit. N.B. red-didō, where the red- has been abstracted from the pres. *re-didō > reddō; note also *k̂red-dhē- 'put heart (trust) in' > PIt. *kresē- / *krest-, remodeled to --> *kres-d- (by analogy to the other compounds of dō-, dhē) and thence > *krezdō > crēdō. See Leumann 1977.168.

[4] -tinuī in the MSS but assured by meter.
[5] For which see Leumann 1977.595

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b) stō and sistō, stetī; ad-, con-, in-, ob-, per-/ dē- prae-, re-; the disyllabic preverbs form ante-, inter-, super-stetī (not **stitī) by analogy to the unreduced vowel of the simplex.

c) bibō, bibī; con-, ex-, in-, per-.

As is clear from these examples, the perfects which always show reduplication have monosyllabic bases and hence disyllabic perfects. They therefore were never subject to the sync. 2/4 of the deredupulicated perfects. Thus, ad-di-dī [6], but *at-te-tolai > *at-ttulai (sync. 4/2) > attulī.

6) There are, however, two forms which are not monosyllabic, but which nevertheless show consistent reduplication: discō, didicī and poscō, poposcī. Discō, a reduplicated -sḱō present with zero-grade of the root < *di-dḱ-sḱō, builds a perf. *de-dok-ai > di-dicī[7]. In compounds we would expect: *con-di-dicai > *con-ddicai > **condicī. However, all compounds of didicī (ad-, con-, ex-, per-; dē-, prae-) show the reduplication, that is fail to show syncope. These are clearly cases of analogic restoration to the simplex, itself remodeled to the present.

7) The same holds true for poscō, poposcī (peposcī Val. Antias ap. Gell. 7.9.9). Again, we have a -sḱō present *prḱ-sḱe/o- already simplified to late-PIE *prsḱō (see §5.1) > PIt. *porscō > poscō. The perfect would be *pe-pork-ai which was remodeled to the pres. --> peposcī --> poposcī, (see §3).

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[6] Again, because the conditions under which various syncopes took place are difficult to formulate, it is hard to say why re-didō shows sync. 3, and *re(d)-de-dai does not. The high vowel -i- might be more subject to syncope than -e-. Analogic restoration is always, of course, a possibility in all these forms.

[7] Instead of the historical -e- reduplicating vowel that of the root syllable is used analogically when this is the same in both the perf. and pres. Thus, from a model such as pendō, pependī were created discō, didicī; poscō, poposcī, mordeō, momordī, etc. (see note 8). Forms with the original -e- are preserved and have already been noted (memordī, etc.).

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In compounds we should expect dereduplication: *re-pe-poscai > *re-pposcai > **repóscī. Again the reduplication is restored, perhaps to keep the 3. sg. forms of the present and the perfect from falling together[8].

8) Compounds of currō, cucurrī (cecurrī: Gell. 6.9) show both inherited -currī (e.g. *sub-ce-currai > *suc-ccurrai > *succúrrī > sucurrī), and analogic -cucurrī. Most compounds: ad-, con-, dē-, dis-, ex-, in-, ob-, per-, prō-, re- (anteclassical), super- (no attested perf.), trans-, are attested in both forms, frequently in the same author (Livy especially seems to show this variation). Thus: dis-cu-currī (Livy 25.25.9), but dis-currī (Livy 3.7.32); ex-cu-currī (Plaut. Most. 359; Livy 1.15), but ex-currī (Livy 25.30), etc. Only three forms show no variation: succurrī and intercurrī[9] always appear in the historically correct form, while praecucurrī always shows restored reduplication.

9) The Latin situation seems clear: syncope 2/4 followed by **Geminate Simplification** (for preverbs ending in a vowel) or the Mamilla-Gesetz (for heavy root syllables) lead to the loss of the reduplicating syllable in compounds, whence it spread to the simplex. However, the situation is complicated by the fact that there are examples of dereduplication in Italic. The process, though, need not necessarily be a shared Italic innovation, and in fact the evidence points to independent operation in each branch, caused by its own peculiar syncopes. The building of reduplicated perfects, as well

[8] Though no such disambiguating remodeling has taken place in bibit, (re)-pendit, (re)-tendit, or (re)-vertit.
[9] Tmesis at Lucr. 5.1375 may point to a late and analogical creation, though the tmesis itself may be artful.

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as dereduplication, proceeded branch by branch[10], each showing independent innovations (which have been marked by a + sign in the table below; subj. and fut. pf. are not specifically singled out).

10) Examples of reduplication in Oscan and Umbrian:

Oscan	Umbrian	Latin
deded	dede, teřust, dirsust	dedī
(+)fefacid, fefacust[11] fifikus 'decreverit'(?)[12]	fakust	fēcī, Ven. +facxsto

[10] Indeed a similar process occurs in OIr., where reduplicated suffixless preterites (primarily from the PIE perf.) lose the overt reduplication after the preverb ro-, though this is due to dissimilation and not syncope: canid 'sings', pret. cechan-, *for-ro-che-chan > for-roíchan 'you have taught'.

[11] Latin reflects the original root aorist, with no trace of reduplication (cf. feced on the Duenos Inscr.; the fhefhaked of the Manios Inscr. is a forgery). Ven. appears to have a s-aorist, which is a claer innovation beside the Lat. root aorist, while Osc. has a reduplicated perfect, which may be either a preservation of a possible Italic innovation, or else an Oscan innovation.

[12] This is the only example in O.-U. of the frequent Latin innovation where the root vowel seems to be used as the reduplicating vowel (note 7; Buck 1904.170). Buck (ibid.) mainains etymological connection with fingō, cf. Umb. pres. imp. fiktu, but more likely this represents an analogical fut. perf. from fēc- (so also Watkins 1969.154). However, contra Watkins and Lejeune (1955.148 ff.), I do not believe this can represent a *fe/i-fēc-. Reduplication with -i- is unknown outside the present (indeed this led Brugmann in the first ed. of the Grundriss 1888.II.1240 to propose that fifikus was a pres., an idea rightly refuted by von Planta 1897.330, and by Brugmann in the 2. ed. 1916.25) and U. dirsust (fut. pf.; Latin alphabet) beside teřust (epichoric) therefore seems more likely to be a reflection of a Mittelaut before ř than extension of pres. reduplication to the perf. (von Planta 1897.259; the vocalism of dirsust is not discussed by Buck 1904 or Poultney 1959). Nor can it be the regular reduplication in -ĕ-, since -ĕ- shows up as -e- in Osc. (fefacid). Rather, it must represent an analogical *fē-fēc-, on the model of de-de-d, as occurred in Latin (spopondī), with the regular *ē > ī in Osc., which appears to have both *fe-fak- (perhaps inherited) and an innovated fē-fēc- (perhaps with semantic specialization).

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fuid, fusíd	+fefure[13]	fuī
	pepurkurent	poposcī
	peperscus, peperscust[14]	
dicust	dersicust < *de-dik-	dixī[15]

11) Thus forms such as O. dicust cannot be claimed as good examples of dereduplication, since the U. dersicust may represent an innovation, and the Osc. an inherited form. Even for cases where a reduplicated form exists next to a nonreduplicated form in the same branch, the reduplicated form could reflect an innovation. Thus Osc. ανaSακετ next to fefacid, fefacust is likely to represent dereduplication, but that is not necessarily the case.

12) There are two clear examples which indicate that dereduplication proceeded by syncope in Oscan: prúffed < *pro-fe-fed, and aa-mana-ffed < *man-fe-fed (the -a- of mana- by anaptyxis) next to manafum (1. sg. pf., showing no trace of the syncope). Further examples of dereduplication in compounds are Osc. da-did (next to dedid), and Umb. pro-canurent (cf. Lat. cecini, Olr. cechan-). Note especially that Osc. here shows syncope and dereduplication in precisely the case where Latin does not, that is in compounds of *dhē-.

13) There is also a possible single example each from Volscan, Paelignian, and Venetic. Volscan shows an atahus presumably from *ad-te-tahus 'attigerit' (von Planta 1897.327). Paelignian has afõed < *afdeõed

[13] The U. form is a clear innovation.

[14] These presumably represent the same root as the forms above, having here in Osc. undergone the same extension of the -sc- of the pres. as L. poposcī, and with a semantic specialization to 'posuerit' (so Buck 1904.164; von Planta 1897.328). The reduplication may be inherited (Skt. papracha).

[15] Latin shows the s-aorist *deik-s-, Osc. an unreduplicated perf., Umb. a reduplicated (inherited?) perf.

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(ibid. 326 -9). Venetic shows an atisteit. While the -ei- recalls the -ei-d of the Latin perf. (> eit > Īt > it), it is more likely, since final stops appear to be lost in Venetic (Lejeune 1974.146), to reflect the primary ending -ti with apocope. Though Latin -it itself reflects a spread of the primary ending, the examples of a 3. sg. preterite (s-aorist) : vha.g.s.to, dona.s.to 'donavit' (44x) and doto 'dedit' point to -to as the ending. Lejeune therefore takes atisteit to be the 3. sg. pres. of a thematic (ibid. 79-80, 331), and a pres. also accords well with what can be understood of the tombstone:

vineti karis vivoi oliialekve murtuvoi atisteit:

"the love of Vinet(i)us remains for [Fontus] alive (as well as?) dead" (ibid. 224-5; no. 75 ter). However, that Venetic maintained the preterite (aorist) and perfect as separate categories is possible and a perf. "has remained" will also fit the sense of the epithaph.

14) The conclusion then is that a similar process of dereduplication occurred in various branches of Italic. However, it cannot be considered a Proto-Italic change. Rather since it was conditioned by the syncopes which occurred independently in each branch, dereduplication is an independent innovation in each language. The clearest evidence for the independence of the treatment of the perfects are those which show innovation in adding reduplication: Latin fēcī from the strong stem of the root aorist, Umb. facust from the weak stem, but Osc. fefacid, fifikus with a reduplicated formation built to both stems; Lat. fuī, Osc. fuid, but Umb. fufure. In addition, not only is there clear evidence that the reduction depended on the syncopes in Oscan and Latin, and presumably in the other branches as

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well[16], but also that the syncopes were quite different ones. Thus in Oscan, where the greater number of examples occur and syncope is more pronounced than Umbrian, the syncope was that of the even vowel in trisyllabic words, so: *pro-fe-fed > prúffed, man-fe-fed > (aa)-man(a)ffed (Buck 1904.170, 57) and most interestingly deded but *da-de-dēd (subj.) > dadid, with dereduplication, where Latin keeps the reduplicating syllable (dēdidit), and whose dereduplication seems dependent on the syncope of the second syllable in four syllable words..

[16] For syncope in Venetic, see Lejeune 1974.111-25, esp. 120-4; for Osc., Umb. and the other Italic languages von Planta 1892.213-26. Note in particular the Pael. pperci, Ptruna, etc., which von Planta takes as syncope in initial syllable, though the spelling of CV- syllables with the initial letter alone (i.e. the letter is to be read as the letter name) is seen elsewhere, cf. DCUMIUS (CIL 12.1445, Praeneste), GMINIA (CIL XIV.3142), PTRONIO (CILXIV.3210).

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Chapter 4: CSC-Loss; N + Spirant Lengthening

1) We can turn now to a new rule which feeds, and so provides more examples of, several of the rules previously discussed. For Latin we have a rule that "Stop + s becomes s before another consonant, and if this is voiced the s (which in this case may also come from ns) is lost." This is Buck's formulation (1933:153) and needs to be examined closely. We'll establish a rule called **CsC-Loss** where C > Ø /_ sC and give a more precise formulation later. Let us first examine what consonants are lost, taking examples from as many morphological categories as are available, beginning with clusters closed by voiceless consonants.

2) K (+ s + voiceless consonants)

a) -ksk-:

*eks-kuilīnos (cf. inquilīnus) > Esquilīnus

*seks-kentom > sescentum

*eks-skandō > escandō (see below)

*mik-skō > *miscō --> misceō (Leumann 1977.203)

b) -kst-:

*seks-tijcs > Sestius

*in-lukstris > illustris

*leukstrom > lūstrum[1]

[1] W.-H. s.v.; Osc. luisarifs points to a *lois(s)- as its immediate preform. However, context (one of the iovila dedications) suggests connection with Lat. lūsus, lūdus < *loid- (a festival celebrated with games) rather than lūstrum (so Buck 1904.248, von Planta 1892.150).

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These reductions are subject in the morphologically analyzable cases to analogical restoration from the antevocalic (or rather non-anteconsonantal) cases. So, beside Sestius we have Sextius, (CIL 1^2.677, 106 B.C. = Ernout 1957.49) after sex; textus after texō; exscindō (beside escendō). All the cases of -ksp- are in compounds of ex- and restored (ex-pandō, etc.; Leumann 1977:302). Note, however, that -kst- that arises by syncope[1], at least between s and t, remains. So *deksiteros > dexter (cf. §11), which has no analogical source for restoration of a lost k (i.e. no associated antevocalic forms **dex-V-).

3) If the consonant following the s is voiced, after the loss of the initial consonant, the s voices and disappears with **Compensatory Lengthening (CL)**[2]. That is, we have the order:

i) CsG > sG (where G is any voiced consonant, including resonants)

ii) s > z /V _ G

iii) ṼzG > ṼG **(CL)**

Examples are:

a) -ksl- :

*akslā (cf. OE eaxl, Mod. E. axle) > *asla > *azla > āla
*tekslā (cf. texere) > *tesla > *tezla > tēla

[1] It is uncertain whether the sync. here is 2/4 i.e. *deksiteros > *deksteros and then > dexter, by the sync. of -rV̆, see §2.23, 9.9), or whether *deksiteros first became *deksitṛs > *deksiter(s) and then sync. 3 applied. The former seems more likely in view of the relatively late date of rV̆ sync.

[2] So Buck. There is no reason to assume that -ksl- > -gzl- before the initial consonant is lost, so Leumann 1977.207, Kent 1945.126. Rather it seems clear that the s was voiceless, and indeed devoiced the preceding stop. See above (Ch. 2, § 21) on *mag-isamos > maximus, below § 4 on *magis-u̯olō > *maksu̯olō > māu̯olō > *māolō > mālō. Further, in Ch. 5 it will be argued that since r is included in **CsC-Loss**, the case of *kerssna > *kessna > *kesna > *kezna > cēna, is decisive for ordering **CsC-Loss** before the voicing of s to z. For nsG, see §26, 5.9.

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*pagslom (cf. pangere) > *pakslom > *paslom > *pazlom > pālum

To each of these we may compare a diminutive:

*akslela > *aks|$_2$la > axilla

*teksela > *teks|$_2$la > texilla

*pakslela > *paks|$_2$la > paxilla

We can say then that the syncope of -le- > -|$_2$- in 3 syllable words (sync. 3-|) bleeds (precedes) **CsC-Loss**.

4) Further examples of ksG:

-ksm-:

*iougs-ment-om > IOUXMENTA (neut. pl. CIL 1^2.1) > iūmentum

*seks-mēnstris > sēmēnstris

*leuks-men > lūmen (here the Compensatory Lengthening applies vacuously)

*eks-ag-s-men > exāmen (to agō; see §6.19).

-ksn-: *leuks-na[4] > *lousna > lōsna (Praen. LOSNA CIL. 1.2.549 = Ernout 1959. 6, 30) > lūna (again Comp. Length. applies vacuously)

*seks-noi > sēnī 'six at a time'

-ksd-: *eks-deuko > ēdūcō

-ksu̯-: *seks-u̯iroi > sēvirī

Here there is a case after syncope (2/4):

*magisu̯olō > *maksu̯olō > * masu̯olō > *mazu̯olō > *māu̯olō > *māolō (u̯ > Ø / _ o, see §7.2, 4) > mālō.[5]

Therefore, while dexter points to sync. 2/4 (/ s _ t) occurring after **CsC-**

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[4] The OPr. lauxnos does not necessarily point to an o-grade, since *eu can become ou in BS (*teutā > Lith. tauta).

[5] From the 2. sg. *magis-u̯īs (for vīs see §1.5) > *maksu̯īs > mazu̯īs > māvīs, a new 1. sg. māvolō and paradigm is formed. Plautus shows both forms frequently, with māvolō, māvelim only verse finally.

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Loss, mālō shows sync. 2/4 (/ g _ s) occurred before and fed **CsC-Loss**. Though syncope is more likely after or between voiced segments (Leumann 1977.96), this again illustrates the difficulties of using syncope as a rung in building a relative chronology. However, the case of *magisyolō > *maksyolō > *masyolō > *mazyolō > māyolō > mālō does allow us to say that s had not yet voiced to z nor disappeared with compensatory lengthening to yield a **magīvolō.

5) The case of IOUXMENTA seems to show the earlier stage preserved in an extremely archaic inscription (at least as old as 500, see §11). However, as the lost consonant is subject to analogical restoration, perhaps we have such a form here. There are two objections to this view. The first is that the other analogical forms are the later and regular ones (Sextius, cf. Sestius; textus etc.); the second, that the analogical source of an -x- is rather far away here. Whereas Sestius can be made Sextius by comparison with sex, sex-ennis, etc., *testus --> textus after texō, etc., the nearest -x- in the derivatives of the root *jeug- are in coniū(n)x, iūnxta and the pf. iūnxī (see below §19 f., for -nks-). While connection with iungō must have been obvious to a native speaker (cf. Cornelius Nepos. Tim. 4: vectus iumentis iunctus), the pf. is not a likely place for analogical influence. In addition none of the other derivatives in -men, -ment-um show such a restoration: lūmen (cf. lūx), ex-āmen (< *eks-ag-s-men, cf. agmen), subtēmen (cf. texō), etc., and restoration where it does occur is only before voiceless consonants. This places the rule of **CsC-Loss** rather late in the prehistory of Latin, or more accurately, rather early in historical Latin. The fact that **CsC-Loss** is after sync. 3-į accords well with this fact, and as we shall see below several changes can be ordered before **CsC-Loss**. For the Oscan and Umbrian cases see §§10 ff.

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6) Loss of Labials

For PsC examples are:

-psk-: *sups-kapiō > suscipiō

*subs-kue > susque

Opscos (Enn. Ann. 296) > Oscus

-pst-: *obs-tendō > ostendō

*subs-tolai > sustulī

-psp- *abs-peldō > aspellō

*abs-spernō > aspernō (see below)

Here, too, restoration is common, particularly with the preverb abs-. Thus, we have etymological sustulī beside restored abstulī, susque beside absque. Indeed the restored abs- shows a new pattern of distribution, so that we have:

abs /_ [-voice]: abs-tulī, etc.

ab / _ [+ voice]: ab-ēgī, ab-latus

as /_ p: aspellō

ā /_ m,u,b: āmoveō, āvellō, ābītō

i.e., we have a new rule of dissimilatory loss of p:

p > Ø /_ s [+labial],

with subsequent voicing of s > z if the labial is voiced and then **Compensatory Lengthening** of a preceding vowel.

7) For cases of PsG we have only one case, after sync. 3:

-ps(e)m-: *sups-emō> *supsmō > *susmō > *suzmō > sūmō. Here, sync. 3 (of e / s _ m) feeds (is ordered before) **CsC-Loss** and sync. 3-ļ bleeds (is also ordered before) **CsC-Loss.**

8) The treatment of dentals is, of course, different. An original -ts- became -ss- in Plt., so *claud-sai > *clautsai > *claussī > clausī, as did Plt.

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-ts- < PIE -t^{s}t- [6] < dental + dental, e.g. *met-to- (cf. metō) > *metsto- > *metso- > messus. However before r, the PIE -tst- rather than losing the last -t-, instead lost the first and became -str- in Plt. It seems that before r, the first of the series of four dentals -tstr- was lost[7], while in -ts(t)- the last was dropped in Plt., Gmc., Celtic (i.e. West IE, see Brugmann 1897:624 f., Szemerényi 1970.96-7, Strunk MSS 25.113-29)[8]. So to the root of rōd-ere 'gnaw' there is built a *rōd-trom > PIE *rōtstrom > rō(s)strom > rōstrum 'beak, gnawer' (cf. Leumann 1977:197). For Osco-Umbrian , cf. *ɥert-to- > PIE *ɥertsto- > Plt. *ɥersso- > Osc. Fερσοπει, Umb. trah-uorfi (where the f shows *-rss- > -rf-), Lat. versus (see §5.3-4). The only putative O.-U. example before -r- is doubtful. If O. castrous, U. castruo can be compared to Latin castrum, and if castrum comes from *kat-trom with a

[6] Preserved e.g. in Hitt. e-iz-du /etstu/ 'let him eat' < *ed-tu

[7] Or, perhaps retained in Latin, and then underwent -ts- > -ss- with **Geminate Simplification** before the -r-, see §3.23.

[8] Not with Cocchia (see Gl. 18.252f.) PIE tt > ϐϐ > ss; followed by Niedermann (1953.162) who imagines a *ssr > s(s)tr, similar to the epenthesis in Gmc., e.g. *srou-mos (cf. ῥεῦμα) > PGmc. *straumaz > Mod. Eng. stream; nor with Kent 1945.117-8, who holds that PIE -tst- became -st- everywhere, but that the influence of roots ending in dentals rather caused the second -t- to be dissimilated: "while *patstos (< *pat-tos), ptc. to patior, would normally have become *pastos, the influence of the the root pat- in patior caused rather the second -t- to disappear". How exactly this process is supposed to be envisioned, I do not know; perhaps *patstos > *pas-tos (by regular sound change) --> *pat-tos (by analogy to the root), which then undergoes again the PIE change of -tt- > -tst-? Kent does not state, but appears to imply, that the -tst- stage is analogically retained; but this would involve reanalysing *patstos as *pat-stos (thence to *pats(t)os > *patsos > passus) thus creating a new suffix for the pf. part. Kent imagines the same process for Celtic and Gmc.

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root *kat- 'save, rescue' (said to be seen in cassis,-idis[9] 'helmet': von Planta 1892.422-3, Buck 1904.236 n. 1), then O.-U. might show the same development of -tstr- as Latin. However, O. castrous, U. castruo seems to mean 'fundus, landed property' or more probably 'head/person' (Buck 1904.15, 236 n.1; Poultney 1959.300). Nor is the etymology of L. castrum clear. W.-H. (1938.180) and E.-M. (1959.104) take it < *kas-trom (cf. Skt. śastram 'knife'), formally identical to the intrumental noun which underlies castrāre, i.e. 'a position cut into the ground', and while such a semantic change from 'cutter, knife' to 'thing cut, camp' seems unlikely, since -tro- nouns can also indicate the result of an act (so Skt. śro-tram, Av. sraoθram 'hearing' as well as 'ear'), a change from 'cutting' to 'camp' is possible and might accord with 'fundus (land marked out by cutting with a plow)', if that is the meaning of the Oscan and Umbrian forms. Thus, O. castrous and U. castruo are at best uncertain evidence for the development of -tstr- in Oscan and Umbrian.

9) For Latin, we can say that in groups of stop + s + consonant (which may be voiced or voiceless, including resonants) the stop before the s was lost. There is no need to specify the stop as [-coronal] (or some other feature to exclude dentals) as the ordering of the rules:

1) tst > ts (West IE)

2) ts > ss (PIt)

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[9] The etymology is far from certain and the -idis gen. (found elsewhere only in lapis) looks non-IE (so W.-H. 1938.177, s.v. 2. cassis). If PIE, the phonology of relating this word to Skt. chad 'cover' (Leumann 1977.313) or a supposed PIE *kadh (perhaps in OHG huot, etc.; not outside of Gmc.; see W.-H.) is dubious.

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3) CsC > sC (**CsC-Loss**; Latin only ? Plt.? Shared innovation?)

will generate the proper outcomes. Let us take a look then at CsC clusters in Osco-Umbrian.

Loss of Consonants before sC in Osco-Umbrian and Celtic.

10) For loss of velars in Oscan and Umbrian we have:

-kst- : *seks-tant-āsijo- > U. sestantasiaru[10] (g. pl.) as if L. *sextantarius

*deksiteros > O. destrst, U. destram-e vs. L. dexter (so continuing after O.-U. syncope; see, §2).

For labials:

-pst- : *ops-tendito > U. ostendu

11) It might seem then that O.-U. also had a rule of CsC > sC. However, as the cases of vexilla < *ueksle-la (§3) with sync. 3-l and the attestations of IOUXMENTA and (perhaps Opsci) show, this rule occurred in early historical Latin (the Forum Inscription is dated to c. 500 B.C. by Leumann 1977.6 and may be somewhat older, "Late 6(?) cent. B.C." in Gordon 1983.78). We may then say that the O.-U. cases point to an independent innovation of **CsC-Loss**. However, there is another possible explanation for the O.-U. cases, which has important implications for Latin.

12) In O.-U. there is a clear development of -kt- > χt, and for -pt- > -ft-; in Umbrian, this f (as well as f < PIE *dh) became χ > h . For k > χ > h, cf.

[10] The lack of rhotacism is these forms is a longstanding difficulty. Buck (1904.74) for plenasier, urnasier (also to be compared to the L. -ārius suffix) suggests that the non-rhotacised forms show a by-form -āsio- and that the consonantal i prevented intervocalic rhotacism. However, that such a form of the suffix existed, or if it did that the i would have survived unvocalized until after rhotacism in U. is very unlikely.

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*ek-tro- > O. ehtrad, U. ap-ehtre (cf. L. extra), O. Uhtavis (L. Octāvius); U. uhtur L. auctor. The -h- is very weak in U. and may be omitted entirely in writing in both alphabets, thus: U. speture, L. spector. For -kt- arising by syncope, O. keeps it unchanged, while the spirantization continues in U. and the χ passes to ĵ (as in French, see below). So: *fakitōd > O. factud, but *agitōd > *aktod > *axtud > U. aitu vs. O. actud. For -pt- > -ft- (> -ht- in Umbr.), cf. O. scriftas (nom. pl.), U. screhto (L. scrīptus); O. ufteis (g. sg.) 'will' < *opti-. For -pt- arising by syncope, there is the single example *hapitōd > *haptōd > haftōd > haxtōd > U. ha(h)tu, which shows the same development continuing in Umbrian (cf. the treatment of secondary -kt-).

13) A similar change of k > χ and p > f is likely before s as well. A natural rule of assimilation of spirants can account for the outcome of both as -ss-. The antevocalic cases provide indications, but not conclusive proof, that this was the case (see Brugmann 1897.681, who notes the lack of clear examples). If so, the loss of labials and velars before sC might be due to a development of psC > fsC > ssC > sC, and ksC > χsC > ssC > sC rather than the direct disappearance of the initial consonant of the cluster due to a rule of CsC-Loss. For the sequence:

1) p > f / _ {t, s}

2) fs > ss

possible antevocalic examples are: U. essuf which is probably to be connected with ipse[11] (see Buck 1904, 143, and additional notes of 1928 p. 360). The meaning is "reasonably certain" (ibid.) as ipse, though the O.-U.

[11] Note that in Latin VpsV remains: ipse < *is-pse,[1] a type of haplology (Leumann 1977.231, 471); for original PIE -ps- there may be a regular (?) metathesis in vespa < *wopsa cf. OE wæps, wæfs (but already wæsp whence standard ModE. wasp), OCS osa, Lith. vapsà. So too crispus < *kripso-. There is no reason to suppose an endingless *i- as the stem for ipse (so E.-M.) in the face of *is-dem > īdem.

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preform would have to be *e(s)-pse (and not the *is- /eyo- thematic which underlies L. ipse and results in U. n.m.sg. ere-k, f.a.sg. eam etc.; Buck 1904.54). Another possible example is O. osiiíns 'adsint' (ibid. 79), and although this might reflect *ops-sī-nt and not VpsV, and therefore might be due to a CsC-Loss in O. and not to assimilation, an *ops-sī-nt could equally have become *ofssint > o(s)ssint. Secondary -ps- remains in Osc. (upsed, etc.) but it continues to change to -ss- in Umb. (osatu < *opesāto; ibid.). Neither of these examples proves a passage of -ps- > -ss-, but they are suggestive.

14) In addition, intervocalic -ks- may become -ss-. Final -ks, both original and secondary becomes -ss (Osc. n.sg. meddíss, Mar. medis). One example of intervocalic -ks- is: abl. O. eksuk 'hoc', exa-c 'hāc', U. es(s)u, f. esa. The etymology is not definite. The pronoun is said to be formed from a particle ek plus the pronominal *e-/i-, and so/ā in the cases outside the n./a. sg., n. pl. (von Planta 1897.211). Whether this -ks- represents an original -ks- or a secondary one is uncertain; however, the retention of -ks- in O., but -s(s)- in U. looks like the behavior of secondary -kt-: retained in O. but with spirantization continuing in U. (and in final position in O.: nom. pl. medíss < *med-dik-es)[12]. The change of χ > s would be followed by the natural rule of χs > ss (the assimilation of two spirants, parallel to that of ps > fs > ss; see Brugmann 1897.681). There is additional evidence for this rule (so Brugmann loc. cit.) in Pael. usur, if it does mean 'uxor' (so von Planta 1893:376). If indeed ks > χs (> ss), we may have the following parallelism of spirantization for O.-U. intervocalically:

[12] If so, the stem is more probably *e-ke-so-, with the e-ke- seen in Grk. *eke-ieno-s > ἐκεῖνος, cf. the -so- in the oblique cases of the O.-U. is pronoun.

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kt > χt (O. Uhtavis, U. rehte)

pt > ft (O. scriftas, U. screhto)

and before s the additional rule of assimilation takes place:

ks > (χs) > ss (certain examples only in final position; but see above)

ps > (fs) > ss (O. essuf, if cogn. with ipse; see above)

While before C:

ktC > χtC (*ek-tro- > O. ehtrad, U. ap-ehtre; cf. L. extra)

ptC > ftC (no good examples[13])

and with s, the additional changes of assimilation and simplification of geminates:

ksC > χsC > ssC > sC (secondary: O. destr-st, U. destram-e)

psC > fsC > ssC > sC (U. ostendu 'ostendito'; *ops-sint > osii[ns])

15) This set of changes has a curiously exact typological parallel in Celtic.

pt > ft > χt (*septm > Olr. secht; in W. χt > χθ > ith: seith[14])

kt > χt (*ok̂tō > Olr. ocht; W. wyth)

Before s, with assimilation[15]:

[13] U. ah-trepuřatu < *ap-tripodato, is not a good example since it is also spelled with a- and aha-, which would seem to indicate a separate prefix ā-, cf. the preposition ē in Oscan and Umbrian. These almost certainly had their origins in a form of vowel lengthening accompanying the loss of the spirant, but conditions are difficult to formulate (see Buck 1904.49).

[14] With progressive assimilation and then χ > i as in Umbrian. Gaulish SEXTAMETOS 'seventh' shows the spirantization was Proto-Celtic.

[15] See Thurneysen 1946.136, Lewis-Pedersen 1961.20; Gaulish preserves the χs stage, spelled X or XS: *deks- > DEXSIVA, *ups- > ὑξέλλον, UXELLODUNUM 'high-camp'

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ps > χs > ss (*lap-s- > Olr. lassar 'flame' cf. λά-μ-π-ω; W. llachar 'bright'[16])

ks > χs > ss (*uksō > Olr. oss 'deer'; W. ych 'ox': also word final: W. moch, Olr. mo < *mos(s), seen in mos-riccub-sa 'i shall come soon': Lewis-Pedersen 1961.64)

Before C:

ptC > ftC > χtC > tC (*oup-tro- 'upper' > *ōftro- > *ōχtr > Olr. óchtar, uachtar[17], but W. *uχθr > uthr 'marvelous')

ktC > χtC > tC (example?)

With s and assimilation:

psC > fsC > χsC > ssC > sC (*teps-tu > Olr. tes 'heat')

ksC > χsC > ssC > sC (*mik-sk- > Olr. mesc-aim)

This spirantization of labials and velars before s, with assimilation and simplification of geminates can account for their disappearance, while dentals before sC would become ssC and simplify to sC; further, if the consonant in the combination is t, the resulting st undergoes the usual assimilation to ss, except before r: that is -T-s-t- > sst > st (by the regular geminate simplification) > ss, but -T-s-tr- > sstr > str (showing the geminate simplification, but with the r blocking the assimilation[18]). This midstage of -ss- for both -ks- and -ts- is neatly shown by the fact that both velar and dental stems in the s-subjunctive share the same development. Thurneysen (1946.391) writes:

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[16] In W. the outcome of -ps- and -ks- is ch. Since we have no examples of -ss- in Celtic (Lewis-Pedersen 1961.18), we cannot know whether the W. ch is the final stage of -ss- from -χs-, or if the same progressive assimilation as in χt > χθ occurs, so χs > χ(χ).

[17] So that the change of new r_2 in final position to -ar in Olr. bleeds the loss of χ/ _ CC. See Lewis-Pedersen 1961.27.

[18] Cf. the Latin situation with str preserved, e.g. mōnstrum; see §18.

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> In the 3 sg. and the deponent 2 sg. the personal ending is added directly to the -s of the stem. (s)st has become ss (2 sg. depon. -sser < -s-ter), except in the old group -str- of the 3 sg. depon. ; the vowel in estir, ·fester is a secondary development. The development of -ss+t to -ss- is regular only where the final -ss- of the stem goes back to -ts- ; where it has come from -ks- (-χs-) one would expect rather -cht- ; cp. echtar 'outside of' from *eks- ;

However, since both velar and dental stems share the -ss- midstage the development is regular for both. Thurneysen's counter-example echtar comes rather from an s-less *ek-. So: guidid 'prays', but 1. pl. of the s-subjunctive ·gessam from *ged-s-o-men, but 3.sg. ·gé from *ged-s-t > *gesst > *gest > *gess > gé[19], and cf. midithir 'judges' and the depon. 2. sg. mes(s)er from *med-s-ter > *messter > *mester > messer, but 3. sg. mestar from *med-s-tro > *messtra > *mestr > mestar[20]. For velar stem, cf. tíagu 'I go', 1. pl. ·tíasam from *tēg-s-o-men > *tēχsam > tessam, and 3. sg. ·té(i) from *tēg-s-t > *tēχst > *tesst > *test > *tess > ·té.

16) In summation, although for O.-U. and Celtic the result of CsC > sC is the same as Latin CsC > sC, the process may have been quite different, due to the possibility of a midstage with a spirant. For Latin, as we have no spirant from -ps-, -pt-, the CsC rule is simply one of disappearance: C > Ø / _ sC.[21] It seems likely then that **CsC-Loss** is a Latin rule, after sync. 3-l at least, and not occurring till historical times, while the O.-U. shows a

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[19] Vowel lengthened in final position in stressed monosyllables; Thurneysen 1946.32.

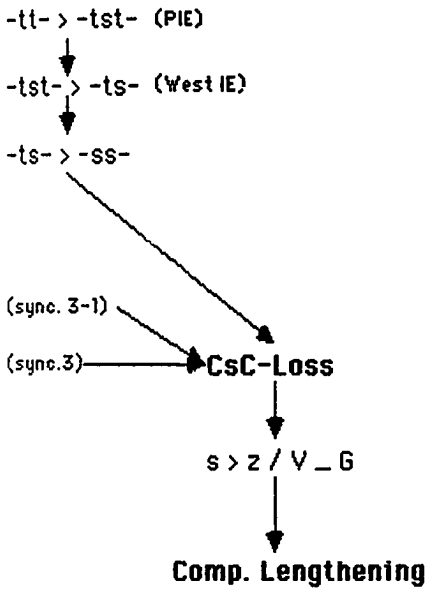
[20] Cf. n. 15. The r prevents the assimilation, and the vocalization of -r* occurs after the change of -st- > -ss- is over.

[21] However, we do have a spirant from k in nkt, nks, (and as will be shown below nkn as well), i.e. k > χ / n _ [+apical]. Since this change to spirant occurs only after a nasal, the cases we have been examining of VksC > sC are due to a simple cluster simplification (**CsC-Loss**) and not to the process outlined for O.-U. and Celtic (see §21).

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similar outcome, but with different midstages. We can summarize our findings thus far:

CHART II



N plus Spirant Lengthening

17) This spirantization in O.-U. and Celtic gives a typological parallel which can be very valuable in describing another important Latin sound law. It is clear that "vowels were regularly lengthened before ns, nf, nx, and nct as in cōnsul, īnfra, iūnxi, iūnctus" (so Buck 1933:94) [22]. The cases of -ns-,

[22] Sommer-Pfister 1977:100 notes V̆ > V̄ /_ ns or nf, and /_ nc + C (though examples are given only for s and t) as separate cases; Maniet 1975:101 lumps together V̆zC > V̄C and V̆ > V̄ / _ ns, nf; Niedermann 1953:76-77 notes only ns, nf; Leumann 1977:112-3 notes ns, nf and nct, nx as separate cases.

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-nf- can clearly be combined as a law where V̆ > V̄ / _ n plus a spirant. Though disputed by Collinge (1965.237-8), the testimony of ancient sources (Cic. Or. 159) and inscriptions with I-longa, or apex (cónficiunt CE 98.6, tránslata CIL.XIII.1668.I.27, etc.) are undeniable. We can set up a rule of **N + Spirant Lengthening** (abbreviated **NS**): V̆ > V̄$^{(n)}$ / _ [+nasal] [+spirant]. The long vowel is presumably nasalized, as typological camparison with Skt. (aṁś-, etc.) might indicate. The grammarians, however, do not speak of a nasal vowel here (Leumann 1977.112), though early Latin inscriptions frequently omit n before s (so COSOL, CESOR in the first line of the L. Cornelius Scipio epitaph = CIL 1^2.8 and COSENTIONT in the second part, beside CONSOL, CENSOR), as does Vulgar Latin (MESES, Leumann 1977.145) and Romance (Fr. mois, It. mese, Sp. mes). The n was written and probably pronounced in careful or refined speech, though Vel. Long. (7.79.1) reports that even Cicero preferred the pronunciations foresia, hortesia. The n is, however, subject to frequent restoration. Buck's statement, that "It is probable that nothing more subtle is involved here than a conflict between total omission of the n in common speech and its plain pronunciation (as before any other consonant) in careful refined speech, in which case it would be an example of spelling pronunciation" (1933:150) receives typological confirmation from Polish, where the nasalized vowels are usually rendered as vowel plus homorganic nasal, or omitted finally, in the standard speech (urban Warsaw dialect). Examples of -n- restored in spelling, but with the long vowel showing the effects of its original disappearance occur in many morphological categories: -ns-: cōsul, vs. cōnsul; -nst-: mōnstrum (but N.B. (Fabula) Mōstellaria, where the connection with mōnstrum is obscured); -nf-: Īnfāns (analogy to other compounds with

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in-); -nks-: in pf. sanxi; -nkt-: *sanktos > saŋχtos > sā^{n}(χ)tos --> sānctus (see below, §20).

18) Many languages show such an n plus spirant law, for example the Ancient Greek second compensatory lengthening, in Modern Greek before θ, φ, χ; in Gmc before χ (PGmc. *þaŋχta > Goth. þāhte, where OE þōhte shows that the vowel was still nasalized); for Oscan and Umbrian, there are examples before ns and (n)χt < *kt (for which, see §12-16), also perhaps nf: O. keenzstur[23], saahtúm (L. sānctum), U. śihitu (cīnctōs), alanfehtaf[24] (cf. Ínfec.ïu 'uncooked'). It appears that the nasalization was lost completely in O.-U. when the spirant was followed by a consonant, though the number of examples is small. Celtic also shows a lengthening before n plus s or ch (< *kt, *ks, see §15): *kent-t- > *kenss- > cés(s)aid 'suffers'; *ro-ink > r-ic(c) 'reach', but verb. noun *ro-ink-tiō > *rinχtu > ríchtu, s-subj. ro-ink-s- > *rinχs- > rinss- > ríss- (Thurneysen 1946.128; Lewis-Pedersen 1961.51).

k > χ / _ [+apical]

19) The cases of lengthening in Latin before -ks- and -kt- can be easily brought under the law of **N + Spirant Lengthening**. We need only postulate a stage at which k had become a spirant before t and s. For this change there are a number of typological parallels, both within and outside Italic. Besides the O.-U. and Celtic cases detailed above, there is clear evidence that k > χ before t and s in Romance, and this χ went on to i in

[23] *kens-tor > *kēnstor (n + spir.; Buck 1904.47) > *kēntstor (ns > nts, ts spelled <z>; ibid. 72) > kentsstor (with doubling of s before t; ibid. 99), a nice bit of relative chronology in Oscan.

[24] As Buck (1904.47) notes, the line division al|a diminishes the evidential value, since dittography is a possibility.

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French[25]. So for -kt-: L. lacte > laχte > Sp. leche, Fr. lait; octō > oχto > Sp. ocho, Fr. huit. For -ks-: L. laxāre > laχsare > OF laisare > laisser.

20) Both Celtic and Osco-Umbrian furnish typological parallels for k becoming a spirant before the dentals t and s, as well as for lengthening before nasal plus such a spirant. A similar change in Latin allows the different cases of lengthening before ns, nf, nx, and nkt to be treated as the single rule of **N + Spirant Lengthening (NS)**. This idea, though found in none of the handbooks, is clearly stated by Allen in Vox Latina (1965.66-7):

> The same regular lengthening of vowels takes place before nct, and has a similar explanation. It is probable that in this environment the c was first reduced to a fricative [χ] (like the German ach-Laut), and before this fricative there occurred the same loss of n, with nasalization and lengthening of the preceding vowel, as before the fricatives s and f. Thus for example, quinctos became quiŋχtos, thence quīχtos; subsequently the [χ] was lost, and since the ī was now followed by a plosive and not a fricative, the nasalization was in turn replaced by n--whence quīntus, the attested form (similarly, the loan-word spinter from Greek σφιγκτήρ). In all other cases, however, the lost c was restored by analogy with other forms, but with retension of the long vowel--hence e.g. sānctus, cīnctus, fūnctus after sancio, cingo, fungor; quīnctus was also restored after quinque (which by a complementary analogy lengthened its first vowel to quīnque), but scarcely occurs except in the derived names Quīnctus, Quīnctius... A long vowel is also marked in coniúnx, coniúnxit. If, as is probable, this represents a regular phonetic development before nx (= ncs), it presumably has the same causes as before nct.

Thus, we can set up the ordering:

1) $K^{u} > K / _ C$ (*k^{u}iŋk^{u}tos > *kuiŋktos)

2) $k > χ / n _ \{t, s\}$ (*kuiŋktos > *kuiŋχtos)

3) **NS** (*kuiŋχtos> *kuī^{n}χtos)

[25] Cf. χ > i in Umbrian, §12.

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4) χ > Ø (/ _ C) (*kuīⁿχtos > quīntus)

For the last rule a simple disappearance before t is to be imagined. However, before s there is the possibility of assimilation of the spirants: *sank-sai > *saŋχsai > *sāⁿs(s)ai --> sānksī, with the -ss- simplified after the long vowel or after the restored k.

21) Further, while O.-U. and Celtic show k > χ (and p > f) before t and s whether or not a nasal precedes, the only evidence for a spirant from k in Latin is the long vowel that results from **NS**. No grammarian, or any other source, says that <c> was pronounced as anything other than /k/ even before t or s. Therefore, we can either postulate that k > χ before dentals everwhere and leave the preceding environment unspecified, in which case we must imagine a Rückverwandlung of χ > k everywhere[26] except after nasals, or else specify that k > χ only in the environment / [+nasal] _ C. The latter is clearly preferable. This means that the cases of k disappearing / V _ sC are due to a rule of **CsC-Loss** and not to the passage of k to χ and subsequent assimilation as in O.-U. and Celtic[27]. Further, the restored k shows no tendency to re-spirantize, and so this rule is completely over by the time of the analogical restorations. There is a single example known to me of -nksC-: *tenksmō (cf. OHG dīhsala < *þeŋχslā; ON tīsl, OE tīxl < *teŋχslō) > *teŋ(χ)smō > *tē(ⁿ)smō > *tē(ⁿ)zmō > tāmō 'wagon pole'. Note that the outcome does not allow one to decide whether k, after the n, had passed though a stage of χ or not (i.e. whether the k had disappeared due to **CsC-Loss** or to k > χ), since the vowel length could equally be due to the following s. However, we can make an interesting observation: since the e is not raised to i, the form *teŋksmō must have lost its k and so the

[26] Leumann 1977.113, suggests something of the sort as a possibility.
[27] See above, §10 f. and n. 18.

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homorganic velar nasal, either by ks > χs > (s)s (assimilation or simple loss) or by **CsC-Loss**, before the raising of {e, o} to {i, u} before ŋ (for **Rais-iNG**, see below).

22) If then k > χ / _ t and s in Latin (as well as in Romance, Celtic, and Osco-Umbrian) what of the remaining dental: -n- (the combination -kd- being phonotactically impossible)? Again the evidence of vowel lengthening before n + spirant shows that k became χ before n, as well. Thus, we have: *kᵘenk(u)-noi > kuiŋknoi (with e > i / _ ŋ; **Rais-iNG**, §6.16) > *kuiŋχnoi > *quīχnoi (with **NS**) > quīnī; here it is clear that the raising of e > i must precede the loss of the nasal with lengthening[28]; *kom-kneigᵘh-eiĉ > koŋkneiχueō (here the o is not raised to u / _ ŋ presumably by analogy to other compounds in con- [29]) > *cōχneiχueō > cōnīveō. Further examples are perhaps ānus 'ring' (< *anknos < *h₂(e)nk-n-o-, cf. ἀγκών 'bend of the arm, elbow') from a root *ank- seen in ancus 'bent' (< *h₂(e)nk-o-, the simple thematic; from the idea of 'bent', the secondary meaning 'servile'; also 'one with twisted arms' Paul. Fest.). So Leumann 1977:218. The dim. ānulus (Naev.) might argue against this as we might expect *ankn-elos > *ankŋlos > *ancenlos > **ancellos, that is if sync. 3 with samprasāraṇa of ŋ preceded k > χ and **NS**[30]. However, if sync. 3 does not occur (is bled?), ānulus can be regular: *ankn-elos > *aŋχnelos > *ãᵑχnelos > *ānoɬos > ānulus. No ordering

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[28] For a case after syncope, cf. *kon-ki-to-s (concieō) > conctos (Arval Song) > cūnctus, where the o has been raised; see §6.21.

[29] No transparent compound of con-c- shows raising (though, see n. above). One cannot rule out the possibility that the preceding k in some dissimilatory fashion blocked the raising in koŋ-k-. Cf. the o / _ ŋ in longus somehow? Another possibility for longus, is that since we have a retardation of the change of o > u / u _ (e.g. volt), a velar ɬ may cause the same effect in a different law of raising, though the failure of o to raise after v may be in part orthographic.

[30] As sync. 3-ļ precedes **CsC-Loss**.

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can be based on this, since the conditions of the syncope are uncertain and since ānulus can be built directly to ānus (by analogy of the focus, foculus type). The OIr. áinne, ánne seems to point to an immediate preform *ānn-; if it were due to compensatory lengthening, one would expect an én-[31]. It has been claimed as a loan-word (Vendryes 1902:111); however, the fact that ānus in the sense 'ring' had dropped from the language early (Varro Ling. 6.8, Plaut. Men. 9) and only its diminutive ānulus was the current form, argue against this (so Pedersen cited by W.-H. I.55). Another possible case is pānus if from *tu̯ank-nos (Lith. tviñkti 'swells up': Leumann 1977:187). However, the etymology is not certain. See Walde-Hoffman (II. 248) for further possibilities.

23) We can establish therefore a rule that k > x / n _ [+apical] and this must be ordered before the rule of v̆ > v̄ / _ n [+spirant] (**NS**). Further as *kom-gnōskō > cognōscō shows no such lengthening as occurs in *kom-kneigh- > cōnīveō, we can say that -ŋgn- passed to- ŋŋn- > -ŋn- (written <gn>), and that the spirantization before dentals applies only to k.

24) The treatment of original PIt. x (< *gh, *ĝh, *gu̯h) is quite different from that of secondary x (< *k / _ [+apical]). Intervocalically x > h early, so that it was not subject to the voicing and change to stops that affected the other spirants *θ < *dh and *f < *bh , *dh (/ r _, / _ r, / _ l, / u̯ _): PIE *u̯eĝh- > vehō, but θ > ð > d (PIE *medhios > PIt. *meθios > *meðios > medius), and f > ß > b (PIE *nebhelā > PIt. *nefela > *nebelā > nebula). After

[31] However, as we have no other -nkn- cluster in Celtic, as far as I know (none of the Latin words has a Celtic cognate), we cannot be certain how nkn would develop. One might expect *ank-n- > *échn-, and as chn also lengthens a preceding vowel > **én-. But, since stops are regularly lost between nasals and another consonant (-ndn- > -nn-, see Thurneysen 1946.93, 112), the development to -nn- is conceivable. The long ā would have to be due to the unlenited (geminate) nn; ibid. 32.

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a nasal, however, original χ did voice and become a stop: PIE *sningʷh-e-ti > Plt. *niŋχuet > niŋɣuet > ninguit. These changes must have been over by the time secondary χ arose, since it does not voice even in a voiced environment (*kʷenk(ʷ)-noi > kuiŋknoi > *kuiŋχnoi > *quīχnoi > quīnī), nor does original χ show **NS** (nĭnguit). Thus, we can set up the ordering:

1) χ > h

2) **Spirant Voicing**

3) **Spirant Stopping**

4) k > $χ_2$ / _ [+apical]

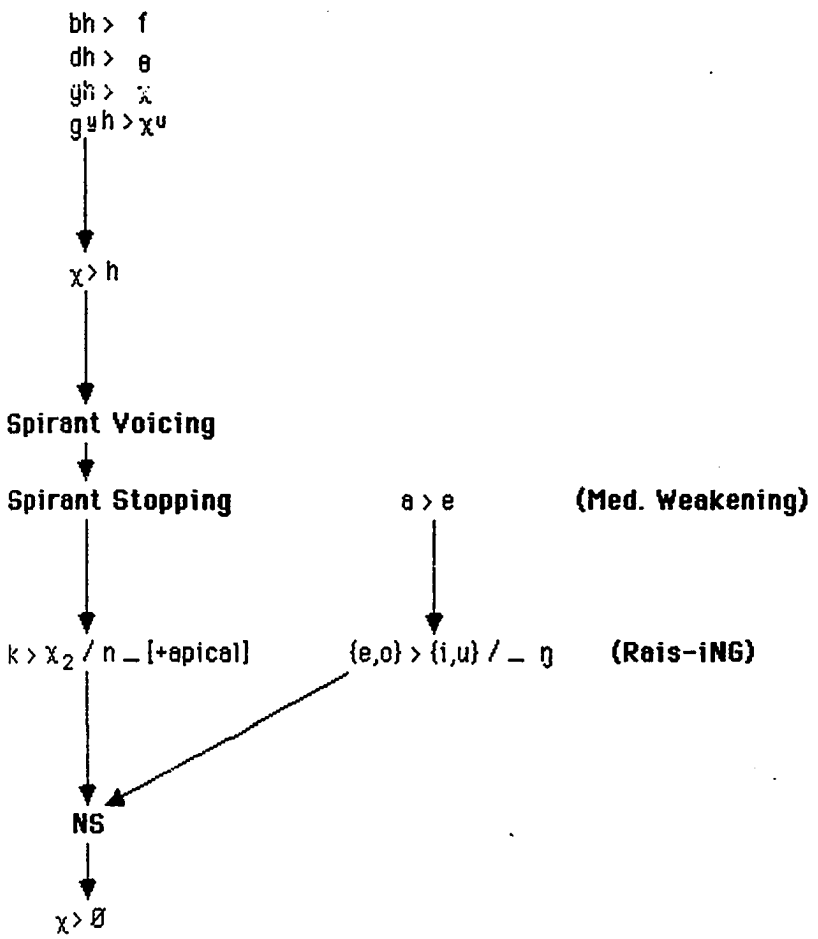
5) **NS**

6) $χ_2$ > Ø / _ C

25) However, this rule of **NS** occurs after e > i / _ ŋ (**Rais-iNG**; see §6.16), e.g. kweŋχtos > quīntus, and this latter rule also applies to e from medially weakened a: tangō but *ad-tangō > *atteŋgō (Medial Weakening) > attingō; *ad-scand-tos > *as(s)cens(s)os (Medial Weakening) > ascēnsus (see §8.11-12). So, the findings can be summarized as:

CHART III

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26) Descriptively (synchronically) we can note a difference in the treament of Vns before voiced and voiceless consonants: trādō, trādūcō, trāiciō, trāloquor, trānatō, trāvehor, etc. but *trant-s > *transs > trā^{n}s(s) >

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trāns[32], trānsferō, etc. with the nasalization still present. Further examples are: *skand-slā (cf. scandō) > *skanssla > *skāⁿs(s)la > *skāⁿzla > scāla, *tonslēs > *tōⁿslēs > *tōⁿzlēs > tōlēs[33] 'goiter', but the dim. *tonsle-lā > *tonslla > tōnsilla[34] 'tonsil' (cf. Leumann 1977.208). In the cases before the voiced consonants there has been a regular loss of the nasalization, and it is not subject to restoration[35]. Diachronically, this can be most easily accounted for by Buck's formulation (1933.153), quoted above: "Stop + s becomes s before another consonant, and if this is voiced the s (which in this case may also come from ns) is lost [with Compensatory Lengthening]". In terms of ordered rules this gives the sequence:

1) V̆nsG > V̄ⁿsG (**NS**)

2) V̄ⁿsG > V̄ⁿzG (s > z / V _ G)

3) V̄ⁿzG > V̄G (**CL**)

That is, when **Compensatory Lengthening** applies, the nasalization of the already long vowel is lost. This might be viewed as reflecting the phonetic impossibility in Latin of the sort of three-morae vowels that existed in Proto-Germanic.

---

[32] Note that **NS** cannot be ordered with respect to **Geminate Simp.**, since the vowel could have nasalized before s as well as ss. Contrast *trant-s-d-> *transsd- > *trāⁿs(s)d- > *trāⁿsd- > *trāⁿzd > trād-, with *kerssna > *kessna > *kesna > *kezna > cēna, where **CsC-Loss** must be before **Gem. Simp.**

[33] Isidore 11.1.57 claims the word as Celtic. However, there is no similar word in attested Celtic. For the root cf. Lith. tesiù 'stretch', Goth. at-þinsan 'drag hither', Brugmann 1897.371 (but see 321), W.-H. s.v.; the original meaning is clearly 'lump, stretched out place, swelling'.

[34] Note that **NS** cannot be ordered with respect to sync. 3-l, since it does not bleed it.

[35] trāns can be added to completely new analogical creations: tráns-latus (CIL XIII.1668.1.27), vs. the regular trālatus.

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27) Leumann (1977.204), however, states: "trans-d- > trązd- > trād-, die Länge ā stammt erst aus der Ersatzdehnung, nicht aus der Dehnung von Vokal vor ns". That is, Leumann holds that the necessary ordering is:

1) VnsG > V̆(n)zG (nasalization, but without lengthening as a co-articulation, or simple loss, and then s > z; order not specified; simultaneous?)

2) V̆(n)zG > V̄G (CL)

I am not sure why Leumann insists on this view. There are difficulties with his formulation. First, a new rule has been assumed which creates short nasalized vowels when the nasal disappears. However, all the cases that we have seen show the vowel lengthened as a concomitant of the nasalization. It is possible to split the process into two parts: first the vowel is nasalized and afterwards becomes lengthened, with **Comp. Lengthening** intervening, but it is difficult to see what descriptive advantage would be gained or what evidence might confirm the conclusion. Secondly, since we have a well established rule of n disappearing, with lengthening of the preceding vowel, before voiceless spirants (-ns-, -nf-, -nχ-), and since s does not voice even in the voiced environment / n _ V (cōnsul, cōsul; see above §17), this argues that **NS** occurred first, then when the preceding nasal was removed, the s now in the environment / V(n) _ G could voice, and so compensatorily lengthen the preceeding vowel, which already being long and nasalized lost its nasal co-articulation. It would be possible to rewrite the rule of **NS** to include lengthening before z as well as s (i.e. leave the rule unspecified as to voice), and to assume that s would voice / n _ G (but not / n _ V), so that *translatos > *tranzlatos > *trā(n)zlatos > trālatus, but as the other cases feature only voiceless spirants, it seems better to leave assume the n was lost before s had become z The voiced spirants other

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than z (*ß < *f < *bh; *ð > *θ < *dh) had already become stops by the time **NS** applies. It might therefore be argued that the fact that **NS** seems to apply only to the voiceless spirants f and χ_2 is an historical relic, not part of the Structural Description, and that **NS** could apply to z, the only voiced spirant extant when it occurred. However, arguing against **NS** applying to voiced spirants is the case of *kuenk-noi > *kuiŋχnoi, with the spirant followed by a voiced consonant, > *kuīχnoi (**NS**) > quīnī, and not *kuiŋχnoi > *kuiŋxnoi (χ voicing after n, see §24 for this change with original χ) > *kuīgnoi > **quīgnī, phonotactically an unexceptionable form. That is, since **Spirant Voicing** and **Spirant Stopping** were over with by this time, the loss of n must have occurred before the voiceless spirant χ. There is no reason then to add a special exception to the rules which have already been well established, to allow n to disappear before z, when the unnecessary assumption neither adds to descriptive simplicity nor aids in the explanation of any crucial cases. Thus we have the ordering:

1) **NS**

2) s > z / V _ G

3) ŭzG > ūG (Comp. Length.)

28) Here, we can note an interesting sound change. While z before the other voiced consonants is lost and lengthens a preceding vowel (e.g. *is-dem > *izdem > īdem, *eksdeukō > *esdūkō > *ezdūkō > ēdūcō, *trans-dūcō > trā$^{(n)}$sdūcō > *trā$^{(n)}$zdūcō > trādūcō), a z before the voiced velar g becomes r and so bleeds **Compensatory Lengthening**, while -sk-, of course, simply remains. So *mesgō > *mezgō (cf. Lith. mazgóti 'wash', Skt. majj- 'dunk') >

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mergō.[36] Therefore, *ious-gland-s > iūglāns, *dis-gerō > dīgerō, are younger creations, as must be the pf. mersī < analogical *merg-sai (due to **RkC-Loss**; see §5.19 f.), not the purely lautgesetzlich *mesk-sai > **mexī ? (cf. *is-pse > ipse, see n. 11; no example of -sks- is known to me; none of the other -sk̂ō pres. in Latin form an s-perf.[37]; see Leumann 1977. 205). Thus, we have the ordering:

1) s > z / V _ G

2) z > r / _ g

3) V̆zG > V̄G **(CL)**

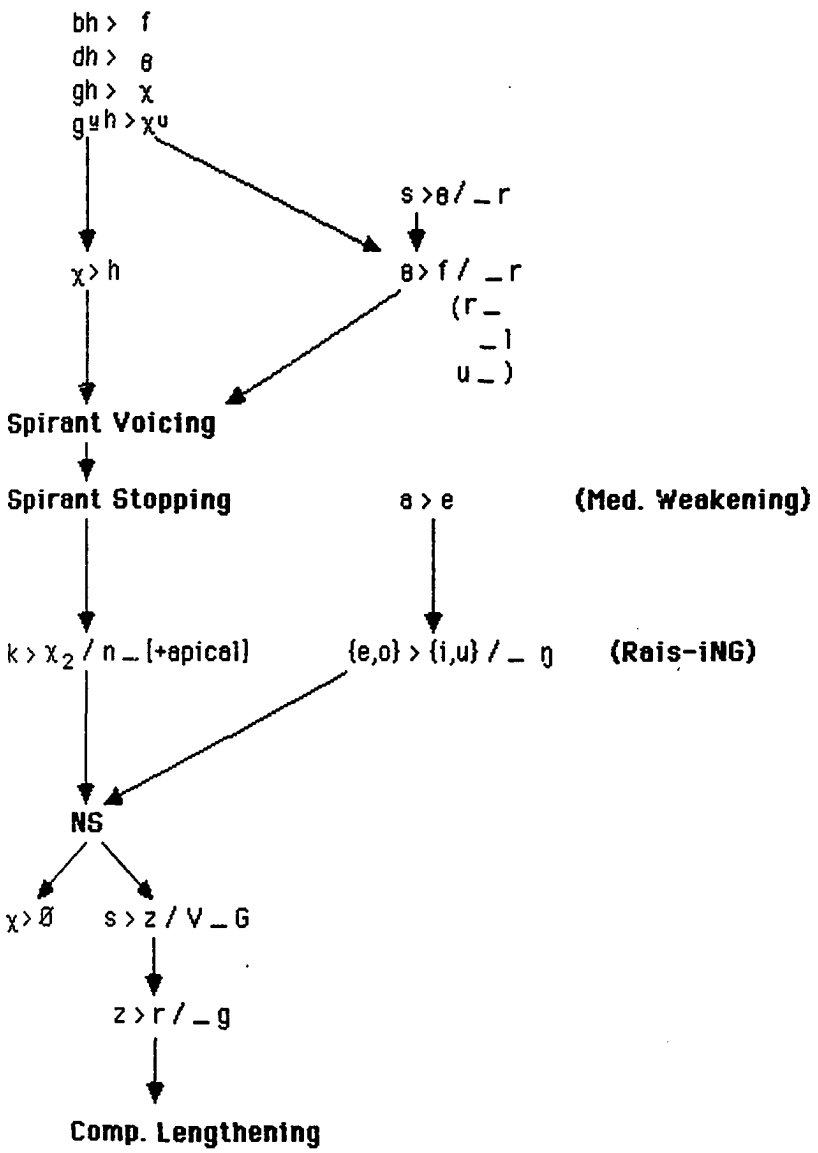
One would like to have a word with the cluster -nsg- or -Csg-, to see how this rule interacts with **NS** and **CsC-Loss**, but there appear to be no examples, and mergō appears to be the only example of z > r / _ g. We can add these rules to the Chart given above:

CHART IV

[36] For the form carmen, Leumann 1977.231 proposes a dissimilation of n ... n (or perhaps n ... m?): *can-men > carmen, similar to one in *gen-men > germen.

[37] That is, characterized -sk̂ō presents do not form s-aorists in PIE, except secondarily.

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Chapter 5: Loss of r before sC; RkC-Loss

1) We have established a clear rule of **CsC-Loss**, whereby stops are lost before sC. A similar loss occurs of r before sC. We need to see if these two rules, usually treated separately in the handbooks[1] can be combined.

Examples are:
*tristis (i.e. originally 'third party'[2]) > *trstis (cf. for structure the thematic m. pl. O. trstus < *tristo-[3]) > *terstis > testis.

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[1] So Buck 1933.154, Leumann 1977:211, Kent 1945.139, Meillet-Vendryes 1968.88. Pisani 1974.51-2 takes these cases as a single rule: "Un groppo di tre (o più) consonanti, di cui la prima o la seconda sia s, si semplifica in s + la terza consonante, purché questa sia un'occlusiva sorda", but offers no examples of s as the first member of such a group. Maniet 1975.97-8 as a single rule (at least descriptively). Monteil 1973.82 holds that s between two consonants prevents "normal" syllabification (i.e., though not stated, VCsCV would syllabify as VCs $ CV; the idea goes back to Juret 1913.27; in general Monteil follows Juret) and under these conditions the initial consonant assimilates to s, even in cases where such assimilation does not occur intervocalically, and the geminate ss then simplifies before the final C: "*eks-dūco > *egz-duco > *e(z)z-duco > ē-dūcō [where an additional stage of voicing is assumed]" or "*ter-stis > *te(s)stis". Besides the fact that syllabification of -VCCCV- as -VCC $ C-, appears to be the normal Latin division at least in the historical period (see Leumann 1977.21-3 and litt., Kent 1945.62-3, Hermann 1923.234), it is difficult to see what descriptive advantage is gained by positing these unparalleled assimilations. In addition, it seems unlikely that -rs $ C- will assimilate when -rss < *-rts (ars, etc.) does not, nor do -ks*, -ps*, (see Hermann 1923.214-5). Juret (1938.53-4), operating with his own idiosyncratic system of phonology, has *sekskent- (syllabified ks $ k) become sescent- through dissimilation, but sextus (ks $ t), dexter, etc. remain. The k in illūstris < louc-stris (likewise ks $ tr), however, is lost "parce que, placé après une diphtongue ou, il contribuait à former la syllabe trop longue"; so too with quīntus. The group ksG is said to become gzG and the g then disappears (whereas a k would not). The cases of loss in sustinere, etc. are explained by a separate preverb sus-. No mention of the distribution of abs-, syncope, analogy.

[2] Buck 1904:60

[3] But note O. trístaamentud, with no syncope.

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*trs-to-s > Plt. *torstos > tostus (cf. o-grade *tors-eio- > torreō; §2.7)

*prk̂-skō > Late PIE *prskō[4] (cf. Skt pṛcchāmi, OHG forscōn, etc.) > Plt. *porscō (cf. U. persclo, persnihimu[5]) > poscō

*kert-snā > Plt. *kerssna (O. kerssnaís) > *kessna > *kesna (by **Gem. Simp.**) > kezna[6] > cēna

2) In the last example we note especially the difference in treatment between *kerssna with -ss- (voiceless by definition) so that r is lost, and *persna > *perzna > *perrna > perna (see §3.25) where the single s has voiced and assimilated to the preceding R. So then: 1) s must have become z (at least in the environment / R _ G) before rsC > sC, since we do not have *persna > *pesna > *pezna > **pēna: that is the voicing of s > z here bleeds **CsC-Loss**, which needs a voiceless s in its Structural Description; 2) the loss of r occurs before the simplification of geminates next to another consonant, since we do not have *kerssna > *kersna > *kerzna > *kerrna > **cerna; and so 3) rss meets the structural description of r > Ø / _ sC (i.e. C = s) and so of **CsC-Loss**. Thus cēna necessitates the ordering:

1) s > z (/ R _ G)

2) **CsC-Loss**

3) **Gem. Simp.**

4) s > z (/ V _ G)

5) V̆zG > V̄G

---

[4] N.B. therefore that the reduction of ksk > sk is not an example of an Italic sound law, so taken (or not distinguished from the latter) in Kent 1945.139, Meillet-Vendryes 1968.109, Pisani 1962.61-2, Sommer-Pfister 1977.192. The different periods of application are rightly and explicitly noted by Brugmann 1897.442, 457, 637; Buck 1904.91; Juret 1938.53; Leumann 1977.33, 57.

[5] With loss of k /C _ C (through a midstage of χ? See §4.11-12, 5.22)

[6] Attested as cesna, Fest. 222.

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3) This loss of r before ss is exemplified by the lautgesetzlich forms:

dossum beside dorsum (cf. doss-ennus 'hunchback', doss-uārius 'pack animal', DOS(S)UO CIL 1^2.170)

contro-vosias (Sententia Minuciorum: CIL. 1^2.199 = Ernout 1957:90 f., c.119 B.C.) beside contro-vorsiae (same inscription; where both forms are frequent)

prōsa[7] (orātiō) beside prōrsus

deossum (Sent. Minuc.) beside deorsum.

4) These clusters of -rss- > -ss-, usually arising in past participles to roots in r plus dental, are subject then to restoration of the -r- from the root[8], and this -rss- becomes -rs- by the later rule of **Geminate Simplification**. So *uorssos (< *urt-tó-s) > vossus --> *vorssos (by analogy to vortō) > vorsus > versus. Note that these analogical forms support the ordering of rsC > sC before C_1C_1 > C_1 // C. So then for most roots in dentals there is the development as in: *ard-sai > *artsai > *arssī > *assī --> *arssī (analogic to ard-eō) > arsī.

5) Note that this pattern fits in with the regular outcome seen in torsī to torqueō, ursī to urgeō, etc., so that there is a synchronic rule of "drop the consonant before the -s- of the perfect" within the Sprachgefühl. So:

*torkᵘ-sī > torsī

*urg-sī > ursī

*ard-sī > *arssī > *assī --> arssī > arsī

6) Since we have a rule of stop + s + C > sC and a rule of r + s + C > sC it is hard to see any reason not to combine them as a single sound change. The

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[7] *prouorsa > *prousa > prō$_2$sa --> prōsa (for the syncope and analogical development to ō, not ū, see §7.54)

[8] Cf. the frequent restoration of lost stops: sextus, etc. (§4.2).

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fact that O.-U. retains these rsC clusters,e.g. O. trstus, U. persc-lo vs. L. testis, posco, while it shows TsC > sC (see above, §§4.10-14), e.g. U. sest-entasiaru, L. Sestius should present no problem. As we have seen the O.-U. change is probably due to an entirely different process (change of {p, k} > {f, χ} and subsequent assimilation to a following s), and the Latin rule is crucially ordered after the two Latin changes of sync. 3-l (-lV̆- > -l-) and the resulting -l- becoming -il- : *ueksla > *uesla (**CsC-Loss**)> *uezla > vēla, but *ueksle-lo-m > ueksllom > vexillum (§4.3). It is also crucially ordered after the Latin syncope of rV̆ > r > er : *tristāmentom (O. trístaamentud) > *trstāmentom > *terstāmentom > testamentum.

7) If we wish to combine TsC and rsC into a single rule of **CsC-Loss**, what of the other resonants? For lsC with a final voiceless[9] stop, there seems to be no certain example. For cases of -lss-, we have only the outcome -ls-. The cases are: (*peld-to-s >) *pelssos > *peɬsos > *polsos > pulsus, mulsus (see §2.4 f. on mulsus and mel, mellis), pulsus, (per)culsus. This perhaps represents the analogical restoration seen for rss: versus, etc. Note, however, that forms such as pulsus confirm the ordering of **Geminate Simplification** as late and after Rz > RR, as we do not find *pelssos > *pelsos (**Gem. Simp.**) > *pelzos > **pellus. For the cases with nasals: -ms(C)- > -ns(C)- is a PIE change (cf. aor. *é-gᵘem-s > Skt. agan). For -ns(C)- as we've seen (§4.17 ff.) the n undergoes the different rule of **N + Spirant Lengthening**: cō(n)sul.

8) We can now turn to the question of whether **CsC-Loss** and **NS** can be ordered with respect to one another. It is clear that they are very similar processes, involving the loss of a consonant before s, but are quite different in structure and outcome. **CsC-Loss** necessitates a consonant following

[9] See above §3.26-7, for the case of alnus with -lzn- > -lln- > -ln-.

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the s, so that -ksC- > -sC-, while -ks- remains, while **NS** nasalizes the preceding vowel even if followed by a vowel (cō(n)sul). Further, **NS** results in a compensatorily lengthened vowel while **CsC-Loss** does not. However, the similarity of the losses before s makes a relative ordering difficult.

9) There appear to be no crucial cases of overlap between the two rules. We might wish on general principles to state **CsC-Loss** as broadly as possible, i.e. to include loss of all consonants[10], as we have seen that it includes both stops and resonants (at least r; no examples for l). In this case we would wish to order **NS** before **CsC-Loss** lest we have **CsC-Loss** bleed **NS** (*monstrom > **mŏstrum), that is to have the cases of n before s removed by **NS**, with resulting long vowel, and **CsC-Loss** can then apply to all consonants. However, it is easy enough simply to rule nasals out of the application of **CsC-Loss** and in generative terms we need only specify CsC-Loss as [+cons., -nasal] > Ø / _ s [+cons.], and so no ordering is necessitated by their Structural Descriptions, which never overlap. Since, s does not voice / n _ V, though it does / R _ {V ,#} (Rz > RR), that is we have cō(n)sul and not *conzul > **cōnul/**cŏnnul?, it might be argued that **NS** is ordered before a broader rule of s > z / [+sonorant] _ V. But as was stated above (§2.23), the form of the rule that we have been employing includes only the liquids r and l, and so no ordering is possible.

10) Since **NS** is shared with Oscan and Umbrian, can it be considered early? Because of the possibility of independent innovation, it cannot be automatically placed as Proto-Italic. In fact, since **NS** is crucially ordered after **Medial Weakening** (a > e; not shared with Faliscan: peparai) and

[10] But not the semi-vowels u̯ or i̯. Since in aestus, etc., the semivowel simply remains as part of the diphthong, the rule of **CsC-Loss** applies only to segments that are [+consonantal], and not to those that are only [-vocalic].

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raising of {e,o} to {i,u} / _ ŋ, **NS** is a Latin rule, and so the similar **NS** in O.-U. is an independent innovation. The inscriptional record is not copious enough to let us state any ordering.

12) We need words where the Structural Descriptions of both rules would be met in an ordered application, that is sequences of n + C + s + C. The case of *un(k)sra > *unsra > *unθra > umbra might be interpreted as showing **CsC-Loss** (to remove the k) before s became θ > ß > b, changes ordered before **NS**. This, however, seems impossible given that **CsC-Loss** is late and s > θ / _ r is early and probably PIt. Further, the protoform is doubtful, and the k, if present, may have disappeared by processes different from **CsC-Loss** (§18).

13) The case of *teŋksmō > tēmō might show that the k disappeared by **CsC-Loss** (> *tensmō) before the raising of *e > i / _ ŋ, a change ordered before **NS**. However, since in the sequence -VŋksC- the k can have disappeared by becoming χ before the s (> *-VχsC-) and then assimilating to the s (-VnssC > -VsC-) or by χ > Ø / _ C, with **NS** at any point after k > χ, no ordering is possible.

14) What, therefore, is required is a sequence of vowel + nasal + labial + s + C (since k can pass to χ, and ts > ss), which might pass by **CsC-Loss** to V̆nsC and then feed **NS** to yield V̄nsC, or if the order were reversed to yield a surface cluster V̆nsC[11]. Unfortunately no cluster -V̆mpsC- appears to exist, nor, given the structure and phonotactics of root and suffix, is one likely to.

15) Comparing CHART II (for **CsC-Loss**) with CHART IV (for **NS**), we can see that there is no clear case of any rule crucially ordered before one and

[11] The fact that there appear to be no such cases of short vowel before ns, though in the absence of any direct statements by the grammarians we could not know the vowel length, might argue that **NS** was a comparatively late rule.

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after the other. **Geminate Simplification** will not serve, since both -ns- and -nss- meet the Structural Description for **NS**. Both rules, however, are ordered before s > z / V _ G (by removing the consonant, which makes it possible for s to voice) and before **Compensatory Lengthening** (the loss of z).

16) There is another rule that we can, however, order with respect to these consonant clusters rules. That is: s > θ / _ r: i.e. basically a lisp of s.[12] This θ before r gets the same treatment as the θ from dh and passes to f, and if in voiced environment undergoes the usual changes of the spirants to β and then to b (Buck 1933.149; Leumann 1977.189). Examples for initial position:

*srīgos (ῥῖγος) > *θrīgos > frīgus

*srāg-o-m (ῥάξ, ῥαγός) > *θrāgom > frāgum (Leumann ibid. and litt.)

Examples for medial:

*founes-ris (cf. fūnes-tus) > *fūneθris > *fūnefris > *fūneβris > fūnebris

*con-suesr-īnos > consueθrīnos > consuefrīnos > consueβrīnos > cōnsobrīnus.

17) An interesting example for our ordering is:

*mēms-ró-m (Grk. μηρός, Olr. mír showing the -ro- stem: mír < *mírr < *mēms-ro-, see Lewis-Pedersen 1961:22, contra W.-H. s.v.; Skt māms-ám, cf. Goth. mimz, OCS męso) > *menθrom > *memfrom > *memβrom > membrum, which shows that s > θ also bleeds **NS**. In addition, of course, s > θ is before the voicing of s > z / V _ G and the subsequent **Comp. Length**. A further case of **NS** not occurring until after original spirants had become

[12] Cf. the LateAv. change of s > θ / _ r: Skt. śrótas-, LAv. θraotah-, etc.

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stops before a voiced consonant is *londhuos (OHG lentī, OCS lędv-ije[13], Russ. ljádveja) > PIt. *lonθuos > lonfuos > lumbus[14].

18) The case of umbra is quite complicated, and its protoform is unclear. If we compare Skt. andhás 'blind', ándhas 'darkness', we can set up a PIt. *ondh-rā, a substansivized -ro- stem adjective. Here, since the zero-grade (and not o-grade) is usual, one might postulate a *h$_3$(e)ndh- as the root for the Skt. and the Latin forms. However, much closer semantically is Lith. ùnk(s)-nā,'shadow'. This would allow us to set up a protoform: *unks-ra (so W.-H. s.v.; Sommer-Pfister 1977.192; Leumann 1977.206, though with no statement of the Lith. data or the preform), presumably with the development *uŋksra > *unsra (by **CsC-Loss**?) > *unθra (by -sr- > -θr-) > *unfra > *umβra > umbra. However, the other putative examples with similar structure (R + velar + s + R) show as outcome a long vowel plus the final resonant. So perhaps: *man-terg-slo-m > *manterslom > *manteslom (**CsC-Loss** applied iteratively?) > *mantezlom > mantēlum[15]; *tenksmō perhaps to *tēⁿxsmō > *tēⁿssmō > *tēⁿsmō > *tēⁿzmō > tēmō 'wagon pole' (see §13). The problem here is how to get rid of the -k- for umbra. One way is make use of the following s. The ordering which this derivation seems to necessitate (**CsC-Loss** before sr > θr) is not only counter-intuitive, it also runs against our findings that **CsC-Loss** is comparatively late in Latin

[13] With the meaning 'spirit, soul'

[14] Cf. the irregular raising before m in umbō, umbilicus (cf. ὀμφαλός), umerus < *omesos (§6.23); see Sommer-Pfister 1977.59; lumbus is the only good example of internal -dhu- (but perhaps plēbēs if connected with πληθύς; see W.-H. and litt.; see also Exc. II §22), no other examples for -Cdu-.

[15] From manus plus tergeo: Sommer-Pfister1977.192; Leumann 1977.207; E.-M. 385; Walde-Hofmann 1954.32 and litt. The etymology (Varro 6.85) is not certain (with tero 'rub'?) and we might wonder indeed if instead -rksl- would yield -rsl- > -rzl- > -rrl- > -rl- > -ll- (?).

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(occuring at the beginning of the historical period). In addition, there is one case in O.-U. which points convincingly to a similar change of -sr- to -θr-: O. tefúrúm 'burnt-offering', U. tefra, tefru-to 'ex rogo' < *tesro- < *teps-ro[16] (cf. Skt. s-stem tápas-, L. tepor; Buck 1904.78, Brugmann 1897.763; von Planta 1893.476-8), and there is no reason not to consider the change Proto-Italic. The k might have disappeared / n _ s through a midstage of χ. However, since in in all the other examples this results in a lengthened vowel[17], and since ordering the late k > χ$_2$ / _ s before the apparently Proto-Italic s > θ / _ r is implausible, this sequence must be ruled out. We might consider a series: *unksra > *unkθra, and then an ad hoc rule to simplify the unparalleled cluster -nkθr- (bzw. -nχθr-) to -nθr- (the immediate preform) without having to invoke the late loss of consonant before an s. This seems a reasonable solution. However, it may not be necessary to invoke it. There is a pronounced tendency in the Baltic languages for an epenthetic -k- to develop / _ [+sibilant] (C)[18]: añkštas 'narrow' < añštas < *anž-tas < *h$_2$nĝh-to-s (cf. L. ang-ustus, ἄγχω), cf. Lith. aĩksnis, L. alnus above (§3.26-7). The Lith. ùnk(s)-nā 'shadow' is not therefore by itself good evidence for the presence of a -k- in the protoform. As umbra has no other close cognates[19], we cannot be certain what its

[16] Through a midstage of *tes(s)ro- < *tefsro-, with p > f / _ {s, t} perhaps, see §4.10 ff., so also Brugmann 1897.674.

[17] It could be argued that that an original long vowel would have been shortened by Osthoff's Law in any case. However, as the previous examples show, the lengthening due to **NS** is preserved (i.e. is after) any Latin shortening even in Oshtoff's Law environments. Though a reapplication of Osthoff's Law occurs sporadically in Romance (L. īnfra, It. inferno, Fr. infernal; but Fr. enfer < *ĭ-), the uniformity of Italian ombra, Fr. ombre, Rum. umbra is prima facie evidence of a ŭ-.

[18] See Senn 1966.83, Stang 1966.108-9.

[19] Even ùnk(s)-nā itself has no cognates outside Lith.

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protoform was, and a sequence *unsra > *unθra > *unfra > umβra > umbra is equally likely. In any case it does not provide evidence for ruling **CsC-Loss** before sr > θr. The only cases of -Csr- appear to be membrum and umbra, both with nasals before the -sr-.

RkC-Loss

19) Let us now look at a new set of consonant clusters. Again, we begin with the formulation of Buck (1933:154):

> A stop is lost between r or l and another consonant in most such groups. So in rct, lct (but rpt, lpt remain), rdn, rcn, rcm, rpm, rbm, lcm, lgm, lpm, also rcs, lcs (but these remain when final, rx, lx); apparently also in rts, lts, but here the change belongs under **193** [i.e. rts > rss > rs[20]; but see above §§1-4, for rss > ss --> rss > rs].

However, as Buck's own exceptions show, this is not a simple rule of C > ∅ / R _ C. Leumann (1977:216) states: "Hinter r l m n: V[er]schl[uß]l[aut] vor Vschll. oder s. In dieser Stellung bleibt p erhalten, c wird meist ausgedrängt." We need to examine all the evidence on this law, organizing our data by the consonant that is lost.

I. Cases after r or l:

Labials:

-lpm-: *pelp-ment-om > *polmentom > pulmentum (cf. *pelpa > pulpa)

-rpm-: *sarp-ment-om (cf. sarpō, sarptum below) > sarmentum.

but -Rpt- stays, so:

carptus

insculptus

[20] Curiously Buck does not consider the behavior of *rss < *rts and *rss < *rtt together, even though he treats both on the same page.

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sarptus, etc.

It looks then like the cases of p > Ø are not to be explained simply as the outcome of a general RTC > RC rule. Since a labial stop becomes a velar before a velar: ob-cipiō > occipiō, and becomes m by assimilation before a nasal: summus < *sup-mos (cf. sub-, ὑπό), somnus < *svepnos (cf. Skt. svapna-, ὕπνος), we can have the development:

*pelpmentom > *polmmentom > pulmentum.

i.e. assimilation followed by **Geminate Simplification**. This will explain why the cluster -pt-, which doesn't assimilate, simply remains after R.

Dentals

-rdn-: *ōrdon-ā-se > *ōrdnāre (cf. ōrdō) > ōrnāre

-rtc-/-rdc-: two not completely certain examples

a) *Mārt-ko-s > Marcus (cf. Mārs, Mārtis[21])

b) *kord-kelom (with the -kelo- dim. suf.) > corculum 'little heart'[22]

Again these cases seem best explained by an assimilation of dn > nn (ad-notō > annotō) and (-dk- >) -tk- > -kk- (ad-cipiō > accipiō) respectively; so Leumann 1977:219. For cases of *Rts/*Rtt there is a first stage *Rss. The sequence of *rss > *ss --> *rss > rs has been dealt with above (§§1-4). Again for the dentals the loss need not be due to simple rule of RTC > RC.

20) Velars

For velars, however, there does appear to be a genuine loss, as shown by cases where the two consonants after the resonant do not assimilate. So:

-rkt-: *tork$^{(u)}$-tos > tortus

[21] Speculations on proper names are always uncertain. The relation of Mārs, Māvors, Marmar (C. Arval.) O. Mamers, is not certain.

[22] This may, however, be built straight to the nom. cor; Leumann 1977:217).

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-lkt-: *ulk-tos (cf. ulc-Tscor) > ultus

-rks-: *tork$^{(u)}$-sai > torsī

*orksos (cf. Skt. rkṣa-[23]) > ursus

-lks-: *mulg-sai (cf. mulgeō) > mulsī

-rkn-: *urk-na (cf. urc-eus) > urna

*kuerk$^{(u)}$-nos > quernus

These cases are all of k before a dental and it is possible to imagine that the k has disappeared through a midstage of χ, though as noted above (§4.2) the only evidence for k > χ comes after nasals. However, -rkm-, -lkm- and -lgm- also lose their velar, here in cases where no assimilation[24] or spirantization occurs:

-rkm-: *tork$^{(u)}$-mentom > tormentum

-lkm-: *fulk-mentom (cf. fulcio) > fulmentum 'heel, prop'[25]

-lgm-: *fulg-mn̥ > fulmen

21) There is also clear evidence from O.-U. for these processes. So for labials lost by assimilation: *pelp-m(e)n-eis > U. g.sg. pelmner. For pm > mm, cf. U. somo 'summum' < *sup-mo-. For dentals rtk > rkk > rk in *pert(i)ka (cf. L. pertica) > U. percam, O. perek (with Oscan anaptyxis). For tk > kk, cf. O. púkkapíd 'whenever' < *pod-k-; *ad-u̯ok-ā-tos > *adokātos (du̯ > d) > adkātos > O. akkatus[26]. But for velars, again a genuine disappearance:

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[23] The thorny problem of the PIE form of this word is not relevant to the concerns of the present discussion. The PIt. form would have no analogic source to interfere with a lautgesetzlich outcome.

[24] Other than in voice: km > gm, e.g. *sek-mentom (cf. secō) > segmentum.

[25] For fulcrum, see below n. 36.

[26] N.B. these assimilations continue to occur after purely Oscan syncopes. Note too a nice piece of rel. chron. pointed out by Buck (1904.58): du̯ > d precedes the syncope, otherwise we would have *adu̯okatus > **adukatus with samprasāraṇa of the u̯.

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-lkt-: O. molta, U. muta[27] (L. multa 'fine', cf. mulcō)

-rkn-: U. g.sg. Urnasier[28] (as if L. Urnārius) < *urk-no-

-rkt-: (O. fortis, if L. forctis is genuine: XII Tables apud Fest. 348).

Again, these examples are before dentals and we might invoke a midstage of k > χ (as in O. saahtúm, etc. §4.12). However, the velar also disappears in rkm > rm, where km does not become χm nor assimilates to mm. There is secondary -km- in O. g.sg. Dekmanniúís, as if L. Decumanius. There are no certain examples of primary intervocalic -km- in Oscan[29], though we might expect anaptyxis and can compare kn > kVn in abl. akeneí, g.pl. akunum but U. acnu; tn > tVn in Patanai (< *pat-na-, cf L. Padna; Buck 1904:53). For rkm > rm (and then with anaptyxis), compare:

O. turumiiad 'torqueatur', a denominative with immediate preform *tormiio- from *tork$^{(u)}$-m-iio-[30], cf. L. tormentum.

22) Buck (1904:91-2) claims this simply as a rule whereby k > Ø between any consonants. However, the only two examples without an immediately preceding r or l involve the apparent root *persk- 'ask', whose ultimate connection with *prḱ-sḱ- is clear, but details of the formation of this new e-grade are uncertain. In any case we can only reconstruct for O.-U. a preform *persk (not back to a supposed **persk-sk; cf. above §2 for *prsḱ- already in Late PIE). For *persk-, cf. U. persclo, etc. above, with no loss of the k since it is not immediately after a resonant. An apparent simple

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[27] Again a bit of rel. chron.: lkt > lt feeds the U. loss of l before t.

[28] For the lack of rhotacism, see §4.10 n. 10.

[29] The evidence that the velar disappeared from intervocalic velar plus nasal in O.-U. is handled by von Planta 1893.380-4. The cases are individual suggestions by various scholars and are based on incorrect or at least dubious etymologies.

[30] So Buck 1904.32, 51,91; Brugmann 1906.248, not with von Planta 384.384 n.3 "möglich" < *torp-m-, cf. torpeō.

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disappearance of the k / s _ C in the new root *persk is seen in U. sg. impv. depon. persnihimu 'precātor' < *persk-nī-mōd[31] and U. impv. perstu < *persk(e)tōd. Here in the clusters rskn, rskt one might suspect a midstage of rskn > rsxn, rskt > rsxt and then assimilation to the preceding s and simplification of the geminates. In any case, it seems unlikely that rsk + dental is behaving in the same way as RkC.

23) Leumann (1977:216), quoted above, has left open the possibility that this loss of velars happens not just after r and l, but after m and n as well. The data argue against this. Again, they are arranged by the consonant lost:

Labials: (with homorganic -m-):

Only a few (mostly in ambi- with syncope):

-mpk-: *ambi-caput-is > *ambcapets > ancipes; see §9.16.

*ambi-kuolos > *ambkolos > anculus

So am-putare, am-plector, etc. (Leumann, loc. cit.)

However, as P + K > KK (oc-cidō, etc.) as well as P + N > mN (somnus, etc.), this loss of labials seems also to be due to assimilation. Thus, we find mpC retained in precisely those cases where the p does not assimilate :

-mps-: secondary in sum-p-sī (to sūmō)

-mpr-: im-probus, etc.

-mpt-: original in temptō (see Sommer-Pfister 1977:175); secondary in *em-tos > *en-tos --> *emtos (by analogy to emō) > em-p-tus.

24) The evidence for a regular change of mpt > nt is not secure. For the early λαμπτήρ --> lanterna (Plaut.), Etruscan mediation is a possibility (Ernout-Meillet 1959:340; Sommer-Pfister loc. cit.). Later Latin shows mpt > mt > nt (see Leumann 1977:216-7 for inscriptional evidence) but also

[31] Fourth conj. denominative from an i-stem *persk-ni-; see Buck 1904.164, 191.

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frequent cases of -nt- >-mpt- (volumptas, etc.) A conflict between a substandard pronunciation and spelling pronunciation is to be recognized here.

Dentals:

For the most part, their loss after nasals is explained by the Double Dental rule, with subsequent **NS** (see above §§4.8, 17):

word-final: *-V̆nts > *-V̆nss > *-V̄ⁿss > -V̄s

*trants-C > *transs-C > *trāⁿs(s)-C : trāns-cendō

*trants-G > *transs-G > *trāⁿs-G > *trāⁿz-G > trā-G : trā-ducō

*skand-tos > *skanssos > *skāⁿssos > scānsus.[32]

Again for clusters that do not assimilate the cluster remains: mō(n)strum, etc.

25) Velars:

Here, two processes are at work. All the cases where voiceless -k- disappears are before dentals, so that there is first the change of k > χ and then **NS**. These have been treated above (§§4.17 ff.):

-nkt-: *kᵾenk(u)tos > *kᵾeŋχtos > *kᵾiŋχtos > quīntus

-nkd-: *kᵾenk(u)dekem > *kᵾiŋχdecem > quīndecim[33]

-nkn-: *kᵾenk(u)noi > quīnī

-nks-: restored in sānxī, etc. (see above).

26) For voiced g the cases are of -ngn-: Here the g assimilates in its order (as did the labial stops) to become a velar nasal before the nasal n:

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[32] Where ss > s after a consonant (**Gem. Simp.**) or after a long vowel.

[33] not with Sommer-Pfister 1977.190 "aus quíŋqu(e)decem, *quīŋgd-", which leaves the ī unexplained. One might claim that the ī was by analogy with quīntus as the ī in quīnque clearly is. However, if Sommer-Pfister's ordering is correct the sequence *kᵾenk(u)noi > *kᵾeŋgnoi > *kᵾeŋŋnoi > should produce a **quĭgnī (see next para.). Note, therefore that k > χ and **NS** must precede the voicing of k > g and the assimilation of g > ŋ.

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*gn > ŋn, spelled <gn>. The velar -ŋ- is shown by the raising of e > i: *dek-nos > *degnos (voicing assimilation) > *degnos (assim. to nasal) > dignus <dignus> (see §6.18-19). So for *ngn there is assimilation to *ŋŋn, and then *ŋŋn becomes ŋn (again spelled <gn>) by **Gem. Simp**. Thus: *congnatos > *coŋŋnatus > coŋnatus[34] <cognatus>. See Leumann 1977.200, 218.

27) Thus, the cases of a consonant disappearing between a nasal and another consonant are due to either the spirantization of k, or various assimilations of the vanished consonants. We are left then with genuine cases of k > Ø after r or l. We can establish a new rule: k > Ø / R _ C, a rule which will be called **RkC-Loss**.

28) To this rule there are a small number of apparent exceptions. Beside the regular mulsī (mulgeō), there is also mulxī; beside fulsī (fulgeō), fulxī; both are attested by Priscian (2.486.9, 539.2) and are clearly analogical (and artificial) creations to differentiate these perfects from the identical perfects of mulceō and fulciō. Also analogical is the perf. pass. participle mulctus beside multus, for both mulceō (rare) and mulgeō, to avoid homophony with multus 'many' (see Leumann 1977.217-8). Two words attested by Paulus ex Festo and probably showing the archaic form are ferctum, beside fertum 'sacrifical cake'[35], and sarcta 'repaired' (to sarciō), though this last could simply be analogic restoration. See §21 for forctis. The forms fulcrum, mulctrum and mulctra 'milking pail' also violate **RkC-Loss**. Here, we might say that r does not meet the Structural Description of **RkC-Loss**. The examples have had s, t, d, n and m as the consonant following k, which seems to show that the rule is activated not just by

[34] The o not raised by analogy to other compounds in con-, see §§4.22 and nn. 28, 29.

[35] Also attested in Cato RR 134.2; for ferctum there seems no possibility of analogical restoration; see Vine 1986 (HSCP 90, forthcoming).

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dentals (m) and not just by stops (n, m). If we rule out r, we might have a regular development of *fulk-tlo-m or *fulk-lo-m (see Leumann 1977. 311 and 313) > Plt. *fulk-klum > fulcrum (with dissimilation to l ... r) and no **RkC-Loss**. However, since many of these -crom, -culum (< *-tlo-) forms are built to the perf. pass. participle, fulcrum could have been built to fultum; so sepul-crum to sepultus, and mulctra to the analogical mulctus (Leumann loc. cit.). Since these forms do not necessarily continue the historical development, no definite chronological observations can be drawn from them[36]. The only other form whose surface form violates **RkC-Loss** is pulc(h)r-. The etymology is uncertain and nothing certain can be deduced from it. The oblique could in any case be easily analogical to the nom., as if *polcros > pulc(h)er, on the model niger : nigr- :: pulc(h)er : X = pulc(h)r-, if rV̆-syncope bleeds **RkC-Loss**, as it seems to from the inscriptional record (since -ros > -er occurs soon after the Forum Inscription c. 500, while forms before **RkC-Loss** appear still to have early attestation; see above), though there are no crucial examples (that is pulcr- is the only form showing -Rcr-).

29) To review briefly, we have three rules which might involve an R as the first member of a cluster of three or more consonants:

1) RsG > RzG > RRG > RG **(Rz-Assim.; followed by Gem. Simp.)**

2) RsC > sC **(CsC-Loss)**

[36] That is if we were sure that **RkC-Loss** included l and r, and that fulcrum were strictly lautgesetzlich, we might argue that a *fulkklum did undergo CsC-Loss to *fulklum, and therefore that **Gem. Simp.** must be after **RkC-Loss** lest *fulkklum > *fulklum (**G.S.**) > **fullum (**RkC**). Further, if **RkC** includes r and l, we might argue that anaptyxis bleeds **RkC**, i.e. *mar-tlo-m > *marklom > *marcolom > marculum (see §2.15) and not *marklom > **marlom (**RkC**) > **mallum.

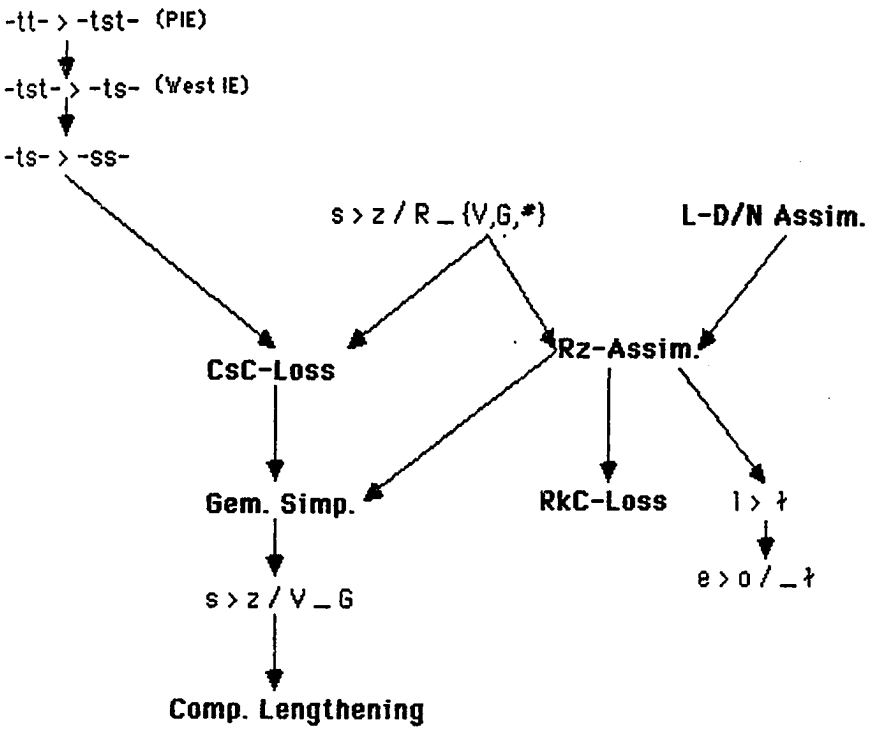
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3) RkC > RC **(RkC-Loss)**

The first two are clearly related. The quality of the final consonant, voiced or voiceless, determines whether the s will voice and so whether the z assimilates to the preceding R or the R is lost before the following voiceless s. That is, the voicing of s > z / R _ G bleeds **CsC-Loss**. Further, the first and the third are clearly related since **RkC-Loss** produces new surface -rs- (torsT, etc.) and -ls- (fulsT) which do not assimilate to RR. That is **RkC-Loss** is after (counter-feeds) s > z / R _ V and **Rz-Assim.**, not in itself an unexpected outcome. We can sum up our findings in a new chart, incorporating some earlier orderings:

CHART V

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Part II: Vowels

Chapter 6: Changes of Short Vowels in Initial Syllables

1) We turn now to changes in the vowels. The procedure in this section will be to state briefly first changes of short vowels in initial syllables, and secondly changes of diphthongs in initial syllables, i.e. rules independent of stress and Medial Weakening. Once these are established, we can then discuss Medial Weakening and relate both new rules and those already seen in initial cases, for which Medial Weakening creates new input, that is rules fed by and so ordered after Medial Weakening. Changes conditioned by consonants, many previously examined, will allow us to fit our new findings in frameworks already established. Changes which do not lend themselves to relative chronological observation, in particular sporadic or possibly dialectal changes[1], will be passed over in silence.

High Vowel Backing

2) The first change we wish to examine is naturally that of e > o / _ ł (see §1.1-3). A few examples already discussed are:

*u̯eł̄ō > volō, but velim, etc.

*ĝhelhɜ-o-s > *helos (see n. 2) > *hełos > holos, but *heliu̯os > helvus (§3.8 ff.)

*seluu̯ō > solvō

ἐλαίϝα --> *elaiu̯a > *oł̄e̯u̯a > oīva
ἔλαιϝον --> *elaiu̯om > *oł̄e̯u̯om > oleum (see below §7.4, 8.5, 13)
As has been stated before, *seluu̯ō > solvō, etc. do not allow us to order e > o

[1] For example o > u / _ rC: *r̥ksos > ursus, furnus beside fornus; e > i / _ mp, mb: simplex but templum; imber, but membrum. See Leumann 1977.45 ff.

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/ _ l̥ relative to syncope, since the l could have become velar because of the dark vowels before syncope occurred (§3.3 n. 3, 2.4). Indeed *heliuos > helvus, though isolated, argues that this was the case, since it seems to show that l > l̥ and e > o / _ l̥ were over by the time of the syncope of V̆ / V̆R _ u̯ at least (§3.15). More importantly the case of olīva shows that e > o / _ l̥ occurred at least before the the raising of the high monophthong ē to ī (see §6.1-4 and cf. Goth. alew borrowed after ei > ē and e > o / _ l̥), since the high vowel would have prevented l from becoming dark (§1.1-2, 3.17).

3) Besides helvus, there are a few exceptions where e is retained before an l which descriptively (synchronically) should be velar. These are celer, celōx, celsus, gelū, scelus, and their derivatives. The explanation of Sommer-Pfister (1977.56) and older works is that these nominatives are by analogy to forms with an e following a consequently palatal l': thus celōx after celer, scelus after the oblique sceleris. However, this cannot be correct since e equally produces a velar l̥[2]. Curiously, even Leumann (1977.47), who had correctly pointed out the evidence that l before ē (at least) was velar, writes "richtiger nimmt man wohl analogische Restitution des e nach celer gen. sceleris gelidus an." Analogy to gelidus might play a part in gelū, gelāre, while analogy to celer is clearly possible in celōx, and analogy to cellō is possible in the p.p.p. celsus (so Leumann 1977.141, presumably on a model currō : cursus :: cellō : X = celsus; N.B. however, closer at hand is pellō : pulsus, and esp. per-cellō : perculsus). This, however, still leaves scelus, and celer itself to be explained. The correct explanation for these forms was first seen by Havet (MSL 5 [1872] 46 n.1),

[2] Accordingly helus in Paul. Fest. is explained (Sommer-Pfister loc. cit., W.-H. s.v.) as analogical to an oblique *heleris. It is rather an archaic form.

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that the sound preceding the e in all these exceptions was velar[3]. There is clear evidence for a more palatal pronunciation of the velars before the front vowels i as well as e in the distribution of the three signs for a velar in Old Latin: Q, C, and K. Thus, Q is regularly resticted to the position before o or u, K regularly before a and consonants, and C before e or i.[4] So in the Forum Inscription (CIL 1^2.1): QUOI but SAKROS, KALATOREM but RECEI. Traces of this distribution are seen in the letter names kā, cē, qū; the restriction of g to the position before u in classical Latin, and the abbreviations K. for Caeso, Kal. for calendae, etc. See Buck 1933.74, Leumann 1977.10, Allen 1965.15. Thus a palatal pronunciation of a preceding velar inhibited the rounding of e to o before a velar ł, and this effect was aided in some cases by an analogic conspiracy. Since perculsus, however, shows no such inhibition, we may speculate that the palatal quality of the preceding velar was effective only if in addition the vowel was stressed. Futher in color the ō of the second syllable seems to have undone (by o-umlaut?) the hindering effect of the preceding velar (cf. Leumann 1977.101).

4) We need now to examine a development touched on above (§2.14). The dim. PIE *pō-tle-lom > PIt. *pōklelom > *pōk|lom (sync. 3-|) > pōcillum, with -i- retained in a medial closed syllable shows that the regular outcome of *$|_2$ was -il-. So too in the superlative to facilis: *faklisamos > *fak|zamos > facillimus. From the same stem we have the abstract *fakli-tāts > *fak|tās > *fakiltās. This -i- from -|- should also have been retained

[3] Noted by Leumann (loc. cit.). Buck 1933.82 says they "admit of various explanations".

[4] This is of course an allophonic and not phonemic difference in Latin. In English also the pronunciation of velars is distinctly more forward before front vowels: Ladefoged 1975.49.

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in a medial closed syllable, but instead we find *fakiɬtās > facultās, where it seems impossible to derive the u from i by medial weakening, i.e. *faciɬtas cannot become **faceɬltos which could then go on to **facoɬtas > facultas. That is, the dark -ɬ- must be considered as backing the -i- directly to -u- without an intervening stage of -e-. In the place of the familiar rule from the handbooks of e > o /_ ɬ, we must expand it to include {i,e} > {u,o} /_ ɬ. There appear to be no certain examples of original i before ɬ in initial syllables[5]. We can write a new rule of **Front Vowel Backing** (FVB). In generative terms:

$$\begin{bmatrix}+\text{voc}\\-\text{back}\end{bmatrix} > [+\text{back}] \;/\; _ \begin{bmatrix}+\text{son.}\\+\text{lateral}\\+\text{back}\end{bmatrix}$$

Thus in the vowel triangle there is a movement before ɬ:

CHART VI

i -------> u

e ---> o

a

4) The same change can be seen in the old adverb, facul, where the original neut. sg. *fakli > *facl$_2$* (by apocope) > *faciɬ*, which in a closed syllable (-VC*) should simply remain (e.g. i-stem navis, etc.; 2. pl. *-tes > -tis). So again, the i is backed directly to u with no intervening stage of e: *faciɬ > facul.

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[5] silva (see §3.7f), siler, silicia/siliqua, stil(l)ago, (as plant names likely to be non-Latin), silex, pilus/pila, and gilvus are without etymologies, as is sileō; bi-lūstrum, etc. are analogical new formations; hilarus and derivatives are loan words from Grk. ἱλαρός. For anomalous i before l in open medial syllables, see §8.17.

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5) The other nouns in -ul probably do not reflect this backing of -i- to -u-. Thus for praesul 'celui qui saute en avant' (Ernout-Meillet, s.v.) from salio 'leap'. Here the nom. is unmarked by the -s of stop-stems (dux, e.g.). Whatever the inherited pattern may have been for such a deverbative noun, Latin often shows a nominative backformed from the oblique stem (before Medial Weakening applied) for stems in a resonant. Thus gen. *cornu-can-es --> nom. *cornucan > cornicĕn; gen. pect-en-es --> nom. pectĕn (cf. mellis --> *mell > mel, see §2.12, 21, cf. 9.9 n. 30). So for gen. prae-sal-es a new nom. *praesal is created which undergoes development to *praeseł (**A-Weakening**) > *praesoł > praesul.

6) Walde-Hofmann (1938.432) and Ernout-Meillet (1959.27-8) both claim that exul shows the same root seen in ambulo and that the root is *al-, and the preform *eks-al-s, comparing Grk. ἀλάομαι. If U. amboltu does mean 'ambulato'[6] then we must have not a root *al- but *ol-,[7] which E.-M. (27) hold to be from a root *el-, seen in MidW. el, 3. sg. subj. 'he may go', and perhaps, they claim, also in ἐλ-θεῖν.[8] In amb-oltu we have presumably either the o-grade or zero-grade(though neither grade has reflexes claimed that I know), and can have nothing to do with alacer (E.-M. loc. cit.).

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[6] Vetter (on vi b 52) believes the passage demands a meaning more like 'surgito'. This unfortunately forces him to take amboltu from an unattested root *(an)-bol-.

[7] W.-H. 39 cites von Planta 1898:239 for -al- weakening to -ol- in Umbrian. However, as von Planta makes clear, this vocalic interchange is sporadic, and the best examples show -a- alternating with -u- not -o- in both alphabets (praefucus beside facus). See also Buck 1904:55-6.

[8] Of course, ἦλυθον tremendously complicates this simple view.

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Phonologically any root *V̆l- could underlie ambulo. However, if we do have *eks-al-, we would expect the nom. to be *exal[9] > *exel > *exol > exul.

7) The only statement I have seen of midstages for the development of *fakltas > facultas[10] is that of Pisani (1958.162-4) who derives facul from *facli with an apocope that resulted in a velar -ļ, which then presumably, though not explicitly stated, became -ul (bzw. -ol?), while -ļ- (palatal, in Pisani's phonemics) became -il-. Can this be ruled out? There is no a priori objection to multiplying in this way the allophones of ļ, and if consonantal l exists beside ɫ, their vocalic forms could also retain or acquire such a distinction. However, as the example of *faklisamos > *faklzamos > facillimus shows, no such dark allophone could have arisen till after ļ > il since the assimilation of -lz- > -ll- bleeds the darkening of l to ɫ, i.e. not *faklsamos > *faculsamos (bzw. *facolsamos) > **facullimus. Thus the direct change of iɫ to uɫ has to be assumed. See above (§2.14 ff.) for the ordering i) sync. 2/4, ii) ļ > il, iii) l > ɫ, iv) e > o /_ ɫ, and now i > u /_ ɫ.

8) The apparent counter-examples to final -l being velar (vel, mel, fel, semel, and subtel) have already been discussed (§2.16). There are three possible counter-examples to the expanded version of **High Vowel Backing**. These are vigil 'watchman' or as adj. 'watchful', pugil 'boxer' and mugil 'mullet'[11]. The origin of this l is obscure. Leumann (1977:360) rightly points out the gen. pl. of all three in -um apparently rules out any i-stem of the form *pug-li-s, etc. (so Kent 1946:44, et. al.). However, as the

[9] Not a marked nom. *eks-al-s with Solmsen IF 26.105, or eks-el-s with Vendryes BSL 47.23, either of which would give **exel(l), like *vel-s(i) > vel(l).

[10] Usually simply left at that, e.g. Leumann 1977: 97, who, however at 144 notes "eine Entwicklung *facli > *facļ > *facil (> facul)" but does not explain how this final stage is to be considered.

[11] nihil is from nehilom, freq. attested in OL.

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gen. pl. apum (the usual form since Livy, who shows both) besides apium, partum (Enn.) beside partium, marum (Naev.) beside marium show an i-stem can be reinterpreted as a C-stem, and an i-stem is attested for vigilium[12] (Livy 10.33.6) and Varro (Men. 22.2) cites a pugilis.[13] However, these are just as likely to show an expected spread of -ium. What kind of ablaut pattern we might expect is therefore equally obscure. The nom. is likely to have been *-ils > -ill > -il (so Leumann 1977:94; Cowgill 1978:40). This will give the attested outcome. Leumann 1977:94 states, "In vigil, pugil bewahrtes i wie e vor l wohl aus -ll, d.h. mit -il(i)s > -ll > -il." This, however, is opposed to his view (360) that vigil, etc. are not i-stems. Nor is the -s the inherited -s that marks the nom. of stems ending in stops, but rather is likely, given the paucity of words in this class, to be analogical. Such an analogical nom. in -s exists in sanguīs < *sanguin-s, backformed from the gen. sanguin-is, and replacing neut. sanguen. Cf. also hiems built to gen. hiemis. So too perhaps for these words, with a new nom. *vigil-s built to the gen. vigilis.[14] A similar process occurs in Oscan n-stems: fruktatiuf, úittiuf < *-ion-s, statíf < *-in-s[15] (see Buck 1904.130). This new analogical nom. in il-s would give -ilz > -ill (> -il) and so bleeds the velarization of the l. Thus vigil, pugil and mugil do not constitute counterexamples to the expanded version of **Front Vowel Backing**.

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[12] The gen. pl. VIGULUM (CIL 5. 2998, 14.3626) points not to a nom. **vigul, but is the phonologically regular outcome, without paradigmatic leveling.
[13] The form mugilis is late, first attested in Juv. 10.317.
[14] The odd shape, resembling syncopated nouns in Oscan, such as famel < *famel(o)s, makes me wonder if, in spite of the lack of attestation, these words for 'watchman', 'boxer, and a type of fish, are not borrowings.
[15] The short vowel -í-, beside the usual -ī- = /ī/ of the oblique, is probably due to a secondary Osthoff's shortening.

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9) An o, from PIE *o, *l > ol, and *e > o / _ ɫ, is subsequently raised to u before velar ɫ in a closed syllable, i.e. followed by a consonant other than l or a word boundary. Thus there are the contrasting pairs volō, *kᵘel-ō > colō (open syllable), but vult, *kᵘl-to-s > *koltos > cultus (closed syllable). Further standard examples for initial syllables are: *solkos (cf. ὁλκός) > sulcus, *pelpmentom (cf. U. pelmner, §5.19) > *polmentom[16] (with e > o / _ ɫ) > pulmentum.

10) There are several examples where the older form with o is attested:

colpa (Gram. II.27.12)

MOLTA (CIL 1².366, c. 241: Warmington 1940.154, #17)

MOLTATICO (CIL 1².2442)

MOLTATICOD (CIL 1².383; Ernout 1957.43: 3. cent.)

POLCIER] (CIL 1².640 = D 459, c. 131; polcher: Prisc.)

FOLVIUS (CIL 1².615, c. 189: M. F. Nobilior; also CIL 1².635, c. 135; CIL 1². 643-4, c. 123; CIL 1².825, c. 2. cent.).

The earliest example known to me of o > u / _ ɫ in initial syllables is MULTA (CIL 1².11, L. Scipio, c. 160)

11) This change also occurs medially, *insalsos > *inselsus (**A-Weak.**) > *insolsos (**FVB**) > insulsus (see §8.15 ff.), already CONSULTO (CIL 1².584.4, c. 117: Sent. Minuciorum), and word finally: CO(N)SOL, SEMOL (CIL 1².1531, c. 150: Vertuleii dedication), beside consul, simul, etc. This change seems well under way by 120. Prinz (1932.25-17) dates the start of the change to c. 150.

[16] The loss of the consonant, whether by **CsC-Loss** or as here by assimilation and **Geminate Simplification**, cannot be related chronologically to **Velar Backing**, since the l could have become velar at any point.

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12) It seems from solvō and volvō, with o before łC, that the change of o > u before ł{C,#} was over by the time that *sołuuō and *uołuuō syncopated to solvō and volvō (not **sulvō, **vulvō)[17]. This seems unlikely since V̆ > Ø / V̆R _ u is an early change, with only doubtful exceptions[18], while o > u / _ łC is comparatively late and well into the historical period. However, it cannot be claimed that -lv- does not meet the form of the rule for raising, since there is vulva beside earlier volva, fulvus and fulvius beside earlier Folvius. Rather, the o of solvō, volvō must be analogical, probably to the participles solūtus, volūtus where the o in an open syllable remains (whereas vulva, fulvus lack any analogical model).

Thus, we can set up the ordering:

1) l > ł

2) {i,e} > {u,o} / _ ł (**FVB**)

3) o > u / _ łC

13) It is clear that a preceding u in some way hindered the change of o to u, both before a velar ł and in the change of o to u final syllables (-om, -os, -ont). Thus regularly volt, volnus, volgus in Plautus (and servos, equos, vivos, mortuos, g.pl. divom, bovom; volvont, etc.). Since there is a strong possibility that spelling conventions were a factor (the desire to avoid VVLTIS and the like, so VOLTIS still in CIL 12.1423.9.11, 78 A.D.), whatever the original phonological basis may have been, this vexed problem will be passed over. Because of the uncertainty of the data, they cannot be used in establishing either an absolute or a relative chronology. See in chief, Prinz 1932.46 ff., Buck 1933.83, Leumann 1977.49, Sommer-Pfister 1977.60.

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[17] This seems to be the ordering implied by Leumann 1977.47 and 48.
[18] The scansions volŭō at Enn. Ann. 174 and solŭō at Lucil. 48 are poetic rather than archaic, as solŭō clearly is at Lucr. 1.559, Cat. 66.38, etc.

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14) There are several examples of e rounded to o after u̯. One class is cases of *su̯e- > *su̯o- > so-, that is the e is rounded to o / _ u̯, and the resulting o then causes the u̯ to disappear (§7.3-5, 19). Standard examples are: *su̯ekuros (Skt. śvaśura-, ἑκυρός, OHG swehur) > *su̯okeros > socer (§7.5); *su̯epnos[19] (Skt. svápna-, OE swefn < *su̯epnós) > *su̯omnos > somnus; *su̯esōr (Skt. svasā, Lith. sesuõ) > *su̯osor > soror. There appear to be no medial examples of -su̯e- in Latin[20], so we cannot say whether the rule applies only initially, though it is usually treated so.

15) A rounding of initial u̯e > u̯o is shown also in the single item *u̯emō > vomō. Leumann 1977.101 maintains that the o of vomō can be an inherited o-grade, and omits it from his list of examples of e > o (ibid. 47). However, in view of the unanimity of the cognates ἐμέω[21], Skt. vamiti, Lith. vemiù, an e-grade is correct. Leumann (loc. cit.) also invokes the type of o-Umlaut seen in *duenos > bonus, but *duenēd > bene, *duene-lo-s > bellus. However, this seems less likely in the face of the paradigm *u̯emis, *u̯emit, etc. A further case is perhaps: *kᵘekᵘō > *ku̯oku̯ō > coquō (with u̯ lost before o in the first syllable, but restored after coquit, etc. in the second). Whereas the rounding after su̯- occurred irrespective of the following consonant (before k, p and s), these two roundings seem to occur only in initial syllables before labials (and perhaps helped by the following o-vowel). We might then tentatively propose for the last two cases, despite the few examples, a broad rule that e > o / (C)u̯ _ (C)[+labial], that is e > o in a heavy labial environment. In *kᵘeɬō > colō, *u̯eɬō > volō, the rounding is due to the

[19] See Schindler Sprache 12 (1966) 72-75, Jamison KZ 96 (1982/3) 6-19.
[20] For -su̯-, cf. *prusu̯īnā (Skt. prusvā) > pruīna. The compounds of these words: insomnis, cōnsobrinus, are too easily analogical to be evidential.
[21] Though lack of an initial digamma is a well-known difficulty, see Chantraine 1968 s.v.

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following ł, and is not conditioned by a preceding u̯ alone[22]. For *eks-kuol-īnos, in-kuol-īnos (o-grade, cf. in-cola, agri-cola; Leumann 1977.280, 326) > Esquilinus, inquilīnus, with o weakening to e (ESQELINO, CIL 12.416) > i, see §8.21.

Rais-iNG

16) There is a clear rule, already mentioned above (§4.22), of the raising of e to i before the velar nasal ŋ. The allophone [ŋ] occurs only before velar consonants and is therefore in complementary distribution with[m] and [n][23]. Since ŋ necessarily occurs only in closed syllables, changes which it conditions cannot be related to vowel changes of open syllables.

17) That n before velars was velar is clear from Accius' proposal to follow Greek orthographic conventions and write aggulus for angulus, agceps for anceps, etc.[24] Cf. further the hybrid-Greek form CUNNULIGGETER[25] from Pompeii. However, since other than these notices the grammarians are quiet on this subject, it is clear that nasal before velar had by that time ceased to have any specifically velar allophone, or more precisely that this is a case of spelling pronunciation (Buck 1933.147).

18) It is also clear that g before n was assimilated to its homorganic nasal ŋ. Clear evidence, beside the raising of e to i itself, of the pronunication as [ŋ] are spellings of the velar nasal as NG, such as DINGNISSIME (CIL XIV.1386), SINGNIFER (CIL VI.3637), etc. Again, since the

[22] So no rounding in vetus, venia, etc. Other cases of vo- are from o-grades: vocāre, vorāre, and compounds (carni-)vorus (cf. βορός), with medial o retained by analogy.
[23] And so not a separate phoneme, with Safarewicz (Fest. Pisani 1969.871) who operates with an unusual phonemic system.
[24] Mar. Vict. 6.8.11K, 6.19.11; Prisc. 2.30.12; cf. Nigidius Figulus apud Gellius 19.14.7; collected in an appendix in Allen 1965.
[25] CIL IV.4699, where -eter = -ητήρ; see Leumann 1977.15.

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grammarians are silent, one sees the effects of spelling pronunciation (Buck ibid., Ward Lang. 20.73).

19) Standard examples for this raising are:

*dek-nos (cf. decet) > *degnos (assimilation in voice) > *deŋnos (assimilation to nasal) > dignus

*teŋgō (nasal infix, cf. τέγγω) > tingō

*ŋgᵘēn̥ (cf. ἄδην) > Plt. *eŋgᵘen > inguen

*kᵘeŋkᵘe > *quinque --> quīnque (by analogy to quīntus)

*kᵘenkᵘ-to-s > kᵘeŋxtos > kᵘiŋxtos > kᵘĩⁿxtos > quīntus (see §4.22, 24)

The exception iuvencus is probably analogical to iuventus, iuvenis (itself influenced by senex; §8.29), though there is also the possibility of dialectal influence (cf. O iuengar, see Sommer-Pfister 1977.54), since this raising does not occur in O.-U. The title of a priest of Hercules cupencus is attested only in Verg. Aen. 12.539 and (in imitation) Silius 4.537. Servius (ad loc.) calls it Sabine. The earliest datable example known to me is DIGNOS in CIL 1².586.7 (Cornelius ad Tiburtes)[26], c. 156.

20) Thus, we can say that 1) the assimilation of k to g before voiced consonant feeds 2) the assimilation of g to ŋ before n, which feeds 3) the raising of e to i. For assimilation in voice cf. *sek-ment-om (cf. sec-tō) > segmentum; for assimilation of original g, cf. *teg-no-m (cf. tegō) > *teŋnom > tignum. Contrast the development of the labials which share the assimilation in voice, but assimilate to a nasal before both m and n; so:

[26] The INSIGNE of CIL 1².10 is in the first line, which was added later. The Dedication of Mummius (con. 145; CIL 1².626), which shows SIGNU, may be a later copy of an archaic inscription, but in general its language seems to reflect the old date; see Ernout 1957.73

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*supmos > *submos > summus (vs. retained gm) and *sυepnos > *sobnos > somnus.[27]

21) It is also clear that o was raised to u before ŋ. Standard examples are:

*oŋkos (cf. ὄγκος) > uncus

*hom-ke > hoŋke (HONCE[28]) > hoŋk (HONC[29]) > hunc[30].

There are two exceptions to this rule: longus (see §4.22 n. 29) and a tongeō 'know', quoted from Ennius by Paul. Fest.[31] (presumably o-grade, cf. Goth. þagkjan, ModGer. denken). Sommer-Pfister (1977.59) suggest "scheint der Labialisierungsprozeß durch den vorhergehenden Zahnlaut verhindert worden zu sein." However, the process was not one of labialization but raising, and a preceding dental did not hinder it, as tingō makes clear[32]. The easiest explanation for tongeō is that Ennius simply preserves the older form and that tongeō had dropped from the active vocabulary before raising applied.

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[27] The evidence on the outcome of tn and dn is conflicting. Goth. aþna points to a *atnos > *adnos > annus, vs. *pat-no- (cf. pat-ere) > pandō; Skt. budhnas, OE boden point to *bhudhnos > Plt. *fuθnos > *funδos > fundus; *udn- (r/n stem) > unda (apparently with metathesis); also cf. πίτνημι : pandō. For this long standing difficulty, see Thurneysen 1883.301-304 for first statement of the problem; Leumann 1977.201 and litt.

[28] CIL 1².366, c. 241

[29] CIL 1².9 (L. Scipio), c. (240-) 200; for dating see most recently Gordon 1983.81

[30] The difference may not be strictly chronological but dialectal since HONCE comes from Oscan Spoletium and HONC is Roman. However, these inscriptions do show that apocope preceded the raising of o > u.

[31] Fest. "Aelius Stilo ait noscere esse, quod Praenestini tongitionem dicunt pro notione," so a dialectal explanation is not inconceivable. See W.-H. s.v. I have been able to find no examples of -ng- in the Praenestine inscriptions, except for CIL 1².1471: LONGU.

[32] Anticipating the assumption that e > i and o > u / _ ŋ are a single rule; see below.

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22) It is clear that these two rules of raising of midvowels before ŋ should be combined as a single rule {e, o} > {i, u} / _ ŋ, which we may name, by way of a mnemonic, **Rais-iNG**. In generative terms:

$$\begin{bmatrix}+\text{vocalic}\\-\text{low}\end{bmatrix} > [+\text{high}] \;/\; _ \begin{bmatrix}+\text{nasal}\\+\text{back}\\+\text{high}\end{bmatrix}$$

noting that the rule is well motivated since the vowel becomes [+high] before ŋ which is [+high][33]. Buck (1933.81 and 82) treats the two rules separately, but handles together the raisings of o to u before "nc, ngu, mb, and l + cons."; so too Leumann 1977.45 and 48-9; Kent treats together the raisings of o > u "before ŋ and m"; Brugmann 1897.122 and 142 "unc ungu umb"; Meillet-Vendryes 1968.111 and 112 (o > u "devant nasale, sans règle bien claire"); Niedermann 1953.62 (no mention of o > u); Monteil (1973.76 and 95) has no mention of o > u, and states e > i only before <gn>, which he claims was pronounced /ŋŋ/, though no evidence is adduced to support this. The generalization that the raisings before ŋ are a single rule is drawn by Sommer-Pfister 1977.59, Maniet 1975.65, Pisani 1974.14 and 15, Juret 1938.69 ("ṅ = nasal palatale d'avant ou d'arrière"; he does not specify what the difference is, what is supposed to condition it, or which of his examples show which allophone).

23) The raisings of o to u before m and mb are sporadic and not paralleled by similar changes of e. Thus while o usually remains before mV in initial syllables (domus), there is a raising to u in humus (cf. χθον-, U. hondra 'below') but homō 'Erdling', umerus < *omesos (U. onse 'in umero'),

[33] See Sommerstein 1977.102, 112.

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Numidiae <-- Νομάδες.[34] Sommer-Pfister 1977.59 (following Sommer 1914. n. 14) claim perhaps the initial dental hindered a regular process of raising. However, there is equally no raising in homō, compounds of com-V (where the prep. cum might be thought to aid such a raising if it were regular), coma, glomus, vomō (though here the preceding v- could be thought to have impeded a regular raising, however, there is never **vumō)[35]. Also to be considered a separate phenomenon from the rule of **Rais-iNG** is the change of o to u before mb in umbilīcus (cf. ὀμφαλός), lumbus < *londhuos (see Exc. II §22), and perhaps before mp in rumpia (Ennius, quoted by Gell. 10.25.2), beside rhompheia from ῥομφαία ' a type of spear'. However, there is no raising in bombus (onomatopoetic), compounds of com-b, rhombus. These are not strong counterexamples. However, due to the infrequency of -mb- in Latin, neither are there any other examples of this raising. Whatever the nature of these changes they cannot be fitted into a relative chronology, and are clearly separate from the regular raising before ŋ. Likewise the sporadic raising of o to u before rC is dialectal: furnus beside fornax, curvus but corvus, etc.; see Leumann 1977.48.

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[34] Sommer-Pfister 1977.59 claim numerus as a borrowing of νόμος; I am very doubtful. See W.-H. for the range of suggestions. The lack of syncopation shows the suffix must have been -V̆s-o-s, but what this represents derivationally is uncertain.

[35] The strongest argument for the raising before m as lautgesetzlich is put forward by Maniet 1975.68, who holds that o is raised before m plus vowel or any consonant, but is blocked by a preceding dental. The rule would have to specify that the preceding consonant was a dental stop, though none of the authors who propose this rule note this fact, since the raising is not blocked by the preceding n of Numidiae. It is hard, however, to see how a dental stop specifically would have hindered such a raising. Maniet argues that the o in several of the apparent counterexamples (homō, vomō) is not original, but if the rule is supposed to apply to the borrowing Numidiae, why not to coma < κόμη (since Ennius)?

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24) There are two examples which might seem to indicate that this raising before ŋ includes also the long vowels ē and ō, thus: *nōmi-capāre (denominative from *nomi-cap-s) > *nōŋcapāre (by sync. 2/4) > nuncupāre; *sēmi-caputs[36] > sēŋcaputs > sinciput. However, these certainly reflect a re-application of Osthoff's Law[37] with a shortening of V̄/ _ RC after sync. 2/4. If the preform of ulna was *ōlena, there was a reapplication of Osthoff's Law after sync. 3 as well (§3.2-3). Besides the general stability of long vowels in all positions in Latin, that **Rais-iNG** applies only to short ĕ and ŏ is shown by the fact that ē does not change /_ ŋ in sēgnis /sēŋnis/[38]. The form cūnctus < *con-citos, p.p.p to concieō, therefore must reflect an interesting series of changes: *kom-kitos > *koŋktos (sync. 2/4) > *kuŋxtos (**Rais-iNG**) > cūnctos (**NS**, with the c restored see §4.20, 22), which confirms the ordering of **NS** after **Rais-iNG** which includes both e to i (*kʷenxtos > *kʷiŋxtos > quīntus) and o to u (see §4.24-5). Thus we have:

1) sync. 2/4

2) Osthoff-II

3) **Rais-iNG**

4) **NS**

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[36] Leumann's suggestion (1977:97) of sʷīno-caputs accords well with the fact that sinciput, where it can be checked, does refer to hog jowls, but the *sʷīno- of the pre-form is hard to justify in view of the simplex su(ʷ)īno- itself.

[37] Thus Brugmann 1897.149; Pisani 1974.21; Meillet-Vendryes 1968.110; Leumann 1977.49; Sommer-Pfister 1977.51.

[38] The traditional etymology (W.-H.), first proposed by Froede BB 16 (1892) 192 and Osthoff IF 5 (1896) 297, derives sēgnis from < *sēk-ni-s, comparing ἧκα, ἧσσων < *sēk-iōn. However, since the Greek means 'low, gently' and 'less, weaker', these forms are not close semantic matches for sēgnis 'inactive, lazy'. Alan Nussbaum has suggested to me the possibility that sēgnis represents the privative sē- (sē-cūrus, etc.) plus *gnH-i- to be seen in beni-gn-o-, mali-gn-o-; cf. OIr. do·gní 'does'.

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25) Further on Osthoff-II

The reapplication may not apply to r + C (g in any case). So parallel to *nōmi-capāre > nuncupāre are:

*ious-agāre > iūrigāre (cf. iūrigum; iúrgia: CIL IV.7698c; iurigandum: Plaut. Merc. 119) > iūrgāre

*pūr-agō > pūrigāre (cf. purigant Plaut. Aul. 753, purigans: Cist. 384)[39] > pūrgō[40] (ū shown by Italian purgatorio etc.)

Leumann claims ărdeō beside āridus (1977.107), but it is not clear to me how the quality of the a in ardeō can be known. Equally uncertain is why Ernout-Meillet (1959:45 s.v. āreō) confidently write ārdeō, ārsum[41]. We have direct evidence from the grammarians that the i of princeps < *prīmo-caps < *prīs(a)mo-caps was short, at least by the time of Servius (Gram. V 426.34). There is the possibility that Osthoff-II applied to the outcome of syncope between nasal and k, but was over by the time of the syncope between r and g, which appears to be later, both from the attestations and since it is after rhotacism.

26) There was yet another application of Osthoff's Law in Vulgar Latin as shown by the differing outcomes in various branches of Romance. Thus while we have *oino-decem[42] > ūndecim for Latin and the long vowel is reflected in Italian undici, while French onze < ŭndecim shows the Vulgar

[39] OCT reading. Both restored for metrically faulty MSS. purgita-

[40] Though here restoration by analogy to the simplex pūrus is possible.

[41] There was no qualitative difference between ă and ā, as shown by the undifferentiated outcomes in Romance.

[42] Leumann gives *oinoz-decem (1977.107) as the preform. Not only is a nominative as first member unnecessary, but phonologically the expected outcome would be **ūnōdecim.

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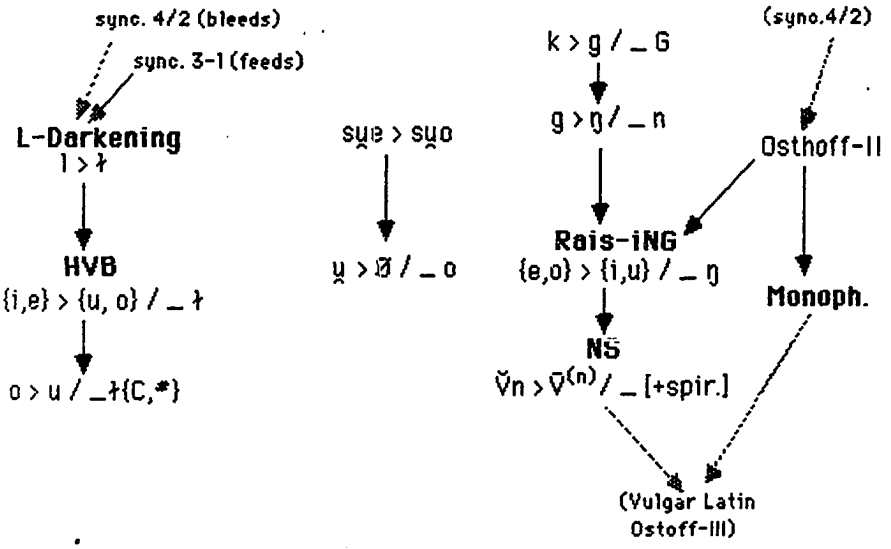
Latin change[43]. However, after Osthoff-II we can place the origins of those long vowels which are preserved Latin before RC[44]. Thus the monophthongization of oi > ū (see §7.14 f.) and of (*oue >) *ou > $\bar{o}_2$ > ū (see §7.42 ff.) in *nouentios > nountios[45] > nōntius[46] > nūntius, must be after Osthoff-II, but before this late third application of Osthoff in Romance, since Fr. annoncer[47] < *annūntiāre, but Italian annuncio < annūntiō. Equally after Osthoff-II then must be the new long vowels of **N + Spirant Lengthening**. This confirms the ordering of **NS** after **Rais-iNG** which is after Osthoff-II.

We can sum up the finding of the changes in initial syllables in a chart. The series of changes will be related to each other later. Thus, we have:

CHART VI

[43] Leumann (loc. cit.) ignores the Italian evidence, as does Sommer-Pfister 1977.102 (citing Sommer 1914 §84, who claimed this change as Latin), though the Ital. may reflect an analogical restoration from ūnus; see n. 47.
[44] The evidence for the length of the vowel, apart from apex or I longa in inscriptions, direct statements of grammarians, etc., will be precisely this kind of differing outcome in various branchs of Romance.
[45] Attested by Mar. Victor. GL VI 12.18; cf. NOUNDINOM (SCB).
[46] NONTIATA in CIL 1^2.586.5 (Cornelius ad Tiburtes; c. 156)
[47] As Sommer-Pfister (loc.cit.) rightly note "doch frz. annoncer beweist als Lehnwort wenig."

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27) Discussion of the lowering of i to e and u to o before r (*sisō > serō; *fu-se > fore) is postponed to Chapter 9. For **N + Spirant Lengthening**, see above §§4.17 ff. For eu > ou before consonants or vowels, see §7.31 f. (under diphthongs).

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Chapter 7: The Diphthongs in Initial Syllables

1) We can begin with the i-diphthongs, arranging them in the usual vowel triangle:

ei oi

ai

Let us begin with ei. There is clear evidence for a midstage of a long high monophthong ę̄ which then was raised to ī. The inscriptions up to c. 150 spell this sound <EI> or occasionally <E>, but it is still distinct from original ī, spelled consistently <I>. After 150, this new monophthong ę̄ was raised to ī and became identical in sound with original ī. The archaizing spelling <EI> still persists in many cases and indeed there are examples from the Augustan period, but reverse spellings using <EI> for original ī show that the sounds had merged. Thus from the inscriptional record it is not possible to give an absolute date to the beginning of the monophthongization of ei > ę̄, but the raising of ę̄ to ī can be dated to c. 150. Examples are numerous, but the Senatus Consultum de Bacchanalibus (abbr. SCB; CIL 1^2.581 = Ernout 1957.60, *126) of 186 provides an excellent terminus post quem, with etymological <EI> preserved:

DEICERENT, SEI, SIBEI

but original ī in:

LATINI and other gen.sg., SCRIPTUM, AUDITA, etc.

However, COMPROMESISE with E for classical ī shows that ei had in fact already started to monophthongize but had not yet been raised. Examples of unetymological <EI> for original ī, showing the raising of ei to ī, are:

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COGENDEI, DISSOLVENDEI, gen.sg. but PROMISERAT (CIL 1².632, Dedication of Mummius, c. 145)

AFLEICTA[1] (CIL 1².1531, Dedication of the Vertuleii, c. 150)

2) Besides the inscriptional evidence, the midstage of ę̄ is shown in the paradigm of the thematic *deiu̯-o-s:

n. sg. *deiu̯os > *dę̄u̯os > *dę̄os > dĕus.

That is the diphthong first monophthongizes, then the non-inital u̯ disappears before the o and the new long monophthong ę̄ is then shortened in the resulting hiatus to ĕ before it can be raised to ī, so that we do not have *deiu̯os > *dę̄u̯os > dīu̯os (raising) > **dius.[2] Cf. the gen.

*deiu̯ī > *dę̄u̯ī > dīvī

where the ę̄ is raised to ī, since there is no loss of u̯ and so no shortening. So we have the sequence of changes (all of which will be further specified in the following paragraphs):

1) ei > ę̄	**(Monophthongization)**
2) u̯ > Ø / {C, V} _ o	**(WO-Loss)**
3) V̄ > V̆ / _ V	**(Shortening in Hiatus)**
4) ę̄ > ī	**(Long High Raising)**

There is attestation of the early stage in gen. pl. DEIVO (CIL 1².455, c. 300)

[1] Classical flīgō, cf. Aeolic φλίβω; see W.-H., E.-M. s.v.

[2] So Buck 1933.87, 134; Leumann 1977.64-5 (but see below). Not with Lindsay 1894. 244, Meillet-Vendryes 1968.79: *deivos > *deivus > *deyus (loss of v before u) > *deus (the loss of y that Lindsay compares in eum, eō is prehistoric).

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and of the monophthongized stage in fem. dat. pl.[3] DEVAS (CIL 1^2.975, date uncert.).

3) The u̯ disappears either before o̲ (**WO-Loss**) or between like vowels (**VWV-Loss**; examples below). Therefore, the g. sg. dīvī shows that there was a midstage ę̄ distinct from ī, and that the rule that u̯ > Ø / V̲₁ _ V̲₁ was also over before the raising of ę̄ to ī. Thus we have *deiu̯ī > *dę̄u̯ī > dīvī and not *deiu̯ī > *dīu̯ī > *dīī > **dī for the genitive. However for the plural we do see the disappearence of u̯ beween like vowels: *deiu̯oi > *dę̄u̯ę̄ (-oi > -ei > -ē > ī, see §8.23) > *dę̄ę̄ (u̯ disappears while still at the ę̄ stage, before the raising) > dī [4]. So too for the d. pl.: *deiu̯ois > *dę̄u̯ę̄s > *dę̄ę̄s > dīs. The later nom. pl. deī and d. pl. deīs are analogical to the nom. deus (Buck 1933.182).

4) For u̯ > Ø / _ o̲ (**W-O Loss**), cf.:

*elaiu̯om (ἔλαιϝον) > *oɬeiu̯om > *oɬēu̯om (with the long ē stage attested by Goth. alēw) > oleum, but

*elaiu̯a (ἐλαίϝα) > *oɬēu̯a > olīva.

This loss does not occur, of course, in absolute Anlaut (though it does /*C_). So: voveō, volō, etc. As we have noted above, the raising of ę̄ > ī must follow the darkening of l > ɬ. We can have no evidence on when the backing

[3] So Ernout 1957.56. Taken as gen. sg. by Leumann (1977.63; but argues for dat. pl. at 421), Blümel (1972.39), Warmington 1953.78. However, the inscription: DEVAS CORNISCAS SACRUM argues rather for a dat. pl. The usual case with sacrum is dat. (e.g. CIL 1^2. 47, 365) and the Corniscae were a collective of goddesses (Paul. Fest. 56L. cited by Ernout; the dedication to a masc. singular CORONICEI in CIL 1^2. 976 does not affect the case).

[4] Note, however, that we cannot tell if *dę̄ę̄ first became *dę̆ę̄ (with shortening in hiatus) or *dę̄, with the like vowels contracting and then was raised to dī, or if the raising occurred first and then *dīī contracted. One's natural supposition is that the contraction occurred immediately on the disappearance of the u̯. That is **VWV-Loss** cannot be related directly to **Shortening in Hiatus**.

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took place relative to the monophthongization of ei > ē, as l becomes ɫ before both ĕ and $ē_1$ (see §1.1-3) and there is no reason to suppose it could not have darkened before ei and $ē_2$ as well. From these two divergent forms deus and dīvī two paradigms are created. To the nom. deus, plus deum, deō, deō(d), an analogical gen. deī is created, semantically nominal, while to the inherited gen. dīvī a new nom. dīvus is formed for the adjective (Buck 1933.87, 135). Parallel to the creation of these new paradigms we may compare acc. *paruom > parom > parum, but gen. *paruī > parvī to which a new nom. parvus is created, leaving parum (beside parvum) as a relic form. The effects of u̯ > Ø / _ o were largely eliminated by various sorts of similar paradigmatic leveling. Thus *ekû̯os > ecus (frequent) --> equus back formed form the gen. *eku̯ī > equī, etc. Can this loss be dated absolutely? It is later than the weakening of ai > ei (**A-Weakening**, §6.4 ff.) and **Monophthongization** (deus, oleum) and the syncope of ou̯ŭ > ou (see below), which we can date after the Forum Inscription (c. 500) but before the SCB of 186, and before the raising of ē > ī (c. 150; not *olīuom > **olium). Further, we have attestation of the sequence -u̯o- in GNAIVOD, QUOIUS (CIL 12.7, c. 200), and QUOM (SCB.18). Despite the possibility of paradigmatic leveling in quoiius (influence from quoi > quei > quī, though not in the later cuiius), GNAIVOD but later Gnaeus seems good evidence for the u̯ still existing c. 200. Sommer-Pfister (1977.125 and litt.) give a rough dating as "vor Beginn des 2. Jh. v. Chr."

5) Further examples of u̯ > Ø / _ o, cf. *suepnos[5] (Skt. svapna-) > *suomnos > somnus. That is, the rule rounds e to o after u̯ (§6.14), and then that o causes the u̯ to disappear. So too in *sueĥuros (Skt. śvaśura-, OHG swehur) > *suokeros > socer (§6.14), *suesōr (Skt. svasā, Lith. sesuõ) >

[5] See Schindler Sprache 12.72-5.

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*s̥u̯osor > soror. A loss of *u̯- in absolute Anlaut before -u̯- has been claimed for Latin (Buck 1933.135; Sommer-Pfister 1977.125; Leumann 1977.138 "unsicher") based on the single example urgeō. Here the preform is taken as *u̯ŭr̥ĝ-, showing a different outcome of the zero-grade *u̯r̥ĝ-, or a special reduction vowel *u̯ᵘr̥ĝ-[6] (as opposed to the usual development in *u̯r̥t-to-s > vors(s)us, versus). However, here rather than invoking a-secundum or the like, one might assume that Latin has undergone a resyllabification, generalizing a *u̯rĝ-, rather than the *u̯r̥ĝ- seen in OHG wurgen, Skt. vr̥jánam.[7] In addition, the preform *u̯r̥ĝ-eio- is itself odd, with a zero-grade where one would expect o-grade in a causative, cf. varjayati. Perhaps then if we assumed the expected o-grade *u̯orĝ-eio-, we might invoke the dialectal raising of o > u of the fornus/furnus, corvus/curvus type.

6) Further examples of the loss of u̯ / V_1 _ V_1: cf. *sei-u̯eis (for *u̯eis from *u̯el-s see above) > *sēu̯ēs > *sēēs > sīs; *oblīvitus > oblītus. Again, we cannot be sure whether *sēēs became (*sĕēs) > **sēs > sīs , or **sēēs > *sīīs > sīs.

Thus we have:

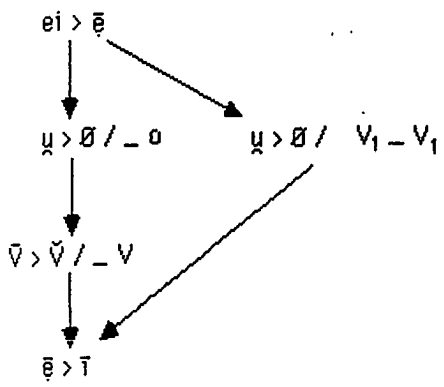
CHART VII

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[6] So Hirt 1921.98; Brugmann 1897.453 "u̯-Färbung"; see bibliography in Schwyzer 1959.352. Szemerényi's idea (KZ 71 (1953) 200-1; repeating ideas of his 1941 Budapest diss.) that the regular outcome of r̥ and l̥ in Latin are ur, ul, with or, ol due to analogy to o-grade forms, is difficult to maintain in the face of po(r)scō, cor, etc., with no obvious source for analogy. See Leumann 1977.57-8.

[7] Admittedly this is ad hoc, when Latin seemed to have no trouble over a root shape *u̯r̥t- (versus) or *u̯r̥s- (verrēs). However, the sequence *u̯r̥K- does seem subject to a certain amount of reshuffling in IE. Compare the difference in treatment of PIE *u̯l̥ku̯os (Skt. vr̥ka-, OCS vlŭkŭ), but with a metathesis of sonority *lúku̯os > Lat. lupus, Grk. λύπος.

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(Monoph.)

(WO-Loss) (VWV-Loss)

(Shortening in Hiatus)

(Long High Raising)

7) Parallel to *deiu̯ī > *dēu̯ī > dīvī, with u̯ between different vowels, we have *sei-ue > *sēue > sīve. Likewise we should expect *kei-ue (cf. ἐ-κεῖ) > **cīve; however, we find only the apocopated form *keiue (> *kēue) > ceu; that is: the u̯ now tautosyllabic forms a diphthong with the e. This seems to indicate that sīve has been remade to sī, while ceu remains alone since *kei > **cī has dropped from the language. We might suspect that *seiue had monophthongized before the apocope, but since *seiue > *seiu̯ > *sēu̯ > seu is not impossible, we cannot state this as certain. However we can say that the apocope, when it occurred, was before and prevented the raising of ē to ī, and that the resulting eu diphthong arose after the change of *eu > ou (§7.31 f.).

8) An exception to the raising of ē to ī is seen in lēvis. From the preform *leiu̯is (cf. λεῖος, see Exc. II §3) we might expect *lēu̯is > **līvis, parallel to dīvī. Sommer-Pfister 1977:65, following Hermann (NGG 1919.240), propose a rule that -ēiu̯- > -ēu̯- after an initial l-, that is, though not so formulated, the medial stage of ē is retained, prevented from raising by the environment *l _ u̯. Phonologically, we might expect that the initial l was

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already velar. Perhaps then the ē̯ between two back resonants was prevented from moving forwards (and higher). However, since the other labial consonants do not block the raising (līmus 'mud' < *lēmos < *leimos < *loimos, līber < *lēberos < *leiberos < *loiberos; see §20), it seems that the environment is limited to before u̯. The specification of initial l is necessary, though Sommer-Pfister do not cite the reason for it, as shown by (*elaiu̯a > *oɬeiu̯a >) olēu̯a > olīva, while in (*elaiu̯om > *oɬeiu̯om >) *olēu̯om > oleum, the change of ē̯ > ī is prevented by the loss of u̯)[8].

9) Leumann 1977.64 maintains that ē̯ was preserved before u̯ regardless of the preceding consonants and cites the perfects crēvī and lēvī as examples. He offers as further evidence for the retention of ē̯ before u̯: seu, ceu, deus, oleum. However the reason that these forms show evidence of a midstage ē̯ is not because ē̯ was retained before u̯ longer than before any other consonant, or did not rise before u̯, but rather because u̯ is the only consonant that, by disappearing or forming a new diphthong, can show such evidence. Leumann acknowledges the many counter-examples, some already mentioned (see §20). He explains sīvī/posīvī as perhaps by analogy to its present sinō[9] and to distinguish it from sēvī < *seh$_1$-u̯ai, pf. of serō, but offers no explanation of dīvus, olīva, etc., or Ἀχαιϝοί--> *Ac(h)aiu̯oi > *Achaiu̯ei > *Acheiu̯ē̯ > Achīvī, which seems immune to analogy.

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[8] A further example is oblīvīscor, to be connected with lēvis. The original suggestion by Pott (1833-6; cited in W.-H.) to connect it with linō, i.e. 'smear over', makes excellent semantic sense; however, it leaves the morphology obscure, in particular the -v- in the present (certainly not built to a perfect in -v- which the deponent lacks). Rather with lēvis (so first Sommer 1914.602) so 'smooth over, forget'. Thus *ob-leiu̯-itus > *oblēu̯itus > *oblīvitus > oblītus, to which the present is built (type nānctus : nancīscor; profectus : proficīscor; see Leumann 1977.536).

[9] Rather the phonologically regular sinō/sīvī provides the model for linō/lēvī --> līvī.

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10) If then the rule only preserves ē̯ / *l _ u̯, two questions still remain, both concerning u̯-perfects. Does linō, pf. lēvī represent another example of the rule, and what then is the explanation of the long ē of cernō, pf. crēvī?

11) The situation for linō, lēvī is more complicated than it might first appear in the older handbooks, since for linō, we have an original seṭ root *leih$_2$-, cf. Skt. *li-ne-h$_2$- > linā́ti, O.Ir. as-lenaimn 'soil'[10]. Thus the original paradigm of

*li-né-h$_2$-ti/*li-n-h$_2$-énti

has been remodeled to a thematic *lină- --> linō[11], cf. *tl̥-n(e)h$_2$-ti --> *tolnă- --> tollō. The p.p.p. *lih$_2$-to-s > **lītus, would then be remodeled to lĭtus parallel to lĭnō (on the analogical model of sĭnō : sĭtus). The pf.[12] could be equally analogical: *lei-u̯ai > lē̯u̯ī (as *sei-u̯ai > *sē̯u̯ai > sīvī), with the ē̯ retained. However, this may not be the case, and Warren Cowgill (1982) has suggested the following analysis. In most cases it is impossible to say any definite about the ablaut of the -vī perfect. However, the few more easily analyzable cases seem to indicate a full-grade, which points to a basis in the root aorists, spreading from those ending in a long vowel in Latin to derived verbs in long vowels, rather than the PIE perfect. So seh$_1$-u̯ai > sēvī, beside the participle *sh$_1$-tos > sātus. For *leih$_2$- we might

[10] Further, if the meaning was extended in PIE to 'stick to, adhere': OIr lenaim 'follow'. Grk. λίναμαι· τρέπομαι (Hsch.) and λιάζομαι 'turn aside' are quoted as cognate (W.-H., s.v.) but I do not understand the semantics.
[11] With the generalization of the zero-grade -nh$_2$- (> -nă-) of the infix throughout the paradigm; see §2.5, 8.7 n. 3; Watkins 1965.184, 188.
[12] The history of the -vī pf. in Latin is complex and varied, especially for seṭ roots and it is difficult to say what formation one might expect. Thus l̥h$_2$- shows a reduplicated o- or zero-grade, but *str̥h$_3$- shows the common pattern with strāvī built to the p.p.p., see below. Not with Godel Gl. 57 (1979) 233 who maintains a regular development of -eu̯z- to -āv-.

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therefore reconstruct a *leih$_2$-u̯ai > *leja-u̯ai > *lĕă̯ai (early loss of i̯) > *lēu̯ai (contraction) > lēvī.

12) For cernō, crēvī, crētum, we begin with a root *k̂rei 'separate' (and hence 'distinguish', 'judge'). The original nasal infix present

*k̂r̥-né-i̯-ti / *k̂r̥-n-i̯-énti

has been extensively modified in all the daughter languages (e.g. κρίνω < *κριν̯ω). In Latin a pres. *kri-nō > *kr̥$_2$nō > cernō (with rV̆-sync.) has been built (like sinō : situs) to the original past part. *kri-tos (> *kr̥$_2$tos > certus, preserved only as an adjective). The pf. then might be considered to rest on a *krei-u̯ai (parallel to *seiu̯ai > sīvī), but (as sīvī shows) ē is not the regular outcome. Rather, a new stem *krē- has become mixed into the paradigm. Thus, beside the lautgesetzlich crī-brum 'sieve' < *k̂rei-dhlom, i.e. 'separator' and dis-crī-men, we find dis-crē-tus, dis-crē-tio, sē-crē-tus, etc. These forms with crē- all occur before -t- suffixes, i.e. they are all built to the new perfect passive participle crē-tus. The one exception, not before a t, crē-vī, therefore, is more likely to be built to the perfect passive participle, a freqent model, than the perfect passive participle is to be built to an e-grade perfect active indicative (as claimed by W.-H.). Semantic confusion with the root crē- of crē-scō 'grow, become firm' (also pf. crēvī, crētum), creō, crēber 'thick' < *k̂reh$_1$- seems unlikely. However, if we look at the five nasal presents which have -vī perfects:

sinō	sīvī	situm
linō	lēvī	litum
sternō	strāvī	strātum
spernō	sprēvī	sprētum

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cernō crēvī crētum

it is clear that crētum (beside inherited certus), the source of the analogical perfect crēvī, is itself analogical and that perhaps sprēvī has provided the model, while in turn it seems that cernō (where the -er- is phonologically regular, by rV̆-sync.) has influenced the vocalism of spernō.[13]

13) An additional exception to ei > ē > ī is that ei was retained before i, that is the cluster -eii- remained. Thus:

*e-sio-s (Skt. asya) > *ezios > eiios > eius

This rule is also fed by assimilations of di, gi and ix to ii. Thus we have:

*ped-iōs (cf. *ped-isamos > pessimus) > peiior

*diou-es > Iovis

*mag-iōs (cf. mag-nus) > maiior.

*ag-iō (cf. ad-ag-ium) > aiiō

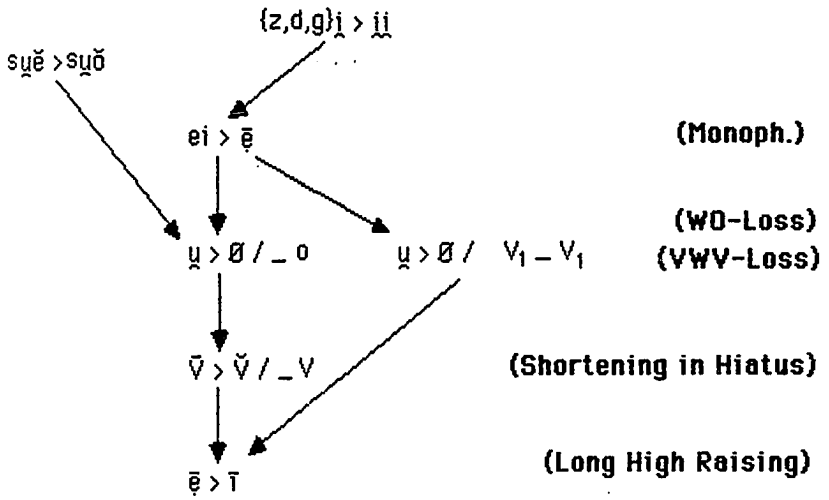
*meiĝhō (cf. Skt. mehati, Grk ὀμείχω) > meixō > meiiō > meiō

Note that the ei in ei-ius is no less tautosyllabic than that in *dei-kō > *dēkō > dīcō or *deiuos > *dēuos (> dēos > deus). Therefore the changes of zi, di and gi to ii precede the passage of ei to ē, as does the change of ix to ii. For oii > uii in huius, etc. see below §28. For -eii- retained in medial syllables, see §8.13. We may sum up our finding thus far in:

CHART VIII

[13] *spr-né-h_1-ti in Skt. spṛṇáti, and thematic sphuráti. Note the accent on Lith. spìrti. Here therefore we expect perf. (in somewhat anachronistic terms) *spreh$_1$-uai > sprēvī and the p.p.p *spr̥h$_1$-tó-s > **sprātus --> sprētus is formed after it. Spurcus and spurius are of doubtful connection here and appear to be Etruscan loan-words in the religious sphere (cf. the Etr. proper names Spurinna, Spurinus, Spurianus, etc.); see W.-H., s.vv.

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14) Moving across the vowel triangle, for oi, the picture is more complicated, but can be straightened out. The usual, that is unconditioned, outcome of oi in initial position is ū. Here the midstage was almost certainly a monophthong ō̯, which becomes ū, like the ō̯ that comes from ou (see below), and parallel to the midstage of ei > ē̯ (> ī). We have evidence of this ō in CORAVE[RU]NT in CIL 1^2.2661 (c. 171) where we have the stem *koi̯s-ā- seen in Paelignian coisatens = cūrāvērunt. The U. form kuraia, etc. is said by Ernout-Meillet (s.v. cūra) to be borrowed from Latin. But as U. (in contrast to O.) also shows monophthongization this is not necessary. So, beside O. múíní-kú (= com-mūnis), we have U. muneklu 'sportulam' < *moine-klom, (Buck 1904.45). A form CORAVERON is attested at Praeneste (CIL 1^2.59), but as always with Praenestine it is difficult to distinguish the archaic from the dialectical. Cf. further OINO (CIL 1^2.9, c. 200), OINVORSEI (CIL 1^2.581, SCB, c. 186) > *ō̯nos > ūnus,

15) However, in words of a legal or religious nature, the oi was retained as an archaism and this becomes oe, parallel to the change of ai to ae (see

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below, §29; Buck 1933.87, Leumann 1977.65-6, Cowgill 1982). Thus, we have the well-known contrasts of Pūnicus, the everyday adjective, vs. Poenus the official state term < *poin-o-s, -ikos (Φοῖνιξ); pūniō, vs. poena < *poin- (ποινή), moenia 'shared things, walls' but mūnia 'shared things, duties'. The oe is also retained in foedus 'bond, treaty (*bhoidh-os, cf. e-grade in πείθω, *feidō > *fēdō > fīdō), foedus 'foul, hateful' (o-grade *bhoiH-ido-, cf. zero-grade *bhiH- > Ved. bhī- 'fear, dread'), and in amoenus (medial syllable; doubtful etymology). All these examples are after a labial (other than u; see §18). However, as the pairs Pūnicus/Poenus, poena/pūniō show this is not a regular sound change, but rather a learned archaism. Sommer-Pfister (1977.67) and Maniet (1975.73-4) explain these pairs phonologically with a stage oe retained after p or f when an i follows in the next syllable. However, this seems special pleading for a poorly motivated rule. It appears ad hoc that p and f activate this rule, but m[14] be excluded, as it must to avoid the pair mūnia, but moenia, which cannot be explained on any purely phonological basis and which Maniet is forced to acknowledge as an archaism. Cf. further Greek loans Phoebus (Φοῖβος), as well as Croesus (Κροῖσος), etc., showing the treatment oi > oe, usual in Latin at a later date; proelia (etymology uncertain); while with ū even after a labial m: mūnia, mūnicipium, mūria.

16) The change of (oi >) ō > ū can be dated to c. 170, comparable to ei > ē > ī. Thus OI attested in OINO CIL 1^2.9 (L. Scipio, c. 200); OINVORSEI, COMOINE, OITILE (CIL 1^2.581.11, 9; SCB, c. 186), but UTIER (CIL 1^2.10; c. 170), USURA (CIL 1^2.632; c. 145). After this are the new cases of

[14] Maniet calls m "plus faible que les autres [p, f], parce que sonore et nasale". No examples for b-.

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secondarily arisen oe from contraction of bisyllabic o + i or e: *co-itus > coetus, and *co-aptos > coeptus (see Leumann 1977.67).

17) Various midstages between oi and ū have been proposed. Thus with different assimilations of the second vowel to the first: oi > oe > ȯ > ü > ū (Sommer-Pfister 1977.66; exact phonological meanings of the symbols not given); oi > oe > oö > oo > ou > ū (Kretschmer KZ 31 [1892] 352 f.); oi > oe > oo > o̅ (contraction[15]) > ū (Niedermann 1953.69); oi > oü > ou > ū (Leumann 1977.66)[16]. However, these seem unnecessarily complicated[17], and the most natural assumption is that oi monophthongizes to an o̅, of the same height as (or higher than) ŏ, which then is raised to ū, parallel to the development of ei > e̅ > ī its companion directly across the vowel triangle.[18] Thus:

1) ei > e̅, oi > o̅ **(Monophthongization)**

2) e̅ > ī, o̅ > ū **(Long High Raising)**

This can be represented on a vowel triangle as:

CHART IX

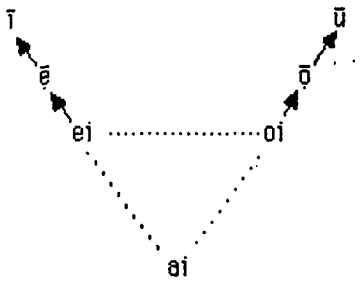
[15] But in fact the contraction product of ŏŏ does not raise: *co-opia (cf. in-ops, in-opia) > cōpia; an important point. See §47, 54.

[16] Who however, notes the difficulties in a midstage of ou for oi, and notes the collapse of oi and ou to o̅ in Praenestine.

[17] Why the blanket assumption of assimilation of the second member to the first in the light of the behavior of ou?

[18] Besides the inscriptional evidence and the typological comparison from Praenestine (§14), which is likely to be simply an older state than that shown by Latin, cf. the similar monophthongization in Umbrian *oinom > unu.

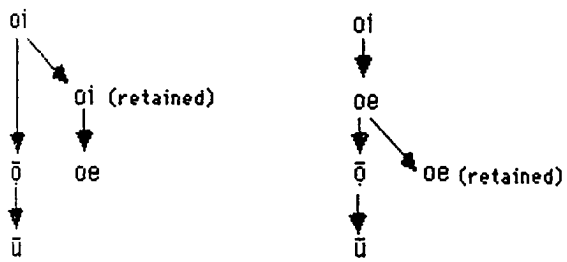
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The question remaining is: before the stage of monophthongization did oi pass through a stage oe, a diphthong with a lowered second member, distinct from both oi and ō, as seen in foedus, etc.? That is, do we have:

CHART X

or



Sommer-Pfister, Kretschmer, Niedermann, Maniet (locc. citt.), Meillet-Vendryes (1968.117) assume that oi passed first to oe (which then may have become ū variously). However, there is nothing particularly to recommend this stage, since no descriptive advantage is gained (oi and oe can monophthongize equally to ō), and there is no inscriptional evidence to support it. The LOEDOS and COERAVERE (CIL 1^2.678) which Maniet (1975.73) cites as evidence for a midstage of oe date to 106, and the same inscription shows MURUM already monophthongized, while CIL 1^2. 675 (two years earlier) shows LOIDOS, COIRAVERUNT next to MURUM; cf. COERAVERUNT (CIL 1^2.672 (c.111/2), noted by Sommer-Pfister (loc. cit.) and rightly criticized

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by Leumann 1977.65, 66. These are clearly archaizing spellings, well after oi had become ū. Rather, the most likely sequence of events is that the diphthong oi where it is retained then became oe, after oi had become ō bzw. ū, with a lowering of the second member of the diphthong (as in ai > ae, §29).

18) From oi in two labial environments, we have new sources for ei, which then follows the development of original ei. The first is after an initial u̯: *u̯oiĥos (οἶκος; etc.) > *u̯eikos > u̯ēkos (CIL 1^2.1806; also 388, 391 from central Italy) > vīcus. This change is shown by this single, but certain example. Other comparisons which have been cited are more problematic. Latin vīnum is probably not to be compared directly to οἶνος, as if from *u̯oinom, since Faliscan vino (Vetter no. 244), which does not show monophthongization, U. vinu, Volscian vinu, O. Viínikiíís, would all need to be loans from Latin (as maintained by Buck 1904.21; Poultney 1959.40 makes no mention of the phonology of vinu). Rather οἶνος, vīnum, Hitt. wiyanas, even the Semitic words, all represent independent loans from some ultimate "mittelmeerländische" source (see Leumann 1977.61; W.-H. s.v.; Buck 1949.390, § 5.92 holds that "the source could still be some IE language of the region"). Nor is vīdī to be compared directly to οἶδα, (as does Buck 1933.87), since the PIE morphological perf. already had present meaning 'I know' (noted by E.-M., s.v.), but rather reflects the PIE e-grade athematic aorist *u̯eid- (cf. Ved. pass. védi, RV 4.16.4; Old Lith. weizdmi, OCS impv. viždь; see Watkins 1969.152-3). As U. vuku-kum (Ib1, 4), vocu-com (VI b 43,

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45) 'ad aedem'[19] with the Umbrian monophthongization of oi to ō shows, this process of dissimilatory unrounding of the first member of the diphthong oi after the round/labial u̯ is a Latin innovation.

19) Sommer-Pfister (1977.67) claim that sūdor shows that the u̯ must be in absolute Anlaut, since we have: *su̯oidōs > *soidōs (loss of u̯ / _ o) > sōdōs > sūdor, and not, so presumably their argument would run, *su̯oidos > ** sueidos (> **svīdor ?). However, the comparative evidence does not support this conclusion. PIE shows an o-grade thematic λόγος-noun *su̯óid-o-s in Skt. svéda-, Av. xᵛaēδō, OHG sweiz, OE swāt, etc. There are also traces of an s-stem in W. *su̯it-s-o- > chwys, εἶδος (Hsch.), Ion. ἶδος[20], and Latin sūdor, -ōris, itself[21]. It is usually claimed that sūdor represents a mixture of the thematic *su̯óid-o-s and the s-stem *su̯eid-es- (W.-H. II.635, E.-M. 663), to explain the o-vocalism of the (immediate) preform *su̯oidōs. This type of change of vocalism does occur. So the o-stem modus influences the s-stem *mod-es-, seen in modes-tus, moder-āre; o-stem *pondos seen in adv. pondō, influences pondus, ponderis (Leumann 1977.378). However, it is not necessary to invoke the spread of o-vocalism seen in these two cases of -us, -eris nouns for sūdor, since in other examples of original s-stems changed to -ōs flexion in Latin the inherited e-grade is retained; so: tepor,

[19] So with Buck 1904.46. Poultney 1959.40 (et al.) connect it with lūcus, "by a slight margin of preference." However, note that all the certain examples of l > u̯ in U. are before a labial, so vutu (< *lau̯itōd, cf. lavatō), vapeř-e (cf. lapis), Vuvçis < *Lou̯ki-; see von Planta 1892.285-9, Buck 1904.68-9, Poultney 1959.71, though this is not a conclusion they draw. To these examples may be added U. Vofione < *loubh-iōnei, see §41. The change of U. l > v appears to be a conditioned change.

[20] With psilosis. The long ι vs. expected (ϝ)εἶ- is a difficulty. See Chantraine 1968 s.v.

[21] I.e. a part of Caland system, though admittedly a weakly established one. Also traces of an -r-stem (ultimately from the -ro-adj.?) in Arm. k'irtn < *su̯id-r- (with metathesis), Latv. swiêd-r-i, and (with change to -ōs) ἱδρώς.

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-ōris (Skt. tápas-); tremor, -ōris (τρέμω, ἀ-τρεμ-ής); decor, -ōris beside decus, -ŏris (Skt. denominative daśas-yáti), etc.[22] Thus, sūdor is likely to retain the e-grade, and show the regular change of e > o after *su̯- (§6.14): *su̯eid-ōs > *su̯oidōs > *soidōs > *sōdōs > sūdor. The development of sūdor therefore would be parallel to that of *su̯en-ōs (Skt. svánas-) > *su̯onōs > sonor. The possibility should be raised that sūdor could be a deverbative from the ā-stem iterative sūdāre < *su̯oid-ā-, where o-grade is regular, and so also for sonor, beside sonāre. However, since the class of deverbatives of this type in Latin is very small (only error/errāre, amor/amāre, clamor/clamāre) and in view of the close comparative evidence for both sūdor and sonor, this seems less likely. Note that this line of development, *su̯eid-ōs > *su̯oidōs > *soidōs > *sōdōs > sūdor, argues against the connection of sīdus with the root seen in Lith. svidėti 'shine', as if *su̯eid-es- > *seid-es- (with u̯ lost somehow) > sīd-er- (so W-H., s.v.). This etymology is in any case problematic, since there is no explanation for the disappearence of the u̯ after the s-, and Ernout-Meillet, s.v., are properly sceptical: "Même si l'on arrivé à en montrer la possibilité phonétique, le rapprochement avec le groupe de lit. svidù, svidėti «briller» est sans grand intérêt. Terme technique dont l'étymologie est incertaine." The other two putative examples of *su̯- > s- in Latin (besides cases of *su̯-/*s- alteration in PIE: Lat. sex, sē, vs. Cret. ϝέξ, Pam. ϝhέ) are equally uncertain.

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[22] There are a number of apparent o-grades. That of color is due rather to the i̯ < *ḱel-os (Skt. śaras-) and perhaps o-umlaut, see §6.15; odor shows an initial h₃-: < *h₃ed-es-, cf. Arm. hot (see Beekes 1969.120-1, 131, 139; cf. the pf. ὄδωδα < *h₃de-h₃od-h₂e). There are no good cognates for honor. Since -ōs is rare in PIE and Greek but very productive in Latin, the ablaut grade is not necessarily inherited if the word is a secondary formation and not directly from an older s-stem. So secondary zero-grade in clu-or (to clueō, vs. Skt. śravas-), rubor (to ruber), etc.

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Thus, sermō, if it is to be compared to O. sverruneí 'speaker' (< *su̯er-es-ōn-, cf. Eng. swear, etc.), where the zero-grade occurs in *su-sur-eso- > susurrus; sērius, if connected with Goth. swērs, etc. See W.-H. and E.-M. s.vv.; Leumann 1977.189.

Since we have only one certain example for the unrounding after u̯-, we cannot say if the lack of unrounding is due to the rounding nature of *su̯- preventing such a change (a phonological explanation), whether it never applied except after absolutely initial *u̯- (a structural explanation), or whether it applied only to original and not to secondary oi (a chronological explanation). The last seems to me the most likely[23], that the unrounding of oi to ei did not occur until after the u̯ of su̯- had disappeared.

20) The second environment which shows unrounding of the first element of the diphthong oi is between l (not necessarily in absolute Anlaut) and a labial[24]. So:

*kloiu̯os (Goth. hlaiw 'grave mound') > *kleiu̯os > *klēu̯os > clīvus[25].

*loimos (OHG leimo 'mud, slime') > *leimos > *lēmos > līmus.

*loibā-i̯ō (denominative, cf. λοιβή; λοιβᾶται· σπένδει Hsch.) > *leibō > *lēbō > lībō 'make a libation'.

But if no labial follows:

---

[23] In any case we do not have *su̯eidōs > *su̯oidōs (rounding) >*su̯eidōs (unrounding) > **svīdor.

[24] No mention in Buck 1933.86-7, Juret 1938.64, Niedermann 1953.69-72, Meillet-Vendryes 1968.118, Monteil 1973.108-9; Brugmann 1893.187-8 (līquī as only example), Leumann 1977.61 state as oi > ī after u̯- and l-; but Leumann notes lūdus as an exception. Sommer 1914.76, Kent 1945.91, Sommer-Pfister 1977.67 correctly as conditioned by labial. So first Solmsen KZ 38 (1905) 453 = Kl. Schr. (1933) 467.

[25] u̯ restored after the oblique; N.B. that clīvus is another counter-example to Leumann's assumption of a simple retention of ē before u̯ in all environments and in addition shows that if ē is retained / l̂ _ u̯, the l must be in absolute Anlaut.

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*loidos[26] (LOIDOS CIL 1^2. 364, c.200) > *lōdos > lūdus.

For the case of līber, see §40-1. A further possible example of this rule is: *loikᵘai > *leikᵘei > *lēᵘē > līquī. However, it is sometimes suggested that līquī is rather abstracted from (re)-līquit where oi > ī is claimed as the regular development (i.e. presumably that oi weakens medially to ei, with monophthongization and raising, as in final position; so Kent 1945.91). However, we cannot be sure what the regular outcome of medial oi was (see §8.33). Influence from the compound to the simplex is not the normal pattern (though not unknown, see §59), and since most other examples of medial oi show ū (commūnis, etc.), this is either the regular outcome or else analogical to the initial cases. In either case the ī of (re)-līquit has to explained phonologically by an unrounding rule if it continues a PIE -oi-. Leumann (1977.560) has suggested that relīquit is not a compound at all but rather continues the PIE reduplicated pf. *le-loikᵘe > reliquit, with oi > ī > as the regular change and dissimilation of l...l to r...l and subsequent identification of re- with the preverb. This is attractive as explaining the identity of semantics between linquit and relinquit, though it is only a weak arguement However, it is possible that līquī owes its lack of reduplication not to dereduplication in compounds or abstraction from a compound (see Excursus) nor to Leumann's dissimilation, but to līqu- being the old uncharacterized root aorist *leikᵘ- (so Cowgill 1982).

These two rules are of a dissimilatory unrounding of the first element of a diphthong. It would seem then that while initial u̯ is sufficiently labial to motivate the unrounding, an i̯ (certainly at this time dark before the oi)

[26] Zero-grade in Grk. λίζει· παίζει (Hsch.) < *lid-i̯ō; also nasal infix: λινδέσθαι· ἁμιλλᾶσθαι (Hsch.). Pokorny, for no reason I can see, wishes to combine these and take λίζει from a *λινδi̯o-.

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needs a following labial to dissimilate the oi > ei. It is difficult to know whether these two similar changes represent a single rule or not. It is perhaps better to keep them notationally separate. Thus we shall identify a rule of:

Woi-Unrounding: o > e / u̯ _ i

and a rule of:

First Element Unrounding (FEU): o > e / (C)ɬ _ i [+labial].

We would like to have an example of *loiu̯- to see if the resulting ei would be retained at the monophthong stage, comparable to ē from original ei in lēvis, but none exists.

21) An unusual development of oi is to be seen in nōn, beside nūllus. Both derive from nĕ[27] plus a form of *oino- 'one'. Thus apparently: *ne oinom > noinu(m)[28] > nōn, but *ne oinelos > *noinlos > nūllus. There are three theories which have been proposed to account for the difference in development between ū or ō. We must first dismiss the suggestion of Ernout-Meillet (1959.444): "Pour le passage de oe a ō entre deux n, cf. nōnus de *noenos." This envisions a sequence *nouenos > *noenus (with the u̯ disappearing) > nōnus (somehow conditioned by the flanking n's). The parallel cited in nōnus is incorrect. The sequence nou̯enos with -ou̯e- is not at all equivalent to neoinos with -eoi-. Further, Ernout-Meillet are mistaken in presuming a midstage of *noenos (unattested) for nōnus. There

[27] Italic shows both nē, the PIE prohibitive particle (Skt. mā́, μή) and nĕ. Thus Latin nē, O. ni, ni-p 'nēvē', Marrucinian ni. The nĕ (full-grade of the *n̥- neg. prefix; cf. Hitt. na, natta; Skt. ná, OCS ne, Goth. ni), is not found isolated in Italic but only in compounds or reinforced: *nĕ-i (the deictic particle) in L. nei > nī, U. nei-p (cf. Av. naē-čiš, Lith neĩ); *ne-kᵘe > neque, nec, O. ne-p, etc., and is frequent in composition: ne-sciō, ne-fās, etc. and so presumably underlies nūllus, and nōn. For the semantics, cf. Gmc. *ni ainaz > OHG nein, ON nei, OE nā.

[28] Caecilius 48 (Ernout 1957.250); Enn. Ann. 314 ; Plaut. Aul. 67, etc.

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is no rule which accounts for u̯ > Ø / o _ e, while novem itself shows that this is precisely what does not happen, nor is it clear how two flanking n's could condition either such a change or that of oe > ō. Rather, the development of *nou̯enos is to *nounos by the syncope of ou̯V̆ > ou (see below). Thus *nou̯enos > nounos, which then becomes *nǭnos (regular monophthongization) and the ǭ/ō$_2$ becomes ō (and not ū) by analogy to novem, etc.[29]

22) More seriously to be considered is the idea of von Planta 1892.I.152; Brugmann IF 6 (1896) 79f. They attribute the ō of nōn (vs. the regular ū of nūllus) to the development of oi in an unstressed syllable, i.e. -n'oinom developed with ō, while n'oín(e)los developed with ū. However, this suggestion is not without its difficulties: primarily it involves an arbitrary assumption that *noinom was enclitic. There is no evidence for this and arguing against it is the fact that Plautus uses both nōn (Pseud. 803, etc.) and noenum (Aul. 67, etc.) verse initially, which seems to indicate that noenum is an archaizing form, used for its metrical convenience. If nūllus represents the usual development of oi in an initial syllable we have: *ne oinelos > *noin(e)los (with elision of the first vowel in sentential sandhi) > nǭllos > nūllus, (cf. oinos > *ǭnos > ūnus). Another possibility is that nūllus represents a late combination of ne plus ūllus.[30] Supporting this might be the lack of attestation of any form other than nūllus, i.e. no *noellus, etc., parallel to noenu(m). This would show an outcome different from ne-uter > neuter where such a late diphthong remains. However, this is not a precise

[29] Cf. perf. pass. part. *mou̯e-tos > *moutos > mōtus by analogy to moveō, etc. That is ǭ becomes ō if there are related forms with ō; if not then the regular raising of ǭ to ū occurs; see §§42 ff., 52.

[30] Solmsen KZ 44 (1911) 205. Leumann seems to accept this on p. 67 but takes nūllus < *ne-oin(e)los on p. 119.

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parallel, as neuter reflects ĕ + ŭ, not ĕ + ū, and is further complicated by the fact that uter itself is late and abstracted from necuter, re-segmented as nec-uter, and then recombined with ne, cf. ubi but alicubi (see Leumann 1977.149-50 and litt.[31]). Similarly beside quisquam (< kᵘi-) is umquam from *ne-quom-quam > *necumquam, re-analysed as *nec-umquam. Here, however, the new negative is ne umquam > numquam, so that nūllus could well be a late creation following this pattern. In either case it is still necessary to account for the difference between ūnus, ūllus, nūllus on the one hand and nōn on the other.

23) The major problem with considering the ǭ of nōn as the outcome of oi in an unstressed/medial syllable is that we have no further examples to support such a development, while we do have: il-lūdere (lūdus), sē-cūrus, impūne, etc. These, however, are too easily built to the simplex to be good evidence for oi > ū as the regular medial change. The pf. re-līquī may represent ī from oi: *(re-)le-loikᵘ-ai, but here we have a heavy labial environment, and might expect *loikᵘ > *leikᵘ > līqu- in any case (**FEU**, §20).

24) The only cases where we have ō in Latin initial or medial syllables are where it reflects PIE *ō or else the outcome of ouV̆ > ou > ǭ/ō retained under analogy to forms with o. Even if medial ōi/oi passed through a stage ǭ in nōn, which seems doubtful, there are no o-forms to which it could be analogical. If not analogical, it is also unlikely that while ǭ from all other sources becomes ū (e.g. also in an unaccented syllable: *ab-douket > ab-

[31] The closest cognates show not kᵘu- but kᵘo-: O. púterei-píd (with O. anaptyxis) 'in utroque', U. podruh-pei 'utroque'< *kᵘo-tro-; πότερος, Skt. katara- < *kᵘo-tero- (no Brugmann's Law as the o is non-apophonic). Leumann explains the u as the regular outcome of medial -kᵘo-: *ne-kᵘo-tros > necuter (at least in closed syllables).

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dǭket > abdūcit; §8.13), the development of oi in nōn should stop at ǭ as its lautgesetzlich outcome[32].

25) With these phonological difficulties in mind, we turn to an early suggestion that *ne oinom contracted to nōinom (so first Thurneysen in his dissertation 1879; see Solmsen loc. cit.; Leumann 1977.67). The same sorts of problems occur here. If *ne oinom contracted to nōinom, why does *ne oinelos contract apparently to nŏinelos?[33]

26) There are no parallels for the contraction of e + oi to either ōi or oi. In general it seems that the quality of the first vowel predominates. However, the details and processes make this uncertain. Thus while we have dē-agō > dēgō, co-agō > cōgō, does this reflect a simple contraction of $V_1 V_2 > \bar{V}_1$, or a contraction after medial weakening: dē-agō > dēĕgō > dēgō, co-agō > coegō > cōgō? Other forms with ne- are not helpful here, cf. neuter, numquam, above. The first ō of nōlō does not show a contraction of e + o, as if *ne-u̯elō[34] > *ne-u̯olō > *neolō (loss of u̯) to nōlō (Brugmann loc. cit.), nor of *nou̯olō (eu̯ > ou̯ and e > o / _ ŧ) > *noolō (u̯ > Ø / V_1 _ V_1) to nōlō, but rather *nou̯olō > *noulō (ouV̆ syncope) > *nǭlō --> nōlō by analogy to volō (see §54). The other potential source of contraction of ĕ has been eliminated by the generalization of red- before vowels. There are further

[32] Kent (1945.91) says that oi > ō before tautosyllabic consonant, and offers nōn as the sole example; so Sommer-Pfister 1977.87. However, this would have to mean that noinom had become *noin while noinelos still had an open syllable, which, as the attestation of noenu(m) beside nūllus shows, was not the case.

[33] Sommer-Pfister 1977.213, following Brugmann IF 6 (1896) 79, therefore take nūllus as < *ne ūllus. However, this simply restates the same problem: why should *ne oinom have contracted early, but *ne oinelos remained two separate words until quite late?

[34] N.B. that it does show that u̯ > Ø / V_1 _ V_1 is not before both eu̯ > ou̯ and e > o / _ ŧ applied, i.e. we do not have *neu̯elō > **nēlō. Not a surprising conclusion.

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difficulties, since if *ne oinom contracted to nōinom the ō of nōn would represent the treatment given to ōi in absolute Auslaut, for example the thematic dat. sg. -ōi > -ō, and still does not explain why ne oinelos, which should equally contract to nōinelos becomes nūllus, not **nōllus.[35] We have no absolutely sure examples of long diphthongs in Latin in anything other than word final position, and the difference in treatment between -ōi* (ā-stem dat. sg.) > ăi > ae (but after original -ai > ei > ę̄ > ī as in *ce-caid-ai > cecidī) and -ōi > ō (with the off-glide simply lost) should make us wary of arguing from analogous final long dipthongs. The treatment of the dat. pl. -īs, which is preconsonantal as is *nōin(om), cannot be used as a parallel as it may come from the loc. -oisu with apocope, or, if from -ōis, reflects an Osthoff's Law shortening (-ōis* > -ois > -eis > -ę̄s > -īs). Whatever the phonology of ōi > ō, there appears to be no way of explaining the difference between nūllus and nōn except by an arbitrary assumption that *ne oinelos contracted to noin(e)los but *ne oinum became nōinom.

27) If we wish to refer nōn but nūllus to such a difference, it can be made somewhat less arbitrary by proposing that the first part of the compound was not, as has been assumed, identical in both cases, and perhaps while nūllus represents n(e) oinelos, nōn might be from ne-i 'not', since Latin and Italic show ne beside ne-i; thus: *neioinom 'not one'. For ne-i- in compounds, cf. nīmīrum, O. nei-p, etc., above. We might then imagine an early assimilation of *neioinom to *noioinom (similar to the type of assimilation in ne-hilom > nihil) > *nooinom > nōinom > nōn. Then noenu(m)

[35] Juret (1938.64) proposed that "Ne + oin > *nōin > *nōen > nōn, où l'ō a pu être conservé dans ōe parce que le mot est monosyllabique; *ne + oinolos > nōinlos > nŏinlos > nūllus présente le traitement normal dans un polysyllabe." This is interesting, if ad hoc; however, it is difficult to see why *nōin would not represent an Osthoff's Law environment as well as *nōinlos.

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could reflect an archaizing representation of the nōjnom stage, since a *n(e)oinom ought to become **nūn parallel to nūllus, with which it co-exists.

28) Another unusual development of oi is seen in huius, cuius. Here it appears that oii, unlike eii, has not simply been retained but has been raised to uii-. So in kᵘoiios > cuius, hoiios > huius, kᵘoiiei > cui, *hoiiei-ke > huic. The earlier stage is attested in QOIUS (CIL 1^2.7 Scipio Barbatus; 583.9 Lex Repet., c. 123, archaizing), has been raised in QUIUS (CIL 1^2.1297, end of second cent.), CUIIUS (CIL II.1964); HOIUS, HOIUSCE (Lex. Repet.) and HUIIUS (CIL II.1964). These attestations seem to indicate that the change of oii to uii was after the change of -os > -us, itself comparatively late. Kent 1945.105, Sommer-Pfister 1977.87 claim that oii was simply retained, and that cuius, huius, represent the change in unaccented position. Again, this seems an arbitrary assumption. Leumann 1977.477 says, "Zur Rechtfertigung des Wandels o > u in huius cuius und huic cui läßt sich wenig sagen; vielleicht steht er als *hoi-ius > huius in Verbindung mit oi > ū, trotz der verschiedenen Bedingungen". However, this seems to envision a sequence hoi-ius > *hō-ius (with monophthongization in a closed syllable) > hūius, but the forms were huiius, cuiius, eiius, and the evidence is clear that the initial diphthong did not monophthongize. It seems better simply to state a rule that o was raised to u before ii while eii remained, based on these few examples, which are the only cases of the environment in the language.

29) The diphthong *ai in initial position remained essentially unchanged except for a lowering of the second element of the diphthong from i > e. A

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similar change has occurred in Hindi[36], where the Skt ai is now pronounced /ae/, while the secondarily arisen diphthong /ai/ is spelled <āī>. This is parallel to the change seen in the lexically retained oi which becomes oe (foedus, etc.; §15, 17). There are numerous early attestation of ai: GNAIVOD, AIDILIS (CIL 1^2.7), AIDILIS (CIL 1^2.9), etc. The earliest attestations of ae are AESCOLAPIO (CIL 12.27, 28; undated, c. 200 ?), AETOLIA (CIL 1^2.616: M. F. Nobilior; c. 189)[37] and AEDEM in the SCB (186 B.C.). We can date the change of ai > ae to the earlier part of the second century.

30) There is also evidence of a pronunciation of ai as a long low monophthong ę̄, in substandard speech and outside of Rome[38]. There are also inscriptional examples PRETOR (CIL 1^2.365, Faliscan area), CEDERE, CEDITO (CIL 1^2.366, Umbrian area)[39]. This "rustic" pronunciation, continued in Romance (Bourciez 1956.44), is shown in certain Greek borrowings. So σκηνή /skę̄nę̄/ borrowed as scaena beside scēna, σκῆπτρον ---> scaeptrum beside scēptrum[40].

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[36] And indeed in English where the sound of <bye> is usually [baɛ], while only certain speakers of Scottish and Canadian English end the diphthong with a high front i. See Ladefoged 1975.70: "The diphthong [ai] as in "high, buy" moves towards a high front vowel, but in most forms of English it does not go much beyond a mid front vowel".

[37] Though perhaps recut in Augustan times? So Warmington 1940.79.

[38] Mar. Vict. 6.32.4-6; Varro Ling. 5.97, 7.96. Cf. Lucilius 1130 (= Ernout 1957.243): Cecilius pretor ne pretor rusticus fiat.

[39] And in Umbrian itself: kvestur, etc. Buck 1904.44. See Leumann 1977.67-8.

[40] That is, the word is first borrowed into an outlying dialect and the urban Roman scaena represents a hyperurbanism; so Niedermann 1905.6 cited in Leumann 1977.68. No reason to assume Etruscan intermediacy (so Ernout 1954.67). Etr. ai represents η (Doric α) in only two examples, which may themselves reflect a reverse spelling: Calaina < Γαλάνα (Orvieto 5.- 4. cent.) and Zimaite < Διομήδης (second half of 5. cent), but eight times: Ziumiθe, Zimite (de Simone 1970 II.25).

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31) We can now turn to the u̯-diphthongs:

eu̯ ou̯

au̯

The two diphthongs eu̯ and ou̯ are usually said to have fallen together in Proto-Italic[41]. The change of eu̯ to ou̯ indeed appears to be a shared innovation in most branches of Italic. Standard examples are:

*neu̯os (Skt. náva-, νέFος, etc.) > novus, O. Núvellum

*teut-éh$_2$ (Goth. þiuda, Lith. tauta) > O. touto, U. totam, Marrucinian toutai.

However, Venetic, in the earliest attestations (of the Padua area at least), seems to show eu̯ unchanged, and later ou̯/o̯. So *leu̯dheros > lo.u.dera.i., lo.u.derobo.s. (see §40-1), but teut[a], teu. (see Lejeune 1974.110-1).

32) Several pieces of evidence point to eu̯ > ou̯ as independent in Latin/Faliscan. Two words borrowed from Greek indicate that the change of eu̯ > ou̯ did not begin until (or at least was not over by) the time of early contacts with the Greek colonies[42]. For ouārē the most plausible etymology connects it with εὐάζω (Leumann 1977.46, E.-M., W.-H. s.v.) and Pollouces is borrowed from Πολυδεύκης (§2.10-11)[43]. Later Greek borrowings, even as

[41] So Brugmann 1897.197, Buck 1933.81, Kent 1945.92. Leumann 1977.70 calls the change "gemeinitalisch", but notes evidence for eu in early Latin (see below). Maniet (1975.66, 69) treats together ewV > owV and ełV >ołV (with novus as his example of the first), and astonishingly says of O.-U. "Aucun changement correspondant"; the "diphtongue ew" (which, judging from the Latin examples, means ante-consonantal) is said to become ou in O.-U. but no examples. I am following Lejeune (1974.163-173, esp. §191) in holding Venetic as most likely to be a branch of Italic, on a level with O.-U. and Latin-Faliscan; see Leumann 1977. Allge. Teil 31-2, and litt..

[42] Very approximately c. 500; so Stolz-Debrunner-Schmid 1966.51.

[43] The importance for Latin chronology was first pointed out by Schulze GGA 1895.547; so too Sommer-Pfister 1977.69

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early as Plautus, show eu-: eunuchus, Europa, etc.[44] (so too with euhoe, where the cry itself has been borrowed later than the verb). The reverse cases, early Greek spellings of Latin words, are less reliable evidence. Thus while there are attested λεύκιος (D 148; 171 from Delos, c. 150; 183 from Rhodes, third cent.), λεύκολλος (Polyb.), λευκανοί (Isoc.), since Greek ου had long since become monophthongized to /ō̯/, ευ might simply represent the closest phonological approximation in Attic Greek to a Latin /ou̯/, perhaps aided by etymology to λευκός (see Leumann 1977.70-1).

33) In addition two forms provide somewhat uncertain attestation of eu still unchanged in the earliest Latin. The fragments of the Carmen Saliare (Ter. Scar. 28.11) preserve a Leucesie, which is quite likely to be a vocative to a god of light (as if the ancestor of a *Lūcerius, cf. Iuppiter Lūcĕtius[45]). However, the tradition and preservation of the Carmen Saliare are not strong enough for us to place great reliance on its accuracy, though the intervocalic s , and the preserved -i(i̯)e (vs. -ī in later Latin and in O.-U.) might point to a reasonable antiquity. The Ardea Inscriptions (late fourth/early third cent; CIL 1^2.455 = Vetter 1953.362) record NEVEN : DEIVO[46] (= novem deorum) and NEUNA FATA (= nōna fata, which is attested to by Varro ap. Gell. 3.16.11-16). Here the problem is that the forms are

[44] A very fragmentary inscription (CIL 1^2.613) refers to booty taken from Leucas (presumably by L. Quinctius Flamminius in 192), but perversely breaks off in mid-dithphong: Le]ucado or Lo]ucado?

[45] Naevius apud Gell. 5.12.6, Serv. ad Aen. 9.570, Macr. 1.15 (quoting also from a Carmen Saliare). Cf. also the tribal name Luceres (OCD s.v. Tribus; Paul. Fest.; Varro Ling. 5.55, 81; Liv. 1.13.8, 1.36.2; Ov. Fast. 3.131), though this is claimed as Etruscan in some of the ancient sources.

[46] Warmington 1940.74 takes as ne ven. for "ne vendas or the like". The other pieces all feature the name of a god in the gen. Note in particular the inherited final -n < -n̥. The importance of these inscriptions for Italic has been reiterated most recently by Hamp Gl. 59 (1981) 228 f., esp. n. 1, who rightly criticises Leumann's and others' assumption of "gemeinitalisch".

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dialectal, rather than purely archaic (Vetter loc. cit.), but still indicate that eu > ou is not Proto-Italic. In particular since neuna must come from *neu̯enā (cf. Lat. nōnus < *nou̯nos < *nou̯enos < *neu̯enos, cf. U. nuvime; see §42 f.), it shows a different syncope and treatment than the Latin, which does not show ou̯V̆ > ou until after eu has become ou.[47] However, if dialectal, it is likely to represent a preservation of an older state of affairs, and not a new unrounding of ou̯ back to eu̯, and so provides some evidence that eu̯ to ou̯ was not a PIt. change[48].

34) Note in particular that there is no advantage in describing the change of eu̯ > ou̯ before a consonant (*deu̯kō > *dou̯kō; sometimes with eu written to distinguish it from eu̯) separately from that of eu̯ > ou̯ before a vowel (*neu̯os > novus). This is a single change of rounding of the first member of a diphthong, whether ante-vocalic or ante-consonantal. Rather the difference in treatment applies to the resulting ou̯. So Leumann (1977.70)[49] : "Die Wandel eu > ou und eu̯ > ou̯ (idg. ew > lat. ov) sind phonetisch

[47] So IOVESTOD on the Forum inscription (see §44-5), which also shows a fragmentary]EVAM of uncertain meaning. There is also a cozeulodorieso from a Carmen Saliare in Varro Ling. 7.26, but the proper interpretation is doubtful. See Altheim 1954.406.

[48] Safarewicz (Eos 47.101-3) explains the e of NEVEN as by paradigmatic analogy to decem, as is the case in OCS devętĭ after desętĭ. Note that this, even if so, is still more likely to imply a PIt.*neu̯em retained by analogy than a PIt. *nou̯en changed back to neu̯en.

[49] Who nevertheless does treat them separately, though cross referenced. So "idg. ew" on p.46-7 with changes of ĕ, "idg. eu" on p. 61 with changes in diphthongs. Similarly Buck 1933.81, 92; Meillet-Vendryes 1968.111, 161; Sommer-Pfister 1977.55, 69-70; etc. Again rightly criticized by Hamp (loc. cit.) for failing to capture the generalization.

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gleichartig als Labialisierungen des e zu gerundetem o durch Angleichung an das folgende gerundete u bzw. ṷ".[50]

35) If eu > ou is not a Proto-Italic change, can it be combined with e > o / _ ŧ as a single rule? Both are roundings before back, round consonants. Though there are no other rules in the language which would necessitate or be better explained by these two rules being combined for initial position at least, perhaps we are missing a significant generalization by treating them separately.[51] However, the change of original eu̯ > ou̯ is very early, so already IOUXMENTA and IOVESTOD on the Forum inscription (CIL 1^2.1, c. 500), while the rounding of e > o / _ ŧ seems to take place considerably later (so HERCELE CIL 1^2.551, 3. cent., see §1.3, 8.30) at least in medial syllables. In addition, it is clear that the change of eu̯ to ou̯ occurred again with new e_2 from a in medial closed syllables (by **A-Weakening**, see §8.14), so *dēfraudō becomes *dēfreudō (**A-Weakening**) > *dēfroudō (a second application of *eu̯ > ou̯) > *dēfrōdō > dēfrūdō, and that the rounding of e to o

[50] Rather than "gleichartig" it might be more accurate to write "gleich". Despite the cross-referencing, a significant generalization is being missed (due principally to the handbook format): that these are not two parallel (and therefore possibly chronologically distinct) changes, but the same change. It may be necessary therefore to belabor the point that there is no phonological difference in the diphthong eu̯ in *n-eu̯-os and the diphthong eu̯ in d-eu̯-ket. In both cases the sequence consists of a full open syllabic followed by a nonsyllabic glide. There is a phonological distinction in the structure of the word, rather than the diphthong itself, in that the words have different syllabic structures: ne-u̯os but deu̯-ket. Thus to write neu̯os but douket may be a useful graphic convention in emphasising that douket contains a tautosyllabic diphthong, but should not be taken as showing a difference in the actual phonetic nature of the sound so represented. See Sommerstein 1977.30-33.

[51] Maniet 1975.66 treats the changes together but he is not making an historical claim, but merely describing all changes of ĕ before consonant together. Meillet-Vendryes 1968.111 do seem to be taking the two as a single rule.

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/ _ ɬ also applies to e from medially weakened a, so *insaltos > *inseltos (**A-Weakening**) > *insoɬtos (**HVB**) > insultus, as well as to i in closed syllables (facultās). It is therefore perhaps at this point that we should combine the two rules of rounding in medial syllables. There are then two distinct phases: a very early but still Latin rule of e_1 > o / _ u̯ (perhaps to be combined with e_1 > o / _ ɬ) and a second rule that rounds e (also from a) and i to o and u before the back resonants u̯ and ɬ (in open or closed medial syllables), that is an expanded version of **HVB**. This point is argued more fully below under the heading of Medial Weakening, see §8.15 ff.

36) After the change of eu > ou comes the simplification of xu̯ and gu̯ to u̯ (Leumann 1977.47, Buck 1933.89). The two examples which show unchanged eu are *mrexᵘis > brevis, and *lexu̯is > levis (for these preforms, see Exc. II §10, 12). Further examples of the simplification are: initially *gᵘm-iō > *gueniō > veniō, medially gen. sg. *snigᵘh-es > *snixu̯es > nivis.

37) There is a second application of rounding eu to ou in brūma 'winter equinox', which is clearly a form of the superlative of brevis, meaning 'shortest day' (Varro Ling. 6.8: quod brevissimus tum dies est). The usual preform is given as *brevimə (E.-M., W.-H.)[52], that is after the change of gu̯, xu̯ > u̯, a syncope occurrred giving *breumə, and this eu now tautosyllabic presumably undergoes a reapplication of eu > ou which then follows the usual development of monophthongizing to ō and raising to ū (so explicitly Sommer-Pfister 1977.69). In the main, as to the development of the initial syllable this seems correct, but as Cowgill (1970.126-7) points out the

[52] Leumann 1977.498: "eher alt mreg'humā > brehumā > breumā > brūma, als jung *brevimā > breumā". However, there is no evidence that Lat. ever had anything but the i-stem, and the superlative would not have been *mreĝhu-mā in any case. A positive, meaning 'the short one' would not be impossible, but such a -mo- adj. is without comparative support.

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superlative formant -(m̥)mo- belongs only to stems which show -(t)ero- comparatives. The superlative ending for brūma would have to have been -is-m̥mo- (with the development of R̥RV > aRV). We can start with a Latin pre-form *mrexu̯-isamo-. Cowgill (loc. cit.) writes:

> So if we are to hazard a preform for brūma, the best guess but a weakly supported one is *mrég'h-ism̥mā. Assuming mutual approximation of superlative and positive, we can get pre-Lat. *mrexwis, mrexwisemā. Now the latter will become brūma if we can get rid of the the x in time to take advantage of the sound law *ew > *ow (as in novus < *newos), which would then set things up for the change owi > ū (as in prūdēns < *pro-widen-s), after which the rest is easy *mrexwisemā > *brewisemā > *browisemā > *brousemā (*broisemā?) > *brousmā (*broismā) > brūma.
>
> The difficulty is that *ew from *exw is not supposed to turn into ow in Latin, but the evidence for this is precisely the two problematic items brevis and levis ... It seems legitimate to suggest that brūma shows the regular outcome of exwi, at least in a tetrasyllable, and that brevis, levis whose vocalism cannot be original anyhow, either attest a different outcome in disyllables or had not yet acquired their e at the time of the change ew > ow.

38) Cowgill rightly takes the preform as *mrexu̯isamā. He invokes analogical influence from the e-grade comparative to explain the vocalism of brevis. However, for brevis, levis continuing an e-grade in the Latin stems *mrexu̯i-, *lexu̯i-, see Excursus II, §10. Therefore brevis does show that the rule of eu̯ > ou̯ was over when gu̯, xu̯ > u̯, and so brūma cannot show the regular treatment of *exu̯i (as opossed to an analogic brevis), but rather already -eu̯i-, which remains in brevis, but in the tetrasyllable *breu̯isamos undergoes a syncope (2/4), and a new rounding of the now tautosyllablic eu̯ to ou̯. The only problem with deriving brūma from *mre(x)u̯isamā is that it will need to have undergone a second syncope after the reapplication of eu̯ > ou̯. Cf. the structurally similar superlatives:

*fakl-isamos > *fak|zamos > facillimus,

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*mag-isamos > *maksamos > maximus

which remain trisyllabic. So one might expect a *mrexuisama > *breuisama > *brousama > **brūr(i)ma. But this should present no difficulties; as the rhotacism in the hypothetical outcome shows the forms are not in fact parallel and syncope in the environment Vs _ m might occur where it did not in the environment Cs _ m.[53] Nor should a reapplication of eu > ou be considered unlikely. We have good evidence for the second rounding of eu in medial syllables: dēfraudō > dēfreudō (**A-Weak.**) > dēfoudō (eu > ou again) > dēfrōdō (**Monoph.**) > dēfrūdō (see above) and the reapplication in brūma, though in an initial syllable, is parallel to this law (see §8.14). Thus we may set up the development: *mrexuisama > *breuisama > *brousama (sync. 2/4) > *brō(z)ma (sync. 3) > brūma,[54] and the sequence of changes:

1) eu > ou (with original e)

2) gu, xu > u

3) reapplication of eu > ou (with new e and new u; part of **HVB**)

4) **Monophthongization**

5) **High Long Raising**

39) To recapitulate: the ou resulting from the change of eu > ou (and its reapplication) remains unchanged under the accent if it stands before a vowel (unless that vowel syncopates; see §42 ff.); so:

*neuos > novus

---

[53] I.e. no syncope which would create new and phonotactically disallowed consonant clusters: ksm, llm; see Rix 1966 = 1973.99 f.; cf. n. 27.

[54] Nothing, of course is being implied about the chronology of mr > br or of compensatory lengthening with loss of z which applies vacuously to the long vowel. Note too that it is not impossible that the syncope 2/4 preceded the change of xu to u; that is we could have a *mrexuisama > *brexŭsama (sync. 2/4) > *breusama (with the loss of h seen in *liĝhēn > liēn) > *brous(a)ma (eu > ou again) > brūma (sync. 3, etc.).

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*neu̯n̥ (Skt. náva, Goth. niun) > novem

*diēu̯-es > Iovis.

However, if it is tautosyllabic, or becomes tautosyllabic by syncope, it monophthongizes to ō̥, thus falling together with the ō̥ from oi, and is then raised to ū. Standard examples are:

*deu̯kō (Goth. tiuhan) > *dou̯cō > *dō̥cō > dūcō

*eu̯sō (Skt. óṣati, εὔσω) > *ou̯sō > *ō̥rō > ūrō

40) This is the usual development. However, ou (like oi) is also subject to an unrounding in a labial environment (but this time of the second element of the diphthong) to oi, which still in a labial environment has the first element unrounded to ei (FEŪ, §20) which is then subject to the usual monophthongization to ē and finally raising to ī.[55]

41) The whole process is beautifully illustrated by the one example of the change--namely līber, with pleasant attestation at every stage, and several important chronological implications. By comparison with Grk. ἐλεύθερος we can begin with the preform PIE *leúdheros which becomes *louθeros (cf. the Venetic m. dat. pl. lo.u.derobo.s, f. dat. sg. lo.u.dera.i). This becomes *louferos, with the change θ > f // u̯. The ou can not yet have changed to oi as the u̯ of the ou is still necessary to change the θ to f, and it is now this f which gives us the labial environment (ŧ _ f-) which is necessary to unround the second element of the diphthong. The *loufero-stage is indirectly[56] attested by O. gen. Lúvfreís (LīberĪ 'of the god Liber'), U. Vofione (with U. monophthongization of ou > ō; as if Latin LībiŌ, dat. LībiōnĪ) and Pael. loufir. This ou then undergoes a **Second Element**

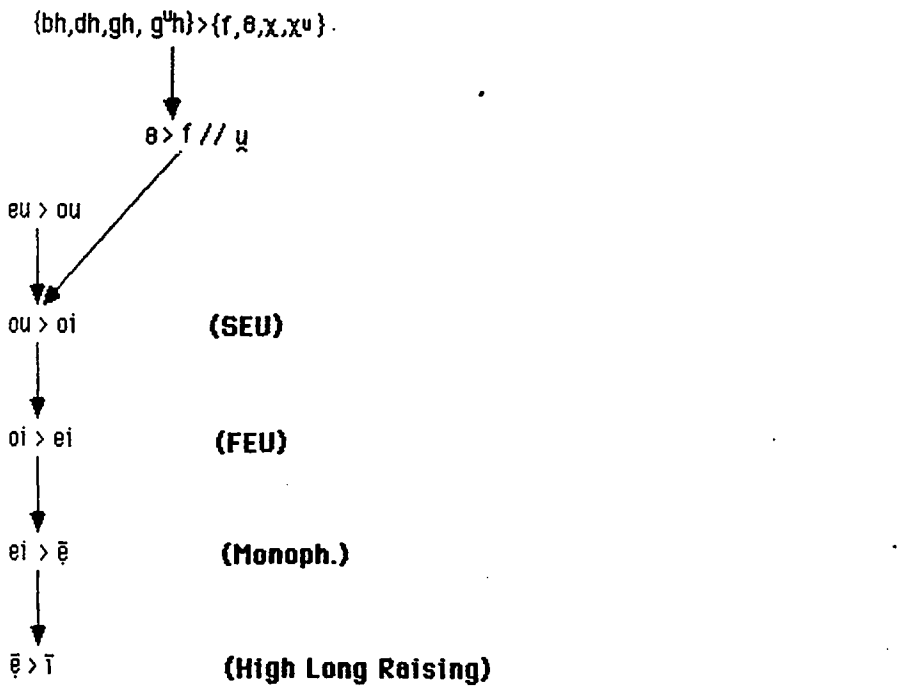
[55] Cf. a similar change of u̯ > i̯ in Greek: aor. *é-ue-uk̑u̯-o-m (cf. Skt. ávocam) > εἶπον with dissimilation of u̯ ... u̯ to u̯ ... i̯; Rix 1976.216, Lejeune 1972.228.

[56] Since in O.-U. θ > f in all environments.

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Unrounding in Latin and Faliscan only: *loiferos, attested in Fal. loifir-ta 'liberta, freedwoman' (Vetter 1953. nr. 276a); this *loiferos (along with **Spirant Voicing** and **Spirant Stoppage**) undergoes a change of **First Element Unrounding (FEU)** to give Latin leiber[57] (note that the syncope of -ros > -rs > -rr > -r has already occurred); with monophthongization lēber (LEBRO CIL 1^2.381) and finally līber. The diphthong is, as it were, slowly unwrapped: *eu > ou > oi **(SEU)** > ei **(FEU)** > ē **(Mono.)** > ī (raising). This can be illustrated by:

CHART XI

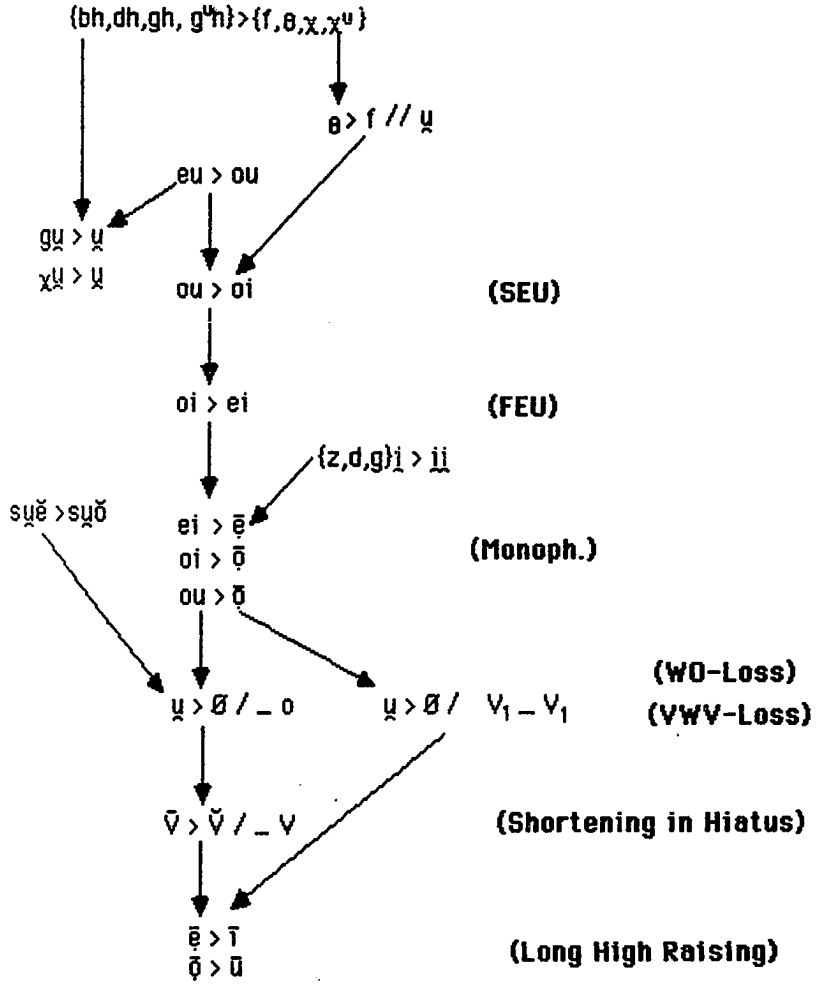


[57] LEIBER (nom. CIL 1^2..563, Praeneste), LEIBERO (CIL 1^2.2440), LEIBERUM (CIL 1^2.2235, Delos, 2. cent.), LEIBEREI (nom. pl. CIL 1^2.614, Aemilius Paulus,189), LEIBEREIS (CIL 1^2.1531, Vertuleii, c.150; though here false spellings show ei was already ī, see §1).

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We can combine these changes with previous ones. Thus:

CHART XII



We would like to have words of the shape *lougu̯- / *louχu̯- or *loigu̯- / *loiχu̯- to see if the simplification of gu̯, χu̯ > u̯ feeds either **Second**

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Element Unrounding or **First Element Unrounding**, but there appear to be none in Latin.

42) The diphthong ou̯ tautosyllabically (i.e. before a consonant) became ǭ, which, like the ǭ from oi̯, was raised to ū. In general ou̯ remained before vowels, i.e. heterosyllabically. However, certain sequences of ou̯V̆ undergo a syncope of the short vowel. This results in a tautosyllabic ou̯ which then undergoes the usual monophthongization and raising, resulting in either a regular ū or under analogy ō[58]. Thus for cases with ū:

*pro-u̯idēns > *prǭdens > prūdens[59]

*nou̯en-din-ai 'period of 9 days' > noundin-um (SCB 23; 186) > nōndinai (CIL 1^2.583.31, c. 125) > nūndinae

*nou̯o-par-os > *nouperos > *nǭper > nūper[60]

As the SCB example indicates, we have a change of ou̯V̆ > ou, following which is the usual reduction to ǭ and then to ū.

43) Several points must be noted. First, this syncope does not occur in disyllabic words. So:

*h$_3$eu̯is > ovis

*neu̯os > novus

*dieu̯es > Iovis

*g^{u}ou̯-es > bovis (with dialectal *g^{u} > b), bovem

---

[58] The nature of the change in these cases and the reason for the different outcomes of ō and ū are disputed. I have simply presented my view (which agrees in the main with Buck and Leumann) on the matter first, and have dealt, I hope adequately, with other scholars' opinions below.

[59] prō-videō, prōvidentia, are later re-formations, though the connection of prūdens was obvious (Cic. de nat. deorum 2.58).

[60] The etymology (W.-H., E.-M. s.v.) which derives it from nū- (νῦν) is unlikely as Latin shows no trace of the long ū form: num, *num-ke > nunc, could be from a long ū, but cf. especially nŭ-dius in compound.

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*neu̯m > novem[61]

There are two counter-examples[62]. Syncope seems to have occurred in iūs, and rūs:

nom. *ieu̯-os (s-stem), but

gen.sg. *ieu̯-es-es > *iou̯eses > *iou̯ses > *i̯ōses > iūris;

nom. *reu̯-os (cf. Av. ravah- 'space'), but

gen.sg. *reu̯-es-es > *rou̯eses > *rou̯ses > *rōses > rūris.

However, these two nominatives could easily have been built to the oblique cases[63], and in particular note that of the disyllabic forms, these are the only ones with singular obliques that are longer than two syllables[64] (since they are s-stems and the others i-, C- or o-stems). For a model cf. mōr- : mōs :: mūr- : mūs; so iour- : X, rour- : X; the nom. could have been backformed at any point. Sommer-Pfister (1977:127) suggest that the syncope is regular in all cases and that ovis could be backformed after other i-stems. However, it is difficult to see what the nom. could be backformed from. If Sommer-Pfister are correct, the paradigm would have been:

---

[61] The list is, I believe, complete.

[62] Benveniste's suggestion (1935.54) that plūs can be accounted for by supposing a "contamination" of a comparative *plē-is by a neut. substantive s-stem *plewos which develops like *yewos > iūs, cannot be maintained. Not only is this not the development for iūs, but the neut. s-stem does not exist. An explanation of plūs, if one is possible, is beyond the limits of this study, but the PLOUS of the SCB is not good evidence for syncope from a *plewos (bzw. a reformed comp. *plew-is).

[63] So Leumann 1977:134; not with Meillet-Vendryes (1968.402) who write "des primitifs *rouos, *iouos, avec un traitement pareil à celui de deus; rous, ious ont finalement abouti à rūs iūs". However, the treatment they propose is not parallel to that of deus. If it were one would expect *rou̯os > *roos > **rōs.

[64] only the dat./abl. pl. would be trisyllabic for the i- or C-stems: ovibus, etc., but this is clearly analogical to the non-syncopated sg. The gen. pl. ovium is not subject to this syncope as it occurs only before a consonant §48).

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*o̯u̯is > **ūs

*o̯u̯eis (> **o̯u̯īs) --> *ovis > **ūs

*o̯u̯eie̯i > *o̯u̯ēi > ovī or ove̯ii (loc.) > ovī

*o̯u̯im > *ūm > **um

*o̯u̯īd > ovī (Varro Ling. 8.66) --> **ove (> **ū?)

---

*o̯u̯ie̯ies > ovēs

*o̯u̯iōm > ovium

*o̯u̯ibos > **ūbus

*o̯u̯ins > ovīs (if the syncope is later than the lengthening, otherwise

*o̯u̯ins > *ouns > **ūs?)

It seems unlikely that a dat. or abl. sg., even conjoined with the pl. forms, could have remade an entire paradigm (though the gen. sg. equī, vs. ecus, ecō, etc., has restored the the qu to the sg.). Cf. in particular the paradigm of bōs, bovis, bovem; bovēs, boum, būbus; where the unsyncopated gen. sg., far from being analogical, is the source of analogy for bovem, also unsyncopated, replacing *gʷo̯u̯m > PIE *gʷōm (Skt. gām, Av. gə̨m, βῶν) > It. *gʷōm (U. bum).

44) In words of more than two syllables, the vowel following the u̯ is lost. This behavior argues strongly that the change of o̯u̯V̆ > ou is due to syncope. Further arguing for syncope is the fact that this change does not occur except word initially, that is the vowel immediately following the accent is syncopated: *pró̯u̯idēns > *proudēns > *prŏdēns > prūdēns, but medially *oínou̯orsus > oinuu̯orsos (OINVORSEI; SCB) --> ūniversus (with medial weakening, but not syncope; §31-2). Some examples of the syncope in words of three or more syllables are (using the disyllabic stems above):

*o̯u̯i-piliō > *oupiliō > *ŏpiliō > ūpiliō (see below)

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*nouentios > *nountios > *nōntios (cf. NONTIATA, CIL 1^2.586, ad Tiburtes, 156) > nūntius

*nouen-dina > noudina (cf. g. pl. NOUNDINUM, SCB.23) > nōndina (CIL 1^2.583.31, c. 125) > nūndina

*diou-es glān(d)s ' Zeus' acorn' (διὸς βάλανος) > *iouzglāns > iūglāns

iou-es-to-s (Forum Inscr.) > *ioustos > *iōstos > iūstus

ioues-ā-t (Duenos Inscr.[65]) > *iourat > *iōrat > iūrat

*rou-es-tikos > *rousticos > *rōstikos > rūsticus

Cf. also:

*co-uir-ia > *couria > *cōria > cūria[66]

45) Here we have one of our few pieces of absolute chronology. From the IOVESAT of the Duenos Inscription (CIL 1^2.4, 7. - 5. cent.?; see Gordon 1983.77) and IOVESTOD of the Forum Inscription (CIL 1^2.1; late 6. cent.; ibid. 78-9) we can date this syncope after 550 B.C. or so, and from the IOUSISENT, CONIOURA[SE, NOUNDINUM of the SCB before 186; not a very precise window of time. Further we can note that the epigraphical evidence shows that the syncope occurs only after eu > ou, as we would expect since the syncope also applies after etymological o (prūdens, etc.), so that the syncope seen in the Ardean NEUNA is dialectal (§33)[67].

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[65] If IOVESAT is one word; see Gordon 1983.77-8 for the epigraphical and dating problems.

[66] So first Pott 1833/6. I.123; cf. the Vol. coverhiu (abl., Vetter 222). Accepted by W.-H.; E.-M. cautiously calls it "n'est pas impossible" and speculate on a non-Italic borrowing. However, no candiate is forthcoming.

[67] Leumann (1977.134) gives as a possibility that eue syncopated first to eu: "Das ū geht sicher zunächst auf ou zurück, das durch Syncope aus -eue- (wegen Neuna) oder aus -oue- -oue- (wegen iouestod) entstand". But we have no other examples and descriptive economy as well as the attestations argue clearly for a syncope of ouV̆ fed by the change of eu to ou.

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46) Rix's explanation (Krat. 11 (1966) 157 ff. = 1973.91) is that this ū arises from ou̯e not by syncope, but with the intermediate steps of ou̯ < *ou̯ŭ with u̯ disappearing before the ŭ. This envisions the sequence *nou̯en-dina > *nou̯ŭndina > *nŏŭndina > *nou̯ndina > *nǭndina > nūndina. This is unlikely, as it necessitates an unparalleled weakening of e (> o?) > u̯ in a closed syllable and before nonlabials[68]. So too then for other examples with closed syllables, where the vowel following the u̯ will not have weakened: *rou̯orsos > *roursus > *rǭrsos > rūrsus; *nou̯entios > *nountios > *nǭntios > nūntius.

47) The handbooks generally present this change as applying only to ou̯e and ou̯i.[69] There is, however, a clear example, just mentioned, with the same outcome from etymological -ou̯o-: *re-u̯rt-tos > *róu̯orssos > *roursos > *rǭrsos > rūrsus (see §5.3-4 above on the change of rss to ss and the

[68] It is possible, as Rix appears to do, to make ou̯V̆C(C)- > ou̯ŭC(C)- an ad hoc rule conditioned by the preceding u̯, but even after u̯ an e shows no weakening in medial closed syllables: convexus (which seems safe from analogy), etc. It seems profitless to propose such a rule when syncope is a common Latin phenomenon and the distribution of the loss (only post-tonically in words of three or more syllables) would favor syncope as an explanation. Rix argues that syncope occurs only before single consonants, except for sC. This does seem the case for many examples of sync. 3 and sync. 2/4, but there is no reason to suppose that the exceptionless syncope in *nou̯endinai is the same as that in cal(i)dus, or facillimus. Further, the limitations on syncope are not dependent solely on the shape of the following syllable but the preceding one as well, and on the resulting consonant cluster (Rix ibid. 99). I see no a priori reason to exclude syncope before CC. The fact that syncope occurs before sC (auspex, etc.) is probably due not to an exception to a general morphophonemic law, but to the fact that sC makes up a large portion of the limited number of medial V̆CC clusters in the pre-language.

[69] So Buck 1933.89, Maniet 1975.74-5, Leumann 1977.134; Sommer-Pfister note that it also occurs in ou̯o. Niedermann 1953.39 gives a single example of this type in prūdens lumped together with all other cases of syncope. For Meillet-Vendryes see n. 43.

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analogical reintroduction of r). Here *r̥ > Plt. -or- and the o will remain in a closed syllable. Note too that the syncope of ouo > ou occurred before the loss of u̯ before o[70], that is we do not have the development: *roui̯orsos > **rŏŏrsos > **rōrsus. For prōsa (also with syncope of ouo / _ CC), nōlō, see below. The other cases of oui̯o > ō: *noui̯o-paros (thematic compound) > nūper, cannot be considered as good evidence, since presumably the ŏ could have weakened in the open syllable. So too in *nogu̯odhos > *noui̯odos > *noudos > *nŏdos > nūdus (§49; not *noui̯odos > *nŏŏdos > **nōdus). There are no cases known to me of etymological oui̯aC. Even if there were, we could not rule out the a having weakened to e (even in a closed syllable) before syncopating. There appear to be no cases of original oui̯ŭC(C). Though we have examples only of e, i and o, since these do not seem to form a clearly natural class (not all high or front vowels for example), it seems best to write the rule as broadly as possible to include a syncope of all short vowels after oui̯.

48) Further, this syncope did not occur unless the second vowel stood before a consonant. Thus voveō, moveō, foveō, do not syncopate; and in other morphological categories: fovea, flovius, noviēs, ovium, etc.

49) The case of *u̯oue-tom > vōtum, *u̯oue-u̯ai > vōvī (for the ō, see below) shows that this syncope occurs also with new u̯ < gu̯, xu̯. So

[70] That this rule applies to the sequence Vu̯ŏ is clear from *deiu̯os > *dēu̯os > dĕos > deus.

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*(h_1)u̯oĝ$^{u̯}$h-ei̯ō[71] > *u̯ox$^{u̯}$eō (cf. U. vufetes < *u̯ox$^{u̯}$e-to-; vufru; Vufiune, Uofione 'god of vows'; Buck 1904.94) > voveō. Cf. also cōntiō < *cou̯entiō < *co-gu̯en-tiō < *ko(n)-g$^{u̯}$m-. The case of nūdus is somewhat complicated by the variety of changes, some due no doubt to taboo, in the various daughter languages. Comparison of Skt. nag-ná- < *nog$^{u̯}$-nó-, Av. mayna-, mayna-nta- (with dissimilation of the initial n-, or taboo), Olr. nocht, W. noeth < *nog$^{u̯}$-to-, Hitt. ne-ku-ma-an-za /negumants/ < *neg$^{u̯}$-mo-nt- < *neg$^{u̯}$-no-nt-[72] (with dissimilation of n ... n to n ... m, unless Skt. shows rather an assimilation of a *neg$^{u̯}$-mo- to *neg$^{u̯}$-no- and Av. a metathesis of *neg$^{u̯}$-mo- to *meg$^{u̯}$-no-), allows us to set up a root *nog$^{u̯}$- with a variety of

[71] Causative with o-grade, a "State II" root from the zero-grade *ug$^{u̯}$h-; cf. Skt. vāghát-, Av. (rāstarə)-vayantī; with "State I" root *eug$^{u̯}$h-o- > Skt. óhate, εὔχομαι, *eug$^{u̯}$hto > Proto-Indo-Iranian *augdho (Bartholomae's Law) > Av. aogadā. Rix (1976.48) reconstructs a *$h_1$eug^{h}etoi (with plain velar) > εὔχεται, but if Latin voveō is to be connected here, as it should, the root will have contained a labio-velar aspirate (as a possibility already in Brugmann 1897.579) and the Grk. is due to the regular pre-Myc. loss of the u̯-component next to a u̯: *eug$^{u̯}$h-e- > *eukh-e- (cf. *k^{u}ék^{u}los > *k^{u}úk^{u}los > κύκλος). Beekes (1969.90) rightly notes that there is nothing which necessitates an initial laryngeal (except the assumption that CVC(C)- was the regular root pattern); nor, however, is there anything which argues against it. Polomé (1965.26 n. 118) in criticizing his own earlier views about the development of *Hu̯- in Arm. writes "Arm. gog 'say': Lat. voveō 'vow' (*a_1w- on account of Gk. εὔχομαι 'pray')"; however, there is nothing to indicate that εὔχομαι was zero-grade, and Skt. óhate points to a full-grade. Further, as he himself points out, *u̯- without laryngeal regularly becomes *gu̯- > g-.

[72] Not with the PIE possessive -ment-; so Sturtevant-Hahn 1951.80 (which they acknowledge would be unique in the language).

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suffixes[73]. Comparison of nūdus with Goth. nagaþs, nagad-, OE nacod indicates a preform *nogᵘodhos[74] > *nogu̯odos > *nou̯odos, which by syncope becomes *noudos > *nō̯dos > nūdus. To recapitulate: the rounding of eu > ou is over by the time that gu̯, χu̯ become u̯ (*mreχu̯is > brevis; §36). The simplification of gu̯, χu̯ to u̯ feeds the syncope of ou̯V̆ to ou, which in turn feeds monophthongization and raising. Thus, we can set up the ordering:

1) eu̯ > ou̯

2) gu̯, χu̯ > u̯

3) ou̯V̆ > ou **(OW-Syncope)**

4) ou > ō **(Monophthong.)**

5) ō > ū **(High Long Raising)**

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[73] Grk. γυμνός is the most difficult (though I also do not understand Arm. merk apparently < *megᵘ-ro-). Cowgill (1965.156 n. 31, and 1982) and Lindeman (RHA 32 [1965] 32) suggest that a *nogᵘ-nos became *nugᵘnos (under the influence of the labio-velar) which then went to *nugnos (regularly; see κύκλος above). This *nugnos could then undergo a dissimilation to *mugnos, similar to that in Av. and then a metathesis, perhaps motivated by taboo deformation, to gumnos, but this is of course highly speculative. Alan Nussbaum has suggested that the simplest explanation for Grk. is two successive metatheses (due to taboo?): *negᵘ-mo- > *gᵘenmo- > *gᵘemno-. The e-vocalism that Hitt. has in ne-ku-uz g. sg. 'night' (Schindler KZ 81 [1967] 290-303) and nekumants, against the rest of the IE languages, suggests that in *negᵘ-nó- (bzw. -mó-) Hitt. has preserved the original vocalism (e- and o-grades both existing for the root, with each branch generalizing one is less likely in a derived -nó-adj.; Cowgill 1968. §74), while the rest of Indo-European (Indo-Iran. is ambiguous) has undergone a change to o (regular / n _ Kᵘ-C ?: see Cowgill loc. cit., though note that Lith. núogas has apparent extended-grade ō with gᵘ-V-; for a somewhat different analysis, see Szemerényi 1964.402). Not with Winter 1965.196, a change of o > e before Kᵘ in Hitt. For our purposes a root *nogᵘ- for Latin is assured.

[74] Not *nogᵂedos, with plain d, so Buck 1933.89, Juret 1938.38, E.-M.; W.-H. indicate "*nogᵘedhos (oder -odhos)", but the Gmc. points only to -o-.

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50) However, beside the outcome ū, there are several words which show a long ō as the product of original ou̯ŭ. Thus:

*fou̯e-mentom > fōmentum

*fou̯e-kolom > fōculum

*fou̯e-u̯ai > fōvī

*fou̯e-tom > fōtum

*u̯ou̯e-u̯ai > vōvī

*u̯ou̯e-tom > vōtus

*nou̯enos > nōnus

*cou̯entio > cōntiō

*mou̯e-mentom > mōmentum

*mou̯e-u̯ai > mōvī

*mou̯e-tom > mōtum

(For other examples, see §53-4, below[75]).

51) This second development has engendered several attempts to find a lautgesetzlich solution to the forms with ō versus those with ū. Debrunner (IF 54 [1936] 115) took ū < ou̯i, but ō < ou̯e (so too Pisani 1974.23). This, however, is falsified by many examples: *iou̯es-tos > iūstus (never

[75] tōmentum 'stuffing' presumably from a *tou̯e-mentom points to a lost *tou̯eō, with which one can plausibly connect tōtus 'vollgestopft' hence 'complete, whole' < *tou̯e-tos (so Sommer-Pfister 1977.126, Leumann 1977.133). This *tou̯eō would be the o-grade causative *tou̯h$_2$-eio- of the root *teu̯h$_2$- (h$_2$ on the basis of σάϝος > σῶς) seen in Skt. tavīti, zero-grade *tuh$_2$-i- > tuvi-, hence 'make strong; fill up/out'. However, whether to connect this with tŭmeō (op. citt.) is difficult to know. Skt. shows forms with ŭ: *tum-ló- > Skt. tumrá-, Lat. tumulus. These cannot simply represent a *tuh$_2$-C-, cf. *tuh$_2$-m-: OHG dūmo, OE þūma; perhaps a new quasi-root from mis-segmentation of ante-vocalic cases : *tuh$_2$-o- > *tuu̯-o-, recut as *tu-u̯o- or the like, or simply an anit byform?

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**iovistos)[76], etc. Solmsen (1894.82) wished to have oue in the Allegroform undergo syncope to ou > o̯ > ū, but in the Lentoform to remain oue with subsequent (though unparalleled) rounding to ouo, thence to o͝o͝ (with loss of u̯ / _ o) contracting to ō, citing VOOTUM (CIL 1[2].365)[77]. This, however, fails to take into account rūrsus where the outcome of even original ouo is ū. It also fails to explain why e/i should become o in medial syllables open or closed. Thirdly, the Lars Cotena inscription is hard to assign to Latin or Faliscan and the oo though "fremd" (Vetter loc. cit.) follows the tendency to write long vowels double (freq. in Oscan; in Latin from c. 135 - 75; see Leumann 1977:12); VOOTUM simply spells vōtum. Monteil (1975.70) has a variation on this idea, where owe loses its w regularly before e (no examples) to become oe which contracts to ō, but oui changes to ou by syncope and thence to ū. Sommer (1914.159) also held that ō was due to a contraction, and was the regular outcome. He explained the ū of nūndinae, nūntius as due to a shortening. Diom. 1.433.18 states that the vowel length (by the late fourth cent. A.D. in any case) was cŏntiō. This, however, shows the late secondary Ostoff's Law (see §6.24), and cannot be used as an explanation of the u in nundinae and nuntius, where Sommer imagines *nōndinae > *nŏndinae > nundinae by the irregular raising seen in fruntes for frontes, frundes for frondes[78], which even if it were correct does not explain the ū of other cases (e.g. prūdēns in a non-Ostoff's

[76] Paul. Fest. 93.12 lists a ioviste, compositum a Iove et iuste, but this of course is a folk-etymological pseudo-archaism.
[77] Lars Cotena; D175, Ernout 1957.37, Vetter 320. Cf. Solmsen in IF 31 [1912] 480 n. 2; so Stolz IF 13 [1902] 111 ff.; Rix 1966 = 1973.91 follows this for the outcome in ō; for the cases with ū, see above.
[78] Sommer here is following Thurneysen, see Sommer-Pfister 1977.59 and litt. So too in Juret (1938.70) who mentions nundinae as his only example showing u as the outcome; see following.

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environment). Rather, nūndinae, nūntius show the regular development of ou > ō > ū. Juret (1938.37) idiosyncratically imagines a regular loss of u̯ before short vowel plus apical (r, t, n, l), followed presumably (though Juret does not mention it) by a contraction of the vowels, lumping together a variety of changes[79]; Juret believes that d was not apical[80] and therefore had no effect on the w (the example given is pavidus)[81]. He does not mention here any examples of the outcome with ū (other than nūdus, which he does not explain), nor of loss before non-dentals (mōvī, etc.).

52) The correct explanation had been given as long ago as 1899 by von Planta (IF Anz. 10.56). The forms with ō owe it to analogy to other related words with o in the paradigm. Thus nōnus to novem, mōtus and mōvī to moveō, vōtus and vōvī to voveō, etc. (see above n.29). The process is that of retention of a stage of development under analogy to other forms. Thus the regular development is seen in: *nou̯en-dinai > noundinai (NOUNDINUM, SCB 186) > nōndinae (NONDINIUM, CIL 1^2.582.31, c.125) > nūndinae; but *nouenos >

[79] His examples are prōsa; deorsum; quorsum; Mārtis < Māvortis; nōlim, nōlem, nōlam < ne-velim, ne-vellem, ne-volam (how nevelim became nōlim is not answered); nōnus; nōnaginta; dītis < dīvītis, but dītior < dīvitior (not by loss of u̯ between like vowels); and concludes with a phonologically, morphologically, and etymologically outlandish "peut-être ātrium < *e$_2$ew-et-r-".

[80] Following (though no references are given) a number of testimonia of grammarians (Ter. Maur. 6.331.199-203; Mar. Vict. 6.33.24), who in the absence of the idea of 'voice' are attempting to explain the difference between t and d (and c/g, b/p) in terms of wholly illusory tongue positions. See Allen 1965.

[81] How exactly an apical (but not a laminal) is supposed to condition a loss of w is not stated. Nor how *nogu̯edos > *novedos is supposed to become nūdus. Nor does he mention any other examples of owV becoming ū.

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nounos (Paul. Fest. 1.76) > nǭnus --> nu̯ǭnus by analogy to novem, while nūndinae has lost its overt semantic connection with novem.[82]

53) The case of cōntiō is somewhat more complicated. It is usually taken from *co-u̯entiō, (cf. COVENTIONID, SCB.22) with ǫ due to other compounds in co- (so already Lindsay 1894:163). However, since the SCB shows that the syncope has already occurred, COVENTIONID must be an etymological spelling due to its easy analyzablility as a compound in co- (so in Paul. Fest. 34.1). Leumann (1977.133) says that the spelling shows rather co(n)ventio, perhaps then with a long ō. However, on p.226 he holds that co- is regular before u̯, comparing co-u̯iria > cūria. The SCB itself shows COMVOVISE[83], which must also be a re-compounding (Leumann loc. cit. 139). So too for the later conventum, etc.

54) This syncope rule may be stated as,

OW-Syncope: V̆ > ∅ /*(C)(C)ou̯ _ C(C)V,

that is a short vowel (we have examples for ĭ, ĕ, ŏ) is syncopated in words of three (or more) syllables after an initial -ou̯-. The syncope occurs only after an initial accented o, since the o of a medial -ou̯V̆- is not syncopated: *oino-u̯or(s)sos > oinu̯orsos (nom. pl. OINVORSEI, SCB.19), with regular weakening of V̆ > ŭ before u̯ in an open syllable (see §8.31 f.) --> ūniversus (by analogy to other compounds with ūni-). This syncope, which appears to be completely regular, is ordered as we have seen after eu > ou, and before ou > ō (i.e. it creates the closed syllable for monophthongization), and so before the loss of u̯ / _ o (see §7.2-5). This means that for prōsa (prōsa ōrātiō; see §5.4) and nōlō the change was not simply one of u̯ disappearing

[82] Cf. in English fourteen, fourteenth but fortnight < Mid.Eng. fourte(n)niht < OE fēowertīene niht.

[83] Which may spell u̯ou̯u̯isse, or since monophthongization has already occurred in one example (§7.1), u̯ōu̯isse.

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between like vowels (or before o in some accounts) and contraction (as if *neuelō > *nouołō > *nŏŏlō > nōlō) as is usually stated[84], but rather:

*pro-uorssos > *proursa (syncope)[85] > prōsa --> prōsa (by analogy to prō)

So too for nōlō:

*neuelō > *nouelō > noulō > nōlo --> nōlō (by analogy to volō).

Unfortunately one cannot tell if the syncope took place at the stage *nouelō or *nouołō (after e > o / _ ł) as both e and o were syncopated. However, for medial examples, where there was no syncope, the forms do show simple contraction: *intro-uorsum > intrōrsum, *retro-uorsum > retrōrsum (but *re-uorsom > *rouorsom > *rourśom > *rōrsom > rūrsum; reversus a new formation after the surface etymological connection of rūrsus to re- and vorsus is lost).

55) Analogy to forms with o explains the pair ōpiliō (to ovis) beside ūpiliō (the lautgesetzlich outcome). Ernout (1909:46) suggested that ūpiliō was urban, ōpiliō rustic. This may be correct, but there is no evidence specifically pointing to it[86], and positing a dialectal basis for ō/ū is unnecessary.

56) There are a small number of exceptions to this rule, unsyncopated forms that are clearly analogical or new formations. Thus Novembris, novendiālis will be analogical to novem (septem : Septembris :: novem : X); novissimus, novellus, novitās, etc. to novus (bonus : bonitās :: novus : X; or recompounded after the syncope was over and underwent regular weakening:

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[84] So E.-M. and W.-H. s.vv.; Meillet-Vendryes 1968.119, Monteil 1973.111, Maniet 1975.92; Buck 1933.275 'probably'; Kent 1945.182 "contraction" in *reworssos > rūrsus, but no statement of midstages elsewhere; Leumann 1977.119 but notes *rouorsos > rūrsus, see above. For Juret, see §51.

[85] Again of o in a closed syllable.

[86] The fact that Virgil (sua voce) uses ūpiliō, while Plautus puts ōpiliō in the mouth of a prostitute is not good evidence.

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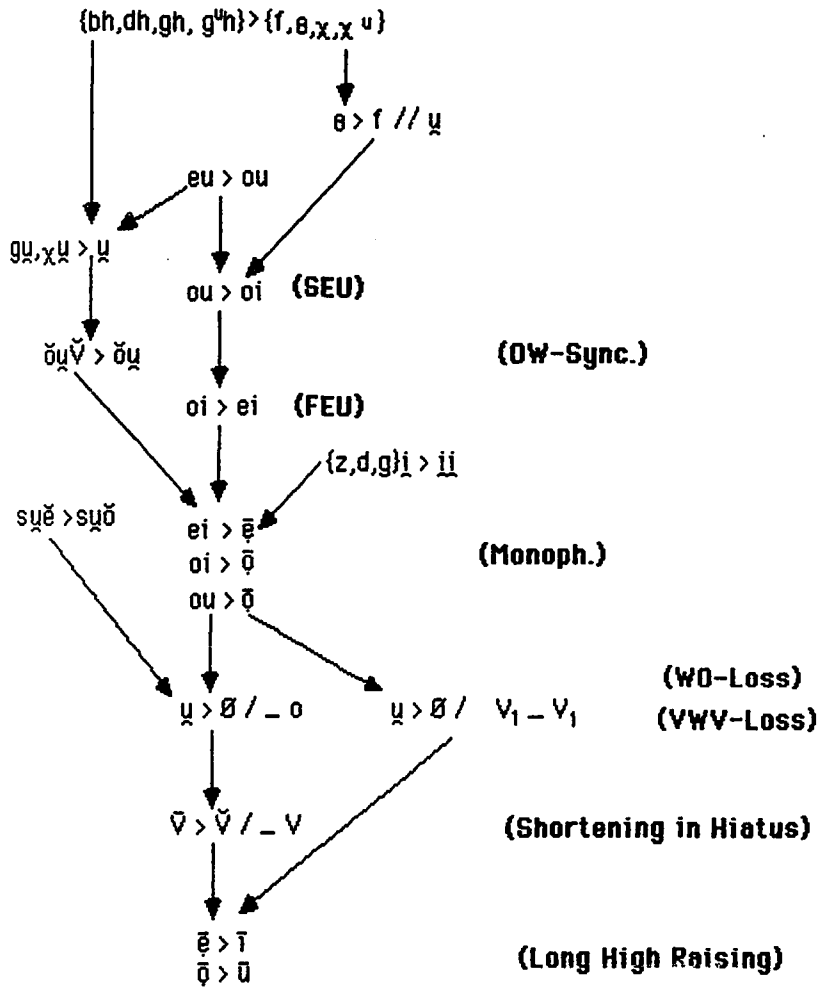
*novo-tās > novitas). Noverca is of unknown etymology. A connection with novus 'new' is likely for a word meaning 'stepmother', but the formation -erca has its only parallel in Lupercī and what it represents is unclear.[87]

57) We can now add **OW-Syncope** and its related rules to the previous chart and sum up diagrammatically:

CHART XIII

[87] See most recently Gruber Gl. 39 [1961] 274 who derives Lupercī from *lupo-sequo- 'wolf-followers' with dissimilation of p ... qu to p ... c. This is unlikely and even Gruber is forced to admit that this explanation will not work for a hypothetical *novo-sequa. Benveniste 1935.29 explains as nov-er-ca an expansion of a thematic *neu-r-o-, comparing νεαρός. Phonological problems aside, νεαρός shows a purely Grk. productive suffix, derived from cases of -ro- adj. built to set roots, or thematic adj. to old r/n stems: ὑδαρός, πιαρός. See Chantraine 1933.226.

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58) Parallel to ou̯V̆ > ou̯, Sommer-Pfister (1977:126) suggest that nōram < nōveram, nōstī < nōvistī could be the result of a ōu̯i/e syncope rule. However, these short forms of the perfect occur widely and in environments

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totally unrelated to VuV̆C and are morphologically motivated[88]. So not only nōrunt but amāvistī/amāstī, amāverunt/amārunt, and especially dīxistī/dīxtī, etc. The only other examples which might point to syncope after ō are compounds of pro-: *pro-uidēns > prūdēns, *pro-uorssa > prōsa. However, since Latin inherited both prŏ- and prō-, neither of these necessarily shows an original ō [89]. Leumann (1977.119) wishes to have *rouorsom > rūrsum by the regular syncope, but prōsa < *prō-uorsa with no syncope after ō and then contraction, but this explanation is not necessary to account for the difference in development. The numerous compounds of prō- (prō-vehō, -veniō, -videō) do not, of course, provide evidence that ōuV̆ did not syncopate. There is no a priori reason why ŏuV̆ should syncopate and ōuV̆ remain, but in the absence of any good examples, the rule will continue to be written without the vowel length indicated, and a long vowel to be the marked case.

59) A small number of cases could be considered as showing a change of ouV̆ in initial position, when not syncopated, to uuV̆ and thence to uu if the following vowel was o (with loss of u before o). So: CONFLOVONT, FLOVIUS (CIL 1^2.584, Sent. Minuc.) but regularly Classical fluō, fluvius; IOVENT (CIL 1^2.364, Faliscan), but Classical iuvō. However, comparison with moveō, etc., shows that this is not the regular development and it is clear that these cases are due to the simplex having been rebuilt to (abstracted from) the compound, where in medial syllables this change is regular (Leumann 1977.135; for medial syllables, see §8.31 f.). So fluō from con-fluō (already COMFLUONT in the Sent. Minuc.), etc. This same explanation has been applied

[88] Though a phonological starting point is likely with u > Ø between like vowels: audīvistī > audīstī, etc., and then spreading by analogy. Buck 1933.295.

[89] On prŏ- vs. prō-, see Leumann 1977.560-1 and litt.

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to suus and tuus but early souos (CIL 1^2.364, above;1211 Epitaph of Claudia; 583.50 Lex Repet.; no attestations of *touos), with souos, *touos weakened to suus, tuus by proclitic use[90]. However, the assumption that tuus and suus were proclitic is somewhat arbitrary, and it is not necessary to invoke medial weakening here. Beside PIE *seuos (ἑός, ἐϝός, Av. hava-) > Lat. souos (O. súvad, with ú) and *teuos (τεός, Av. tava-) > Lat. *touos (U. touer), there are also the forms *suos (ὅς, Dor. ϝός, Pam. ϝΗε, Skt. svá-, Av. hva-, x^{v}a-) and *tuos (σός, Skt. tvá-, Av. θβā), which have Lindeman's Law variants *suuos (Ved. suvá-) and *tuuos[91]. These can account for Latin suus, tuus (and U. tuer[92]), since it is clear that Italic inherited both *seuos, *teuos and suuos, tuuos from PIE.[93] The lack of *touos beside souos in OL is easily explained: tuuos was favored over *touos by the form tū, whereas souos/ suuos lacks a *sū. Once tuus is generalized it can then aid in the generalization of suus at the expense of sovos in Classical Latin.

60) PIE *au (mostly < h_2(e)u-) in general remains in Italic. Standard examples are:

*h_2(e)usōs (Grk. *ausuōs > Arg. ἀϝώς, Ion. ἠώς; zero-grade Skt. uṣás) > aurōr-a

*h_2(e)ug- (Skt. s-stem ójaḥ, αὔξω, etc.) > aug-eō, etc.

However, in non-standard Latin au undergoes the well-known

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[90] Leumann loc. cit., Sommer-Pfister 1977.87, Buck 1933.221, W.-H. s.v., first von Planta 1892.199-200, also 1896.233.

[91] Ved. attests only a single form of tvá-: fem. ins. pl. tvā́bhis; however, cf. the thematic compounds scanning tuvā-.

[92] O. suvam, tuvam in the Latin alphabet could reflect either o or u, but suveís on the Cippus Abellanus (native alphabet) is good evidence of u.

[93] Buck in 1904.140 accepted this, but primarily because he did not believe that ou became uu in unaccented syllables. He followed the majority view on ou > uu in 1933.102, but therefore also accepts souos > suus by enclitic weakening (1933.220).

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monophthongization to ō: so Claudius, but Clōdius after the lower-class pronunciation; aulla beside ōlla, cauda beside cōda, etc. Compare also Vespasian's famous hyperurbanism Flaurus for Flōrus (Suet. Vesp. 22). A similar change is regular in Umbrian (U. uhtur: L. auctor; ute, ote : L. aut, etc.; Buck 1904, 46). This change shows up in only a few words in Romance, where the dialectal form had been established in notably olla, coda. while the Romance reflexes show that au was still the regular pronunciation even in Vulgar Latin (Bourciez 1956.44, 155). In addition there are examples of decompositional -ū- in late clūdō (Tac. Hist. 1.33), etc. Not with Muller (Mnemosyne 58.97) reflecting a au/ū ablaut.

61) In summary then:

A) The low diphthongs ai, au remain relatively unaffected, undergoing at most a phonetic but not phonemic change (ai > ae).

B) The diphthong eu becomes ou at an early but still historical date in Latin.

C) This ou, in a labial environment, has its second element unrounded to oi.

D) This oi, in a labial environment, has its first element unrounded to ei.

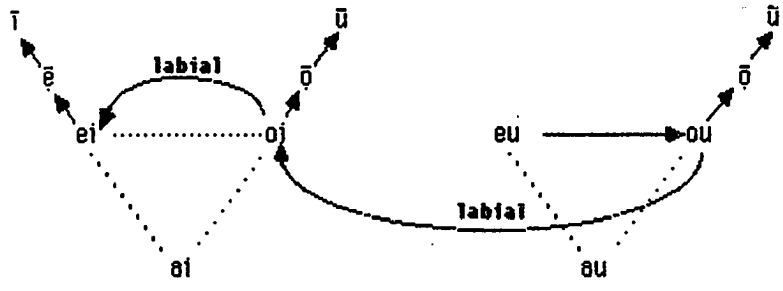
E) All mid-high diphthongs: ei, oi, and ou, of whatever origin, become long high monophthongs, the quality determined by the first element; ĕi̯ > ē, ŏi̯ and ŏu̯ > ō. Exceptions are certain cases of oi retained for lexical (and partially phonological) reasons, which then becomes oe.

F) The long high monophthongs merge with the reflexes of the original high vowels: ē > ī, ō > ū.

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This can be summed up in a type of flow-chart to illustrate the wandering of the diphthongs:

CHART XIV



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Chapter 8: Medial Weakening

1) Nowhere perhaps are the limitations of the handbook format more visible than in the presentation of Medial Weakening. Because of the necessities of organizing the data on a phoneme by phoneme basis, many generalizations are lost and distinct stages in the evolution are described solely by their beginning and end points, as if single processes. Thus one finds the simple statement that a > i in medial open syllables, and that a > e in medial closed syllables is mentioned elsewhere, with no attempt to relate the two[1]. Despite the limitations imposed by the handbook form, some scholars, notably Buck (1933) and Leumann (1977), provide excellent observations, sometimes by means of footnotes and cross-references, on various midstages. So Sommer-Pfister (1977.82), who are not sanguine about the possibility of establishing a relative chronology for Medial Weakening[2], still note, "So ist eine Zwischenstufe e zwischen a und i vor einfacher Konsonanten nur bei der Systematisierung der Schwächungserscheinungen erschlossen". Yet the systematization of the sound changes is precisely the criterion by which one must proceed. In additon, there is rather more inscriptional evidence, though limited, than Sommer-Pfister seem to imagine. I hope in this section further to disentangle the lines of change.

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[1] Niedermann 1953.23 for example, who borrows a poor choice of term from Hermann 1923.216, and calls the change "Umlaut". See Leumann 1977:79 for criticism.

[2] "Eine durchschlagende phonetische Begründung für die Vokalwandel kann nicht gegeben werden, ja es wird sogar ein Einfluß des Akzentes bestritten. Teilweise lassen sich auch nur die Angangs- und Endstufe, aber nicht die Zwischenstufe mit Sicherheit feststellen" (ibid.).

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2) In discussing Medial Weakening, the basic principle must be that of descriptive economy[3]. This is not merely a prejudice for solving problems by relative chronological means, but can be formulated as a type of Occam's Razor: phonological laws and midstages should not be postulated if a sound change is explicable in terms of previously known laws. As an example of failure to adhere to this Occam's razor, Godel in a frequently cited article (1961 = 1973.77) writes, "Pour certaines positions de a, toutefois, il conviendrait peut-être de supposer une fermeture vers arrière: a > å > o, ainsi devant [w]: *ablawō > ablowō > abluō." However, this seems unnecessary, since the changes of a > e and of e > o / _ u are already well established and together are sufficient to explain the passage of a > o (§14). A further example might be Pisani in KZ 67 (1940) 28f. who posits a direct passage of o > i in open syllables (so too Godel loc. cit.). Pisani's solution seems to have the advantage of proposing fewer midstages, but since a midstage of o > e is observable (*hosti-potis > *hospetis > hospes; *piotās > pietās, §21) and since e > i is a well established change, Pisani's proposed law does not meet the requirements for descriptive simplicity.

3) The phenomenon of weakening[4] of vowels in unstressed syllables is familiar from many languages (Eng. syllable vs. syllabic; Russ. nom. galavá vs. acc. gólavu). In Latin, as is well known[5], the PIE mobile accent was shifted to a fixed accent on the initial syllable[6], with effects seen primarily in syncope and weakening in the unstressed syllables following

[3] For recent statements, see Sommerstein 1977.16, 25-6; Dell 1973.266; Hsieh Lingua 1976.1-19

[4] The term is used without prejudice as a calque on the German "Vokal-schwächung." The process is one of primarily raising. See Lass and Anderson (1975) for theoretical considerations using OE data.

[5] First by Dietrich 1852.543 ff. in the first issue of KZ.

[6] A similar change in Czech.

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the accent. I intend to focus primarily on changes in medial syllables. The changes in final syllables are greatly affected by paradigmatic and morphological influences and, though important, because of their position are less useful for establishing a relative chronology.

A-Weakening

4) The first stage in the process of medial weakening was the raising and fronting of the low vowel -a- to -e- in all unaccented syllables, open or closed[7]. This is the only change due to weakening in closed syllables, other than the later raising of o to u. This change can be dated very early on relative-chronological grounds, as the e it creates feeds a number of other laws which have been independently established: **Monophthongization**, **N+Spirant Lengthening**, e > o /_ ɬ, e > o /_ u̯, e > i /_ ŋ (see below). Further, we can "catch" the e midstage of a > e (> i) when it follows an i (§8).

5) We have in addition some indicators of absolute dating. First are Greek words borrowed presumably as early as the first contacts with Magna Graecia, which show that the weakening of a > e cannot be placed much earlier than c. 500 B.C., providing a terminus post quem[8]. Thus we have ἐλαίϝα --> *élaiva > *óɬaiva > *olēva > olīva ; τάλαντον --> *tálantom > talentum; or for an example in an open syllable, μαχάνα --> *macana > *macena > mac(h)ina. Secondly, the weakening of a to e is not found in Faliscan (peparai, Vetter no. 241, cuncaptum CIL 1^2.365). Unfortunately, because of the scarcity of the remains there are no good examples of medial

[7] The communis opinio of the scholars who have given attention to the matter of midstages: Leumann 1977.84-5, Buck 1933.100, Sommer-Pfister 1977.83.

[8] For example the name of the city of Massilia, not founded until c. 600: Μασσαλία --> *Masselia > Massilia. See Leumann 1977.84; Stolz-Debrunner-Schmid 1966.51.

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-a- preserved in the earliest Latin inscriptions. Leumann 1977:84, citing Stolz IF 18 (1905/6) 475,[9] Altheim 1954.302, gives the -cant- of incantassit, occantassit versus the occentassint of Paul. Fest. (196.12 Th.) as examples of medial a still intact in the Twelve Tables, which he dates to c. 450. However, the fact that the regular Classical form is incantāre with the ă unchanged due to compositional analogy to the simplex considerably weakens this as evidence. The concapit of the XII Tables (Paul. Fest. 556.27) versus Classical concipit is perhaps better evidence, but this may also be analogical to the simplex or an archaism of either the text or the tradition. Sommer-Pfister (1977.83) rightly point out that ALIXENTROS (CIL 1^2 557) is not a good example for early a > e. This set of inscriptions is uncertain as to date (4th c. probably, but difficult to be more precise) and are on Etruscan-work mirrors found at Praeneste and therefore may reflect not archaisms, but dialectal peculiarities[10], not least of which is a very open (high) e , often written i before two consonants; so MIRQUIRIOS in the same inscription, or aciptum in the Faliscan Iovei inscription (CIL 1^2.364). Thirdly, the form is likely to reflect not an Italic vowel change, but the Etruscan Alixsantre, as indeed the voiceless -t- would indicate. There are a few early inscriptional examples of a medial a having weakened to e in both closed and open syllables. So for closed syllables: *in-caid- > INCEIDERETIS (SCB.26; c. 186), INIECT[I]O (CIL 1^2.401, date uncert.); for open syllables PROIECITAD (from the same inscription), CONIECIANT (CIL 1^2.583.51, Lex

[9] Earlier still in Lindsay 1894.198.
[10] Though speaking generally the dialectal features of Praeneste seem to reflect simply a more archaic state of the language than that of urban Rome.

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Repetundarum, c.123)[11]. Sommer-Pfister (loc. cit.) write, "Der Zeitpunkt des Lautwandel [a > e] ist schwer zu bestimmen; jedenfalls setzt er den Eintritt des Anfangsakzente voraus, kann aber (2-3 Generationen?) später liegen." However, it is impossible to be precise about the date of the change to initial accent[12], and the assumption of 2-3 generations is arbitrary. Still, to date the onset of Medial Weakening of a > e to after c. 450, but before the inscriptions of the third century would not be unreasonable[13].

6) The weakening of a > e occurs in all medial syllables, open or closed, and is not dependent on the following consonants. Thus, before -ll- (bright l), -ɬC- (dark ɬ), -jj-, and -CC- in general. Also before -iC- and -uC-, i.e. as part of diphthongs (§13). If the syllable is closed the e remains, subject to no further weakening (but does feed various other changes mentioned, §11 ff.). If in an open syllable this e, along with original e_1 and e from the medial weakening of o, i and u in open syllables, is raised to i or to u in labial enviroments. Examples in Closed Syllables:

castus	but	in-cestus
faciō		per-fectus
cantō		con-centus
annus		bi-ennis
armus		in-ermis
fallō		fe-fellī

[11] Here because of the late date, while earlier inscriptions show a > i already completed (ANIMAM: CIL 1^2.586, ad Tiburtes, c. 156) it may be that the text is archaizing, or else that a preceding j hindered the raising of e to i. So apparently Leumann 1977.82.

[12] Leumann 1977.246

[13] The NUMASIOS and FHEFHAKED of the Praeneste Fibula are forgeries: Gordon 1983.75-6 and litt.

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*Pompaiianos (cf. Osc. Púmpaiians) > Pompeiianus.

Examples in open syllables:

agō	*exegō	exigō
canō	*cecenei	cecinī
iaceō	CONIECIANT	coniciō[14]
saliō	*deseliō	desiliō[15]

We can now write the first link in our chain:

1) Shift to Initial Accent

2) a > e /V(C) _ C(C)(V) (**A-Weakening**; c. 400?)

7) For several other stages of weakening, as will be shown below, a following vowel prevents the change, e.g. the regular raising of -e- to -i- is prevented by a vowel in *mon-eiō > moneō (and no further change to **moniō), *aus-eio-s > aureus (and not **aurius). Leumann 1977:81 notes, however, that for -a-, "keine Belege für Stellung vor Vokal." A reason for this might be found in an early contraction of a with the following vowel. For ā there are many examples; so in first conjugation: *cūrā-iō (denominative) > cūrō, etc., or *magisu̯olō > *maksu̯olō > *mazu̯olō > *māu̯olō > *māolō (**WO-Loss**, §7.24) > mālō (§4.4). However, for ă there are no certain examples, though the handbooks usually list the contractions in the form ă + V. The likely sources for such a contraction would be presents of set roots, particularly any examples of the original athematic inflection changed to a -iō present, but none of the possibilities are without

[14] Spelling coniiciō. See Leumann 1977.128. For <u> spelling -u̯u̯-, see §31. Not with Niedermann (1953.39-40), a syncope to conīciō.
[15] With plain l. For velar ɬ, see §15.

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difficulties. The present stō, comparing U. stahu is frequently taken as reflecting a *sth$_2$-(i)e- (Leumann 1977.120, 531-2; Buck 1933.268). However, this is far from certain; for a complete review of the previous literature see Cowgill JIES 1 (1973) 271-303 (who argues for *stă- plus the stative -ē- < -eh$_1$-)[16]. Cowgill (1962) explained the o-timbre of domāre as reflecting perhaps an old iterative/causative and likely to rest on a *domh$_2$-(e)iō > *domaiō > domō, *doma-je-s(i) > domās, etc., comparing *domh$_2$-to-s > domitus. However, this analysis is somewhat arbitrary, and the verb might belong to the type seen in ποτάομαι (Rix 1976.202). For PIE this formation is usually in complementary distribution to nasal-infix presents (seen for this root in Grk. δάμνημι, though of course secondary formations are common) and for a first conjugation verb a factitive (of the novāre, νεάω, newahh-type) is far more likely: *domh$_2$-eh$_2$-(iĕ)-se > domāre. Latin shows no other forms of the root except with an o-vowel. Watkins (1965.185-6) explains the o-timbre as due to spreading within the special group formed by the disyllabic roots of vomō (*we > wo), molō (el > ol), sonō (swe > so) and by analogy tonō and domō (see Cardona's review in Lang. 37 [1961] 413-24 and Watkins's reply op. cit. 188). A further possibility might be arāre, if it reflected the secondary -iō present seen in Olr. airim, Goth. arjan, Lith. ariù, OCS orjǫ. However, arō is as likely to reflect the thematization of the original athematic *h$_2$érh$_3$-ti (cf. the characterized s-aorist in Grk. ἀρό-σαι). The thematization is parallel to that of the nasal infix roots *tolnati > *tollet(i) > tollit; see §2.5 n. 3, 7.11 n. 12). We should expect then *h$_2$erh$_3$-ti > arati > **arit. Watkins (op. cit. 186) convincingly explains the inflection by the preservation of medial a due to the preceding

[16] This idea is not without its difficulties, but a full discussion is beyond our immediate concerns.

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a (the type anatis, etc.; §19). The morphologically aberrant *arat(i) then was taken over into the regular first conj. arāt (Plaut. Asin. 874) > arat.

8) It is sometimes claimed (Leumann 1977.81, Sommer-Pfister 1977.83, Kent 1945.99) that the midstage of a > e (> i), like the midstage of o > e (§21), is preserved after an i (no examples for any other vowel), but neither of the examples given for open syllables are particularly good. Leumann (1977.81) offers variegāre. This is not attested until Apuleius, and it may reflect a new formation *vari-igāre (since -igāre has become a moderately productive suffix; Leumann 1977.550) > variegāre[17], rather than an original *vari-ag-. Kent (loc. cit.) gives hietō as an example. However, there is nothing to indicate the existence of a *hiă- in Latin. Rather, from a PIE standpoint we have a root *ĝhi(e)H-, with a zero-grade seen in *ĝhiH-skō > Lat. hīscō, OE giscian, or the reduplicated pres. Skt. (vi-)jí-hī-tē. There is also a nasal infix *ĝhi-ne-H-, zero-grade *ĝhinH- shown by the ī-forms in Gmc.: Olce. gine, OHG ginen, OE ginian, and the secondary nasal suffix type build to the zero-grade of the root (Skt. krī-ná-ti type) in the Gmc. ī-forms: Olce. gīna, OE tō-gīnan (Strong V); OCS zinǫ. There is also a stem *ĝhiiā-, taken to be the full grade as if *ĝhieh$_2$- (so Leumann 1977.535, W.-H.)[18], which should become Late-IE *ĝhiā-. This would give Lat. hiā-re but not the BS or Gmc. forms and the Latin is far more likely to

[17] Less likely to reflect the weakening of the thematic vowel from a vario-gāre (Kent loc. cit.). Even though -gāre is productive, it arises from the syncopated cases (pūrigāre > pūrgāre) and unsyncopated pūrigāre, iurigandum are still found in Plaut. (Merc. 739, 119; Trin. 68, 70). See §25.

[18] The relation of this, if any, with the i-less root *gh(e)h$_2$- seen in χά-σκω, χάος is problematic (*ĝheh$_2$- with an extension: *ĝheh$_2$-i-, whence a secondary metathesized zero-grade *ĝhih$_2$-, whence a new full-grade *ĝheih$_2$- ?). See Beekes 1969.178, 183. However, it is the existence of these Greek forms with ă that leads to the positing in the pre-laryngeal works of an ā/ă ablaut in this root.

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reflect a factitive in -eh$_2$-. In addition a vowel following the laryngeal seems the only way to explain the intervocalic yod (*ĝhiiā- *not ĝhiā-). So a *ĝhiH-eh$_2$-(iō) > *ĝhiiā- will yield Lat. hiō, Lith. žió-ju, cf. OCS iterative zijajǫ, zijati (not *ĝhiā- > **ž-), while a stative will give the Weak III class of OHG gīen < *giiē- (as if Goth. *gijan, -aida) < *ghiiē- < *ĝhiH-eh$_1$- (cf. Brugmann 1897.285). Thus, there is no source for a *hiă-tō, since *ĝhiH-C > Lat. hī- and the factitive *ĝhiH-eh$_2$-C > hiā-. Again the most likely source for hietāre is from the root hī- plus the productive -itāre: hīitāre > hīītāre > hietāre (§21). Because the e from o is preserved after i (§21), it seems likely that the e from a in an open syllable would be also. Latin simply lacks words of the proper shape. However, clearly showing the midstage of e preserved after i is *ámbi-agnos > amb(i)egnus 'a cow or wether for sacrifice, flanked by two lambs' (Leumann 1977.81). Note that the -e- is not raised to -i- before the velar ŋ, thus a preceding i blocks **Rais-iNG** (§12), but it is not clear from this example whether a preceding i would block the change of e to i in an open syllable, nor whether what prevents the raising is any preceding vowel, or perhaps the palatal quality of i prevents the raising to a further palatal. Yet ambiegnus does allow us to say that **A-Weakening** occurred even after a vowel.

9) The other change which occurs in closed medial syllables usually placed under the heading of Medial Weakening is the raising of o > u. This, however, is not to be treated on the same level as the weakening of a > e, since o > u occurs much later and indeed not until after the Senatus Consultum de Bacchanalibus (186 B.C.), where o in closed syllables is still preserved OINVORSEI (§31), cf. TRIUMPHANS CIL 1^2.626, c. 142 B.C.[19], while

[19] Of Mummius, cons. 146.

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still CONSENTIONT in CIL 1^2. 9 (c. 200)[20]. Further, o is not raised until after the late change of o > e /u _ rC: so OINVORSEI (SCB) but later ūniversus (not **ūnivursus). That is: closed medial weakening affects only -a-. We can therefore specify that a > e occurs in all medial syllables regardless of the following consonants and even if preceded by a vowel. Thus:

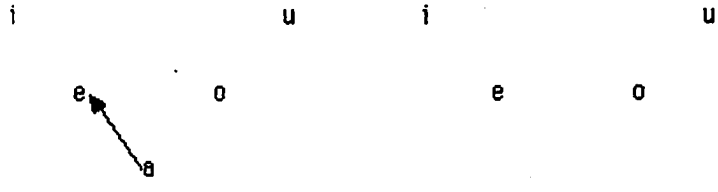
A-WEAKENING: a > e / V(C) _.

Standard examples of e, i, and u unchanged in closed syllables are:

e	congestus
	scelestus
	funebris
i	magister
	antistes
	(ven)-is-se
u	vetustus (see §9.3)
	amb-ustus

10) The first step in Medial Weakening, therefore, is the transformation of the vowel triangle from:

STATE I to STATE II in all medial syllables.



11) We can now order previously established rules which are fed by the change of a > e, i.e. rules which occurred also in initial syllables, and so

[20] The CONSULTO of the epitaph of G. Poplicius Bibulus, trib. 209 is due to a later recarving; see Warmington 1940:8 (no. 12).

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independently of Medial Weakening, but which apply to the outcome of a > e and so are later than **A-Weakening**. First, e > e feeds **N+Spirant Lenthening** (§4.25), that is a medial ă is weakened to ĕ and this ĕ is only then lengthened to ē.[21] Examples are:

*in-cand-tos > *incenssos > incēnsus

*ad-scand-tos > *ascenssos > ascēnsus, etc.

For ă > ā earlier than A-Weakening, see §20.

12) Further, a > e feeds e > i /_ ŋ (**Rais-iNG**):[22]

*ad-tangō > *attengō (closed syllable, so e should remain) > attingō

*con-frangō > *confrengō > confringō,

*con-pangō > compengō > compingō, etc.

As has been noted above (§4.24-5), e > i /_ ŋ (**Rais-iNG**) precedes **N+Spirant Lengthening**, as shown by *kᵘenkᵘ-noi > *quiŋχnoi > quīnī (and not **quēnī). Thus we can expand the ordering:

i) a > e / V(C) _ **(A-Weakening)**

ii) e > i /_ ŋ **(Rais-iNG)**

ii) V̆ > V̄(n) /_ [+nasal][+spirant] **(NS)**

As the example of biegnus shows, the rule of e > i /_ ŋ must be written so that a preceding vowel (at least i) blocks it:

Rais-iNG: e > i /{C,*} _ ŋ; see §6.17, 25.

13) **A-Weakening** also feeds **Monophthongization**. That is -aiC- and -auC- act like any other closed syllable -aCC- and become -eCC-.[23] Then the new e-diphthong created is subject to the same monophthongization as original -ei- and -eu-. Examples with original -ai- are:

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[21] Noted by Leumann 1977.81, 112.

[22] Noted by Leumann 1977.45, 83; Sommer-Pfister 1977.83; Buck 1933.100.

[23] In generative terms both C's are specified as [-vocalic]. Thus a > e even before -ii-.

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*in-caidō > inceidō (INCEIDERETIS, SCB.26) > *incēdō > incīdō

and the familiar pair:

*elaiu̯om > *oɬeiu̯om > *olēu̯om > oleum

*elaiu̯a > *oleiu̯a > *olēu̯a > olīva.

However, as the example of Pompeiianus shows, ei before i was not subject to **Monophthongization** in medial syllables any more than initial syllables (§7.13).[24]

14) For -au- there is a further step interposed before **Monophthongization**. That is, the newly arisen -eu̯- is rounded to -ou̯- in open or closed syllables, and this -ou̯- if in a closed syllable then undergoes monophthongization to -ō- (parallel to ei > ē), together with PIE *eu and *ou > -ō-, and then is raised to -ū- (parallel to ē > ī), a change which has already been established for medial as well as initial position. This rounding is, of course, not the same as the early change of PIE *eu to *ou in Latin and Faliscan in all positions. This new change of -au- to -eu- and thence to -ou- is found only in medial position, i.e. it is created and fed by **A-Weakening**.[25] All examples of original *Ceu- had of course already become ou by the early Latin change of eu > ou. There is no source of new e in initial syllables, but we can see that this second application of eu > ou does not apply to initial syllables from the examples with new u̯$_2$ (< gu̯, gu̯h): brevis, levis. The new u̯ arises necessarily in open syllables (since Ku̯C loses its u̯-component). For the case of brūma with closed syllable, see §7.37-38 and

[24] Cf. the lack of monophthongization in Umbrian of aii, eii: pernaiaf, Teteies. Buck 1904.42-3.

[25] So Leumann 1977.91, Sommer-Pfister 197787, Meillet-Vendryes 1968.126, Niedermann 1953.72, Buck 1933.102 ("probably"), and is a necessary, though unstated assumption of Lindsay 1894.186

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below §18. Examples of eu > ou with e from a by **A-Weakening** include, in closed syllables:

*in-claudō > *incleudō > *incloudō > *inclǫdō > inclūdō[26]

*dē-fraudō > *dēfreudō > *dēfroudō > *dēfrǫdō > dēfrūdō,

and in open syllables (see further, §31 f.):

*dē-paui̯ō > *dēpeui̯ō > *dēpoui̯ō > dēpuviō

Thus, we shall temporarily establish a rule of

New-OU: *e_2 (from a > e) > o / V(C)(C) _ u,

which is necessarily ordered after **A-Weakening** and does not occur in initial position, and shall later explore how it is to be incorporated into the rules that have been established. For the cases of eu̯V > ou̯V, see §31. The possibility should be raised at this point that instead of a monophthongization of -ou̯C- to -ǭC- (> -ū-), the sequence was that of any other -oCC- > -uCC-, that is -ou̯C- (of whatever source) became -uu̯C- > -ūC- by the raising of o > u /_ CC. However, the change of o > u /_ CC is post SCB (186 B.C., DINVORSEI), where monophthongization is already under way (COMPROMESISE). **Monophthongization** then is fed by original -ou̯-, and both early -eu̯- > -ou̯-[27] and new -eu̯- > -ou̯-, by both PIt. ei̯ and new -e_2i̯- (< -ai̯-).

15) The reduction of a > e also feeds another rule of rounding, that of e > o /_ ł. Examples we have already seen in closed syllable are saltō beside *in-saltō > *insełtō > *insoltō > insultō; alō beside *ad-al-tos > *adełtos > *adoltos > adultus; for open syllables *ad-alēscēns > *adelēscēns >

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[26] (ex)plōdō naturally is not an example of the ǭ stage retained, but is built to plōdō, the popular pronunciation of plaudō (§7.60). See Leumann 1977.72.

[27] ABDOUCIT (CIL 1^2.7, c. 200), POLOUCES (CIL 1^2.548), POLOCES (CIL 1^2.549, Praeneste), CONIOURAS[SE (SCB), INDOUCERE (CIL 1^2.586, ad Tiburtes, c. 156).

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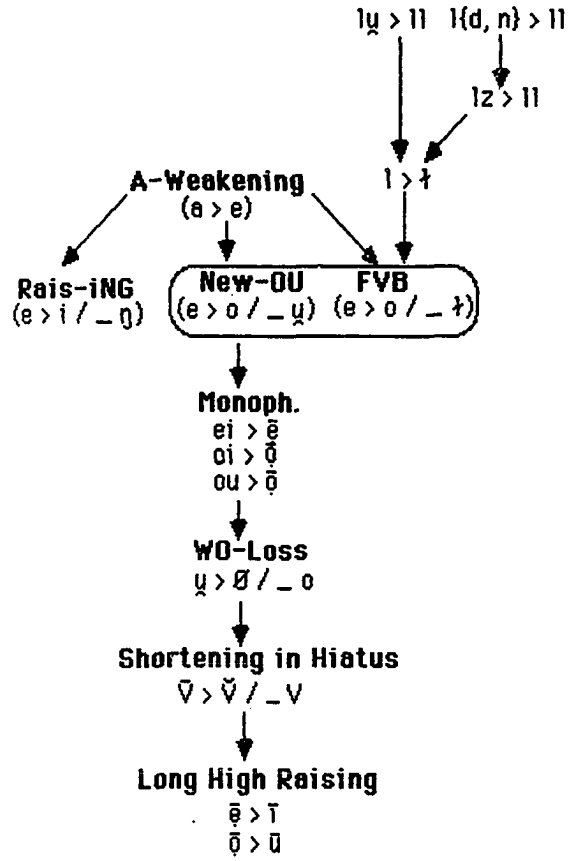
*adonsēscēns > adulēscēns. Leumann comments here (1977:86-7): "die Entwicklung vor ɬK war also entweder alK > elK > eɬK[28] > olk oder direkt alk > aɬk > olK (ohne Zwischenstufe eɬK)."[29] However, this misses the very important point that aɬK does not become olK in initial syllables (altus, calcō, etc.), while eɬK does (*eɬkos > *oɬkos > ulcus) and that the difference in treatment must then be due to the syllable being unaccented and therefore to a medial stage of -e-, which feeds the backing rule of e > o /_ ɬ. Since the various rules of assimilation to -ll- bleed the change of l > ɬ, we can expand the ordering:

CHART XV

[28] There is, and can be no evidence for assigning l > ɬ at this point, since our only knowledge of the velar quality of the l, apart from the grammarians' comments, is precisely this change of e > o. The most that can be said is that the change of l > ɬ feeds (is previous to or still concurrent with) the change of e > o. How long ago ɬ before a consonant became velar, we cannot tell. Generally I have simply specified the ɬ in the stage immediately preceding the change of e to o for the sake of clarity.

[29] This is similar to the idea of Godel criticised above (§1).

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16) We have, beside the inscriptional evidence for the o stage of e > o /_ ɬ (and only later to u), i.e. HERCOLEI (CIL 1².607, 217 B.C., etc.; see §1.2, 8.30), the o stage retained after a vowel (examples for i, and e) and not subject to the later raising of o > u /_ ɬ (see below). Thus, principally in diminutives:

*fīli-elos[30] > fīliolus

[30] For the thematic vowel of the dim. being -e-, see §2.19: cf. the double dim. *pork-el-elos > porcellus

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alveus but alveolos[31] 'little trough'

Puteolī, i.e. 'the little wells'

17) We need now to reiterate a development treated above (§6.4). The dim. PIE *pō-tle-lom > PIt. *pōklelom > *pōk|lom (sync. 3) > pōcillum, with -i- retained in a medial closed syllable, shows that the outcome of *l_2 was -il-. So too in the superlative to facilis: *fak̇lisamos > *fak|zamos > facillimus. From the same stem we have the abstract *fakli-tāts > *fak|tās > *fakiltās. This -i- from -|- should also have been retained, unweakened, in a medial closed syllable, but instead we find *fakiltās > facultās, with no chance of deriving the u from i by medial weakening: *fakiltās > **fake|tās > **facoltās > facultās. That is, the dark -|- backs the -i- directly to -u- without an intervening stage of -e-. Thus, the rule of **Front Vowel Backing (FVB)** is independent of Medial Weakening (§6.2-4). In the vowel triangle there is a movement:

CHART VI (repeated)

i ------> u

e --> o

(a)

There are no examples for original i before | in a open syllable (Cowgill 1977.41). Maniet (1975.127) claims that i was unaffected by a velar |: "pestilēns, mutilus, nūbilus, avec i ancien conservé." A search of the dictionaries will reveal more examples: aquila 'eagle', aquilus[32] 'dark',

[31] While alvus probably comes from a metathesis of *aulos (αὐλός), see §3.6, whence alveus, I'm not certain what the suffix reflects: presumably not not the -éio- material adj. suffix that underlies most Latin words in -eus.

[32] Rare; only in Paulus-Festus 22, Plaut. Poen. 1112, Suet. Aug. 79, Arnob.3, Apul. Socr. 10.

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aquilō 'north wind', aquilex 'water inspector', cartilāgō, iūbilāre, rutilus, sībilus, ventilāre, and their derivatives. A glance at these forms shows that they are a heterogeneous collection and that none of them is absolutely sure to show an original i, which according to Maniet has remained. The anomalous i before l followed by a vowel other than i that they do show, admits equally various explanations and I propose to deal with them very cursorily and refer the reader to the etymological dictionaries for the proposals that have been made. For the most part they seem to represent later (and sometimes opaque) morphological alterations, akin to the spread of the -i- union vowel (e.g. oinuversos --> ūniversus), after ł had ceased to affect vowel quality, and do not in any case represent counter-examples to backing before an old dark l.

aqui-lex: late compound (Tert., Servius) formed after the regular -i- composition vowel from Medial Weakening.

aquilus, aquilō and aquila: unknown etymologies; speculations on their relation with each other (aquilus 'water-y' --> 'dark', aquila 'the dark bird, aquilō 'dark/watery wind' so W.-H.; see E.-M. for a more sceptical position), if any, is pure folk etymology, which may have influenced the forms.

cartilāgō: root etymology and formation obscure.

mutilus: again obscure; there is no trace in Italic of an inherited -ilo- formant (see Leumann 1977.311; though some Latin formants in -ulo- could reflect this phonologically) unless mutilus and aquilus represent it.

sibilō (sifilō), iūbilō are imitative. Neither is likely to carry an inherited i.

nūbilō (Varro 1.13.5, Cato 88.2) to which is built nūbilus (Aug. poetry and post-Aug. prose)[33] and ventilō. These are part of the rustic vocabulary and a

[33] Not the other way around as usually seems to be supposed.

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dialectal explanation is possible. The productive suffix is -ulō, perhaps with Leumann (1977.551) spread from the lexeme speculō, denominative to specula, and ventus: ventilō in the face of nidus: nidulō, etc. is odd. Ventilō seems to be built as venti-lō with the spread of the -i- union vowel.

pestilens (back formation from pestilentus, Leumann 1977.336) in the face of fraus: fraudulentus, ops: opulentus, pūs: pūrulentus, etc. is clearly built to the nom. pesti-s; so too corpulentus (see Leumann loc. cit.)

rutilus is a dialect word apparently showing "Ausonian" t from *dh (cf. *rudhro-, Leumann 1977.171). The tribal name Rutulī seems to have been borrowed early enough to undergo backing.

18) We have two rules: 1) **New-OU**: the reapplication of e > o /_ u for new -ĕ$_2$- arrising out of -e- by **A-Weakening** and 2) e > o /_ ł by **FVB**. Can these be combined in a single rule of backing before back consonants? Above (§7.35), the possibility was considered that **FVB** could be combined with the early change of eu > ou. If so it is necessary to assume that **FVB** has reapplied or continued to apply. There are several problems with combining these rules. The first is that **FVB** as formulated occurs in all syllables. So:

1) initial a) closed: *elkos > *olkos > ulcus

b) open: *uelō > volō

2) medial (with ă > ĕ) a) closed: *insaltos > *insełtos > *insoltos > insultus

b) open: *adalescēns > *adełescēns > *adołescēns > adulescēns

but **New-OU** does not occur initially in open syllables as shown by brevis, levis (with u$_2$ < *gu, *guh), but does in the newly created closed inital syllable of *breu(i)s(a)ma > *breuzma > brūma. There seem to be two

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possibilities for restricting the power of this rule so that it will not apply to brevis, levis, both of which show that **New-OU** has only a partial overlap wiht **FVB**. One is to limit the change of new eu > ou to closed syllables. It will then not apply to brevis and a single rule will account for *breu̯(i)s(a)ma > *breu̯zma > *brou̯ma > *brōma > brūma and *incleu̯dō > *incleu̯dō > *incleu̯dō > *incloudō > *inclōdō > inclūdō. However, if **New-OU** is limited only to closed syllables, it is necessary to propose a separate but structurally similar law, differing only in that it applies to open syllables (see §31 f.):

a: *dēpau̯iō > *dēpeu̯iō > *dēpou̯iō > dēpuviō;

e: *senāteu̯ei > *senātou̯ei > sentātu(u̯)ī;

o: *dēneu̯ōd > *dēnou̯ō > dēnu(u̯)ō;

i: *tridiu̯om > tridu(u̯)um.

The other solution appears to be the better one: to limit **New-OU** to medial syllables (open and closed) only. Thus the rule will not apply to brevis, levis; equally, it will not apply to brūma. However, *breu̯zma (> brūma) is unique in the language, since it is the only case of a new eu̯ (with original e_1 but $u̯_2$) in a initial closed syllable, since $u̯_2$ can only arise in a open syllable (see above). Therefore, brūma can best be accounted for by a rule that applies uniquely to it, and indeed it difficult to see what else a *breu̯zma could have become in Latin.

If then **New-OU** is confined to medial syllables it cannot be combined with **FVB** which occurs in all syllables. However, we seem to missing the generalization of the similar behavior of e > o before u̯ and i̯ in both open and closed medial syllables. Because of the lack of crucial examples in the language the relationship between the rules is difficult to prove, but I would argue that the possibility with the greatest explanatory power is that

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the changes of original eu̯ > ou̯ and eɬ > oɬ in all syllables are to be combined and a later rule (crucially after a has weakened to e) of {i, e} > {u, o} before u̯ and ɬ accounts for the new medial cases. This accords well with the historical data. The changes of initial eu̯ > ou̯ and eɬ > oɬ are early: eu̯ > ou̯ is over by the time that u̯$_2$ arises from g^{u}, g^{u}h (levis, brevis) and the change of eɬ > oɬ is over by the time of the syncope of V̆ / V̆R _ u̯ (helvus). This combination has the advantage of not predicting a change of i > u / {ɬ, u̯$_1$} in initial position, which despite the lack of clear examples seems counter-intuitive, whereas the change of i > u before ɬ in medial closed syllable (facultās, see above) is assured. If we wish to combine **New-OU** and **FVB** as a rule applying to medial syllables, because the backing before dark ɬ affects both front (non-low) vowels, e and i, the same would be expected of the backing before u̯. A good example of i being backed to u seems to be bīduum, trīduum 'period of two/three days'; so: *tri-diu̯om (cf. Skt. tri-divam) > *triduu̯om (directly with i > u) > trīduum[34] (with u before a vowel spelling -uu̯-, see below §31). However, the objection could be raised that trīduum might equally be from *tri-diu̯om > *tri-deu̯om (-i- > -e- in an open syllable) > *tri-dou̯om (new e > o /_ u̯) > trīduum (Medial Weakening before u̯); parallel to *de-pau̯iō > *depeu̯iō > *depou̯iō > depuviō. For arguments against this line of development; see §32 f. What might provide additional help would be either a verb with a stem ending in -gu- with a perfect in -ŭī, so that we could see whether the development had a midstage with -e- > -o- which might remove the -u-, as if there were a **cogu(i)ō, perf. **cogue-u̯ai > **coguo-u̯ai > **coco-u̯ai > **cocuī, or else an i before a u in

[34] With unetymological long ī, spread from postrīdie (so Wackernagel Gl. 2 [1910] 1 n. 1 = Kl. Schr. 1969.883), as well as by conspiracy from analogical trīnī, and forms in bī-.

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a closed syllable (as if a **-civtus) where the u could not have undergone weakening to e (parallel in structure and development to *fakiltās > facultās). Unfortunately the requisite verb does not seem to occur in Latin, and a sequence **-CiuC- does not appear to be phonotactically possible in any stage of Latin. For -eɬC- and -euC- we have good examples (insulsus, inclūdō); for -il{C,*}, the family of facul, facultas; for -iuC-, no examples; for -eɬV-, many examples (Siculus, etc.); for -iɬV-, no good examples (Cowgill 1978:41; see §17); for -iuV-, the only example seems to be trɪduum. Thus, we may write {i, e} > {u, o} /_ ɬ (**FVB**) and e > o /_ u (**New-OU**) together with i > u / _ u (trɪduum) as a single rule for medial syllables merely by removing the specification [+lateral] from the Structural Description of **FVB** (§6.3), and the formulation will cover all the occurring new medial cases, while we can combine the cases involving original e in a rule of e > o / _ {u, ɬ}. We cannot prove that the medial outcomes are due to a single rule rather than very similar processes, all of which, however, must have occured after **A-Weakening**, but not necessarily at the same time. However, the similarity of the results of weakening before uV and ɬV (see below) argue for their being due to a single rule, and we seem to be missing an important generalization by keeping them separate. Therefore, we will write a second version of **Front Vowel Backing**, {i, e} > {u, o} / V(C)(C)_ {u, ɬ} as a single rule ordered after **A-Weakening**:

Front Vowel Backing (II):

$$\begin{bmatrix}+\text{voc.}\\-\text{back}\\-\text{acc.}\end{bmatrix} > [+\text{back}] / _ \begin{bmatrix}+\text{son.}\\+\text{back}\end{bmatrix}$$

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19) There are several well-known exceptions to **A-Weakening**. These can be placed under the headings of Compositional Analogy (Recomposition), Paradigmatic Analogy, and most interestingly, Vowel Harmony[35]. Recomposition directly to the simplex is seen in the noun satis-factiō, and with open syllable the verb satis-faciō (but per-fectus), in-tactus (partially to prevent confusion with intectus), consacrāre (Res Gest. 2.28; but regularly consecrāre), ex-audiō (but inclūdō), etc. A standard example of paradigmatic analogy is Caesar, gen. Caesaris by analogy to the nom. (but frequent inscriptional Caeseris). There are several cases where an a in an initial syllable acts to prevent the change of an a in the following syllable. So: alacer; gen, anatis[36] (but pl. anitēs Plaut. Capt. 1003, Cic. de nat. deorum 2.124), calamitās, farfarus (but farferus Plaut. Poen. 478).

20) In addition to the rules ordered after (fed by) **A-Weakening**, it is clear that any rule which creates a new medial ā < ă will be before (bleed) **A-Weakening**. There are two good examples of the loss of z (**Compensatory Lengthening**, §4.3) lengthening a preceding ă:

*eks-ag-smen > *eksaksmen > *eksasmen (**CsC-Loss**, §4.4) >

*eksazmen (s > z) > exāmen (**CL**)

*con-tag-smen > *contaksmen > *contasmen > *contazmen > contāmen.

Leumann disbelieves that this compensatory lengthening preceded the weakening of a to e (apparently on the basis of hālāre but anhēlāre, see

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[35] See the prefatory remarks of Leumann 1977.79, 83-4 and litt., Buck 1933.102; also Sommer-Pfister 1977.89-90; Meillet-Vendryes 1968.122; Kent 1945.99.

[36] Usually only the gen. is cited. However, the nom. equally shows the effect of this preservation, as ă is weakened to e also in final closed syllables: *anti-stat-s > antistes. Not just in the sequence -ala- as stated by Godel 1961 = 1973.78. For the vocalism of the initial syllable, see Beekes 1969.197 and litt.

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note). He therefore gives the preform as *eks-āgsmen (1977.208)[37] with the ā long by Lachmann's Rule (at p.114). Whatever may underlie Lachmann's Rule[38], a purely phonological compensatory lengthening due to devoicing is phonetically quite unlikely and it is clear that morphological considerations have played a major role (see Kuryłowicz 1968.327-8). A preform *eks-āg-smen would be reasonable only if these formations were built directly to the perf. pass. participle (so āc-tus). However, it is clear that these are primary derivatives (though a limited class in PIE, see Brugmann 1906.243), built to the verbal root, e.g. *leuk-sm(e)n > lūmen, Av. raoxšna- (§4.4). Less certain examples are incānēscō which is too easily analogical to cānus (< *kas-no-, cf. O. casnar, L. cas-cus), and profānus to fānum (if from a *fas-nos, cf. *fēsnā: O. fíísnú). Thus we may order **CsC-Loss** and the subsequent **Compensatory Lengthening** before **A-Weakening**.

OPEN SYLLABLES: **O > E**

[37] Curiously in the section dealing with eksm > ēm. Why he holds that e can be lengthened in this position but a cannot is not clear from his remarks ad loc. However, on p. 208 Leumann derives hālāre from *an-sl- (a denominative to an unattested *an-slā > āla, so W.-H.) and anhēlāre by medial Weakening *ananslā- > *anensl- > an(h)ēl-. However, the root *an- that is supposed to underlie this is taken as the root of animus, etc. (so W.-H., E.-M., Pokorny 1959.38), which, as a set root, would have given *h_2enh_1-slo- > *anaslo- > **(h)anāl-. There appears to be no other putative cluster of Vns + voiced cons. in a medial position to see if perhaps -ansG- might not remain until after **A-Weakening** and **NS** and then a reapplication (or continued application) of **CL**, to the s now / V _ G; see §4.26. Whatever the etymology of anhēlāre it does not constitute a counterexample to ordering the change of **CsC-Loss** before **A-Weakening**.

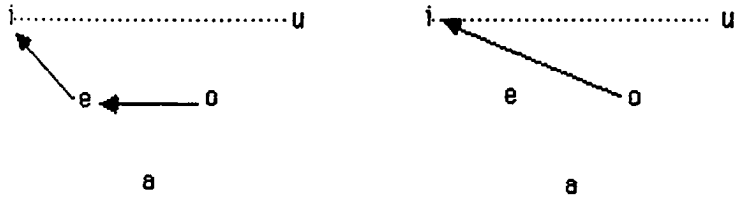
[38] "Regel" (so Saussure MLS 8.246f., Hermann 1923.208, Sommer-Pfister 1977.101, et al.) is better "Gesetz" (so Leumann 1977.114): it is far from exceptionless. See Leumann (loc. cit.) for the extremely rich bibliography; esp. Watkins HSCP 74 (1970) 55-65, and Kuryłowicz HSCP 72 (1967) 295-99.

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21) Chronologically the next stage in Medial Weakening was a change of o > e in open syllables only (O > E). This e then, together with original e, the e from a and from i or u in open syllables (§26) becomes i. Pisani and Godel (see §1) claim a direct change of o > i. Godel (1961 = 1973.79) argues:

> Le passage de o à i suppose apparemment une évolution o > e > i, qui ne laisse pas surprendre ... ce changement ne s'accorde pas avec l'ensemble; o > e ne peut être mis en parallèle avec a > e, puisque o et e ont le même degré d'aperture; et d'ailleurs, en syllabe fermée, a et e n'ont pas convergé (incestus : arbustus, industrius).

Godel's arguments seem strangely beside the point. There is no question of comparing the change of a > e in both open and closed syllables with the change of o > e in open syllables only, and none of the standard handbooks propose it. Godel seems to ignore the fact that a and o do in fact converge in open syllables. There is nothing improbable about a change of o to e, and it seems arbitrary to suppose that a movement in the vowel triangle from o to e (> i) is inherently less likely than a change from o directly to i:



In addition, there is good evidence for an e-midstage. Inscriptional attestation of the o > e stage exists in: APOLENEI (CIL 1^2.368, c. 200), ESQELINO (CIL 1^2.416, c. 220), HEC (from *hoc by proclitic weakening, CIL 1^2.9). An important case for the understanding of Medial Weakening is

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hospes, though its value as evidence is somewhat weakened by the possibility of analogy. We can begin with the preform *hosti-potis > *hos(t)petis (with o weakened to e) > hospets (with sync. of -tis blocking the raising of e > i) > hospes, but gen. *hosti-pot-es > *hos(t)petes (O > E) > hospitis[39] (so Buck 1933.101; Cowgill 1962). Here the syncope of -tis > -ts (only with original i, as the gen. shows) has "caught" the e-midstage, since it occurred after o > e, but before the final stage of e > i,[40] showing the same e-stage preserved before -CC*, as in the examples with a > e: *anti-stat-s > antistes, *arto-fak-s > artifex, and original e: *aui-spek-s > auspex, *prai-pet-s > praepes, etc. Two objections might be raised to this analysis. Perhaps the sequence was *hos(t)potis > *hospots and *-ots > -es(s) is a regular change in final position, as seems to be shown by *ekuot-s > eques. However, it is clear that o in closed syllables, medial or final, was preserved, and then was raised to u by a late change (§8.9): so *medi-tolliom > meditullium, 3. pl. -ont (freq. attested: CIL 1^2.9: CONSENTIONT, etc.) > -unt. Further, since -oC* shows no change until the late raising of o > u in closed syllables, where -VC* counts as a closed syllable (so nom. sg. -os, etc.), then -VCC* is a closed syllable a fortiore. In the cases of in-ops, prae-cox, con-cors, etc., the o is preserved and the final raising to u has been prevented by analogy to the simplex. The second objection is that the nom. of hospes might be analogical to the type *mīlets > mīless (Plaut. Aul. 528) > mīles (so Leumann 1977.94, Sommer-Pfister 1977.116), similar to the analogical spread of the -ex ending (*ious-dik-s > *iūdix --> iūdex, iūdicis by analogy to artifex, artificis, etc). This is clearly the case with

[39] Note that hospes cannot be a back formation from the oblique (for examples see §6.5, 9.9), since this would give *hospit-s > **hospis.
[40] For further syncope before the raising of e > i see §9.13, 16, below under the discussion of **R-Lowering**.

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*ekuot-s --> equos (instead of the expected *equots > **equus, bzw. ecus[41]), *com-i-t-s --> comes (not **comis), and ped-i-t-s --> pedes. However, a gen. in -itis does not necessarily produce a nom. in -es, so capitis/caput, and Leumann and Sommer-Pfister favor an analogical formation because they derive hospes from an immediate preform *hós(ti)pots, assuming (but not demonstrating) that the syncope of -tis must have preceded medial weakening. Since a will weaken to e in a closed syllable also (antistes), the only evidence to whether or not any weakening preceded -tis syncope will be provided exactly by cases of *-otis; forms with -e- retained (seges, etc.) show that the final stage of raising (e > i) in any case had not yet occurred. The military terms equos, pedes and comes are easily influenced by the close semantic connection with miles, but it is unnecessary to propose analogical remodeling for the more distant hospes, when it can simply reflect the lautgesetzlich outcome. As evidence for an e stage of weakening from o, it is frequently claimed that the e stage of weakening is preserved after i , as is original e (Buck 1933.101; Leumann 1977.90) so in thematic derivatives: *socio-tās > societās, *pio-tās > pietās, parallel to paries, parietis, etc. However, there is difficulty in using societās or any noun in -ietās as evidence for -ie- from -iio-(tās), since the -itās suffix is productive, and we would expect any original Italic -iiV̆- in an open syllable to become -Ī-. So in the voc. of -ius nouns: -iie > -Ī; in the fourth conjugation type made from i-stems (finiō, etc.): 1. pl. -iio-mos > -īmus, 2. pl. -iie-te > -īte. Leumann (1977.90,121-2) and Sommer-Pfister (1977.85 and litt.) note tIbIcen < *tIbio-can as a case where -io- seems to become -Ī-. However, the compound built to the noun tIbia should

[41] And a desire to avoid homophony in the nom. may have contributed to the change.

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rather have the preform *tībija-can- and shows the regular outcome of -ijV̆- in medial position. Pisani (1974.28) presents a different analysis, and holds that all vowels proceeded first to i and that the resulting -ii- was, if unstressed under the historical Latin accent, dissimilated to -ie-, a change for which, though not noted by Pisani, there is inscriptional evidence: ADIESE (SCB, CIL 1^2. 581) = ad-i-isse-(nt), though here presumably already accented adiéssent. Pisani (loc. cit.) explains societās as from *socíitās; however, by his own rules (ĪĬ or ĬĬ under the historical Latin accent becomes Ī) he is forced to make the nom. in most cases analogical to the oblique, so societātis. Pisani's rules will not work, however, for the case of pietās, since this dissimilation originally took place only in non-initial position; so Leumann 1977.51 (for initial position: *pí-issamos > pīssimus, with contraction presumably before the fixing of the historical accent). Therefore, *pí(i)o-tāt-s, if the vowels proceeded first to -i-, should have become a **piitās (> **pītās). The treatment of -ijV̆- > -ī- does not apply except in medial position (for -i(i)V̆- in initial position, cf. g. sg. ĝhijem-es > hiemis, and *liĝhen-es > lienis). Further on the phonological nature of this change, see §28. Rather, it seems the -e- of pietās represents the weakening product of o, preserved after -i-. However, pietās might be taken as an analogical creation on the pattern of socius : societās :: pius : X = pietās. Thus hospes and pietās provide evidence for the midstage of o > e, but that evidence is weakened by the possibility of analogy. We have a change in medial open syllables from:

Stage II	to	Stage III

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Medial closed syllables will still retain State II until the later raising of o > u. In addition, we can note that o > e is a phenomenon of open final syllables as well: 2. sg. impv. *sekᵘe-so > sequere. That is o > e in any open non-initial (unaccented) syllable.

22) Here we can raise the question: can **O > E** be related to the backing of ê > o / _ ɬ ? For cases of ouV > uuV see §7.42 f. If **O > E** occurred later than the **FVB**, the chain of development for original o would be *sē-dolos > *sēdelos (**O > E**) > *sēdolos (back to o by **FVB**) > sēdulus. There is nothing inherently impossible about a sound change undoing the effects of a preceding one[42], and this is how Goetze (1923.131) assumes the sequence went; e.g. ἐπιστολά --> *epistola > *epistela > *epistola > epistula. Indeed it could be argued that **O > E** must precede e > o / _ ɬ, since were the order reversed we would have *fīlie-lo-s > *fīlioɬos (**FVB**) > **fīlielus (**O > E**, with the e preserved after the i) or *adalēscō > *adelēscō (**A-Weakening**) > *adoɬēscō (**FVB**) > *adeɬēscō (**O > E**) > **adilēscō. However, this argument is invalid, since all these cases can be taken care of by the natural assumption that o did not weaken to e before a velar ɬ, i.e. an ordered pair of rules where e > o / _ {ɬ, u̯} and the subsequent **O > E** had a structural limitation whereby it did not apply before ɬ or u̯. It is reasonable that the law which had rounded e > o might also keep it from unrounding. Therefore it is impossible to state a direct relative chronological ordering between **O**

[42] The interaction of Grimm's Law, Verner's Law and the WGmc. secondary voicings, or Attic Rückverwandlung, for example; pace the objection of Godel (loc. cit. 79).

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> E and **FVB**. Nor are the inscriptions of much aid. The evidence for **O > E** places it around 220 - 200, but these inscriptions are not precisely dated by reference to a consulship or some similar event. Equally the inscriptions showing o before ł are dated to approximately the same time: POCOLOM (§30), HERCOLEI (CIL 1^2.607, 217 B.C.).

23) Since o weakens to e only in open syllables, what are the outcomes of the o-diphthongs, which before consonants (ouC, oiC) are in closed syllables and not subject to weakening? As has already been discussed, ou (original or from eu or au before a consonant) undergoes **Monophthongization** to ō and thence to ū. For medial -oi- there are no good examples (see above §7.20). The cases of illūdere, sēcūrus, impūne, and the like are too easily analogical to be good evidence, while it is far from clear that (re)lTquT contained an -oi-, and if it did it is possible that it became I by **First Element Unrounding** in any case (§7.20). The word pōmērium shows an aberrant development of oi.[43] In final position it is clear that -oi became -ei (so OINVORSEI, VIREI; SCB, 186 B.C.), which then monophthongized to ē (PLOIRUME CIL 1^2.9, c. 200)[44] and thence to I. Thus, it appears that -oi* acted like an open syllable, with o weakening to e.

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[43] A second example might be the much discussed illīus, if this continues an *illosio-s parallel to kuosio-s > cuius (§7.4, 28). A full discussion is beyond our purpose. In brief: Cowgill (1982) suggested as a possiblity the sequence *ill-osios > *illeiios > *illēios > illīus. Besides the problem of what ill- (archaic oll-) represents, it is not clear that the -īus genitives have an -osio --> -osio-s as their starting point (see Leumann 1977.477 and litt.). Further a weakening of -oii- to -eii- in a closed syllable seems unlikely since -oCC- regularly remains (perhaps then as an ad hoc rule, parallel to the treatment in Auslaut; an assimilation / ll _ ii ?). Even if so -eii- (at least from -aii-, §13), is retained: Pompeiiānus.

[44] The date is uncertain. However, it and the fact that the SCB shows one example of monophthongization medially, suggests that the -EI forms are examples of conservative orthography.

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Perhaps a similar change is to be seen in pōmērium from *pos(t)-moiriom 'a space behind the walls'. Kent (1945.104) maintains that oi > ei > ē > I was the regular development in medial and final syllables, with pōmērium as the only example for medial position. He explains the retention of the ē stage as the archaic spelling of a religious word[45]. Paulus-Festus offers posimirium, which is supposed to show I as the uninfluenced outcome (though the posi- is not explained). Rather, it is most likely to reflect the substandard pronunciation of a retained pomoerium.[46] In the non-literary language oe and ae fell together in a monophthong ē, so besides the frequent MSS confusion (caelum/coelum) there are numerous inscriptional examples from the first cent. A.D. of ē for literary oe and reverse spellings. So PHEBUS (CIL IV 1890, Pompeii; III.2370), PENA (for poena, VI 19596); see Leumann 1977.91. That this was the vulgar pronunciation is shown by It. penale, etc. The earliest inscriptional example of the word I find is POMERIUM (CIL VI 1231, 49 A.D.). Thus the pomoerium of Varro 1.1.5.143 (the earliest literary use) and others "sind nicht lebende Formen, sondern etymologische Spekulationen" (Sommer-Pfister 1977.87; first in Sommer 1914.103, so Buck 1933.87).

24) It is this type of reverse spelling that accounts for oboediō. This form has been much discussed. The usual explanation in the older literature (Schulze KZ 29 [1888] 251 = Kl. Schr. 346, Solmsen 1894.150 f., W.-H., Kent 1945.151, still in Sommer-Pfister 1977.110; see Szemerényi Gl. 38 [1960] 240-5 for a review of the earlier work) derives audiō from a *awizd(h)iō (comparing αἴω, Skt. āví-; the -d- is usually taken as the Italic reflex of the

[45] Or perhaps retained before r, as is claimed for ē; see §9.8.
[46] Compare the misuse of thou, thee, and thine in modern pseudo-archaic language.

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-θ- in Grk. αἰσθάνομαι) > *auzdiō (sync.) > audiō; then after *ob-: *obawizdiō > *obouizdiō (no sync. of the -i-) > *obūīdiō (with sync.of the o and compensatory lengthening) > oboediō. Not only the phonological changes (a series of ad hoc syncopes) here proposed but the preform itself (one would expect a PIE *-sdh- to become Plt. -sθ- > -st-) are problematic. Pisani (Gl. 36 [1958] 145 and 1974.17) derives it from *ob-boidiō, where the root -boid- is supposed to reflect the o-grade of PIE *bheidh (> fīdō, πείθομαι). However, oboediō is clearly a compound of audiō and it is unnecessary to go back further than an immediate preform of *obaudiō.[47] The problem is exactly here, in that one would expect *obaudio > *obeudiō > *oboudiō > **obūdiō. Szemerényi (loc. cit.), who takes audiō from *aus-dō 'give an ear to', proposed at the stage *oboudiō a dissimilation to *oboidiō (and its preservation in religious language), and compared the change of *ou to *oi in līber. So too Cowgill (1982). However, though līber is our only sure example for **Second Element Unrounding** (§7.40), by structural comparison with **First Element Unrounding** (§7.18), it seems clear that **SEU** is a dissimilatory unrounding between two labials, perhaps only in the environment / (#)ŧ _ [+labial]. One might propose a similar unrounding of u̯ > i̯ in the labial environment /obo _, but such a rule occurring before a dental seems overly ad hoc. Rather, the most likely explanation is that the expected *obūdiō is given a pseudo-archaic pronunciation of oboediō (so first Burger Gl. 19 [1941] 236; cf. Godel 1961 = 1973.77 n. 12; Leumann 1977.66). For other examples (cf. the Greek borrowing λάγυνος --> lagoena: Plaut. Curc. 78, etc.), see Leumann 1977.76. Further, cf. the reverse process

[47] None of the etymologies I have seen proposed seems particularly attractive.

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PLOURUMA (CIL 12.1861, c. 165?), COURAVERUNT (D 251, 394), where ū from oi (§7.14 ff.) is falsely spelled OU in initial position.

25) Again, there are analogical exceptions to **O > E**, primarily in the s-stems; so temporis, etc. by paradigmatic spread from the old nom. *tempos (for fulguris, etc. see §9.3). Note that here we have one of few times when something can be said about the history of an analogy: in this case the paradigmatic leveling in s-stems occurred before the raising of o to u in final syllables (seen already in the first Scipio epitaph: CIL 1^2.7, c. 200). For examples of retained o due to compositional analogy: ad-moneō, ar-rogō, commodus etc. Indeed, no compound with o-root vocalism shows **O > E** (see Leumann 1977.82; Buck 1933.102-3). There appear to be no examples of o by progressive assimilation in Classical Latin. However, for Vulgar Latin, cf. tonotru (Appendix Probi), MODORATUS (CIL VIII.10798). Later introduced o in open syllables becomes u; so in *tri-pod-ium > tripudium, repudiō (also to *pod-); Troiu-gena, Graiu-gena (but indi-gena; see Leumann 1977.83; Sommer-Pfister 1977.85).

I, U > E

26) The third change in Medial Weakening was the reduction of the high vowels i and u to e (or else to some neutral a, by typological comparison, though no descriptive advantage is gained thereby[48]). There is some inscriptional verification for this (TEMPESTATEBUS in the Lucius Scipio epitaph, CIL 1^2. 9), but the best evidence for this midstage is the interchange of u and i before labials, where original i appears as both i and

[48] See Leumann 1964.90, 1977.90. The reduction product acts no differently from e, and though the distance from u to e is shorter than to e there is nothing that indicates a over e. Indeed, the fact that the regular weakening product is i, while u is a conditioned reflex in labial environments can be explained mostly easily by the shorter phonological distance from e to i than to u.

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u, and original u appears as both i and u. So for i: vitiperō and vituperō, pontufex and pontifex, testimonium and testumonium; for u: Cornificius and Cornuficius, lacrima and lacruma, inclitus and inclutus. That o, and of course a, had already become ĕ$_2$ before the raising of the neutral e to i and u, is shown by the fact that they also share this behavior. So in thematic compounds: aurifex and aurufex, carnifex and carnufex, etc., and for a: rapere but ērupuit, capere but occupere (while occipere has been given the separate meaning of 'begin'). Analogy and paradigmatic leveling have had a major part to play, primarily in generalizing -i- as the reduction vowel[49], but such an alternation can only be reasonably explained by a midstage where both high vowels had collapsed to a single vowel e, which then is raised to i, or in labial environments[50], to u. This seems also the best way to explain the fact that u becomes i in capitis: *caputis > *capetis > caputis: unless u had first been reduced to the general weakening product e, it is difficult to see why it should not have stayed unaffected[51]. The only other possibility is to imagine a swap across the top of the vowel triangle with i --> u in labial

[49] See Leumann 1977.87-90, for a full review of the data.

[50] Principally before a labial, though a preceding labial or round vowel in the previous syllable seems to have aided the process. Conditions are impossible to formulate completely because of the heavy morphological reshuffling, see Leumann (loc. cit.). The rule, with this caveat, will be specified as taking place simply // [+labial].

[51] The arguments of Janson (1979.51-4) are especially confused. He assumes a reduction of the non-back vowels to "a sound spelled i rather than e. For this would tend to minimize the risk of confusion in perception. If the back vowels [o] and [u] remained intact, the front vowel which is least likely to be confused with them is [i]" (53). It is unclear what Janson imagines this vowel to have been phonetically, or how a spelling convention (if this is what he means) would have aided "perception" (presumably auditory).

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environments and i <-- u in nonlabial environments[52]. Graur (1929.40-74) holds that a > e > i in the usual process of Medial Weakening, and that the final product i becomes u before labials. However, this formulation fails to deal with changes of o > i or u. Further, by Graur's rules u should have remained unaffected, and all cases of original u showing up as i have to explained as analogical (stupeō but obstipuī), but no model presents itself. The usual examples adduced to show that u is unaffected by weakening are too easily analogical to be of any value: tutudī to tundō; coniugem to coniu(n)x, iungō; pecudis to pecus (all before non-labials). Leumann (1977.80), who agrees that the usual development of i is to u in open syllables before labials, offers the isolated arbutus as perhaps arguing for a lack of change in u. However, arbutus lacks an etymology, so it is impossible to say what this u represents. Further, even arbutum shows the form arbitum in Lucretius 5. 941 (E.-M. s.v.), and one suspects the influence of arbor and derivatives (arbuscula 'a little bush', arbustus).

[52] This is similar to the view, once fairly widely held, that the product of weakening was a Mittellaut [ü] in Classical Latin, though in fact a [ɨ] would seem to be phonetically more appropriate,; see Persson 1910.62, Kent 1945.46-7; still to be found in Janson 1979.54, Allen 1965.56-9, Sommer-Pfister 1977.82; Safarewicz (1960 = 1974.177-81) makes this the first stage in the weakening of all vowels. This idea is based primarily on a slightly corrupt passage of Quintilian (1.4.8) and the evidence against it has been throughly discussed by Sturtevant 1940.120f. and Leumann 1977.89-90. Godel (1961 = 1973.72-89) argues for a three stage development: 1) a > e, 2) e, o > i, 3) an indeterminate vowel [ü] for i and u. The first two stages are somewhat close to the position argued here. However, Godel disbelieves the evidence for o > e, and holds to the Mittellaut, which develops after Medial Weakening is over. Buck's conclusion (1933.101-2) is excellent: "It is more probable, though not the usual view, that we have to do with an ordinary u in the early maxumus (as obviously in the persistent occupō) and with an ordinary i in the later maximus, the alleged intermediate sound being imaginary, suggested by the fluctuation in spelling."

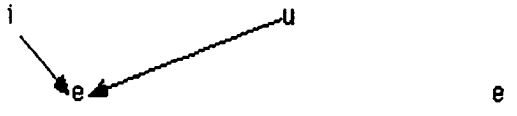
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27) A collection of forms drawn from various morphological categories shows that any vowel in an open syllable can appear in a surface form as i, or as u before a labial. This is only explicable by a common reduction state.

	i	u / _ [+labial]
a	cecinī abigere conficere	eruptī occupere
e	exigō obsidere demidius	monumentum documentum
i	cīvitās aditus	vitupere pontufex
o	hospitis cupiditas armifer	aurufex
u	capitis obstipuī lacrima corniger	cornubus obstupuī lacruma

Thus the next change was from:

Stage III to Stage IV



where in medial open syllables all short vowels have been reduced to a

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common e, and then the final stage, the raising of e > i and to u in a labial environment (subject to frequent analogical change to i):

Stage IV to Stage V



This change of i and u to a common reduction vowel e in open syllables appears also to occur in final open syllables (cf. -o > -e above). Here the data are greatly complicated by morphological influence, but in cases where a final -i is not lost (or perhaps is restored) it appears as -e. Good examples are ante (ἀντί), *mari (cf. pl. maria) > mare. No good examples for -u (cornu by paradigmatic leveling?). However, a, which in open and closed medial syllables becomes e (> i in open syllables), seems to remain unaltered in absolute Auslaut, where one might expect **-e (bzw. -i): ita (cf. Skt. íti < *ith_2), neut. pl. of consonant stems -a.

28) Thus we'll define two new rules **I, U > E** and **E > I/U**. For **I, U > E**:

$$[+\text{high}] > \begin{bmatrix}-\text{high}\\-\text{back}\end{bmatrix} / V(C) _ \begin{bmatrix}CV\\\#\end{bmatrix}$$

For the final stage, **E > I/U**, there are several points to note.

i) As a structural constraint, it does not raise e before o; so: moneō, aureus, etc. Naturally it is after (counter-fed by) the prehistoric loss of intervocalic j and after (bled by) the subsequent contraction: *moneiō > moneō (not **monīō), *moneiesi > monēs (not **monīs), etc.

ii) Chronologically, it is bled by the change of e > o before u and velar ł. Thus there are the classic pairs: Σικελία > Sicilia with regular e > i before

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the plain l, but Σικελός > *Sicoɬos > Siculus; familia (cf. O. famel), but famulus; further *ad-al-ēscēns > *adeɬēscēns (**A-Weakening**) > adolēscēns (**FVB**) > adulēscēns (o > u / ɬ), etc. So in the SCB, where the process of weakening is already complete[53] (FACILUMED, OINVORSEI) we find o still preserved before ɬ in open and closed syllables: CONSOLUERUNT, TABOLAM, OQUOLTOD[54]. The o thus created is raised to u, not of course by **E > I/U**, but by the comparatively late law of o > u / _ ɬ: o before ɬV is frequently attested: POCOLOM (CIL 1^2. 439-52; "before the second Punic War": Warmington 1953.74), AESCOLAPIO (CIL 1^2. 26-8), HERCOLEI CIL 1^2. 61, 62; see §30, 1.2). The earliest datable attestation of u before ɬV is the POPULO of CIL 1^2.586 (ad Tiburtes, 156). The change of o > u before ɬ is itself blocked dissimilatorily after i, e.g. filiolus, violentus, that is the vowels in -ioɬ- remain [+high] [-high, -low], rather than becoming -iul-, [+high] [+high]; see below.

iii) It is also bleed by various syncopes, that of -tis > -ts in *hosti-potis > *hospetis > hospes (§21), and that in *tabelela > tabella (§30). For further syncopes, see §9.9, 13-16.

iv) **E > I/U** does not occur after i. Thus original e (and perhaps from the weakening of o and a, but no certain examples, §8, 21) are retained: *ĝhiem-es > hiemis; pariēs, parietis; abiēs, abietis. Here the possibility should be raised that the change of medial -ii- to -ie- represents not a dissimilation but rather the weakening stage of ĭ̯ > e simply retained after i. This seems an attractive idea and will work for most of the examples found in the

[53] So CAPUTALEM shows not the pre-weakened form but rather recomposition to caput, cf. Leumann 1977.80.
[54] Ernout 1957.61 says simply, "L'ancien o est maintenu en syllabe intérieure" but this is so only before ɬ. Cf. *oino-u̯orsoi > oinuu̯orsei; see below. Note that OQUOLTOD with its false archaizing spelling, shows that **WO-Loss** had already occurred.

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handbooks, which are in open syllables: variegāre, liĕnis, pietās, etc. There appear to be no examples of original medial -iu-, which might become -ie- by medial weakening. If the ADIESENT = adiessent < *ad-i-isse-nt of the SCB represents the regular Latin development and not a dialectical peculiarity, then this dissimilation changes -ii- > -ie- (since there would be no weakening of i in a closed syllable) as well as preventing a raising of -ie- > -ii- and -io- to -iu-. Further, if as seems the case, hietāre and variegāre are from *hī-itare and *vari-igare, with productive suffixes created by medial weakening, then the change must be one of dissimilation and ordered after **E > I/U**. There is a similar dissimilatory change of -ee(i)-, so: MIEIS (CIL 1^2.15 and Plaut. Men. 202, MS. P), and in the paradigm of the pronoun is: EIEIS (abl. pl., CIL 1^2.566 ad Tiburtes, 156), EEIS (n. pl. CIL 1^2.581.5), dissimilated to IEI beside EIS (n. pl. beside abl. pl. CIL 1^2.202, Lex Cornelia, 81). So we can write the rule **E > I/U**: e not immediately preceded by an i in medial syllables before a consonant becomes i, or u before labials. In generative terms:

$$\begin{bmatrix}+\text{voc.}\\-\text{accent}\\-\text{high}\\-\text{grave}\end{bmatrix} > \begin{bmatrix}+\text{high}\\ \alpha\ \text{grave}\end{bmatrix} / \left\{\left[\begin{bmatrix}-\text{voc.}\\+\text{voc.}\\\begin{Bmatrix}-\text{high}\\-\text{grave}\end{Bmatrix}\end{bmatrix}\right\} __ \begin{bmatrix}-\text{voc.}\\ \alpha\ \text{grave}\end{bmatrix}$$

29) Again there are exceptions to the raising of e due to Recomposition, Paradigmatic Analogy and Vowel Harmony. Thus: com-edō, prō-tegō, re-secō, ēnecō (but ēnicō frequent in Plaut.) by recomposition; integer, by analogy to the oblique cases integrī, etc.; seges, segetis, teges, tegetis by progressive assimilation; so too the common inscriptional MERETO(D), even in examples which show completed Medial Weakening (CIL 1^2. 1531, Ded. Vertuleii, c. 150, beside MAXUME). The e of iuvenis (but regular hominis,

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etc.) was correctly explained by Brugmann (ALL 15 [1908] 7) as due to the influence of its semantic opposite senex.

30) About the proper formulation of the rule of anaptyxis we can say little with certainty. An anaptyctic vowel developed between medial -cl- and -bl- (< PIE *-tlo- and *-dhlo-, *-dhli-). We can of course note that the anaptyxis did not occur until after the change of PIE *dhl > fl at least (and almost certainly not until the stage -bl-), but this is an obvious conclusion. Since the change did not occur initially it is difficult to know the nature of the anaptyctic vowel. Anaptyxis seems to be a fairly late change in Latin, as shown by the regular Plautine form of nouns in -clom[55], while the diminutive suffix -colom < *-kelo-m with original non-epenthetic vowel, always shows two syllables. Therefore a cannot be the anaptyctic vowel since **A-Weakening** is long over, and if the somewhat ambiguous data indicate the anaptyxis was a phenomenon of Plautus' generation or only shortly before, then the idea of a single anaptyctic vowel can be ruled out as all the weakenings were over. However, there are examples of early forms in -cl- which show no epenthetic vowel, so: HERCLE (CIL 12.563), FERCLE (.564), as well as early anaptyxis, so the POCOLOM dedications (CIL 1^2. 439-52; "before the second Punic War": Warmington 1940.74), HERCOLE (CIL 1^2. 26-8, 30, 61-2), AISCOLAPIO (.26-8). The data both in the inscriptions and Plautus are greatly complicated by syncope (see Leumann 1977.102. Sommer-Pfister 1977.112-3). Illustrating the difficulties are CIL 1^2. 440: AISCLAPI POC<CO>COLOM showing forms both with and without anaptyxis (as well as dittography) and CIL 1^2. 62: with anaptyctic HERCOLE next to syncopated MERTO. For Plautus a single example might be Capt. 740:

[55] Beside less frequent forms in -colom, which appear more often in senarii. For examples, see Leumann 1977.102; de Groot 1921.

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periclum vitae meae tuo stat periculo. If the anaptyxis was of the Oscan type where the vowel quality is determined by the following vowel, the chain of development might be -V{k, b}lV$_1$- > -V{k, b}V$_1$lV$_1$-, e.g. *sth$_2$-dhli-s > *staflis > stabilis, *fakli-s > facilis, *pō-klo-m > pōcolom > pōculum. This seems to be Buck's assumption (1933.98): "an anaptyctic vowel develops regularly in the case of the groups cl and bl, where it takes on the quality of the following vowel". However, if the anaptyctic vowel were always e or o the outcome would be the same. So for example the development could be, arguing purely on relative chronological grounds, equally *pōklom > *pōkelom (e-Anapt.) > pōcolom (**FVB**) > pōculum, or *stablis > *stabolis (o-Anapt.) > *stabelis (**O > E**) > stabilis by regular Medial Weakening. Only i or u as the anaptyctic vowel before velar ł can be ruled out, since *pōklom > *pōkilom (i-Anapt.) would produce *pokiłom > poculom by **FVB** directly, with no room for the frequently attested midstage pocolom. Leumann's formulation (1977.102) seems sensible: "Der anaptyktische Vokal ist, wie man sieht, u (älter o) vor velarem ł und i vor li, entsprechend dem Vokal beim Vokalschwächung"[56]. That is, the anaptyctic vowel may not depend directly on the following vowel but on the quality of the l and so only indirectly on the following vowel. So perhaps -cł- > -col-, but -cl- (plain l, and so only possible before i) > -cil-. Arguing against this is the form Hercele (CIL 1^2.551, see §1.2), which shows anaptyxis in e. A consistent anaptyxis in e before both ł and l(i), or before ł alone and of i

[56] So Pisani 1974.26.

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before l(i) ordered before **FVB** will account for all the data[57]. However, since this inscription is from Praeneste it may show a dialectal anaptyxis of the Oscan type, though both CIL 1^2. 61 and 62 with HERCOLE are also Praenestine and seem to show the same form after rounding of e > o / _ ɬ. Arguing for an e-epenthetic vowel is the diminutive tabella. This must represent syncope from an immediate preform *tabele-la (compare *porke-le-los > porcellus, §2.19) and would seem to show the epenthetic vowel[58] was e. However, the form might be analogical to the ocellus : oculus type. There are no attested forms of the type *Aiscelapios, but this may be due to the scanty number of sufficiently early inscriptions[59]. In addition, the suffices -bolom, -bilis are never monosyllabic in Plautus or in any inscription, but the earliest inscriptional example I have been able to find is CONCILIABOLEIS (CIL 1^2.583.31, Lex Repet., c. 123). Could it be that the anaptyxis between -bl- (or of -i-?) was earlier than that of -cl-, or do we simply lack early enough attestation? However, if we suppose a consistent -e- for the anaptyxis, this seems to place the change of e > o / _ ɬ extremely late, after a change which is incomplete in Plautus.

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[57] So Kent (1945.148) who simply states, "the new vowel was e" without giving his reasons. Godel (1961 = 1973.79) believes that the epenthetic vowel was identical with its eventual outcome, that is i before l and u before ɬ: "Il est probable que la voyelle d'anaptyxe a toujours été une voyelle fermée [that is i and u]... La graphie POCOLO(M) (CIL 12.439-452) a chance de représenter simplement [pōculom]." There is nothing to recommend this. Why should a u, which has always been u and will always be u, be spelled o?

[58] The etymology of tabula is not fixed (certainly not < tla- to be compared with lātus, so W.-H.), but the PIE -dhl-ā suffix is very probable. Cf. U. tafla, where sync. is possible, but the Latin is still likely to represent anaptyxis. See Leumann 1977.103.

[59] The TABELAI next to TABOLAM of the SCB, and TABELARIOSQUE of the Popilius inscr. (CIL 1^2.638, c.132) are spelling the diminutive tabella.

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31) Medial Weakening before u̯ in an open syllable is usually treated separately in the the handbooks. The resulting vowel is always u̯. This u̯u̯ is usually spelled with the simple sign <u> in early texts, later the spelling VV is used in cases where misreading might result, so principally before antevocalic i̯ and after i̯[60]. There are examples for all vowels except original u̯ before u̯.

a̯	*dē̯-pau̯iō > dēpuviō
e̯	in u̯-stems, e.g. dat. -eu̯-ei̯ > (senāt)-uī
i̯	*tri-diu̯om > trīduum
o̯	*oino-u̯orsoi > OINVORSEI (SCB: oinuu̯orsei) --> ūniversus (with the generalization of -i̯- as the union vowel)

32) The only question is: what were the changes that led to the u̯? Does the ŭ vowel of -ŭu̯V- reflect the usual raising of e̯ to u̯ in a labial environment (E > I/U)[61] ? Leumann (1977.135) states, "Letzte gemeinsame

[60] Not just before i̯ (Kent 1945.101, Sommer-Pfister 1977.87). This is purely an orthographic convention. Thus (mon)-ui̯ spells monuu̯ī, etc., and there is no possibility of misreading as monvī. The signs VV were fairly consistently used in cases where ambiguity might result, begining in the 1. cent. B.C. Thus IVENTA (CIL 1^2.1603, c. 85, in bad hexameters where IVENTA scans ⏑⏑–•) which might be read as iven-, is written IVVENTVTIS (CIL VI 897, end 1. cent. B.C.), so too with IUABAT (misreadable as ivabat) vs. the usual (AD-)IVVABAT. See Sommer-Pfister (1977.130 with a good review of the inscriptional data) and Leumann 1977.135-6. Not with Goetze (1923.112, 122) and Safarewicz (Festsch. Cvikilinski, cited by Leumann ibid.), who misinterpret OINVORSEI as evidence of a syncope: as if *dēnuu̯ō > (*dēnu̯ō) > dēnuō.

[61] Seldom touched on in the handbooks. Buck (1933.102): "au̯ and ou̯ become u̯." Kent (1945.101): "pLat. cu̯ (also from eu̯), au̯, iu̯ before a vowel > uu̯." Niedermann (1953.23): "u̯ vor v" (examples -luō, dēnuō). Pisani (1962.26): "In sillaba aperta... qualsiasi vocale breve passa ad u̯ avanti u̯." Meillet-Vendryes (1968.113): "Devant w (noté u en latin), une voyelle brève intérieure en syllabe ouverte apparaît toujours comme u̯, qui sauf devant i̯ n'est pas noté indépendamment du w consonne qui suit."

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Vorstufe aller dieser -uo- [i.e. -u̯o-, as in dēnuō, comfluont, etc.] in nicht erster Silbe war vermutlich -ou̯o-." This is clearly the case for a, e and original o, which have become u by **FVB**. Thus we have:

*dēpeu̯iō > *dēpeu̯iō (**A-Weakening**) > *dēpou̯iō (**FVB**) > dēpuviō

*sentāteu̯ei > *senātou̯ei (**FVB**) > senātuī

*oinou̯orsos > oinuu̯orsos --> ūniversus

Here we can note again the fact that the first -o- of medial -ou̯o- does not syncopate (§7.44 f.) but remains and undergoes a raising to u. The case of trīduum is instructive. Do we have *tridiu̯om > *trideu̯om (by Medial Weakening of **I,U > E**) > *tridou̯om (**FVB**) > trīduu̯um (spelled <triduum>) or *tridiu̯om > trīduu̯um directly by **FVB**?

33) The principal point to note about the weakening before u̯V is that it is regular, as is the weakening before l̯. That is we do not have forms of the type **dēpiviō, **dēnivō, **monivī or **Sicilus, **adilēscō,[62] as we do for weakening before the other labials (obstipuī and obstupuī, etc.). This suggests what is an obvious conclusion: that the final stage of raising to u before u̯ is not due to **E > I/U**, but rather that **FVB** (which operates in closed as well as open medial syllables) has changed i and e (also from a) to u and o, and that this o is raised by a regular change of o > u / _ u̯ parallel to (i.e. part of) the rule of o > u / _ l̯. That is, **FVB** precedes, and so bleeds, the late raising of e to i and u, and creates new o, which remains unaffected by **E > I/U**, until changed (c. 145?, see §21) by a rule of raising o to u before the back consonants u̯ and l̯. Thus it is clear that we have with original e:

*moneu̯ai > *monou̯ī > monuu̯ī spelled monuī

and not:

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[62] The counterexamples have been discussed above (§17).

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*moneu̯ai > *monou̯ī > *moneu̯ī (**O > E**) > monuu̯ī (**E > I/U**, e raised to u in a labial environment)

It would be easy to write the rule of **E > I/U** to include a specification that e always becomes u before u̯, but is subject to replacement before the other labials. However, the behavior of the vowels before u̯ is exactly matched by the equally exceptionless behavior of the vowels before ł. We can capture that generalization in the weakening before u̯ by ordering **FVB**[63] with a subsequent rule that o > u / _ {u̯, ł}, except when preceded by i or o, a rule we can call by way of a mnemonic **OWL-Raising**. The o created by **FVB** was created subsequently to, or was not subject to the weakening of o to e (§21). Equally the u created by **FVB** (*tridiu̯om > trīduum) was subsequent to, or not affected by, **I, U > E**: that is, we do not have: *tridiu̯om > *triduu̯om (**FVB**) > **trideu̯om (**I, U > E**)[64]. Thus, we can state the developments of:

	eu > ou	**A-Weak.**	**(O > E)**	**FVB**	**OWL** o > u /_ {u̯,ł}
a:					
dēpau̯iō		dēpeu̯iō		pēpou̯iō	dēpuviō
adalēscēns		adeɬēscēns		adoɬēscēns	adulēscēns

[63] Maniet (1975.126) also makes the changes of short vowels before u̯ and ł together, but it is not clear what he is claiming them as a single rule rather than simply listing the changes in medial syllables: "Les voyelles brèves intérieures en syllabe ouverte et non précédées de i ou de e sont devenues : a) u devant w et (sauf pour l primitif) devant l vélaire...". As has been pointed out there are examples for the blocking of the changes in weakening due to a preceded e only for o > e, and for the weakening of a there are no solid examples. His exceptions to i > u / ł have been discussed above (§17).

[64] Though, one supposes that the proper form could be generated by *tridiu̯om > *trideu̯om (**I, U > E**) and then to *tridou̯om (**FVB**) > triduum. But this seems unlikely in the face of the regularity of a rule that backs i to u. Again, because the rule could carry the natural specification that u does not weaken to e before u, we cannot directly relate it to **FVB**.

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but with the alternation of i/u in a labial environment:

occapiō	occepiō	occipio/occupiō	
e:			
senāteuei	senātouei	(senātouei)	senātuī
dēneuōd	dēnouō	(denouō)	denuō
Sikelos		Sikołos	Siculus
fīlielos		fīliołos	fīliolus

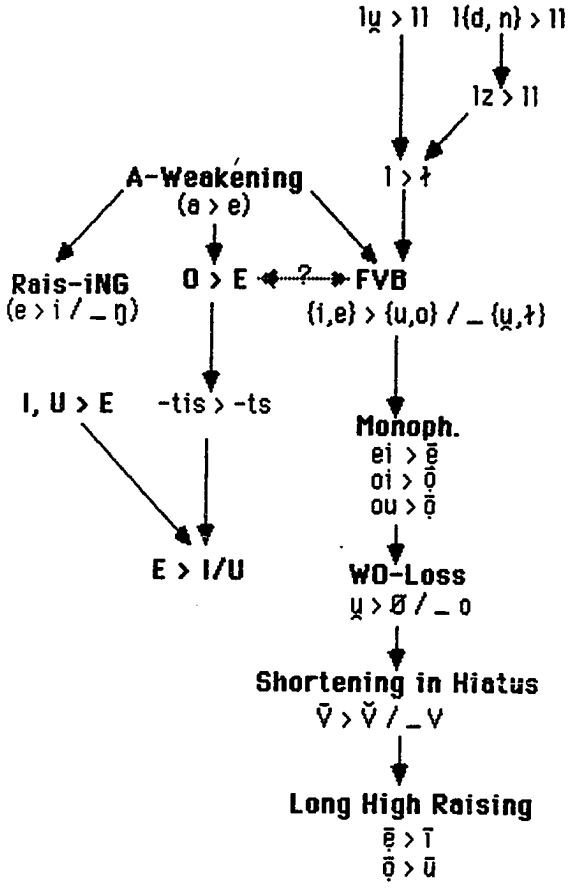
but with e raised to i/u:

monementom	monimentum/monumentum	
o:		
oinouorsoi	(oinouorsei)	OINVORSEI
sēdolos	(sēdołos)	sēdulus
but		
aurofak-s	aurefex	aurifex/aurufex
i		
tridiuom	triduuom	trīdu(u)um

We can sum up the findings to this point in:

CHART XVI

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Chapter 9: **R-Lowering**

1) The history of the weakening before eventual Latin r in open syllables is problematic, and its treatment has been reserved for this point. Many of the observations made on the process of Medial Weakening are a necessary preface for understanding weakening before r. In turn, the discussion of R-Lowering helps elucidate Medial Weakening and provides extra examples and confirmation of the midstages that have been proposed.

2) For the sequence -er- the communis opinio[1] gives three possible origins:

Rule 1: i, before r arising from intervocalic s, becomes e: is / _ V > iz > er, e.g. *si-sō > serō 'I sow' (so Leumann 1977.51, Buck 1933.79, Meillet-Vendryes 1968.111-2). This is said not to occur before original r (Kent 1945.85; by implication Buck 1933.86, Leumann 1977.51, etc.), cf. vir < *wiros (Kent loc. cit.). That is, though not explicitly stated by any of the handbooks (except Kent), the i was lowered before z, not r, and the order of changes was is / _ V > iz > ez > er. This rule is said to apply also medially (Buck, Leumann, locc. citt.), e.g. *kinis-em (cf. κόνις) > cinerem; or in various pf. forms, e.g. *(ēg)-is-ā-m > (ēg)-eram. Sommer-Pfister 1977.58 (repeating Sommer 1914.63 n. 3) doubt this change occurred in initial position, and offer as counterexamples virga, and viridis/vireō. The first (correctly with Walde-Hofmann 1954.798 < *wizga, cf. Olce. wisk 'whisk, broom; wisp, bundle of straw') is not a counterexample, since the lowering

[1] Represented principally by Leumann 1977 and Buck 1933. Also Sommer-Pfister 1977, Meillet-Vendryes 1968. Much less valuable, and in the main derivative, are Kent 1945, Niedermann 1953, Maniet 1975, Juret 1938, Pisani 1974, Monteil 1973, Kieckers 1930. Lindsay 1894 is still useful. I do not claim to be exhaustive in the bibliography on positions taken.

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is said to occur only in open syllables. Viridis lacks a cogent etymology; in any case not, as Sommer claims, to be compared to OHG wisa 'meadow', with which it has no compelling semantic connection.[2] Sommer-Pfister (loc. cit.) and Maniet (1975.128) are therefore forced to explain the vocalism of serō as analogical to (abstracted from)t hat of the compounds, ad-serō, cōn-serō, etc., but this is unnecessary as well as unlikely. The compounds of serō are far less frequent than the simplex, and no other verb of the shape *CV̆r- shows such a change, e.g. pariō does not have its vocalism influenced by peperī or com-, re- periō.

3) **Rule 2**: any short vowel becomes e medially before r arising from intervocalic s: V̆s / VC(C) _ V > er. Clear examples for the various vowels are:

a: dare, but reddere;

e: *gen-es-es > generis (i.e. e remains unaltered before r; so Leumann 1977.81, Buck 1933.100);

i: cinis, cineris; *Falis-inos (cf. Falis-cus) > *Falerinos > Falernus.[3]

o: No good examples. The numerous exceptions are due to paradigmatic leveling, thus tempus, temporis, with the o-grade of the nom. spread to the oblique (after the application of this rule, see §8.25), cf. the paradigmatically isolated adv. temperī, with the original e-grade (Leumann 1977.83, Buck 1933.191, Sommer-Pfister 1977.87-8).

u: There are several apparent exceptions to us > er, and Sommer-Pfister (1977.81)[4] doubt that it occurred at all. They offer as counter-examples

[2] The other Gmc. cognates, Olce vīsir 'shoot', etc., are closer, but connection with Lith. veisiù, veĩsti 'multiply' and veislẽ 'brood' < *u̯(e)is- (so Pokorny 1959.1133) ignores the phonological difficulty of i /_ rV.

[3] Here note that rhotacism and the lowering to e must be complete before the syncope of the third syllable (sync. 3/4).

[4] And apparently Kent 1945.102.

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satura, augurium, luxuria, lemures; and others can be found: fulguris, sulphuris, cicuris, gutturis[5]. However, these are either too easily analogical to a nominative in -ur, comparable to tempus, temporis (so satur, augur, cicur, guttur; cf. Leumann 1977.81), or else a u in the root syllable preserved by assimilation the timbre of the following vowel, cf. the assimilations of anatis, alacer, calamitas (§8.19); so fulgur, fulguris[6]; sulphur, sulphuris; luxuria, with additional pressure perhaps from luxus,-ūs. Rather we should look for forms more isolated and less subject to analogy. An excellent example is peiierāre forswear, swear falsely' (Plaut. Truc. 30) from *ped-ius-ā-se,[7] with the *ped- of *ped-ios > peius and the zero-grade of the stem seen in iūs, iūris < *iəu-os, *iəu-es-es (with syncope of

[5] These last three are of uncertain formation, and may be originally r-stems (Leumann 1977.83, 379; Benveniste 1935.37 and 39), and so belong under **Rule 3**.

[6] The development of fulgur is somewhat more complicated. If we begin with a regular neut. s-stem paradigm: *fulg-os, *fulg-es-es, we would expect phonologically fulgus (Fest.), fulgeris; the first step is that the o-grade of the nom. was extended here as in other s-stems, so *fulgos, *fulgoris parallel to *tempos, temporis. It is this analogical o that is assimilated to the u of the root, so fulgos , fulguris. Finally, the r of the oblique was transferred to the nom., parallel to the change in nouns in -ōs (honōs, honōris --> honor, honōris), so fulgur, fulguris. Traces of the old e-grade are to found in FVLGERATOR CIL VI.377, and of the regular outcome of medial weakening in the MSS fulgeris. See Leumann 1977.83.

[7] Not with Leumann 1977.546 from the comp. peius, and then later given a folk-etymology to iūs. This idea was first proposed by Brugmann 1901.396. If so, as Leumann himself points out, the meaning would have to be factative 'make worse', which to 'swear falsely' (the only attested meaning) is not an impossible change, nor, however, a likely one. Further, these denominatives are made by replacing the thematic vowel with the factitive suffix -eh$_2$- (Hitt. newahh-, Lat. novāre), and in the cases apparently from comparatives: ampliāre, breviāre, satiāre (so Leumann loc. cit.), the ā replaces the os of the comparative -ios. Thus, we would expect a *ped-iāse > **peiiāre, and not peiierāre. For other suggestions rightly rejected, see Walde-Hofmann 1954.274-5.

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ou̯V̆-, §7.44), cf. IOVESTOD of the Forum Inscription (CIL 1^2.1) and iūrāre < *iou̯es-ā-se, with the denominative formant -ā-, cf. IOVESAT of the Duenos Inscription (CIL 1^2.4); periūrāre is a later formation built to iūs, iūris. The evidence for us > er is unmistakable, and even Sommer-Pfister (loc. cit.) are force to state "doch peierō 'schwöre falsch' mit dēierō 'versichere' und ēierō 'schwöre ab' sind von iūrō bzw. einer schwächeren Ablautstufe *iusō kaum zu trennen"[8]. Equally good evidence is vetus, veteris with veternus, etc., which is (pace Sommer-Pfister loc. cit., Leumann 1977.374) a stem in -us-. Balto-Slavic shows a clear thematized form: Lith. vẽtušas, OCS vetŭxŭ < *u̯etus-o-[9]. We have therefore an exact parallel between

*Falis-inos > *Falerinos > Falernus and

*u̯etus-inos > *u̯eterinos > veternus.

4) **Rule 3**: any short vowel becomes e before original r in an open medial syllable: V̆ > e / _ rV. Examples are:

a: pariō, peperī

e: līber, līberī (§7.40-1)

i: lēge-rumpa (cf. lēg-is, lēgi-fer)

---

[8] See W.-H. for earlier suggestions. The connection with a zero-grade of iūrāre seems first to have been proposed by Curtius 1879.77.

[9] This adj. is "vielbehandelte" (Leumann 1977.374), but a methodological rigidity must be maintained in separating the nominal forms, neut. s-stems meaning 'year' from the adjectival forms, -us-stems meaning 'old'. Leumann (1977.269) notes that "Ein Wandel Subst. --> Adj. ist ganz selten und untypisch" and suggests that vetus is abstracted from vetustās. This is untenable, since Leumann cannot explain why there is a u in vetustās in the first place. A reintroduction of the nominative -o-, if it were an e-stem, is conceivable, but no other form shows this, and indeed this points to a -us-stem, since s-stems (as well as stems in -ios and -ōs) form -tās nouns and -tus adj. with the e-grade of the stem (tempes-tās; hones-tās, maies-tās).

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o: compounds in -fer < -foros (cf. o-grade in Grk comp. in -φόρος), e.g. *(armo)-foros > (armi)-fer, etc.[10]

u: gen. *sukur-ī (cf. ἑκυρός) > socerī.

Sommer-Pfister (1977.81) and Kent (1945.102) prefer to ascribe the e of the oblique cases of socer to analogy to the nom., and that nom. to analogy to other kinship terms, in particular gener 'son-in-law', whatever it in turn may be. For their doubts about u > e before secondary r < z, see above. This rule is said not to apply in initial syllables (at least for a, o; e would remain in any case): dare, arō, carō; ferō; forō (o-grade).

5) To recapitulate, we have three rules:

Rule 1: is / _ V > (ez) > er (initially and medially)

Rule 2: V̆s / _ V > (ez) > er (medially)

Rule 3: V̆r / _ V > er (medially)

Clearly some collapsing of these rules, if possible, would be both desirable and necessary. Generalizations are being missed and the principle of economy is being violated.

6) First, the medial treatment of **Rule 1** could easily go under **Rule 2**, whose structural description it meets, and of which it is a subset or particular case. Indeed, there is good reason to think so, since parallel to is > (ez) > er initially is a rule whereby us > (oz) > or initially (vs. the medial treatment > er). Examples are:

inf. *fu-se (cf. fu-ī, Osc. fusíd) > fore

---

[10] The type remained moderately productive as a deverbative, and on the basis of the type vendō : nugi-vendus :: ferō : X = *(armo)-feros > armifer could be created. This is clearly the case with (armi)-ger (< *ges-), etc. However, the exact cognates lūcifer (Varro RR 3.5.17) and λευκο-φόρος, bi-fer (ibid. 1.7.7) and δι-φόρος may indicate an inherited formation.

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*snusā[11] (cf. Skt. snuṣā, OCS snŭxa, νυός) > nora 'daughter-in-law'. Sommer-Pfister (1977.62), who doubt the lowering of i, naturally also doubt the lowering of u and offer nurus[12] as a counter-example, along with furvus (not in an open syllable in any case). However, nora is doubtless the lautgesetzlich form of the inherited thematic. Latin innovates by assimilating the word to the u-stem of socrus, -ūs 'mother-in-law'. This real -u- preserved the preceding u of *snus-us > nurus, -ūs (a type of vowel harmony operating in the opposite direction from the examples in §8.19, 25; Leumann 1977.51, Ernout-Meillet 1959.452; so first Sommer 1900.32, but see above §3)[13]. The o of nora, is explained as analogical to socer, soror, and not inherited, while fore is supposed to show a vowel weakening of u > o in unaccented, i.e. enclitic position, an arbitrary assumption which is apparently supported by no other example[14]. This is impossible, since not only does unaccented u before r become e, but according to Sommer-Pfister themselves, unaccented u should simply remain (see §8.26). That *sisō > serō and *fuse > fore are parallel (i.e. are a single rule) is pointed out by Buck (1933.79 and 80), Meillet-Vendryes (1968.111 and 112), Kent (1945.86), but not by Leumann (1977.50 and 51) who describes the rules sequentially but draws no overt attention to their structural similarity. We can set up, therefore, a **Rule 1'** which lowers high vowels before z in initial

[11] Contra Szemerényi 1964.318, who posits an original PIE u-stem.
[12] Giving as the preform *(s)nusus, with no explanation of why Latin shows a u-stem, when all the quoted cognates are thematic.
[13] Or Alan Nussbaum suggests perhaps after the remodeling of *snus-o- > *noros > norus (u-stem), there was a regressive assimilation of o ... u to u ... u (rather than a preservation of the original u vowel). This would be parallel to the assimilation of e ... i to i ... i seen in cinis.
[14] For verbs in enclitic position (no example for fore), see Szantyr 1972.404-5.

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syllables, and can allow cineris, (fu)-eram, etc. to proceed under the more general **Rule 2**.

7) Secondly, it is obvious that we can combine **Rules 2** and **3** (the medial treatments) in a single rule which applies after rhotacism, that is: a short vowel becomes e in a medial open syllable before r of whatever origin. This is what is implicitly assumed, judging from the use of both original r and r from z in their examples, at least in part by Sommer-Pfister (1977.64 for a and e[15]), Leumann (1977.81 for a, e, o[16]), Buck (1933.100 for a and e[17]), Niedermann (1953.29, "jeder kurze Vokal", but examples only for i, e, a[18]),

[15] For the other vowels: u is doubted, 83; i is assumed to become e only medially before r from z, 82; for o: "Vor einfacher Konsonant wird o wie a, e behandelt": 85; but examples before r are given only for secondary o, temporis, etc., 86. Secondary o normally becomes u, so repudiāre, tripudium, to *pod- 'foot', or Troiu-gena, Graiu-gena (with -o- as composition vowel after the Grk. compounds, vs. older comp. terrigena, indigena), so Leumann 1977.83, Sommer-Pfister 1977.86 (see §8.25). The latter claim that the o failed to become u because of the following r (i.e. apparently that r phonologically prevents raising: see § 4). However, this ignores the fact of paradigmatic leveling, which inserted the o into the oblique cases of s-stems in the first place and keeps it there, and to which the other examples cited were not subject.

[16] With i by **Rule 1** (*the medial component of* sisō > serō), 80, 51; u > e *in* socer, -erī: "Lautentwicklung vielleicht ur > or > er", though further parallel examples are not given and this lowering of u > o is not referred to that in fore.

[17] No example given for o; medial i is elsewhere (79) placed under the rubric of **Rule 1**: no examples for medial u.

[18] No mention is made of lowering in initial syllables (serō, fore). Further, Juret (1938.77), who operates with a highly idiosyncratic system, notes that a, e, i become e before r and "les groupes explosifs [apparently "syllable initial", though never defined] suivants br, pr, tr, gr, cr", a result he seems to think is different from general medial weakening in closed syllables, and offers no examples for i, which might demonstrate the point. Short o and u are said to be retained regularly. Maniet (1975.126): "Les voyelles brèves intérieures en syllable ouverte et non précédées de i ou de e sont devenues...e devant r, sauf sans doute o et peut-être u". Examples include r$_1$ and r < z.

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Kent (1945.99 for a[19]), and others but is stated explicitly by no one. Thus, we would have:

Rule 1': {iz, uz} / #C^0_3 _ V > {ez, oz} (initial; ordered before rhotacism)

Rule 2': $\breve{V}$ > e /VC^1_3 _ rV (medial; ordered after rhotacism).

8) Thirdly, I believe it can be argued that **Rule 1'** and **Rule 2'** are in fact a single rule that applied everywhere, both initially and medially, after both rhotacism and vowel lowering. That is, vowel weakening generates new -i- vowels, which feed "**R-Lowering**", a rule that lowers high vowels before r in any open syllable. This runs counter to the usual view that r prevented raising in the normal course of Medial Weakening. So Leumann (1977.81): "Vor r führt die Schwächung bei anderen [not i] Vokalen nicht zu i, sondern nur bis zu e" and Buck (1933.100): "The retention of e in peperī, contrasted with cecidī, as of orig. e in congerō, contrasted with adsideō, is due to the fact that r often tends to lower a vowel or, as here, to prevent its raising." This new rule, however, has several advantages. One, it captures the structural similarities of **Rule 1'** and **Rule 2'** to give a simpler **Rule 1''** = **R-Lowering**: [+ vocalic, + high] > [-high] / _ rV. Secondly, it explains neatly the anomaly that i becomes e and u becomes o in initial position, but any vowel (including u) becomes e in medial position, since in open syllables all vowels (including u) have become i, and then can be lowered to e before r. That i is the unconditioned reflex of u in medial open syllables, is shown by caput, gen. capitis[20] (§8.26). Other examples (abl. pl. of u-stems cornibus;

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[19] e retained, i by medial component of **Rule 1** (85). No mention of u.

[20] So Leumann 1977.80, Sommer-Pfister 1977.82, Meillet-Vendryes 1968.113, Maniet 1975.126, Niedermann 1953.26. Buck 1933.100f., Brugmann 1897.220f. have no mention of u in medial syllables. Kent (1945.100) mentions u only under the heading of the interchange of i and u before labials (for which see §17), and offers surrupuī (Plaut. Capt. 760) ≈ surripuī as an example.

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corniger, etc.) might be due to the generalization of -ibus, the spread of -i- as the union vowel, but capitis is quite secure. Cases of medial u in open syllables not in labial environments (see above) are due in the main to analogy, so: coniugem to coniu(n)x, iungō, etc.; tutudī to tundō; and so on (Leumann loc. cit.), or else vowel harmony (fulguris, etc; see §3 and n.6).

9) One reason, I believe, this view of the various changes in vowels preceding r has not been put forward before, is that **Rule 1** has always been formulated to specify r/z from intervocalic s. **R-Lowering**, on the other hand, predicts that initial i and u should become e and o before original r as well, to which there are several apparent counter-examples. The only one I have seen given, that is the only explicit argument why lowering in initial syllables should specify r < z only, is vir < *viros (so Kent 1945.85). But vir is clearly not a counter-example, since in its surface form the i, before a word final -r*, is not in an open syllable, and diachronically we need only order the lowering of high vowels / _ rV after the syncope of -rV̆-[21], which as we know from the Forum inscription (CIL 1².1: SAKROS, c. 500) but already in the Scipio Barbatus epitaph (CIL 1².7: VIR, c. 200) was fairly early in the history of Latin. In addition, since -rV̆s with original r is syncopated (*uiros > vir, *suekuros > socer[22], etc.), but -eros < *-V̆s-os is kept (numerus, umerus, etc.), this syncope occurs only with original r and so precedes (counter-feeds) the change of **Rhotacism** (z > r / V _ V), which, we are arguing, precedes (feeds) **R-Lowering**. Analogical to vir are its compounds and derivatives: virāgō, semivir, ūnivira, ēvirō, etc. While virtūs

[21] This syncope seems regular in final syllables, including monosyllables *tris > trs > terr (Plaut. Bacch. 1127) > ter, with merus, ferus, etc. presumably back-formations from the oblique. For syncope of -rV̆- in medial and initial syllables, see examples in §2.18, above. Cf. Leumann 1977.96-8, 142-4.

[22] For which, see §6.14, 7.5.

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may be analogical to vir, it is more likely to show that the syncope of rV̆ in medial syllables also bled **R-Lowering**: *viro-tūt-s > virtūs.

10) There are a number of additional exceptions to **R-Lowering**, which contain an *C(C)(C)irV-. I believe the list to be complete[23]. I do not propose to discuss them in any detail, but simply note that the majority lack any good etymology and refer the reader to the standard dictionaries. The possibility of rr > r (and so not originally an open syllable), by the Mamilla Gesetz[24] is a possibility in several cases. Of the others, many are clearly analogical, a number are loan words, and several are postclassical. None in any case is an adequate counterexample to **R-Lowering**.

bi-rēmis, -rotus: analogical

dir-(imō): by analogy to dis-C[25]

*hir 'hand' ? (not a counterexample in any case; indecl. or gen. hirris ?; probably a grammarian's creation; if a Latinized form of χείρ then presumably a long I: *heir > *hīr; see E.-M. s.v.)

hirūdō 'leech' < ?

hirundō 'swallow' < ?

pirus 'pear': "Mittelmeerlehnwort" (Leumann 1977.51; see W.-H., E.-M. s.v.)

QuirītēS, QuirīnuS < ? (Mamilla?)

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[23] Excepting foreign proper names and words called foreign in our texts. Words appearing in glossaries or only in prose authors where there can be no knowledge of the length of the i or y have been included for completeness.

[24] Simplification of geminates in a syllable before the historical Latin trisyllabic accent, and therefore a relatively late change; certainly later than these under discussion, which depend on the initial syllabic accent. See Excursus I, §3 for examples and Leumann 1977.184.

[25] A weak argument could be made on this basis in favor of **R-Lowering**: that *dis-V > *diz-V > dir-V, which is then kept by analogy to dis-C, is more likely than *dis-V > *diz-V > *dez-V (Rule 1) > *der-V, which is then changed to dir-, by analogy to dis-C.

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quirītō 'wail' < ? (Varro Ling. 6.68; onomatopoetic?; connected with quirritō 'grunt' Auct. Carm. Philom.?, Mamilla ?; see E.-M. s.v.)

quirquir: (Varro Ling. 7.8: meaning and reading uncertain; probably a pseudo-archaism for quisquis; not in an open syllable in any case)

sciropaectēs 'juggler with dice': Not. Tir.

siremps(e) 'similar' < ? (Ī/Ĭ ?; see E.-M., W.-H.)

sirus 'silo' < σιρός, σειρός (Ī/Ĭ ?; Colum. 1.6.15)

tri-rēmis: analogical

vireō/viridis < ? (see §2)

viriae 'armband': Celtic loanword (Pliny N.H. 33.40; see E.-M., W.-H.)

viriculum 'burin': Iberian loanword ? (Ī/Ĭ ?)

11) **R-Lowering** also predicts that there will be no ū in initial open syllables before r. Again, there are a number of words beginning *C(C)urV-, and again they are without etymology, loan-words, post-classical, or otherwise do not provide cogent counterexamples to **R-Lowering**.

furō, furiōsus[26], etc. < ?

muraena < μύραινα

muria 'pickle, brine', muriāticum 'pickled fish' < ?

muriola 'wine made from the lees' < muria ? (Varro frg. 551.25)

spurius: Etr. loanword? cf. spurcus

sur-(empsit): analogical to su(b)s-C[27]

surēna 'a type of fish' < ? (Varro Ling. 5.77: presumably a loanword)

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[26] W.-H. note the difficulties with deriving it from a *dhŭs- (cf. θυΐα, θυστάδες· Βάκχαι Hes.), while the other set of proposed cognates Olr. būrithir 'bellow (of animals, oxen)', būr 'angry', Latv. baũruôt 'bellow' show ū or au and are semantically distant, while the Slav. forms with ŭ are of very doubtful connection: OCS burja 'storm', Russ. burít' 'throw'.

[28] *sups-emō > supsmō (sync.) > susmō > suzmō > sūmō; sur-, then completely analogical to sus-, with rhotacism.

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surīō 'be in heat' < ? (ū/u̯ ?; Apul., Fest.)

suriscula 'kind of drinking glass' (Vitae Patr.; loanword; see W.-H.)

surus 'stake' < ? (ū ?; Enn. A. 525; back formation from surculum ?)

turiō 'shoot' (ū/u̯ ?; Apic. 8.1; Col. 12.50.5)

urium 'a type of earth found in mines': Iberian loanword ? (ū/u̯ ?; Pliny, N.H. 33.75)

12) We can set up a chart illustrating the ordering of the rules, with an example of the various vowels initially and medially. The rules are 1) Medial Weakening, 2) syncope of -ris/-ros > -rs, 3) **Rhotacism**, and 4) **R-Lowering**.

CHART XVII

	*sisō	*u̯iros	*kinisem	*fuse	*peiiusō	*genesem	*forom	*peparisam
1					*peiiisō	*genisem		*pepirisam
2		*u̯irs						
3	*sirō		*kinirem	*fure	*peiiirō	*genirem		*pepiriram
4	serō	vir	cinerem	fore	peiierō	generem	forum	pepereram

13) I can see only two problems with the ordering as it has been described so far. One, it incorrectly predicts the nom. of socer: *socuros > *sociros (MW) > *socirs (sync. and so no longer subject to **R-Lowering**) > **socir. Here it is easy to say that the nom. is analogic to the oblique socerī. It might be objected that in vir, virī the analogy has worked the other way around, but this is no real problem, as many examples can be found of paradigmatic leveling in two directions within a single morphological type[28]. There is, however, a chronological solution which

[28] E.g. the creation of two categories, with semantic specialization from *deiu̯os > *dēu̯os > deus --> new gen deī: gen. *deiu̯ī > *dēu̯ī > dīvī --> new nom. dīvus (§7.2-4).

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accords well with other established (or establishable) facts of Latin phonology and also provides a solution to the second form incorrectly predicted by the current form of the rules. They would give a nom. *armo-foros > *armi-firos (MW) > *armifirs (sync. and so no longer subject to **R-Lowering**) > **armifir. In this case an analogical solution is possible, but the answer lies in the midstages of Medial Weakening, which we have been treating here for the sake of convenience, as a unitary phenomenon. It is clear that Medial Weakening had changed o only to e when rV̆ syncope closed the final syllable and so blocked the raising of **E > I/U**, the final phase of Medial Weakening. That is, the sequence was: *armo-foros > *armeferos (**O > E**) > *armefers (rV̆ sync.) > armifer (**E > I/U**). That rV̆ sync. occurred before the final phase of weakening is shown not just by armifer, but also by forms with original -e-: so *dexiteros > dexter, *pueros > puer, etc.

14) Let us now expand the Medial Weakening rule of Chart XVII, to include the stages determined in the previous discussion. We can set up the order:

CHART XVIII

1) **A-Weakening**: a > e (all syllables)

2) **O > E** (open syllables only)

3) rV̆ sync. (armifer), -tis sync. (*hospotis > *hopetis > hospes)

4) **E > I/U** (open syllables)

after which we still order

5) **Rhotacism**

6) **R-Lowering**

15) It is to this same rV̆ syncope overtaking a midstage of weakening, that we can refer socer. Just as o had become e and been overtaken by rV̆ syncope in *armoforos > *armeferos > armifer, and by -tis sync. in *hostipotis > *hospetis > hospes, so in *sokuros the u had weakened to e:

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*sokeros (**I/U > E**) and then rV̆ syncope intervened, as in the previous cases, blocking the raising of e > i. As we know from puer, etc. rV̆ sync. preceded the final raising of **E > I/U**. So to Chart XVIII we can add the **I/U > E** stage of Medial Weakening, which we will here place after **O > E** (but see §8.22). **I/U > E** is before the rV̆ and -tis syncopes, which are themselves before the last stage of **E > I/U**:

1) **A-Weakening**: a > e (all syllables)

2) **O > E** (open syllables only)

3) **I, U > E**

4) rV̆ sync. (armifer), -tis sync. (*hospotis > *hopetis > hospes)

5) **E > I/U** (open syllables)

6) **Rhotacism**

7) **R-Lowering**

We can run a few of the forms under discussion through this sound mill, to show the sequence of development:

Chart XIX

	caputes	socuros	armoforos	uiros	kinises	peiiusō	fuse	peparisam
1)								peperisam
2)			armeferos					
3)	capetes	soceros			kineses	peiiesō		peperesam
4)		socers	armefers	uirs				
5)	capitis		armifer		kinisis	peiiisō		pepisisam
6)					kiniris	peiiirō	fure	pepiriram
7)	capitis	socer	armifer	vir	cineris	peiierō	fore	pepereram

16) We would like an additional example of syncope preserving e < u or i. No other words with the shape *-urV̆s, *-irV̆s[29], are known to me. The

[29] In *uiros the i was initial and so not subject to weakening.

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situation in which there is only one example illustrating a series of changes, because of the sequence of phonemes necessary to establish an ordering, would not be an unfamiliar one in relative chronology (see Cowgill 1978.33).[30] However, there is, I believe, one further example of e < u, preserved by syncope. Plautus shows ancipes (Rud. 1156) and praecipes (Rud. 671), the earlier forms of the nom. anceps and praeceps, both earlier and later forms with gen. -cipitis.[31] According to Leumann (1977.94), "ist das u von -caput-s zu e geschwächt. Ebenso e aus a in antistes." This is, however, unlikely since while a undergoes **A-Weakening** in all noninitial syllables, so *anti-stat-s > antistes, u in closed and final syllables remains unchanged: manus (and all u-stems), adductus, and caput itself. On purely phonological grounds then one might set up a *prae-caput-is, which would then undergo Medial Weakening (**A-Weakening** and **I, U** > **E**) to *praecepetis

[30] There is only a single example each of **FEU** and **SEU**.

[31] Also bicipes: Prisc. 2.280.16. The nominatives anceps, praeceps are not attested until Cicero. They are clearly later analogical formations but their origin is obscure. The only explanation is that offered by Leumann (1977.451). In some examples third decl. nouns (principally in -is and -ēs) had certain cases of the oblique formed directly from the nom. So: sanguis (itself analogic; gen. sanguin-is) is resegmented as new stem sangu- and the acc. sangu-em (Act. Arv.) is formed, or hērēs (gen. hērēdis) is recut as hēr- and a new hērem (Naev.) is created. It is also clear that the same thing could occur in the small class of nouns in -ĕs. So inscriptional dat. Superstī (NSc 1920.40 nr. 39) or the abl. praecipe (Enn.). If from such an oblique a new nom. were back formed, we would have praeceps, on the model principe : praecipe :: princeps : X = praeceps. The problem with this view is that it requires a two step analogical change, for the first step of which (praecipĕs --> new stem praecip-) there is no exact proportional model, since the class of nouns in -ĕs is so small. This is not a great stumbling block, since it is clear that it did occur at least occasionally. However, this explanation also requires that two analogical changes both of which are intermittent have completely replaced the nom. in the standard literary language. Further, analogy from the abl. is odd, and the inherited gen. prae-cipitis remains undisturbed. Despite these difficulties, I can see no better explanation.

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and then syncope to *praecepets > praecipes, parallel to *hostipotis > *hospetis > hospes. Here again the possibility of analogy might be raised in creating a nom. -es to a gen. -itis, as is the case with *equots --> eques, *com-i-t-s --> comes, *ped-i-t-s --> pedes after *mīlet-s > mīles (see §7.21). However, this analogical nominative fails to be created precisely for caput, capitis. There are in addition good morphological grounds for believing that this was indeed the case. In this type of bahuvrīhi, composed of a prepositon, a negative or numeral and a noun, Latin shows an -is as the compound suffix (Leumann 1977.346, 397). The -is does not substitute only for thematics (though that is its original locus), but is used to form compound adjectives from u-stems (bicornis), n-stems (cognōminis, exsanguis) and, most interestingly from our point of view, from root nouns (concors, concordis). The fact that Latin regularly shows i-stem forms: neut. pl. -ia, g. pl. -ium, abl. sg. -ī) for ancipes, praecipes and concors is indicative that they are, in fact i-stems. Leumann (1977.438) disputes this, noting that beside the genetic i-stems (cōnsors to sors, sortis) similarly inflected are "ursprünglichen Kons.-Stämme mit nomin. nur -s (etwa audāx fēlīx ferōx, vehement-), auch die Komposita mit Kons.-Stamm oder Wurzelnomen als Schlußglied, so an- und praecipit-ī -ia -ium (trotz capit-e -a -um), abl. inopī trotz ope (aber inopum Laber. wie opum), gen. plur. (subst.) locuplētium Cic. rep. 3, 16, simplicium." There is considerable mixing of forms in the adjectives[32] and the original situation is hard to determine; however, Leumann's own examples show that the members of this class are various and the compounds with root nouns should be considered separately. The i-stem forms for other C-stems are irregularly attested (Ernout 1953.57: reg. quadru-pedum, inopum). For the -īx, -āx group

[32] And in nouns, so: OPID (CIL 1^2.2364).

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and the coumpounds of -plex no satisfactory explanation has been given[33]. The simplest explanation for the i-stem forms of ancipes, praecipes and concors is that are in fact i-stems, built with a suffixal -is and Caecilius (109) seems to confirm this with a nom. concordis[34]. There seems to me no other reasonable explanation for the nom. -s. Analogy to the other compound adjectives with an -s would produce ops : in-ops :: caput : X, X = **prae-ciput. Ernout-Meillet's statement (1959.98) that ancipes is "refait sur le génitif" seems less likely since the simplex caput, capitis is not remade to **capes. It is more likely that ancipes and praecipes resemble hospes and superstes not because they are analogic to these forms, but because they share their phonology. Therefore, ancipes and praecipes are supporting evidence for the lowering of u > e and subsequent syncope, which explains *sokuros > *sokeros (i, ŭ > E) > socer.

17) So far, we have been arguing for the placement of **R-Lowering** after **Rhotacism** and Medial Weakening on grounds of descriptive simplicity and accuracy. Besides these relative chronological arguments are there any absolute chronological data to support this ordering? Firstly, we know that **Rhotacism** itself was a fairly late phenomenon in Latin. Cicero in a well-known passage (ad fam. 9.21.2) relates that Papīrius Crassus (dictator 340) was the first to spell his name with an r. Pomponius (dig. 1.2.2.36) says that Appius Claudius (cens. 312) "first used" (invenit) the letter r to spell the names of the Valerii and the Furii. And for what it is worth Martianus

[33] But a base in a heavy syllable, as in the nouns may influence the form (so locuplētum rare beside locuplētium). The participles may belong under the heavy bases rule, though there is a possibility that they were remade in Latin as i-stems. See Nussbaum 1973.207-15.

[34] The form is, however, too easily an analogical creation for metrical reasons on the model (gen.) fortis : (nom.) fortis :: concordis : X, X= concordis to be probative.

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Capella (3.261) preserves a tradition that Appius Claudius "hated" (detestatur) the letter z.[35] This gives us a terminus ante quem of the late fourth century. The earliest inscription showing rhotacism is the epitaph of L. Scipio (CIL 1^2.9, after 260, prob. c. 200). We have a terminus post quem in the IOVESAT of the Duenos inscription (CIL 1^2.2), probably also of the fourth cent. (again, however, the word division is uncertain; see Gordon 1983.77 f.). In addition there are the numerous words preserved by Varro (Ling. 7.27) and Festus with intervocalic s, presumably spelling z.

18) Most importantly, we have the divine name Numisios preserved in two inscriptions (CIL 1^2.32, 33):

32) Ioniius Q. f. | Numisio Martio | donom dedit | meretod.

33) [Nu]misio Mar[tio] | M. Terebonio C. l. | donum dat liben[s] | meritod.

While the NUMASIOS of the Praenestine fibula (CIL 1^2.3) on which the preform *Numasios used to rest is forged, the etymology which connects the name with Numa is very likely[36]. These two inscriptions, on pedestal bases found in the Tiber at Rome, are undated but the language is close to that of the earlier Scipio epitaphs, and c. 200 would be a reasonable estimation. While, like most early Latin inscriptions, they are more likely to be archaizing than archaic, they nevertheless show a > e > i completed[37] before

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[35] "quod dentes mortui, dum exprimitur, imitatur."

[36] W.-H. 2.187, E.-M. 451, s.v. numerus; see Schulze 1904.164, 197. Unfortunately Oscan has syncopated the vowel, which therefore in any case must have been short, in gen. Niumsieís (see von Planta 1892.590, Buck 1904.58). O. Niumeriis is a Latin borrowing.

[37] The medial e of the frequent mereto(d) is due to the preceding e; cf. the similar vowel harmony in anatis, alacer, calamitās, vegetus, segetis, tegetis; see §8.29. Cf. Leumann 1977.100, Buck 1933.102.

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Rhotacism and **R-Lowering** have occurred. The s presumably spells z, and while it might be an archaizing substitution for a current r, to substitute i for a currently pronounced e is the opposite of what we expect and actually observe. Numisios appears then to be confirmation of the form and ordering of **R-Lowering**.

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Excursus II: mollis and the Latin adjectives in -ui-

1) The most commonly accepted etymology of mollis 'soft' derives it from a preform *molduis. This is taken to be an i-extension to the zero-grade u-stem seen in Skt. mṛd-ú-, Hsch. βλαδύς, etc. (Leumann 1977.58, 215; Ernout-Meillet 1959.411; Buck 1933.104, 135; Brugmann 1897.322). It turns out, however, for reasons that will be discussed shortly, that this etymology and the traditional view of the Latin -ui- adjectives on which it depends need to be re-examined.

2) The first question raised by a protoform *molduis is whether this root vocalism and morphological structure can be expected in a Latin adjective in -ui-. The usual view is that Latin inherited the PIE u-stem adjectives with zero-grade root and then added an i-formant as an "extension."[1] The locus for this -i- is commonly taken as the derived feminine in -ih$_2$, e.g. Skt. mṛdvī (Buck 1933.209, Leumann 1977.346, Semerényi 1969.990). This theory of the origin of the i-extension was first criticized by Specht (KZ 65 [1938] 202-3) on the grounds that in the secure cases where Latin continues ī < *ih$_2$, primarily feminines to -tor agent nouns, we find an extension in -k. Thus Latin genetrīx next to Skt. jánitrī < *ĝenh$_1$-tr-ih$_2$. However, there is an obvious fallacy to this argument since a pattern of noun in -tōr : noun in -trī(k) is not comparable to an adj. in -u- : adj. in -uī-. Szemerényi (1976.402) offers three counter-examples: the nouns neptis, pēlvis and the adj. fertilis, where a supposed -ī- stem has been transferred to the i-stems.

[1] N.B. although in Skt. the u-stem adj. regularly build their fem. with accented, ablauting -ī- (<*-ih$_2$-, gen. *-ieh$_2$-s), the zero-grade of the root, though by far the most common, is not without exception. So vás-u-, fem. vásvī (< *ués-u-, cf. OHG Wisu-rīh, and the ā-stem noun Olr. feb 'quality', dat. sg. feib < *uesuā). See Wackernagel-Debrunner 1905.II.2.413.

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For the non-ablauting suffix (the Skt. vṛkī́, gen. vṛkíyas type < *-ih$_2$, *-ih$_2$-es) at least, Lat. neptis (cf. Skt. naptī́-s, gen naptíyas) seems a good example of transfer to an i-stem. So too for pēlvis, skt. pālavī 'a type of container'. One can compare the similar shift of ū-stem to u-stem in *suekrū- (Skt. śvaśrū-) to socrus. However, for the adj. fertilis, there is nothing to show an i-stem and the formation is clearly analogical to the type dūcō: ductilis.[2] For adjectives, in the broadest sense, a trace of the old ī is to be found in Ennius (Ann. 1.30 = Ernout 1959.148): Quos homines quondam Laurentis terra recepit, where the ī of Laurentis must scan long, since -s does not make position in thesis in Ennius. See Nussbaum 1973.207-15, for full discussion; also Skutch 1985 ad loc. We have a remnant of the original fem. participle in -nt-ih$_2$, with transfer to the short i-stem declension and subsequent syncope (-entis > -ents > -ēns). This -ih$_2$ is used to make derived feminines to u-stems in Greek πλατεῖα, Skt. pṛth(i)vī, yAv. pərəθwī, Goth. (acc.) hardja, and Lith. saldi, saldžõs. There is no reason to deny a formation -u-ih$_2$- to the parent language.

3) However, this simplest-seeming view, of an i-extension (of whatever origin) added directly to the inherited PIE u-stem adjective, involves the necessity of explaining all the -ui- forms in Latin as continuing a PIE root zero-grade. This as we will see involves certain serious phonological difficulties. There is, however, another possibility which accords better with the general Latin pattern of adjectival formation. Comparison of Greek λειϝος > λεῖος with Latin lēvis (see §7.8) shows that a Latin -ui- can correspond to to a thematic -uo- elsewhere. This pattern answers well to the general Latin substitution of i-stems for o-stems, especially following

[2] Not with Szemerényi (loc. cit.) and Leumann 1977.347 to an unattested derived feminine agent noun *bher-trí.

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a resonant (Leumann 1977.346, Buck 1933.135, Kent 1946.18, Lindsay 1894.338). So similis (Grk. ὁμαλός), humilis (Grk. χθαμαλός), etc. So already Brugmann (1906.214):

> Ob die lat. tenuis, suāvis, gravis, und dgl. ursprünglich nur F[emina] sind und als Femininformen ... zu beurteilen wären, ist zweifelhaft. gravis d. i. *gʷr̥ə-u̯- vergleicht sich mit gr. τανα-[ϝ]ό-ς, ai. pr̥thi-ví (s.o.), es verhält sich also nebst tenuis zu vielleicht so, wie similis zu ὁμαλός.

4) In addition many u̯-stems have thematic by-forms in -u̯o-: so *tn̥h₂-u̯i- > Ved. tan-ú-, Grk. τανυ- in compounds, but *t(e)nh₂-u̯o-[3] > τανα*ϝ*ός (Myc. proper name) > τανα*ό*ς; man-ú- > μάνυ · μικρόν Hsch., Arm. manr, gen. manu (Pokorny 1959.728), beside *man-u̯ó-s > μαν*ϝ*ός > μανός, beside an e-grade *men-u̯o-s > Olr. menb (Brugmann 1906.200); *sten-ú-, Grk. Στενύ-κληρος (Chantraine 1933.122), beside *sten-u̯ó-s > Ion. στεινός, Att. στενός; *h₁lnguʰ-ú- > Skt. laghú-, Grk. ἐλαχύς (see below), beside *h₁lenguʰ-u̯o-s > Lith. leñgvas (Pokorny op. cit. 660). For further examples, see Brugmann loc. cit. and Schindler 1976.351 for a general discussion of -i- for thematic -o-.

5) So then, if u̯-stems can have thematic by-forms in -u̯o- in PIE, and if -u̯i- can be the Latin correspondent of a -u̯o- adjective elsewhere, the expected vocalism of a Latin adjective in -u̯i- might be judged just as well on the basis of what is shown by the -u̯o- formations as what is shown by the u̯-formations. Since neither -u̯i- nor -u̯o- are necessarily derived directly from u̯-stem adjectives, we cannot insist that they must continue an inherited zero-grade root.

6) When we take a look at what vocalism adjectives in -ŭ- and -u̯o- do show in PIE, we see that the root can appear in full, zero, or even o-grade, with suffixes -u̯-, -u̯o-, and -eu̯o-. Examples are as follows.

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[3] For Cowgill's view of the the development with a full-grade, see §1.3 n. 4.

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A) zero-grade root, accented -ú- ablauting with -éu-:

(clearly originally proterodynamic, but already in the the proto-language the zero-grade of the root had been generalized and the accent "kolumnaliziert." See Rix 1977.149.)

*$h_1$lnguʰ-ú-s > ἐλαχύς

*u̯idh-ú-s > Skt. vidh-ú-s 'lonely'

*man-ú- > μάνυ

B) zero-grade root, thematic -u̯o-:

*s|H-u̯o- > Lat. *salau̯os > salvus (see §2.8).

*man-u̯o- > μανϝός > Ion. μανός, Att. μανός

*u̯rHdh-u̯ó- > Skt. ūrdhvá-[4]

C) zero-grade root, -éu̯o-:

*u̯idh-eu̯e-h_2 > Skt. vidhávā, Goth. widuwo, Lat. vidua, etc.[5]

D) e-grade, thematic -u̯ó-:

*men-u̯ó- > Olr. menb

*ken-u̯ó- > Ion. κεινός, Att. κενός

*$h_1$lenguʰ-u̯ó- > Lith. leñgvas

*sten-u̯ó- > Ion. στεινός, Att. στενός

E) e-grade, -eu̯ó-:

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[4] Note that since there is no simple u-stem attested for this root, ūrdhvá- may be a primary -u̯o- adj. only and not a thematic by-form.

[5] The evidence for the existence of this category is slim. Note first that the forms cited are all nominal 'widow'. However, because of Skt. vidh-ú-s, I am inclined to believe that they reflect a substantivized adjective and not a primary noun formation. Lat. shows the medial weakening before u̯: *u̯idu̯a > u̯iduu̯a, spelled vidua. Olr. fedb, W. gweddw might be taken as showing a *u̯idh-u̯ā (Thurneysen 1946.46) but rather in the face of the unanimity of the PIE data represents a very early Celtic syncope (Cowgill 1978; see further Ringe, forthcoming). It is questionable whether Grk. ἡίθεος belongs here. The initial ἡ- is unexplained and the meaning 'youth' does not agree with its supposed cognates.

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*ken-ey̯ó- > κενεός

*sten-ey̯ó- > στενεός[6]

F) e-grade, -u̯-:

*sten-u̯- > Στενυ- in proper names

G) o-grade, -y̯ó-:

*sol(H)-y̯ó- (see §2.7) > Lat. sollus, soll-ers, Av. haurva-, Skt. sárva-, Ion. οὖλος, Att. ὅλος

*y̯or(H)dh-y̯ó- > ὀρθός, Dor. βορθο-[7]

7) There are in addition parallels outside of Latin for -y̯i- beside -u̯-. Cf. SKt. ghṛ̥svi- 'exuberant' beside ghṛ̥su- (Specht KZ 65 [1938] 201, Leumann 1977.346), Skt. mṛ̥d-ú- beside Arm. mełk < *meldy̯i- (to which we shall return); Hitt. parku- 'high' beside parkui-; and for -y̯i- beside -y̯o- cf. OHG maro 'mellow', gen. murwi, OE mearo 'soft, weak' < *mery̯o-, but Oir. meirb 'weak, flabby', W. merw 'insipid' < mery̯i- (see Lewis-Pedersen 1961.13, Brugmann 1906.203).

8) An additional fact should be noted. Sievers Law does not apply to these -y̯o- and -y̯i- adjectives after heavy syllables: thus Arm. mełk < *meldy̯i-, *y̯ṛ̥Hdh-y̯ó- > Skt. ūrdhvá-, *$h_1$leng^{u}h-y̯ó- > Lith. leñgvas. The failure of these -CC- clusters to condition the long Sievers Law variant -u̯u̯- argues against their creation at a very early date in PIE.[8]

10) Given this variety in the IE languages, what is the situation in Latin? For the -ui- by-forms, all the unambiguous cases point to an e-grade of the root. In the less clear examples a good case can be made for an e-grade,

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[6] Perhaps also ἑτεός beside ἐτυ-μό-ς if the latter is evidence for an original u-stem.

[7] See Beekes 1969.241. Also an example of de Saussure's law, see §2.8 n. 7.

[8] For *y̯ṛ̥Hdh-y̯ó- > Skt. ūrdhvá-, the sequence RHC may not have been subject to Sievers Law. See Schindler Sprache 23 (1977) 60-1.

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when there are any cognates to base an etymology on. The data are as follows.

10) Brevis < *mreĝh-u̯i-, with the structure seen in *men-u̯ó- > Olr. menb and *ken-u̯ó- > κε(ι)νός. The assumption of a zero-grade root to conform with that of the PIE u̯-stem adjectives is, as we have seen, unnecessary and creates grave phonological problems for the derivation. A *mr̥ĝh-u̯i- would have given **morχu̯is > **morvis? One possibility for a source of the -e- seen in brevis and other -ui- adjectives, if they did continue a zero-grade of the root, might be analogical influence from the comparative, where the e-grade is expected, i.e. inherited zero-grade *mr̥ĝh-u̯-is --> *mreĝh-u̯i-s by analogy to *mreĝh-i̯ōs[9]. This is the solution proposed by Cowgill (1970.126-7; 1982; see § 7.37). However, an analogy in such a direction would run counter to the overwhelming trend in Latin, where the comparative is routinely built to the positive. It is difficult to point to any case where the positive has been remodeled to what the comparative would have theoretically looked like.[10] This is not impossible, but to invoke analogical influence from the comparative to give an e-grade to every member of this morphological class of positives and only members of this morphological class of positives is much less likely than the simple assumption of an original e-grade, which accords much better with both the phonology of Latin and the morphology seen elsewhere.

11) *tenh$_2$-u̯i-s (assuming no Siever's Law) > *tenau̯is > tenuis, with the structure seen in *ken-u̯o-s > κε(ι)νός, or tenh$_2$-eu̯i-s with the structure

[9] Note, however, that this form of the comparative is not attested. Grk. shows the zero-grade βραχίων.

[10] Note, too, that this proposed analogy would most plausibly have operated before the Proto-Italic change of *r̥ > -or-, as *morχu̯is (where the m is no longer in contact with the r) --> *mreχu̯is is a much less likely analogical reshuffling.

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seen in *ken-ɥó-s > κενεός. A zero-grade *tn̥h$_2$-ɥi-s would give *tanaɥis > **tanuis (cf. *s|Hɥos > *salaɥos > salvus). If tenuis were to be derived from a *tn̥h$_2$-ɥi-s > *tanaɥis which is then analogically altered by the comparative to tenuis, the objections already made above to mr̥ĝhui- would still stand in the way.

12) levis also seems to show a root vocalism in -e-, but the situation is more complicated. The reflexes in the various languages show three formations.

a) Lith., and Gmc. point to a preform with full-grade of the root and an -n-:

*h$_1$léngʷh o-s > Lith. leñgvas.

*h$_1$léngʷh-to-s > PGmc. *liŋxtaz (with raising and lengthening, §4.18) > Goth. leihts, OE lēoht

b) Grk. and Skt. point to a root with no -n-:

e-grade in *h$_1$legʷh-r̥-io- > ἐλεφαίρομαι[11],

zero-grade in Skt. r̥h-ánt-[12]

c) The Slavic and Celtic forms point to a root without an -n-:

OCS lĭgŭ-kŭ,

Olr. laigu from the comparative *lagiiu 'smaller' < *lagiōs

However, the root vowel Slavic -i-, Celtic -a- is that of the regular reflex of n̥ > Sl. -in-, Celt. -an- (> an before C in Olr.) but with the nasal part of the reflex somehow lost. It is this absence of an n in the reflexes that makes one suspect an originally n-less root *h$_1$legʷh-, with a nasal infix present,

[11] For the formation, cf. its Oppositionsbildung μεγαίρω. The connection of this form with the root of ἐλαχύς has not been noted before, and I owe the observation to Alan Nussbaum. For previous speculations see Beekes 1969.41.

[12] see Mayrhoffer s.v.

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which is found in the thematic present Av. ranjaiti, Skt. rāṅhati, Olr. lingid[13], pointing to a full-grade *h$_1$l-né-g^{u}h-, zero-grade *h$_1$l-n̥-g^{u}h-. This zero-grade nasal-infix might then form the locus for the forms with -n- *h$_1$lng^{u}h- > ἐλαχύς[14], Skt. raghu-, laghu-, and a new full-grade *h$_1$l-n̥-g^{u}h --> *h$_1$leng^{u}h- > Lith. leñgvas, etc. This stem forms a complete Caland system: u-stem in ἐλαχύς, -ī̆ōn comparative in ἐλάττων[15], -istos superlative in ἐλάχιστος, -ró- adj. in *h$_1$lng^{u}h-ró-s > ἐλαφρός 'rapid'.[16] The Latin could represent the same type of development seen in the Slavic or Celtic reflexes: a zero-grade *h$_1$lng^{u}h-ui- > *lenxuis > *lexuis > levis (with the n lost at least before the raising of e > i / _ ŋ; the regular outcome would be *linguis). However, an original e-grade of an n-less root (no forms other than levis from this root appear in Latin), i.e. *h$_1$leg^{u}h-ui- > *lexuis > levis seems likely on both phonological and morphological grounds. In any case, there is no need to invoke analogy from the comparative.

[13] With raising of e > i / _ ŋ: *h$_1$l-n̥-g^{u}h- > *leng- > ling-. See Thurneysen 1946.49.

[14] ἐλαχύς could also be from *h$_1$lg^{u}h-ú-. Note, however, that the similar Caland formations of λανθάνομαι (n. 16, below) lack the -n- of the infixed stem.

[15] The ā is secondary, see Schwyzer 1934.I.522B.

[16] For a similar Caland system built to a nasal infix stem, cf. λανθάνομαι, aor. ἔλαθον, compositional -i- in λαθικηδής, -ro- adj. in adv. λάθρᾳ. See Risch 1974.65-8.

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13) gravis presents similar problems. On the basis of the cases above and comparison with Grk. βαρύς, Skt. gurú- < *g^{u}r̥h_2-ú-[17], Av. gouru-, Goth. kaúrus, we would expect a preform for Latin of *g^{u}reh_2-ui-s[18] which should develop to **grāvis with long ā rather than the ă actually shown. A zero-grade *g^{u}r̥h_2-ui-s would give the same outcome. In short, the *grău- that characterizes the form must itself ultimately originate from a structure of the shape *g^{u}reh_2-u-C. The question is why a preconsonantal form of the u-stem is generalized while *mreĝh-ui-s > brevis shows the expectable prevocalic form.

14) There are parallels in Skt. for words of the shape *g^{u}r(e)h_2-u-. Cf. Skt. *plth$_2$-ú- (πλατύς) > pr̥thú (fem. also), and the derived fem. *plth$_2$-u-ih$_2$ > pr̥thivī (cf. Πλάταια) beside the new fem adj. pr̥thvī, built to the masc. pr̥thú-s (the ante-consonantal form of the stem). So for gravis, cf. the Skt. fem. gurvī built to the masc. : guru-ī- > gurvī.[19] In particular cf. the root *ĝerh$_2$- (Grk. γέρων, γῆρας) whose u-stem *ĝr̥h$_2$-ú- > Skt. jiru- (γραῦς, perhaps). To this ante-consonantal form is added an -i- in jirvi- (AV) but

[17] Szemerényi (1976.402-4) argues against a laryngeal in this root on the basis of the short ă of gravis and Skt. gurvī. However, fem. gurvī shows the laryngeal in the development of r̥ > -ur- and is clearly analogical to the masc. gurú- < *g^{u}r̥h_2-u-. The Grk. is inexplicable except with a laryngeal as **g^{u}r-ú-s > **βρύς (though a Lindeman's Law variant in *g^{u}r̥rus is thinkable) and as the r̥ could not become syllabic anywhere in the paradigm: gen. **g^{u}r-éu-s, etc. Contra Beekes (1969.190-1) who invokes a schwa secundum. Note that a *g^{u}r̥uis would give **gorvis in Latin.

[18] For the CREH-shape cf. Skt. grāvan- 'stone (for pressing soma)'.

[19] Here I reach exactly the opposite conclusion from Szemerényi 1976.402-3. The fact that "this [gurvī] is the feminine formation one would expect from a u-stem adjective anyway" at least to a masc. guru-, argues rather for an analogical than inherited outcome. The same holds true for pr̥thvī. It is pr̥thivī which stands outside the paradigm and so continues the lautgesetzlich outcome.

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metathesized to jívri- 'old' (RV). See Wackernagel-Debrunner (1905/1957) II.463.

15) Suāvis, from an immediate preform sṷāduṷis (dṷ > ṷ / V _ V). The original shape of this root is difficult to determine. However, it is clear that not even the original u-stem adj. in PIE contained a zero-grade (Skt. svādús, Grk. ἡδύς, etc.), and there is no reason to assume one for Latin. The root is usually taken as having built an originally proterodynamic u-stem, the consistent long ā due to the extension of the nom. acc. full-grade throughout the paradigm (Rix 1977.147, 149) or to an anomalous phonological development (Beekes 1969.287, a "reduced grade"):

nom. *sṷéh$_2$d-u-s > Skt. svādús, Grk. ἡδύς (with accent shift)

gen. *sṷ(e)h$_2$d-éu-s > Skt. svādós, Grk ἡδέος < *-éu-os.

However, it is difficult to see how a zero-grade *sṷh$_2$d-éu-s could vocalize as anything other than *sūd-(éu-s)[20], for which there is evidence in Skt. sūd-áya-ti (a causative, where a zero-grade is odd, but the o-grade impossible) and Lith. súduti 'season'. In places where a zero-grade is expected, e.g. the -tó- participle, Skt. shows an analogical full-grade svād-tá-. But there are also forms which show a short -ă-: Skt. svad-a-ti 'make tasty' (RV), Hom. ἑαδότα < *ἑαδϝότα < *he-hṷăd-ṷót-a, pf. part. We seem to have two competing ablaut patterns, that of a set root and a root with a consistent ă, as exemplified by the Skt. forms:

extended- or o-grade	full-grade	zero-grade

[20] So for all CuHC- roots; see Cowgill 1965.149 following Kuryłowicz 1956.195-206). There seem to be no cases of -uh$_1$- or -uh$_3$- in PIE, but for a possibility that -uh$_2$- vocalized in Greek as -ṷă-, cf. δϝαρός (Cowgill 1982, personal communication). However, this could rather reflect a full-grade (for full-grade -ró- adj., cf. *pleh$_1$-ró-, *meh$_1$-ró-, etc.). For auslautend -u-h$_2$, cf. perhaps Hom. δοῦρα < *δορϝα < *dóruh$_2$ (or stem *doru- plus the productive ending -a).

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svād-	svăd-	*sŭd-
svād-	svād-	sūd-

The Latin might represent an e-grade *sueh$_2$d-ui-s. In any case it does not reflect a zero-grade.

16) Pinguis: no acceptable etymology. The various parallels cited are too distant to be of aid. See Ernout-Meillet and Walde-Hofmann, s.v. The quality of the i of the root cannot be determined.

17) Dulcis: the traditional comparison with γλυκύς is possible, given an unparalleled metathesis. Grk. γλυκύς itself has no other cognates and a supposed *δλυκύς involves an ad hoc change of *dl- to *gl-. A Lat. *dlukuis would become **luquis or **dulquis. Either seems phonotactically irreproachable. What has become of the -u- is only part of the difficulty.

18) In summary, PIE admitted a variety of root-grades for derived adjectives in -(e)uo-. The Latin -ui- adjectives in all the clear cases show an e-grade: brevis, levis, tenuis, and an e-grade is a possibility for the other -ui- adjectives which admit of an etymology at all: gravis, suāvis, where a zero-grade in any case seems impossible.

19) After this long but necessary prologue, we can return to the original question of the etymology of mollis. On the basis of what has been determined, if were going to use the root *mld- it should be in the e-grade: *meld-ui-s. In addition, the closest cognate, Arm. meɫk < *meldui- reflects not only the e-grade but also the lack of Siever's Law. A *meld-ui-s therefore seems the only plausible Latin formation of this kind from a root *m(e)ld-, and the traditional *mld-ui-s > *muldvis is unlikely. Ernout-Meillet (1959.411) rightly note the difference in vocalism between the Arm. and the supposed Latin preform.

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20) Yet the pre-form is usually given as *mldʰis > *moldʰis in part because a *meldʰis involves phonological difficulties. To derive mollis from *meldʰis we must do two things: change e to o and -ldʰ- to -ll-, and these two goals seem incompatible. To change e > o we need to invoke the [+back] of a velar ł. But as the examples *peldō > pellō, *(per)celdō > percellō make clear, the assimilations to -ll- preceded the change of e > o / _ ł (see §2.5). We should expect a *meld-ʰi-s to become *melluis > **melvis, parallel to *pērsnā (Skt. pārṣṇā) > *perzna > perrna > perna (§3.25, 5.2), or with reduction (by assimilation ?) of -llʰ- to -ll- a **mellis. Cf. Cowgill 1975.41-41 for parallel phonological difficulties in the derivation of helvus (for which see §3.8 f.). An original o-grade of the root in a form *mold-ʰi-s could give mollis on purely phonological grounds, if in addition we assume that -llʰ- > -ll-. However, there is no comparative evidence for an o-grade here, while there is excellent evidence for the e-grade in Arm. melk, and its assumption merely to explain mollis would be arbitrary. Therefore the preform *meld-ʰi-s must be abandoned and with it any connection to the root *m(e)ld-.

21) We must accordingly turn to another set of proposed cognates which point to a root *meldh-: Skt. mardh-ati 'reduce, lessen', Grk. μαλθ-ακός 'soft', μάλθη 'a soft mixture of wax and pitch used in caulking ships', μάλθων, -ωνος 'weakling', Goth. mildeis, Olce. mildr < *mildjaz < *meldh-io-. The Celtic and Balto-Slavic forms are, of course, not useful for deciding between a plain or aspirated voiced stop. The connection between the two roots *meld- and *meldh-, is a separate problem.[21] The Grk. μάλθη,

[21] As extensions in -d- and -dh- to *(s)mel- 'mill, rub down' as in molō; so W.-H. s.v. Cf. the relation of μαλακός < *melh$_1$- and μαλθ-ακός < *mldh-, and others.

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a substantivized adjective, points to a thematic *mldh-ó-s, as does μάλθων, built like στράβων to στραβός 'squinting', with the -ων, -ωνος suffix which frequently denotes physical characteristics (Buck 1933.321, Chantraine 1933.161). The Gmc. reflexes are e-grade thematics. The zero-grade of the Greek cognates will give the desired -o- seen in the Latin root. With the familiar assumption of an i-stem substituting for a thematic we can suppose a preform *mldh-i-s > Plt. molθis with the usual development of the spirants > *molδis > *moldis > mollis, with assimilation of ld > ll.

22) The only possible objection that I can see to this derivation is that there is no second example of *-dh- after -l- to prove that it behaved like regular internal *-dh- and became d and assimilated. While it seems likely that Plt. -lθ- would become -ld- > -ll-, it might instead theoretically undergo the change to -f- seen before an -l-: e.g. *-dhlom > *-flom > *-βlom > -bulum. So *sth$_2$-dhlom > stabulum, *sth$_2$-dhlis > stabilis, etc. It is clear that *dh becomes f both before and after r. Standard examples are:

/ r _ : *ṷerdhom (Goth. waúrd < *ṷṛdhom) > verbum

/ _ r : *ghladhros (OHG glat, OCS gladŭkŭ) > glaber[22]

The parallelism cannot be shown for u, however. The only example of -dhu- unfortunately has a consonantal u: *londhuos (cf. OHG lentI 'kidneys' < *londhu-In-, zero-grade in OHG lunda 'tallow') > lumbus (with the irregular raising of o > u / _ mb, cf. umbilicus, §6.23), where one could imagine the sequence *londhuos > *loneuos > *lonβuos > londuos (with the unconditioned reflex) > lumbus (with postconsonantal du > b, behaving like initial du) as well as dh > θ > f > β > b. Given the odd distribution of the change of dh > f: / r _

[22] The usually cited *reudh-ró-s > rūber is not good evidence since the preceding u could also have conditioned this change.

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/ _ r

/ _ l

/ u̯ _

these cases may not constitute a single rule, and r, l and u̯ do not form a natural class. There is therefore no reason to suppose that -ldh- became anything other than -ll-.

23) In conclusion: we are forced to reject a pre-form *mld-u̯i-s on morphological grounds and *meld-u̯i-s on phonological grounds, and with them the standard etymology of mollis from a u-stem to the root *meld-. A second set of cognates indicate a root *meldh-. A zero-grade *mldh-, with an i-stem substituting for the thematic formation attested elsewhere accounts for Latin mollis.

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